PATH INTEGRALS
I

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Clase 1

The Action Principle in Quantum Mechanics

More than 80 years ago Dirac [1]-[2] put forward the foundational ideas of what we call today the *path-integral approach* to Quantum Mechanics, Quantum Field Theory, String Theory, ... His motivation was to understand the rôle of the action principle in Quantum Mechanics. Some 15 years later, Feynman [3] elaborated Dirac ideas for non-relativistic quantum systems thus making Dirac proposal operative.

Usually, the action principle is not discussed in Quantum Mechanics courses. One just learns there time-independent variational principles but time-dependent ones are less studied.

In contrast, one does learn in Classical Mechanics courses both variational methods and employs them according to the situation. On the one hand, for static solutions, one stationarizes the Hamiltonian (energy) as a function of p and q, as can be recognized from Hamilton equations

$$\frac{\partial H}{\partial p} = \dot{q} = 0 \ , \qquad \frac{\partial H}{\partial q} - \dot{p} = 0 \tag{1.1}$$

That is, static solutions correspond to

$$\frac{\partial H}{\partial p} = \frac{\partial H}{\partial q} = 0 \tag{1.2}$$

The full, time-dependent equations of motion (the Euler-Lagrange equations) require instead stationarizing the action S as a functional of $q(t)$,

$$S = \int_{t_1}^{t_2} dt L(q, \dot{q}; t) \tag{1.3}$$

$$\delta S = 0 \quad \rightarrow \quad \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}}\right) = \frac{\partial L}{\partial q} \tag{1.4}$$

Note that static variations do not require boundary conditions while time dependent ones do (in general variations $\delta q(t)$ must vanish at t_1 and t_2).

Both static and time-dependent variational principles have their quantum analogues. The former translates into the requirement that expectation values of the Hamiltonian should be stationary. This leads to the time-independent Schrödinger equation. In brief, one minimizes $\hat{H}$

$$\langle\psi|H|\psi\rangle = \int dV \psi^* \hat{H} \psi \tag{1.5}$$

with the constraint

$$\langle\psi|\psi\rangle = \int dV \psi^* \psi = 1 \tag{1.6}$$

The answer is that the time independent Schrödinger equation

$$\hat{H}|\psi\rangle = E|\psi\rangle \tag{1.7}$$

should hold. Concerning the time-dependent variational principle, the proposal of Dirac starts precisely by considering the functional

$$S = \int dt \langle\psi; t| \left(i\hbar \frac{d}{dt} - \hat{H}\right) |\psi; t\rangle \tag{1.8}$$

and then demanding $\delta S = 0$. This yields to

$$i\hbar \frac{d|\psi; t\rangle}{dt} = \hat{H}|\psi; t\rangle \tag{1.9}$$

One recognizes in S the scent of the action (in the sense that the classical action is $\int dt(p\dot{q} - H)$).

A second observation of Dirac is to consider the wave function of a quantum mechanical system (with classical analogue) in the Schrödinger representation,

$$\psi(q, t) = \langle q|\psi; t\rangle_S \tag{1.10}$$

This wave function can be written in the form

$$\psi(q,t) = A(q,t)\exp\left(\frac{i}{\hbar}\Phi(q,t)\right) \tag{1.11}$$

where the amplitude A and phase Φ are real functions which we demand to be slowly varying with q and t. Since ψ obeys Schrödinger equation (1.9),

$$i\hbar\frac{\partial\psi(q,t)}{\partial t} = \hat{H}\psi(q,t) \tag{1.12}$$

one has

$$i\hbar\frac{\partial A(q,t)}{\partial t} - A(q,t)\frac{\partial\Phi(q,t)}{\partial t} = \exp(-\frac{i}{\hbar}\Phi)\hat{H}A\exp(\frac{i}{\hbar}\Phi) \tag{1.13}$$

Now, $\hat{H} = \hat{H}(\hat{p},\hat{q})$ and

$$\begin{aligned} \exp(-\frac{i}{\hbar}\Phi)\hat{q}\exp(\frac{i}{\hbar}\Phi) &= \hat{q} \\ \exp(-\frac{i}{\hbar}\Phi)\hat{p}\exp(\frac{i}{\hbar}\Phi) &= \hat{p} + \frac{\partial\Phi}{\partial q} \end{aligned} \tag{1.14}$$

so that

$$\exp(-\frac{i}{\hbar}\Phi)\hat{H}(\hat{p},\hat{q})\exp(\frac{i}{\hbar}\Phi) = H(\hat{p} + \frac{\partial\Phi}{\partial q},\hat{q}) \tag{1.15}$$

or

$$i\hbar\frac{\partial A(q,t)}{\partial t} - A(q,t)\frac{\partial\Phi(q,t)}{\partial t} = H(\hat{p} + \frac{\partial\Phi}{\partial q},\hat{q})A \tag{1.16}$$

Now, in the classical limit $\hbar \to 0$, the first term in the l.h.s. can be disregarded and eq.(1.16) then reduces to

$$-A(q,t)\frac{\partial\Phi(q,t)}{\partial t} = H(\frac{\partial\Phi}{\partial q},q)A \tag{1.17}$$

or

$$-\frac{\partial\Phi(q,t)}{\partial t} = H(\frac{\partial\Phi}{\partial q},q) \tag{1.18}$$

which is nothing but the Hamilton-Jacobi equation provided one identifies Φ with the classical action S taken as a function of the coordinates and time[1] once one replaces the momentum by $\partial S/\partial q$. That is, one can write eq.(1.11) in the form

$$\psi(q,t) = A\exp(\frac{i}{\hbar}S) \tag{1.19}$$

with S obeying

$$-\frac{\partial S}{\partial t} = H(\frac{\partial S}{\partial q}, q) \tag{1.20}$$

This is what Dirac did in the decade of 1930. Moreover, having in mind that at the classical level the classical trajectory corresponds to an extremum of S but that at the quantum level the notion of trajectory is meaningless, Dirac suggested that quantum motion could in general be described by something related to a wave function of the form (1.18) provided one makes some sort of sum so as to take into account not only the extremum of S but also other trajectories (i.e., other *paths*) which are forbidden at the classical level, but should contribute in the quantum case,

$$\psi(q,t) \sim \sum_{paths} \exp(\frac{i}{\hbar}S) \tag{1.21}$$

It was Feynman who transformed this proposal into an operative tool giving a precise *recipe* to sum over paths.

The Feynman construction

For simplicity we consider in what follows a quantum mechanical system with just one degree of freedom so that phase space is 2-dimensional.

We recall that in Schrödinger picture physical states are described by vectors (in a Hilbert space $\mathcal{H}$) evolving in time,

$$|\psi(t)\rangle_S \tag{1.22}$$

while operators associated with physical observables are time independent,

$$\hat{O}_S(t) = \hat{O}_S(t_0) = \hat{O}_S \tag{1.23}$$

[1] For example by letting $q(t_2) = q$ be an arbitrary point and calling $t_2 = t$.

Concerning the Heisenberg picture, a given physical state at different times is described by the same state vector

$$|\psi; t\rangle_H = |\psi; t_0\rangle_H \tag{1.24}$$

but operators evolve in time

$$\hat{O}_H = \hat{O}_H(t) \tag{1.25}$$

Now, what matters in Quantum Mechanics are expectation values and in this respect, one has the identity

$${}_H\langle\psi|\hat{O}_H|\psi\rangle_H =_S \langle\psi|\hat{O}_S|\psi\rangle_S \tag{1.26}$$

ensuring that, at the physical level, the results one obtains are picture independent. This can also be understood by noting that operator eigenvalues are picture independent.

Let $|q\rangle$ be an eigenfunction of the position operator $\hat{Q}$. In Heisenberg picture

$$\hat{Q}_H(t)|q; t\rangle_H = q|q; t\rangle_H \tag{1.27}$$

(Being the Hesienberg operator time dependent, the corresponding eigenfunctions should be time dependent. The opposite happens in Schrödinger picture).

Given the Hamiltonian $\hat{H}$ (remember that for isolated systems the Hamiltonian, generator of time translations, is time independent and hence $\hat{H}_H = \hat{H}_S$) the equation of motion for $\hat{Q}_H$ reads

$$[\hat{H}, \hat{Q}_H(t)] = -i\hbar\dot{\hat{Q}}_H(t) \tag{1.28}$$

This translates into an equation of motion for states in the Schrödinger picture

$$i\hbar\frac{\partial}{\partial t}|q(t)\rangle_S = \hat{H}|q(t)\rangle_S \tag{1.29}$$

Finally, the connection between the two pictures is given by the relations

$$\begin{aligned} |q; t\rangle_H &= \exp(\frac{i}{\hbar}\hat{H}t)|q(t)\rangle_S \\ \hat{Q}_H(t) &= \exp(\frac{i}{\hbar}\hat{H}t)\hat{Q}_S \exp(-\frac{i}{\hbar}\hat{H}t) \end{aligned} \tag{1.30}$$

Let us consider a quantum system that at time $t = t_1$ is described by the Heisenberg state vector $|q_1; t_1 >_H$. The probability amplitude to find the system at a later time $t = t_2$ in a new state $|q_2; t_2\rangle_H$ is given by

$$Z_{1\to 2} =_H \langle q_2; t_2|q_1; t_1\rangle_H \tag{1.31}$$

Using (1.30) we can write this amplitude in the Schrödinger picture,

$$Z_{1\to 2} =_S \langle q(t_2)| \exp\left(-\frac{i}{\hbar}\hat{H}(t_2 - t_1)\right) |q(t_1)\rangle_S \tag{1.32}$$

Let us consider $t_2 - t_1 = \epsilon$ very small and write for simplicity $q(t_1) = q_1$ and $q(t_2) = q_2$. Also, we shall no more indicate that we are working in the Schrödinger picture. We have

$$Z_{1\to 2} = \langle q_2| \exp(-\frac{i}{\hbar}\hat{H}\epsilon|q_1\rangle = \langle q_2|q_1\rangle - \frac{i}{\hbar}\epsilon\langle q_2|\hat{H}|q_1\rangle + O(\epsilon^2) \tag{1.33}$$

Now, we have

$$\begin{aligned}
[\hat{q}, \hat{p}] &= \hbar i \\
\hat{p}|p\rangle &= p|p\rangle \\
\langle q|p\rangle &= \frac{1}{\sqrt{2\pi\hbar}} \exp(\frac{i}{\hbar}pq) \\
\int |q\rangle\langle q|dq &= 1 = \int |p\rangle\langle p|dp
\end{aligned} \tag{1.34}$$

so that we can write

$$Z^{\epsilon}_{1\to 2} = \int_{-\infty}^{\infty} \langle q_2|p_1\rangle\langle p_1|q_1\rangle dp_1 - \frac{i}{\hbar}\epsilon \int_{-\infty}^{\infty} \langle q_2|p_1\rangle\langle p_1|\hat{H}|q_1\rangle dp_1 \tag{1.35}$$

or

$$\begin{aligned}
Z^{\epsilon}_{1\to 2} = & \int_{-\infty}^{\infty} \exp\left(\frac{i}{\hbar}(q_2 - q_1)p_1\right) \frac{dp_1}{2\pi\hbar} - \\
& \frac{i}{\hbar}\epsilon \int_{-\infty}^{\infty} \exp\left(\frac{i}{\hbar}(q_2 - q_1)p_1\right) h(p_1, q_1)\frac{dp_1}{2\pi\hbar}
\end{aligned} \tag{1.36}$$

where we have written

$$h(p_1, q_1) = \sqrt{2\pi\hbar} \exp(\frac{i}{\hbar}p_1q_1)\langle p_1|\hat{H}|q_1\rangle \tag{1.37}$$

One easily sees that $h(p,q)$ is nothing but the classical Hamiltonian. For example, for a particle of mass m in a potential V, the quantum Hamiltonian is $\hat{H} = \hat{p}^2/2m + V(\hat{q})$ and one has for h

$$h(p,q) = \sqrt{2\pi\hbar}\left(\frac{p^2}{2m} + V(q)\right)\exp(\frac{i}{\hbar}pq)\langle p|q\rangle = \frac{p^2}{2m} + V(q) \qquad (1.38)$$

At the order in ϵ that we are working, we can write instead of eq.(1.36)

$$Z^{\epsilon}_{1\to 2} = \int_{-\infty}^{\infty} \exp\left(\frac{i}{\hbar}\left((q_2 - q_1)p_1 - \epsilon h(p_1, q_1)\right)\right)\frac{dp_1}{2\pi\hbar} \qquad (1.39)$$

Now, for a finite interval (t_i, t_f) we can use the result above as follows. We consider

$$t_0 = t_i < t_1 < t_2 < \ldots < t_{N-1} < t_N = t_f \qquad (1.40)$$

and make

$$t_n - t_{n-1} = \epsilon \qquad \forall\, n \qquad (1.41)$$

Then we have

$$\begin{aligned} &Z^N_{i\to f} = \langle q_f|\exp\left(-\frac{i}{\hbar}\hat{H}(t_f - t_i)\right)|q_i\rangle = \\ &\int dq_1 dq_2 \ldots dq_{N-1}\, \langle q_f|\exp\left(-\frac{i}{\hbar}\hat{H}(t_f - t_{N-1})\right)|q_{N-1}\rangle \times \\ &\langle q_{N-1}|\exp\Big(-\frac{i}{\hbar}\hat{H}(t_{N-1} - t_{N-2})\Big)|q_{N-2}\rangle \ldots \langle q_1|\exp\Big(-\frac{i}{\hbar}\hat{H}(t_1 - t_i)\Big)|q_i\rangle \end{aligned} \qquad (1.42)$$

or

$$\begin{aligned} &Z^N_{i\to f} = \int dq_1 dq_2 \ldots dq_{N-1}\frac{dp_1}{2\pi\hbar}\frac{dp_2}{2\pi\hbar}\cdots\frac{dp_N}{2\pi\hbar} \\ &\exp\Big(\frac{i}{\hbar}\left((q_f - q_{N-1})p_N + (q_{N-1} - q_{N-2})p_{N-1} + \ldots + (q_1 - q_i)p_1\right)\Big) \\ &\exp\left(-\frac{i}{\hbar}\epsilon\left(h(p_N, q_{N-1}) + h(p_{N-1}, q_{N-2}) + \ldots + h(p_1, q_i)\right)\right) \end{aligned} \qquad (1.43)$$

We then define

$$Z_{i\to f} \equiv \lim_{N\to\infty}\lim_{\epsilon\to 0} Z^N_{i\to f} \qquad (1.44)$$

The argument in the first exponential in (1.43) can be written, in the $N \to \infty$, $\epsilon \to 0$ limit, in the form

$$\lim_{N\to\infty} \lim_{\epsilon\to 0} \epsilon \left(\frac{q_f - q_{N-1}}{\epsilon} p_N + \frac{q_{N-1} - q_{N-2}}{\epsilon} p_{N-1} + \ldots + \frac{q_1 - q_i}{\epsilon} p_1 \right) = \int_{t_i}^{t_f} \dot{q} p dt \tag{1.45}$$

Concerning the second exponential, its argument can be written, in the same spirit, as $\int h dt$ so that it is natural to write

$$Z_{i\to f} = \langle q_f t_f | q_i t_i \rangle = \int \mathcal{D}p\mathcal{D}q \exp\left(\frac{i}{\hbar} \int_{t_i}^{t_f} (p\dot{q} - h(p,q)) dt \right) \tag{1.46}$$

where $\mathcal{D}p\mathcal{D}q$ is a shorthand notation for

$$\mathcal{D}p\mathcal{D}q = \lim_{N\to\infty} \lim_{\epsilon\to 0} \left(\prod_{i=1}^{N-1} dq_i \frac{dp_i}{2\pi\hbar} \right) \frac{dp_N}{2\pi\hbar} \tag{1.47}$$

Defined in this way, $\mathcal{D}p\mathcal{D}q$ is what mathematicians call a Liouville measure over all t.

Note that the integrals in (1.45) or (1.46) have to be done taking care of conditions $q(t_i) = q_i$ and $q(t_f) = t_f$. In contrast, there are no conditions on the $p's$ which are to be integrated over $(-\infty, \infty)$. Note also that there is one more integral over p than over q.

One should not forget that computing $Z_{i\to f}$ through eq.(1.46),

$$Z_{i\to f} = \langle q_f t_f | q_i t_i \rangle = \int \mathcal{D}p\mathcal{D}q \exp\left(\frac{i}{\hbar} \int_{t_i}^{t_f} (p\dot{q} - h(p,q)) dt \right) \tag{1.48}$$

implies doing an infinite number of integrals over the phase space (p, q). That is, one is to integrate over <u>all</u> trajectories compatible with $q(t_i) = q_i$ and $q(t_f) = q_f$. Since $h(p, q)$ is just the classical Hamiltonian, we see that the argument in the exponential in (1.48) is related to the Lagrangian in the usual way, $L = p\dot{q} - h$. We shall come back to this important point and discuss it in detail below.

Each integration implied in (1.48) can be thought as follows: the argument in the exponential, which we call S is in fact a sum,

$$S = S_{N,N-1} + S_{N-1,N-2} + \ldots + S_{1,i} \tag{1.49}$$

where each term is given by

$$S_{n,n-1} = \epsilon \left(p_n \frac{q_n - q_{n-1}}{\epsilon} - h(p_n, q_n) \right) \tag{1.50}$$

Then S involves summing over broken paths as those represented in figure 1. Integrating over all possible values of $q's$ implies taking into account all possible paths with fixed extrema. Concerning p integrations, we show in figure 2 the corresponding representation.

A more rigorous mathematical setting would imply restricting the functions $q(t)$ contributing to the "path-integral" (1.48) to Lipschitzian functions of order 1/2, this meaning that

$$|q(t') - q(t)| < K(t'-t)^{\frac{1}{2}} \; , \quad t' > t \tag{1.51}$$

with K some positive constant.

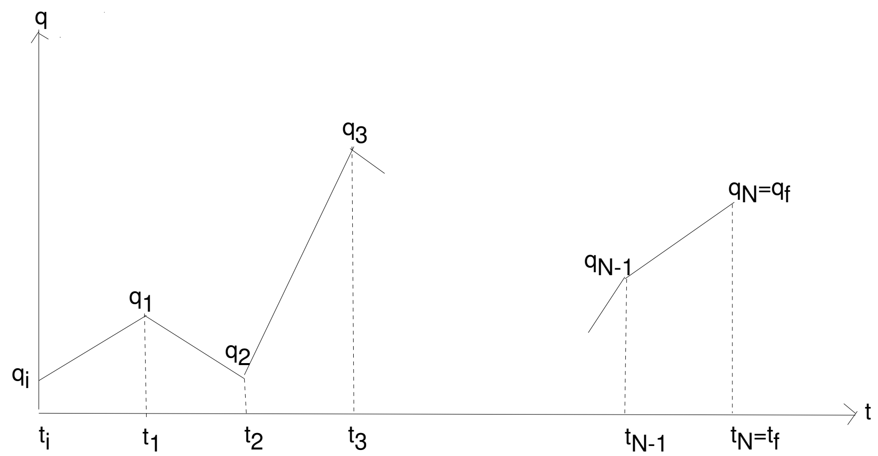


Fig. 1

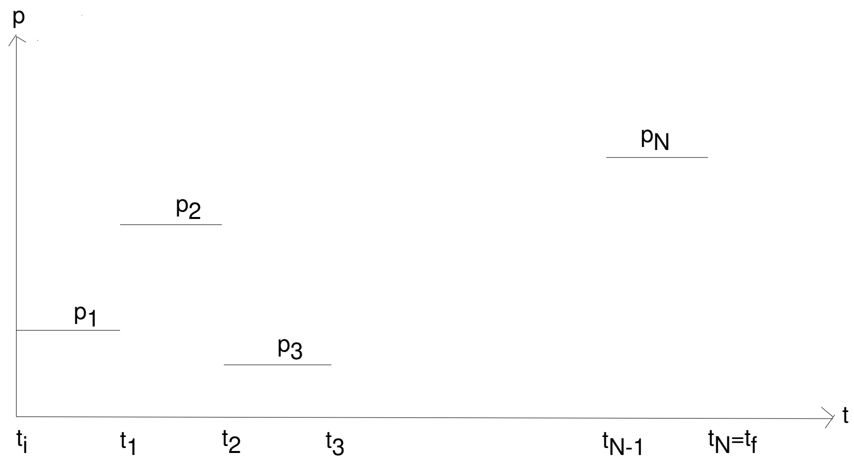


Fig. 2

Examples

1- Consider the simplest case $\hat{H} = \hat{H}(\hat{q})$ (i.e. no kinetic term in the Hamiltonian). Then $h = h(q)$ and one has

$$Z_{i\to f} = \int \mathcal{D}p\mathcal{D}q \exp\left(\frac{i}{\hbar}\int_{t_i}^{t_f}(p\dot{q} - h(q))dt\right) \tag{1.52}$$

Integrations over the p's are trivial

$$\begin{aligned} \int \mathcal{D}p \exp\left(\frac{i}{\hbar}\int p\,\dot{q}dt\right) &= \lim_{N\to\infty}\lim_{\epsilon\to 0}\prod_{i=1}^{N}\int\frac{dp_i}{2\pi\hbar}\times \\ &\quad \exp\left(\frac{i}{\hbar}(p_1(q_1-q_i)+\ldots+p_N(q_f-q_{N-1})\right) \\ &= \delta(q_1-q_i)\delta(q_2-q_1)\ldots\delta(q_f-q_{N-1}) \end{aligned} \tag{1.53}$$

But then we can trivially perform the integrals over the $q's$

$$\begin{aligned} Z_{i\to f} &= \lim_{N\to\infty}\lim_{\epsilon\to 0}\int dq_1dq_2\ldots dq_{N-1}\delta(q_1-q_i)\delta(q_2-q_1)\ldots \\ &\quad \delta(q_f-q_{N-1})\exp\left(-\frac{i}{\hbar}\epsilon(h(q_1)+h(q_2)+\ldots h(q_N-1))\right) = \\ &\quad \lim_{N\to\infty}\lim_{\epsilon\to 0}\exp\left(-\frac{i}{\hbar}N\epsilon h(q_i)\right)\delta(q_i-q_f) \end{aligned} \tag{1.54}$$

Now, writing

$$N\epsilon = N\frac{tf - t_i}{N} = t_f - t_i \tag{1.55}$$

we finally have

$$Z_{i\to f} = \exp\left(-\frac{i}{\hbar}(t_f - t_i)h(q_i)\right)\delta(q_i - q_f) \tag{1.56}$$

This result was obtained computing $Z_{i\to f}$ as a path-integral. But if we remember $Z_{i\to f}$ definition,

$$Z_{i\to f} = \langle q_f|\exp(-\frac{i}{\hbar}\hat{H}(t_f - t_i)|q_i\rangle \tag{1.57}$$

and apply it to a Hamiltonian of the form $\hat{H}(\hat{q})$ one precisely obtains the same result as in (1.56),

$$Z_{i\to f} = \langle q_f|\exp(-\frac{i}{\hbar}\hat{H}(t_f - t_i)|q_i\rangle = \exp\left(-\frac{i}{\hbar}(t_f - t_i)h(q_i)\right)\delta(q_i - q_f) \tag{1.58}$$

2- A more interesting example is that of a Hamiltonian for a particle of mass m in a potential $V(q)$,

$$\hat{H} = \frac{\hat{p}^2}{2m} + V(\hat{q}) \tag{1.59}$$

There is a $\hat{p}$ dependent part in $\hat{H}$ and, correspondingly, in h,

$$h(p,q) = \frac{p^2}{2m} + V(q) \tag{1.60}$$

so that the integrals over $p's$ are now

$$\int \mathcal{D}p\left(\frac{i}{\hbar}\int(p\dot{q} - \frac{p^2}{2m})dt\right) = \lim_{N\to\infty}\lim_{\epsilon\to 0}\prod_{i=1}^{N}\int\frac{dp_i}{2\pi\hbar}$$
$$\exp\bigg(\frac{i}{\hbar}(p_1(q_1 - q_i) + \ldots + p_N(q_f - q_{N-1}) - \frac{i\epsilon}{2m\hbar}(p_1^2 + \ldots + p_N^2)\bigg) \tag{1.61}$$

Let us change from p_i to new variables p'_i

$$p'_i = p_i - \frac{m}{\epsilon}(q_i - q_{i-1}) \; , \qquad i = 1, 2, \ldots, N \tag{1.62}$$

$$dp'_i = dp_i \tag{1.63}$$

As we shall often do, we shall write the N change of variables just as

$$\begin{aligned} p' &= p - m\dot{q} \\ \mathcal{D}p' &= \mathcal{D}p \end{aligned} \tag{1.64}$$

and then we can treat the whole change of variables as

$$\frac{i}{\hbar}\int_{t_i}^{t_f}(p\dot{q} - \frac{p^2}{2m})dt = -\frac{i}{\hbar}\int_{t_i}^{t_f}\frac{p'^2}{2m}dt + \frac{i}{\hbar}\int_{t_i}^{t_f}\frac{m\dot{q}^2}{2}dt \tag{1.65}$$

Then,

$$\begin{aligned} \int \mathcal{D}p \exp\left(\frac{i}{\hbar}\int_{t_i}^{t_f}(p\dot{q} - \frac{p^2}{2m})dt\right) &= \exp\left(\frac{i}{\hbar}\int_{t_i}^{t_f}\frac{m\dot{q}^2}{2}dt\right) \times \\ & \int \mathcal{D}p' \exp\left(-\frac{i}{\hbar}\int_{t_i}^{t_f}\frac{p'^2}{2m}dt\right) \end{aligned} \tag{1.66}$$

Let us call the path-integral in the r.h.s. of eq.(1.66) $\mathcal{N}^{-1}(t_1, t_2)$, a factor that depends on the initial and final time but not on the corresponding conditions for $q(t)$,

$$\mathcal{N}^{-1}(t_f, t_i) = \int \mathcal{D}p' \exp\left(-\frac{i}{\hbar}\int_{t_i}^{t_f}\frac{p'^2}{2m}dt\right) \tag{1.67}$$

Then we can write

$$Z_{i\to f} = \mathcal{N}^{-1}(t_f, t_i)\int \mathcal{D}q \exp\left(\frac{i}{\hbar}\int_{t_i}^{t_f}(\frac{m\dot{q}^2}{2} - V(q))dt\right) \tag{1.68}$$

or

$$Z_{i\to f} = \mathcal{N}^{-1}(t_f, t_i)\int \mathcal{D}q \exp\left(\frac{i}{\hbar}\int_{t_i}^{t_f}L(q, \dot{q})dt\right) \tag{1.69}$$

where $L(q, \dot{q})$ is the Lagrangian of the system.

We have then arrived in this particular example to the specific formula suggested by Dirac (compare with eq.(1.21))

$$Z_{i\to f} = \mathcal{N}^{-1}(t_f, t_i) \int \mathcal{D}q \exp\left(\frac{i}{\hbar} S[q]\right) \tag{1.70}$$

if we identify

$$\text{Dirac} \sum_{paths} \exp\left(\frac{i}{\hbar} S[q]\right) \to \text{Feynman} \int \mathcal{D}q \exp\left(\frac{i}{\hbar} S[q]\right) \tag{1.71}$$

The probability amplitude is computed as a sum over <u>all</u> trajectories (or *paths*) compatible with the initial $q(t_i) = q_i$ and final conditions $q(t_f) = q_f$ for coordinate q. In order to see the rôle of the classical trajectory among all trajectories included in the integration let us note that in the $\hbar \to 0$ (classical) limit the exponential oscillates as q varies so as to wash out all contributions except those for which S is fairly constant. This last corresponds to the condition $\delta S/\delta q = 0$. That is, in the $\hbar \to 0$ limit <u>the classical trajectory dominates the path integral</u>.

In this respect, recall now the *stationary phase method* gives for the case of an integral of the form

$$\lim_{a\to 0} \int_A^B dx \exp\left(\frac{i}{a} f(x)\right) \simeq \lim_{a\to 0} \exp\left(\frac{i}{a} f(x_0)\right) , \quad x_0 \in [A, B] \tag{1.72}$$

$$f'(x_0) = 0 \tag{1.73}$$

If one loosely applies this method to the path-integral (1.70) one gets

$$\begin{aligned} \lim_{\hbar\to 0} Z_{i\to f} &= \mathcal{N}^{-1}(t_f, t_i) \lim_{\hbar\to 0} \int \mathcal{D}q \exp(\frac{i}{h} S) \\ &\simeq \mathcal{N}^{-1}(t_f, t_i) \lim_{\hbar\to 0} \exp(\frac{i}{\hbar} S[q_0]) , \end{aligned} \tag{1.74}$$

$$\frac{\delta S}{\delta q}(q_0) = 0 \tag{1.75}$$

We see that in contrast with what happens in classical mechanics where only the extremum of S was used to determine the "physical state" of the system, in quantum mechanics one needs to know the whole information contained in S.

We leave as an excercise the evaluation of the normalization constant $\mathcal{N}^{-1}(t_f, t_i)$ and just advance its value

$$\mathcal{N}^{-1}(t_f, t_i) = \int \mathcal{D}p' \exp\left(-\frac{i}{\hbar}\int_{t_i}^{t_f} \frac{p'^2}{2m} dt\right) = \lim_{N\to\infty}\left(\frac{mN\exp(-\frac{i\pi}{2})}{2\pi\hbar(t_f - t_i)}\right)^{N/2} \tag{1.76}$$

We only note that being each one of the $p's$ integrals oscillatory, either one adds a convergence factor $\exp(-\eta p_n^2)$ in each integral and then takes the $\eta \to 0$ limit or one considers $-i\Delta t$ as a real parameter, that is, one analitically continues Minkowski time to Euclidean time.

Problems 1

1. Derive formula (1.76) for $\mathcal{N}$.

2. Compute the path-integral $Z_{1\to f}$ for a free particle. Show that

   (a)
   $$Z_{1\to f} = \sqrt{\frac{m}{2\pi i\hbar(t_f - t_i)}} \exp\left(\frac{im(q_f - q_i)^2}{2\hbar(t_f - t_i)}\right)$$

   Hint: make a partition as those above, integrate one by one the corresponding gaussian integrals obtaining for the k^{th} integral

   $$I_k = \sqrt{\frac{m}{2\pi i\hbar(k+1)\epsilon}} \exp\left(\frac{im(q_{k+1} - q_0)^2}{2\hbar(k+1)\epsilon}\right)$$

   (b) Show that $Z_{i\to f} = Z$ satisfies the Schrödinger equation

   $$\frac{\hbar}{i}\frac{\partial Z}{\partial t} = -\frac{\hbar^2}{2m}\frac{\partial^2 Z}{\partial q^2}$$

   $(q_i = 0,\ q_f = q)$

Clase 2

The Superposition Principle

We have seen that for a quantum system with Hamiltonian

$$\hat{H} = \frac{\hat{p}^2}{2m} + V[\hat{q}] \tag{2.1}$$

the probability amplitude for a transition from a state $|q_i\rangle$ at $t = t_i$ to a state $|q_f\rangle$ at $t = t_f$ is given by eq.(1.70),

$$\begin{aligned} Z_{i\to f} &= \mathcal{N}^{-1}(t_f, t_i) \int \mathcal{D}q \exp\left(\frac{i}{\hbar}\int_{t_i}^{t_f}(\frac{m\dot{q}^2}{2} - V(q))dt\right) \\ &= \mathcal{N}^{-1}(t_f, t_i) \int \mathcal{D}q \exp(\frac{i}{\hbar}S[q; t_i, t_f]) \end{aligned} \tag{2.2}$$

with $\mathcal{N}$ defined in eq.(1.76)

$$\mathcal{N}^{-1}(t_f, t_i) = \int \mathcal{D}p' \exp\left(-\frac{i}{\hbar}\int_{t_i}^{t_f} \frac{p'^2}{2m}dt\right) \tag{2.3}$$

In terms of $Z_{i\to f}$, the mathematical expression for the Quantum Mechanics superposition principle reads

$$Z_{i\to f} = \int Z_{i\to t_o} Z_{t_0\to f} dq_{j_0} \tag{2.4}$$

where $t_i < t_0 < t_f$ and $q_{j_0} = q(t_0)$.

We shall now prove the important identity. To this end, let us define the time partition so that t_0 coincides with one of the points in it. Then we can write eq.(2.4)

$$\int \mathcal{D}q \exp\left(\frac{i}{\hbar}S[q;t_i,t_f]\right) = \lim_{N\to\infty}\lim_{\epsilon\to 0}\prod_{j=1}^{N-1}\int_{-\infty}^{\infty} dq_j \exp\left(\frac{i}{\hbar}S[q_i,q_1,\ldots,q_{j_0},\ldots,q_f]\right) \tag{2.5}$$

with an obvious notation for the action $S[q_i,q_1,q_2,\ldots,q_{j_0},\ldots,q_f]$ when written in terms of the time partition. Now, it is evident that we can write

$$\begin{aligned}\int \mathcal{D}q \exp\left(\frac{i}{\hbar}S[q;t_i,t_f]\right) &= \int_{-\infty}^{\infty} dq_{j_0} \Bigg(\lim_{N\to\infty}\lim_{\epsilon\to 0}\prod_{j=1}^{j_0-1}\int_{-\infty}^{\infty} dq_j \\ &\exp\left(\frac{i}{\hbar}S[q_i,q_1,q_2,\ldots,q_{j_0}]\right)\lim_{N\to\infty}\lim_{\epsilon\to 0}\prod_{j=j_0+1}^{N-1}\int_{-\infty}^{\infty} dq_j \\ &\exp\left(\frac{i}{\hbar}S[q_{j_0},\ldots,q_f]\right)\Bigg)\end{aligned} \tag{2.6}$$

Eq.(2.6) can then be written as

$$\begin{aligned}\int \mathcal{D}q \exp\left(\frac{i}{\hbar}S[q;t_i,t_f]\right) &= \int_{-\infty}^{\infty} dq_{j_0} \left(\left(\int \mathcal{D}q \frac{i}{\hbar}S[q;t_i,t_0]\right)\right. \\ &\left.\int \mathcal{D}q \frac{i}{\hbar}S[q;t_0,t_f]\right)\end{aligned} \tag{2.7}$$

In the same vein, one can prove that

$$\mathcal{N}^{-1}(t_f,t_i) = \int \mathcal{D}p' \exp\left(\frac{i}{\hbar}\int_{t_i}^{t_f}\frac{p'^2}{2m}dt\right) = \mathcal{N}^{-1}(t_f,t_0)\mathcal{N}^{-1}(t_0,t_i) \tag{2.8}$$

so that one finally has

$$Z_{i\to f} = \int_{-\infty}^{\infty} dq_{j_0} Z_{i\to t_0} Z_{t_0\to f} \tag{2.9}$$

which is, as stated above, nothing but the mathematical expression of the superposition principle in Quantum Mechanics.

Matrix elements

In quantum mechanics, one is interested in the computation of matrix elements of quantum operators. We now describe how this can be done within the path-integral approach.

Consider the position operator $\hat{q}$. In the Heisenberg picture we have, at a certain time t_0, $\hat{q}_H(t_0)$, where again we take t_0 ($t_i < t_0 < t_f$) as a point in the partition of the time interval where each integration defining "the path-integration" is to be done. As a first example we shall analyze the following matrix element

$$\langle \hat{q}_H(t_0) \rangle \equiv \langle q_f; t_f | \hat{q}_H(t_0) | q_i; t_i \rangle \tag{2.10}$$

We start by writing

$$\begin{aligned} \langle \hat{q}_H(t_0) \rangle = \lim_{N\to\infty} \lim_{\epsilon\to 0} \prod_{j=1}^{N-1} \int dq_j \, \langle q_f; t_f | q_{N-1}; t_{N-1} \rangle \ldots \\ \langle q_{j_0+1}; t_{j_0+1} | q_{j_0}; t_{j_0} \rangle \, \langle q_{j_0}; t_{j_0} | \hat{q}_H(t_0) | q_{j_0-1}; t_{j_0-1} \rangle \ldots \langle q_1; t_1 | q_i; t_i \rangle \end{aligned} \tag{2.11}$$

Now,

$$\langle q_{j_0}; t_{j_0} | \hat{q}_H(t_0) | q_{j_0-1}; t_{j_0-1} \rangle = q_{j_0} \langle q_{j_0}; t_{j_0} | q_{j_0-1}; t_{j_0-1} \rangle \tag{2.12}$$

Then, except for this factor, the integrals in (2.11) are those that define $Z_{i\to f}$. Then, without repeating the arguments leading to the path-integral formula for this last quantity, we have

$$\langle \hat{q}_H(t_0) \rangle = \int \mathcal{D}p \mathcal{D}q \exp\left(\frac{i}{\hbar} \int_{t_i}^{t_f} (p\dot{q} - h(p,q)) dt \right) q(t_0) \tag{2.13}$$

or, for the case of a Hamiltonian of the form $h(p,q) = \hat{p}^2/2m + V(\hat{q})$,

$$\langle \hat{q}_H(t_0) \rangle = \mathcal{N}^{-1}(t_i, t_f) \int \mathcal{D}q \exp\left(\frac{i}{\hbar} S[q; t_i, t_f] \right) q(t_0) \tag{2.14}$$

Note that if we define a *normalized* expectation value, we should write, instead of (2.14),

$$\langle \hat{q}_H(t_0) \rangle = \frac{\int \mathcal{D}q \exp\left(\frac{i}{\hbar} S[q; t_i, t_f] \right) q(t_0)}{\int \mathcal{D}q \exp\left(\frac{i}{\hbar} S[q; t_i, t_f] \right)} \tag{2.15}$$

Let us insist that all quantities conforming the integrand in the r.h.s. are classical objects: the action in the exponential is the classical action and $q(t_0)$ is a classical coordinate. However the l.h.s. corresponds to a quantum expectation value. The natural question to pose is then: Where the quantum character is taken into account in the l.h.s. calculation? The answer is clear: it is the path-integral measure with which one integrates the classical objets the one which gives the quantum character to the calculation.

If instead of the matrix element of just one operator we want that of a product, say

$$\langle q_f;t_f|\hat{q}(t_a)\hat{q}(t_b)|q_i;t_i\rangle \tag{2.16}$$

with

$$t_i < t_b = t_{i_b} < t_a = t_{i_a} < t_f \tag{2.17}$$

we can repeat the procedure above writing

$$\begin{aligned}&\langle q_f;t_f|\hat{q}(t_a)\hat{q}(t_b)|q_i;t_i\rangle \lim_{N\to\infty}\lim_{\epsilon\to 0}\prod_{j=1}^{N-1}\int dq_j\ \langle q_f;t_f|q_{N-1};t_{N-1}\rangle\dots\\&\langle q_{i_a};t_{i_a}|\hat{q}(t_{i_a})|q_{i_a-1};t_{i_a-1}\rangle\dots\langle q_{i_b};t_{i_b}|\hat{q}(t_{i_b})|q_{i_b-1};t_{i_b-1}\rangle\dots\\&\langle q_1;t_1|q_i;t_i\rangle\end{aligned} \tag{2.18}$$

so that in this case we obtain

$$\begin{aligned}\langle q_f;t_f|\hat{q}(t_a)\hat{q}(t_b)|q_i;t_i\rangle &= \int \mathcal{D}p\mathcal{D}q\exp\left(\frac{i}{\hbar}\int_{t_i}^{t_f}(p\dot{q}-h(p.q))dt\right)\\&\quad q(t_a)q(t_b)\end{aligned} \tag{2.19}$$

Now, if instead of this matrix element we were to compute

$$\langle q_f;t_f|\hat{q}(t_b)\hat{q}(t_a)|q_i;t_i\rangle \tag{2.20}$$

we face a problem. The operators appear in the "wrong order" with respect to our partition of the time interval. This means that in fact using the path-integral we are computing always <u>time-ordered</u> product of operators. To see this, it is better to write formula (2.19) in an inverted way

$$\int\mathcal{D}p\mathcal{D}q\exp\left(\frac{i}{\hbar}\int_{t_i}^{t_f}(p\dot{q}-h(p.q))dt\right)q(t_a)q(t_b)=\langle q_f;t_f|T\hat{q}(t_a)\hat{q}(t_b)|q_i;t_i\rangle \tag{2.21}$$

where T defines a time ordering for product of operators

$$T\hat{q}(t_a)\hat{q}(t_b) = \begin{cases} \hat{q}(t_a)\hat{q}(t_b) & \text{if} \quad t_a > t_b \\ \hat{q}(t_b)\hat{q}(t_a) & \text{if} \quad t_b > t_a \end{cases} \tag{2.22}$$

We see that the path-integral, through its recipe "replace operators by functions" automatically orders product of operators from right to left, from the "past" to the "future". In general, for a product of $\hat{q}(t_a) \ldots \hat{q}(t_s)$ operators one will have

$$\begin{aligned} \int \mathcal{D}p\mathcal{D}q \;\; \exp\Big(\frac{i}{\hbar}\int_{t_i}^{t_f}(p\dot{q} - h(p,q))dt\Big)q(t_a)\ldots q(t_s) = \\ \langle q_f; t_f|T\hat{q}(t_a)\ldots\hat{q}(t_s)|q_i; t_i\rangle \end{aligned} \tag{2.23}$$

Adding sources

A practical way of computing the path-integrals defining matrix elements as those above is the following. Consider a "source" $j(t)$ "coupled" to $q(t)$ in a path-integral which we shall call $Z[j(t)]$ defined as follows

$$Z[j(t)] = \int \mathcal{D}p\mathcal{D}q \exp\Big(\frac{i}{\hbar}\int_{t_i}^{t_f}(p\dot{q} - h(p,q))dt\Big)\exp\Big(\frac{i}{\hbar}\int_{t_i}^{t_f} j(t)q(t)dt\Big) \tag{2.24}$$

The idea is that by differentiation with respect to the source one can make descend from the exponential $q's$ so that the resulting path-integral corresponds to one giving matrix elements of $\hat{q}$ in the form defined above. But, to see how this works one has to first define the precise meaning of differentiation with respect to a function like $j(t)$. This is what is knwon as computing a *functional derivative* and is time to give a thumb definition of it.

Quick review of functional derivatives

Consider the following linear functional of $j(t)$, $F[j(t)]$ defined as

$$F[j(t)] = \int_{t_i}^{t_f} K(t,t')j(t')dt' \tag{2.25}$$

In analogy with ordinary calculus, where one can define partial derivatives of a function $f(x_i)$ through the formula

$$df = \sum_i \frac{\partial f}{\partial x_i} dx^i \tag{2.26}$$

one can define the functional derivative of $F[j(t)]$ with respect to $j(t')$ as

$$\frac{\delta F[j(t)]}{\delta j(t')} \equiv K(t,t') \tag{2.27}$$

so that eq.(2.25) in analogy with (2.26) can be interpreted as

$$\delta F[j(t)] = \int_{t_i}^{t_f} \frac{\delta F[j(t)]}{\delta j(t')} \delta j(t') dt' \tag{2.28}$$

For a more rigorous definition of the functional derivative, see for example [4] Eqs.(2.25),(2.27) (or alternatively eq.(2.28)) is all one needs in order to compute functional derivatives. It satisfies all derivative properties.

In particular, note that since

$$j(t) = \int_{t_i}^{t_f} \delta(t-t') j(t') dt' \tag{2.29}$$

one has the nice formula

$$\frac{\delta j(t)}{\delta j(t')} = \delta(t-t') \tag{2.30}$$

which is the equivalent to

$$\frac{\partial x_i}{\partial x_j} = \delta_{ij} \tag{2.31}$$

for ordinary derivatives.

Using (2.30) one has

$$\frac{\delta}{\delta j(t)} \int_{t_i}^{t_f} j(t') q(t') dt = \int_{t_i}^{t_f} \frac{\delta j(t')}{\delta j(t)} q(t') dt' = \int_{t_i}^{t_f} \delta(t'-t) q(t') dt' = q(t) \tag{2.32}$$

so that

$$\frac{\delta Z[j]}{\delta j(t_a)} = \int \mathcal{D}p\mathcal{D}q \exp\Big(\frac{i}{\hbar}\int_{t_i}^{t_f}(p\dot{q} - h(p,q))dt\Big)$$
$$\frac{i}{\hbar}q(t_a)\exp\Big(\frac{i}{\hbar}\int_{t_i}^{t_f} j(t)q(t)dt\Big) \quad (2.33)$$

Hence

$$\frac{\hbar}{i}\frac{\delta Z[j]}{\delta j(t_a)}\bigg|_{j=0} = \langle q_f; t_f|\hat{q}(t_a)|q_i; t_i\rangle \quad (2.34)$$

More generally

$$\langle q_f; t_f|\hat{q}(t_a)\hat{q}(t_b)\ldots\hat{q}(t_s)|q_i; t_i\rangle = \left(\frac{\hbar}{i}\right)^p \frac{\delta^p Z[j]}{\delta j(t_a)\delta j(t_b)\ldots\delta j(t_s)}\bigg|_{j=0} \quad (2.35)$$

The Euclidean Action

In computing certain path-integrals, we shall face the problem of "gaussian" integrals but with an imaginary (not a -1) coefficient. So, these integrals are not really gaussian integrals. One way in which this problem can be overcome is by passing from Minkowski time t to Euclidean time τ,

$$t \to \tau \equiv it \quad (2.36)$$

This can be done by a Wick rotation which is nothing but an analytic continuation. Take for example the Minkowsi Action for a particle of mass m in a potential V,

$$\frac{i}{\hbar}S_M = \frac{i}{\hbar}\int dt(\frac{m\dot{q}^2}{2} - V(q)) \quad (2.37)$$

(Here $\dot{f} = df/dt$).

Let us now proceed to continuation (2.37) to Euclidean time τ

$$\frac{i}{\hbar}S_M = \frac{1}{\hbar}\int d(it)\left(-\frac{m}{2}\left(\frac{dq}{d(it)}\right)^2 - V(q)\right) = -\frac{1}{\hbar}S_E \quad (2.38)$$

with the Euclidean action defined as

$$S_E = \int d\tau L_E = \int d\tau \left(\frac{m}{2} \left(\frac{dq}{d\tau} \right)^2 + V(q) \right) \tag{2.39}$$

Wick rotation of the basic path-integral then leads to

$$\int \mathcal{D}q \exp\left(\frac{i}{\hbar} S_M \right) \xrightarrow{\text{Wick rotation}} \int \mathcal{D}q \exp\left(-\frac{1}{\hbar} S_E \right) \tag{2.40}$$

We see that the Euclidean Lagrangian looks as the Minkowskian energy, a sum of kinetic and potential terms.

Up to here we can summarize the result of passing from Minkowski to Euclidean time by stating that one has to change

- $i/\hbar \to -1/\hbar$
- $S_M \to S_E$ or $L_M \to L_E$
- $d/dt \to d/d\tau$

In order to write the Euclidean action as the difference of kinetic and potential terms one should define

$$V_E(q) \equiv -V(q) \tag{2.41}$$

One can then define an Euclidean Lagrangian

$$L_E = T_E - V_E \tag{2.42}$$

and an Euclidean energy density

$$\epsilon_E = T_E + V_E \tag{2.43}$$

An intuitive way of understanding the effect of the Wick rotation is the following. The Newton second law reads, in Minkowski space

$$m\frac{d^2\vec{x}}{dt^2} = -\vec{\nabla} V_M \tag{2.44}$$

When passing to Euclidean space one changes $t = -i\tau$ while $\vec{x}$ remains unchanged

$$-m\frac{d^2\vec{x}}{d\tau^2} = -\vec{\nabla} V_M \tag{2.45}$$

With the definition $V_E = -V_M$ Newton's law looks unchanged

$$m\frac{d^2\vec{x}}{dx_0^2} = -\vec{\nabla}V_E \tag{2.46}$$

but then the force $\vec{F}_E = -\vec{\nabla}V_E = \vec{\nabla}V_M = -\vec{F}_M$ points in the oposite direction to that of the Minkowskian one.

The relevance of the Euclidean action goes beyond the fact that it allows to handle more properly gaussian actions. As the chemists discovered in the seventies [6] it allows to describe tunneling effect using the path-integral and to develop semiclassical approximations. This, as we shall see, has a lot to do with the fact that one can think that, in passing to Euclidean space, the potential is inverted.

Coming back to the probability amplitude for a transition from a state $|q_i\rangle$ at $t = t_i$ to a state which in the Schrödinger picture was written as $|q_f\rangle$ at $t = t_f$

$$\begin{aligned} Z_{M\,i\to f} &= {}_S\langle q(t_2)|\exp\left(-\frac{i}{\hbar}\hat{H}(t_2-t_1)\right)|q(t_1)\rangle_S \\ &= \mathcal{N}^{-1}(t_f,t_i)\int\mathcal{D}q\exp\frac{i}{\hbar}S_M[q;t_i,t_f]) \end{aligned} \tag{2.47}$$

we see that in Euclidean time Z reads

$$\begin{aligned} Z_{E\,i\to f} &= {}_S\langle q(\tau_2)|\exp\left(-\frac{1}{\hbar}\hat{H}\mathcal{T}\right)|q(\tau_1)\rangle_S \\ &= \mathcal{N}^{-1}(\tau_f,\tau_i)\int\mathcal{D}q\exp\left(-\frac{1}{\hbar}S_E[q;\tau_f\tau_i]\right) \end{aligned} \tag{2.48}$$

where $\mathcal{T} = \tau_2 - \tau_1$. If we now introduce the suggestive notation

$$\beta = \frac{\mathcal{T}}{\hbar} \tag{2.49}$$

and consider periodicity in Euclidean time asking that $q(\tau_i) = q(\tau_f) = q_i$ and sum over possible periodic conditions we can write

$$Z_E = \mathrm{Tr}_i\langle q_i|\exp\left(-\beta\hat{H}\right)|q_i\rangle = \int dq_i\mathcal{D}q\exp\left(-\frac{1}{\hbar}S_E[q;\beta\hbar]\right) \tag{2.50}$$

with Tr_i meaning that we shall sum over all possible periodic trajectories starting and ending in $q(\tau_i)$. We see that Z_E can be identified with the partition function of a system with Hamiltonian $\hat{H}$ provided we identify β with

$$\beta = \frac{1}{kT} \tag{2.51}$$

where k is the Boltzmann's constant and T the temperature. That is why we shall call Z the partition function of the system, even when working in Minkowski space.

We have identified the quantum partition function of a particle with the path-integral over all periodic trajectories. This analogy was presented in a technical way. There should be, however, deeper reasons behind it, connected with the properties of space and time. Although no rigorous explanation exists, arguments coming from gravitation can be invoked to sustain it [9].

Gaussian integrals

In passing from phase space (p, q) to just a path-integral over q we have already performed a gaussian integral (over p) explicitly computing its value in Problem 1.

Now, we shall give some useful recipes that can be derived either formally or by passing through the standard procedure of making a partition of the time interval, proceeding to the change of variables in each of the N resulting ordinary integrals and then taking the $N \to \infty$ limit.

We shall work in the Euclidean formulation described above. Consider the following path-integral

$$I[\hat{A}, \hat{B}] = \int \mathcal{D}q \exp\left(-\frac{1}{2}\int_{\tau_i}^{\tau_f} \Big(q(t)\hat{A}(\tau)q(\tau) - 2\hat{B}(\tau)q(\tau)\Big)\, d\tau\right) \tag{2.52}$$

$\hat{A}(\tau)$ and $\hat{B}(\tau)$ can be (hermitian) differential operators, or just c-numbers. For example, if

$$\hat{A} = -\frac{m}{\hbar}\frac{d^2}{d\tau^2} \tag{2.53}$$

the first term in the integral in the exponential in (2.52) corresponds to the usual kinetic energy term in the (Euclidean) action,

$$\frac{1}{2}\int_{\tau_i}^{\tau_f} q\hat{A}(\tau)q d\tau = \frac{1}{\hbar}\int_{\tau_i}^{\tau_f} \frac{m\dot{q}^2}{2} d\tau \tag{2.54}$$

provided q and/or $\dot{q}$ satisfy adequate boundary conditions.

Consider now the following identity

$$\int_{\tau_i}^{\tau_f} (q\hat{A}(\tau)q - 2\hat{B}(\tau)q)d\tau = \int_{\tau_i}^{\tau_f} (q-\bar{q})\hat{A}(q-\bar{q})d\tau + \int_{\tau_i}^{\tau_f} C d\tau \quad (2.55)$$

valid provided $\bar{q}$ and C ar taken as

$$\hat{B} = -\hat{A}\bar{q} \quad (2.56)$$

$$C = -\bar{q}A\bar{q} \quad (2.57)$$

(We are considering the case in which B is just a c-function)

From (2.56) one has

$$\bar{q} = \bar{q}(\tau) = \int_{\tau_i}^{\tau_f} A^{-1}(\tau,\tau')B(\tau')d\tau' \quad (2.58)$$

where A^{-1} is the inverse of $\hat{A}$. That is, if $A(\tau)$ were just a function of τ, then

$$A^{-1}(\tau,\tau') = \frac{1}{A(\tau)}\delta(\tau-\tau') \quad (2.59)$$

If instead, $\hat{A}(\tau)$ were an operator, like in the example (2.53), $A^{-1}(t,t')$ corresponds to the Green function associated with that operator

$$\hat{A}(\tau)A^{-1}(\tau,\tau') = \delta(\tau-\tau') \quad (2.60)$$

With this in mind, we have from (2.58)

$$C = -\int_{\tau_i}^{\tau_f} B(\tau)A^{-1}(\tau,\tau')B(\tau')d\tau' \quad (2.61)$$

Then using identity (2.55) and relation (2.60) the gaussian path-integral (2.52) can be written

$$\begin{aligned} I[\hat{A},\hat{B}] &= \exp\left(\frac{1}{2}\int_{\tau_i}^{\tau_f} B(\tau)A^{-1}(\tau,\tau')B(\tau')d\tau'\right) \\ & \int \mathcal{D}q \exp\left(\frac{1}{2}\int_{\tau_i}^{\tau_f} (q(\tau)-\bar{q}(\tau)\hat{A}(\tau)(q(\tau)-\bar{q}(\tau))d\tau\right) \end{aligned} \quad (2.62)$$

Now, it is easy to see that the change of variables

$$q(\tau) \to q'(\tau) = q(\tau) - \bar{q}(\tau) \tag{2.63}$$

has a trivial Jacobian

$$\mathcal{D}q = \mathcal{D}q' \tag{2.64}$$

Then, the path-integral in the second line of eq.(2.62) can be written simply as a "normal" gaussian and all dependence on the (linear) $\hat{B}$ term is factored out of the path-integration,

$$\begin{aligned} I[\hat{A}, \hat{B}] &= \exp\left(\frac{1}{2}\int_{\tau_i}^{\tau_f} B(\tau)A^{-1}(\tau,\tau')\ B(\tau')d\tau'\right) \times I_0\,[\hat{A}] \\ I_0\,[\hat{A}] &= \int \mathcal{D}q' \exp\left(\frac{1}{2}\int_{\tau_i}^{\tau_f} q'(\tau)\hat{A}(\tau)q'(\tau)d\tau\right) \end{aligned} \tag{2.65}$$

The procedure we have just described is very usefull for treating path-integrals with a quadratic (gaussian) term but also with a linear term which usually corresponds to a source term. Indeed, consider the case of

The free particle

The generating functional for a free (quantum) particle is

$$Z[j] = \mathcal{N}^{-1}\int \mathcal{D}q \exp\left(\frac{i}{\hbar}\int_{t_i}^{t_f} \frac{m\dot{q}^2}{2}dt\right) \exp\left(\frac{i}{\hbar}\int_{t_i}^{t_f} j(t)q(t)dt\right) \tag{2.66}$$

Working in Euclidean space one has

$$Z[j] = \mathcal{N}^{-1}\int \mathcal{D}q \exp\left(-\frac{1}{\hbar}\int_{\tau_i}^{\tau_f} \frac{m\dot{q}^2}{2}d\tau\right) \exp\left(\frac{1}{\hbar}\int_{\tau_i}^{\tau_f} j(\tau)q(\tau)d\tau\right) \tag{2.67}$$

As explained above, one can rewrite (2.67) as

$$\begin{aligned} Z[j] &= \mathcal{N}^{-1}\int \mathcal{D}q \exp\left(\frac{1}{\hbar}\int_{\tau_i}^{\tau_f} q(\tau)\frac{m}{2}\frac{d^2}{d\tau^2}q(\tau)d\tau\right) \\ &\quad \exp\left(\frac{1}{\hbar}\int_{\tau_i}^{\tau_f} j(\tau)q(\tau)dt\right) \end{aligned} \tag{2.68}$$

so that $Z[j]$ takes the form of $I[\hat{A},\hat{B}]$ in (2.52) with

$$\hat{A}(\tau) = -\frac{m}{\hbar}\frac{d^2}{d\tau^2} \qquad\qquad \hat{B}(\tau) = \frac{1}{\hbar}j(\tau) \tag{2.69}$$

Then, we have from eq.(2.65),

$$Z[j] = \exp\left(\frac{1}{2\hbar^2}\int_{\tau_i}^{\tau_f} j(\tau)A^{-1}(\tau,\tau')j(\tau')d\tau d\tau'\right) Z_0 \tag{2.70}$$

with

$$Z_0 = \mathcal{N}^{-1}\int \mathcal{D}q \exp\left(-\frac{1}{\hbar}\int_{\tau_i}^{\tau_f}\frac{m\dot{q}^2}{2}d\tau\right) \tag{2.71}$$

One easily finds the Green function $A^{-1}(\tau,\tau')$ of operator $\hat{A}$ given in (2.69)

$$A^{-1}(t,t') = -\frac{\hbar}{m}|\tau-\tau'| \tag{2.72}$$

so that one finaly has

$$Z[j] = \exp\left(-\frac{1}{2m\hbar}\int_{\tau_i}^{\tau_f} j(\tau)|\tau-\tau'|j(\tau')d\tau d\tau'\right) Z_0 \tag{2.73}$$

From this formula we can easily compute any "correlation function" for the $\hat{q}$ operator. For example, the "two-point" correlation function is

$$\langle T\hat{q}(t_1)\hat{q}(t_2)\rangle = \hbar^2\frac{1}{Z[j]}\frac{\delta^2 Z[j]}{\delta j(\tau_1)\delta j(\tau_2)}\bigg|_{j=0} = -\frac{\hbar}{m}|t_1-t_2| \tag{2.74}$$

The harmonic oscillator

Another quadratic action which can be easily handled within the path-integral approach is that corresponding to the harmonic oscillator

$$S = \int_{t_i}^{t_f}\left(\frac{1}{2}m\dot{q}(t)^2 - \frac{\lambda}{2}q(t)^2\right)dt \tag{2.75}$$

We leave for a problem the treatment already described above and apply instead a method which will be usefull for cases more involved as for example that of an anharmonic one (i.e. with a quartic λ term instead tof he quadratic one.

Suppose we find a solution $q_{cl}(t)$ of the classical equations of motion resulting from action (2.75),

$$\left.\frac{\delta S[q]}{\delta q(t)}\right|_{q=q_{cl}} = 0 \tag{2.76}$$

with

$$q_{cl}(t_i) = q_i \ , \qquad q_{cl}(t_f) = q_f \tag{2.77}$$

Then, given the path-integral

$$Z_{i\to f} = \mathcal{N}^{-1} \int \mathcal{D}q \exp\left(\frac{i}{\hbar} S[q]\right) \tag{2.78}$$

we proceed to the change of variables $q(t) \to x(t)$,

$$q(t) = q_{cl}(t) + \sqrt{\hbar}\, x(t) \tag{2.79}$$

with

$$x(t_i) = x(t_f) = 0 \tag{2.80}$$

Evidently, the corresponding Jacobian is trivial,

$$\mathcal{D}q = \mathcal{D}x \tag{2.81}$$

so that we have

$$Z_{i\to f} = \mathcal{N}^{-1} \int \mathcal{D}x \exp\left(\frac{i}{\hbar} S[q_{cl}(t) + \sqrt{\hbar}\, x(t)]\right) \tag{2.82}$$

Now, we have

$$S[q_{cl}(t) + \sqrt{\hbar}\, x(t)] = S[q_cl] + \frac{\hbar}{2} \int_{t_i}^{t_f} x(t) \left.\frac{\delta^2 S}{\delta q(t) \delta q(t')}\right|_{q=q_{cl}} x(t') dt dt' \tag{2.83}$$

This formula is nothing but the Taylor expansion of the action $S[q]$ in powers of $x(t)$, around the solution $q_{cl}(t)$. Because of eq.(2.76), there is no linear term in (2.83) and because the harmonic oscilator potential is quadratic, the expansion stops at second order. For a general potential, one can perform such an expansion which will have, after the quadratic term, growing positive powers of $h^{n/2}$. Expansion of the exponential of the action will then generate contributions to the path integral, with growing powers $h^{(n-1)/2}$.

Coming back to (2.82)-(2.83) we have

$$
\begin{aligned}
Z_{i\to f} &= \mathcal{N}^{-1} \exp\left(\frac{i}{\hbar} S[q_{cl}(t)\right) \\
& \int \mathcal{D}x \exp\left(\frac{i}{2}\int_{t_i}^{t_f} x(t)\ \frac{\delta^2 S}{\delta q(t)\delta q(t')}\bigg|_{q=q_{cl}} x(t')dtdt'\right) \quad (2.84)
\end{aligned}
$$

Now, differentiating (2.75) one has

$$
\frac{\delta^2 S}{\delta q(t)\delta q(t')} = -2(\frac{m}{2}\frac{d^2}{dt^2} + \frac{\lambda}{2})\delta(t-t') \quad (2.85)
$$

so that (2.84) reads

$$
\begin{aligned}
Z_{i\to f} &= \mathcal{N}^{-1} \exp\left(\frac{i}{\hbar} S[q_{cl}(t)]\right) \\
& \int \mathcal{D}x \exp\left(-i\int_{t_i}^{t_f} x(t)(\frac{m}{2}\frac{d^2x(t)}{dt^2} + \frac{\lambda}{2}x(t))dt\right) \quad (2.86)
\end{aligned}
$$

The path-integral in the r.h.s. of this last expression depends only on t_i and t_f and not on q_i or q_f. We can then write

$$
Z_{i\to f} = F(t_i,t_f)\exp\left(\frac{i}{\hbar} S[q_{cl}(t)\right) \quad (2.87)
$$

where

$$
F(t_i,t_f) = \mathcal{N}^{-1}\int \mathcal{D}x \exp\left(-i\int_{t_i}^{t_f} x(t)\left(\frac{m}{2}\frac{d^2x(t)}{dt^2} + \frac{\lambda}{2}x(t)\right)dt\right) \quad (2.88)
$$

One can easily compute F. Before doing this, let us explicitly write $S[q_{cl}]$. To this end, we consider the equation of motion

$$
m\frac{d^2q_{cl}}{dt^2} = -\lambda q_{cl} \quad (2.89)
$$

The solution is

$$
q_{cl} = A\cos(\omega t) + B\sin(\omega t) \quad (2.90)
$$

(with the usual notation $\lambda = m\omega^2$). Or, using conditions (2.77),

$$q_{cl} = \frac{1}{\sin(\omega T)} \left(q_i \sin(\omega(t_f - t)) + q_f \sin(\omega(t - t_i)) \right) \tag{2.91}$$

with $T = t_f - t_i$. Inserting this solution into the action we get

$$S[q_{cl}] = \frac{m\omega}{2\sin(\omega T)} \left((q_i^2 + q_f^2)\cos(\omega T) - 2q_i q_f \right) \tag{2.92}$$

so that we have

$$Z_{i\to f} = F(t_i, t_f) \exp\left(\frac{im\omega}{2\hbar \sin(\omega T)} \left((q_i^2 + q_f^2)\cos(\omega T) - 2q_i q_f \right) \right) \tag{2.93}$$

Problems

1. Show that por a free particle

$$Z_{i\to f} = F_0(t_f, t_i) \exp\left(\frac{i}{\hbar} \frac{m\left(q_{cl}(t_f) - q_{cl}(t_i)\right)^2}{2(t_f - t_i)} \right)$$

2. Using that

$$Z_{i\to f} = \int dq_c Z_{i\to c} Z_{c\to f}$$

   show that

$$F_0(t_i, t_f) = F[T]$$

   with $T = t_f - t_i$.

   Find explicitly $F[T]$ for the free particle.

3. Find $Z_{i\to f}$ for the harmonic oscillator following the same procedure as that used in the text for the free particle.

Clase 3

There are many ways for computing the function $F(t_i, t_f)$ appearing in formula (eq.(2.76), that gives the harmonic oscillator probability transition

$$
\begin{aligned}
Z^{osc}_{i\to f} &= F(t_i, t_f)\exp\left(\frac{i}{\hbar}S^{osc}[q_{cl}(t)\right) \qquad (3.1)\\
F(t_i, t_f) &= \mathcal{N}^{-1}\int \mathcal{D}x \exp\left(\frac{im}{2}\int_{t_i}^{t_f}\left(\left(\frac{dx(t)}{dt}\right)^2 - \omega^2 x(t)^2\right)dt\right)
\end{aligned}
$$

$$x(t_i) = x(t_f) = 0 \qquad (3.2)$$

We shall follow a way that will enlarge the number of tools for computing path-integrals. First, let us rewrite F in (3.2) in the form

$$F(t_i, t_f) = \mathcal{N}^{-1}\int \mathcal{D}x \exp\left(i\int_{t_i}^{t_f}\frac{m}{2}x(t)\hat{D}\,x(t)dt\right) \qquad (3.3)$$

with $\hat{D}$ given by

$$\hat{D} = -\frac{d^2}{dt^2} - \omega^2 \qquad (3.4)$$

This is an hermitian operator having a well defined eigenvalue problem

$$\hat{D}\varphi_n = \lambda_n\varphi_n \qquad (3.5)$$

which we supplement with boundary conditions in accordance with eq.(3.2),

$$\varphi_n(t_i) = \varphi_n(t_f) = 0 \qquad (3.6)$$

It is easy to find eigenfunctions and eigenvalues of this problem

$$\varphi_n = N_n \sin\left(\frac{n\pi(t-t_i)}{T}\right)\, , \quad N_n = \sqrt{\frac{2}{T}} \qquad (3.7)$$

$$\lambda_n = \frac{n^2\pi^2}{T^2} - \omega^2 \tag{3.8}$$

with $T = t_f - t_i$ and N_n a normalization constant.

We shall assume that $\{\varphi_n\}$ provides an adequate basis for expanding $x(t)$ (although we are not going to be very precise about the properties of functions $x(t)$). Then, we can write

$$x(t) = \sum_n c_n \varphi_n(t) \tag{3.9}$$

$$\int_{t_i}^{t_f} \varphi_n(t)\varphi_m(t)dt = \delta_{nm} \tag{3.10}$$

Now we arrive to a crucial point that will allow us to path-integrate following an alternative prescription to that already described in the precedent lectures. Indeed, remember that the mathematical expression for the path-integral measure was

$$\mathcal{D}x(t) = \lim_{N\to\infty} \lim_{\epsilon\to 0} \prod_j dx_j \tag{3.11}$$

where N and ϵ were the number of points and the width of the time interval partition respectively. Then, we have seen that $\int \mathcal{D}x(t)$ meant an integral over <u>all</u> trajectories satisfying the boundary conditions (3.2).

Now, given a definite trajectory $x^I(t)$ one has, through the Bessel-Fourier expansion (3.9), a set of coefficients $\{c_n^I\}$. Analogously, for another trajectory $x^{II}(t)$ one has the set $\{c_n^{II}\}$ and so on. Then, we can think that an alternative for integrating over all trajectories with given boundary conditions is to integrate over all possible values of the $c_n's$ coefficients in a Bessel Fourier expansion, in a basis that satisfies the appropriate (same) boundary conditions. That is

$$\mathcal{D}x(t) \equiv \mathcal{A} \lim_{M\to\infty} \int_{-\infty}^{\infty} dc_1 \int_{-\infty}^{\infty} dc_2 \ldots \int_{-\infty}^{\infty} dc_M = \prod_i \int_{-\infty}^{\infty} dc_i \tag{3.12}$$

Here $\mathcal{A}$ is a normalization constant (that we already omitted in the last equality) ensuring that this measure is normalized as the previously defined one.

If we take seriously this definition of path-integral measure and insert the Bessel-Fourier expansion in (3.3), we end with (the $\varphi_n's$ being normalized eigenfunctions of D)

$$F(t_i, t_f) = \mathcal{N}^{-1} \prod_i \int_{-\infty}^{\infty} dc_i \exp\left(\frac{im}{2} \sum_n \lambda_n {c_n}^2\right) \tag{3.13}$$

Or making each one of the gaussian integrals

$$F(t_i, t_f) = \mathcal{N}^{-1} \lim_{M\to\infty} \left(\frac{2\pi}{im}\right)^{\frac{M}{2}} \frac{1}{\sqrt{\lambda_1 \lambda_2 \ldots \lambda_M}} \tag{3.14}$$

The product of eigenvalues in (3.14) can be thought as the determinant of the operator $\hat{D}$ (whatever this means since D is an unbounded operator and hence the product of eigenvalues grows without bound). So, we loosely write

$$\prod_n \lambda_n \equiv \det \hat{D} \tag{3.15}$$

Then, (3.14) takes the form

$$F(t_i, t_f) = \bar{\mathcal{N}}^{-1} (\det \hat{D})^{-\frac{1}{2}} \tag{3.16}$$

with

$$\bar{\mathcal{N}}^{-1} = \mathcal{N}^{-1} \lim_{M\to\infty} \left(\frac{2\pi}{im}\right)^{\frac{M}{2}} \tag{3.17}$$

Coming back to eq.(2.76) we then have for the transition amplitude of the quantum harmonic oscillator

$$\langle q_f; t_f | q_i; t_i \rangle = \bar{\mathcal{N}}^{-1} \exp\left(\frac{i}{\hbar} S^{osc}[q_{cl}]\right) (\det \hat{D})^{-\frac{1}{2}} \tag{3.18}$$

Now, the product of eigenvalues of D is

$$\det \hat{D} = \prod_n \lambda_n = \prod_n \left(\frac{n^2\pi^2}{T^2} - \omega^2\right) \tag{3.19}$$

As announced, the product of eigenvalues of the unbounded operator D is somehow problematic: it diverges!

However, we can write

$$\det \hat{D} = \prod_n \lambda_n = \prod_n \frac{n^2\pi^2}{T^2} \prod_n \left(1 - \frac{\omega^2 T^2}{n^2\pi^2}\right) \tag{3.20}$$

and absorb the first (infinite) product, which is independent of ω and hence of the oscillator properties, in the normalization $\bar{\mathcal{N}}^{-1}$. Moreover, one easily convinces oneself that the whole normalization depends on the difference $t_f - t_i = T$.

The second product in (3.20) can be found in any table (see for example [5], formula 1.431)

$$\prod_n \left(1 - \frac{\omega^2 T^2}{n^2 \pi^2}\right) = \frac{\sin(\omega T)}{\omega T} \tag{3.21}$$

so that we have

$$\langle q_f; t_f | q_i; t_i \rangle = F(T) \exp\left(\frac{i}{\hbar} S[q_{cl}]\right) \left(\frac{\omega T}{\sin(\omega T)}\right)^{\frac{1}{2}} \tag{3.22}$$

We have accommodated all normalization factors in $F(T)$, which *does not depend* on ω but just on T. It must then be the same either for the harmonic oscillator or for the free particle ($\omega = 0$). But for the free particle we know its value (see problem 2 in Clase 2)

$$F[T] = \left(\frac{m}{2\pi i \hbar T}\right)^{\frac{1}{2}} \tag{3.23}$$

The final answer for the transition amplitude for the harmonic oscillator is then

$$\langle q_f; t_f | q_i; t_i \rangle = \exp\left(\frac{i}{\hbar} S[q_{cl}]\right) \left(\frac{m\omega}{2\pi i \hbar \sin(\omega T)}\right)^{\frac{1}{2}} \tag{3.24}$$

Divertissement: The S-matrix

We shall consider now an application of the path-integral method to scattering problems. Consider a particle of mass m interacting with a short-range potential $v(q)$ (for simplicity we consider a one-dimensional problem),

$$\hat{H} = \hat{H}_0 + v(q) \tag{3.25}$$

$$\hat{H}_0 = \frac{\hat{P}^2}{2m} \tag{3.26}$$

The S-matrix for the quantum system can be defined as

$$S = \lim_{t_f \to \infty} \lim_{t_i \to -\infty} U_I(t_f, t_i) \tag{3.27}$$

where U_I is the evolution operator in the interaction picture which can be obtained from that in the Schrödinger picture,

$$U_S(t_f, t_i) = \exp\left(-\frac{i}{\hbar}\hat{H}(t_f - t_i)\right) \tag{3.28}$$

by using the unitary operator $\exp(\frac{i}{\hbar}\hat{H}_0 t)$.

$$S = \lim_{t_f \to \infty} \lim_{t_i \to -\infty} \exp\left(\frac{i}{\hbar}\hat{H}_0 t_f\right) \exp\left(-\frac{i}{\hbar}\hat{H}(t_f - t_i)\right) \exp\left(-\frac{i}{\hbar}\hat{H}_0 t_i\right) \tag{3.29}$$

S gives the probability amplitude for passing from state i to state f.

In the interaction picture asymptotic states ψ_i, ψ_f correspond to free particles since, as it is usually donde in scattering problems, one considers short-ranged potentials. These states are then eigenfunctions of $\hat{H}_0$ which can be accommodated as wave packets which solve the free Schrödinger equations,

$$\langle q| \exp\left(-\frac{i}{\hbar}H_0 t_a\right)|\psi_a\rangle = \psi_a(q; t_a) = \frac{1}{\sqrt{2\pi}}\int dp\, C_a(p) \exp\left(ipq - i\frac{p^2}{2m}t_a\right) \tag{3.30}$$

We call $C_a = \tilde{\psi}_i(p)$ $C_a = \tilde{\psi}_f(p)$ according to the case we consider. The matrix elements of S can then be constructed as

$$\begin{aligned}\langle\psi_f|S|\psi_i\rangle &= \lim_{t_f \to \infty} \lim_{t_i \to -\infty} \int dq_f dq_i \langle\psi_f| \exp\left(\frac{i}{\hbar}\hat{H}_0 t_f\right)|q_f\rangle \\ &\quad \times \langle q_f| \exp\left(-\frac{i}{\hbar}\hat{H}(t_f - t_i)\right)|q_i\rangle\langle q_i| \exp\left(-\frac{i}{\hbar}\hat{H}_0 t_i\right)|\psi_i\rangle\end{aligned} \tag{3.31}$$

or

$$\langle\psi_f|S|\psi_i\rangle = \lim_{t_f \to \infty} \lim_{t_i \to -\infty} \int dq_f dq_i \langle\psi_f|q_f, t_f\rangle\langle q_f; t_f|q_i; t_i\rangle\langle q_i, t_i|\psi_i\rangle \tag{3.32}$$

Now the wave-packet (3.30) can be approximated using the stationary phase approximation. To this end, one finds the extremum of the integrand and replaces the integral by the value of the integrand at this point with appropriate factors

$$\psi_i \sim \sqrt{\frac{m}{t}} \exp\left(\frac{imq^2}{2t}\right) C_i\left(\frac{mq}{t}\right) \exp(-i\gamma) \tag{3.33}$$

with $\gamma = \frac{\pi}{4}\mathrm{sign}\, t$

As mentioned above, in obtaining this expression one uses the stationary phase aproximation for an integral of the type [12]

$$I = \int_{\alpha}^{\beta} \exp(i\nu f(t))dt \tag{3.34}$$

for large positive ν. Such method states that

- *If $f(x)$ has one stationary point in $\alpha \leq t \leq \beta$, namely at $t = \alpha$,*

$$I = \sqrt{\left(\frac{\pi}{2\nu|f''(\alpha)|}\right)}\phi(\alpha)\exp(i\nu f(\alpha))\exp\left(\frac{i\pi \mathrm{sgn}(f''(\alpha))}{4}\right) + O\left(\frac{1}{\nu}\right) \tag{3.35}$$

as $\nu \to \infty$.

With this, we have for S,

$$\begin{aligned} \langle\psi_f|S|\psi_i\rangle &= \lim_{t_f\to\infty}\lim_{t_i\to-\infty}\int dp_f dp_i C_f^*(p_f)\exp\left(-\frac{ip_f^2 t_f}{2m}\right) \\ &\quad C_i(p_i)\exp\left(\frac{ip_i^2 t_i}{2m}\right)\sqrt{t_f t_i}\langle q_f; t_f|q_i; t_i\rangle \end{aligned} \tag{3.36}$$

where we have changed variables

$$q_i = \frac{p_i t}{m} \qquad q_f = \frac{p_f t}{m} \tag{3.37}$$

We can then write compactly

$$\langle\psi_f|S|\psi_i\rangle = \int c_f^*(p_f) S(p_f, p_i) dp_f dp_i c_i(p_i) \tag{3.38}$$

where

$$S(p_f, p_i) = \exp\left(\frac{i}{2m}(p_f^2 t_f - p_i^2 t_i)\right) \langle q_f; t_f | q_i; t_i \rangle \tag{3.39}$$

We see that $S(p_f, p_i)$ (and hence the S-matrix) can be computed using a path integral of the kind we learnt. The integral should be done over all trajectories with boundary conditions

$$q(t_i) \to \frac{p_i t_i}{m} \quad \text{for} \quad t \to -\infty \tag{3.40}$$

$$q(t_f) \to \frac{p_f t_f}{m} \quad \text{for} \quad t \to \infty \tag{3.41}$$

In one dimension conservation of energy implies of course that $p_f = \pm p_i$ but the formalism above can be easily generalized to the many-dimensional case.

The Euclidean Action

In computing certain path-integrals, we have already faced the problem of having to compute "gaussian" integrals but with an imaginary coefficient. We have seen that a way to overcome this problem is by passing from Minkowski time t to Euclidean time, which we shall now call x_0,

$$t \to x_0 \equiv it \tag{3.42}$$

This can be done by a Wick rotation which is nothing but an analytic continuation. We have seen that the Minkowsi Action for a particle of mass m in a potential V,

$$\frac{i}{\hbar} S_M = \frac{i}{\hbar} \int dt (\frac{m\dot{q}^2}{2} - V(q)) \tag{3.43}$$

(Here $\dot{f} = df/dt$). becomes, after Wick rotation

$$\frac{i}{\hbar} S_M \to -\frac{1}{\hbar} S_E \tag{3.44}$$

with the Euclidean Action taking the form

$$S_E = \int dx_0 L_E = \int dx_0 (\frac{m\dot{q}^2}{2} + V(q)) \tag{3.45}$$

where now $\dot{f} = df/dx_0$.

The partition function becomes, in Euclidean space,

$$Z_E = \int \mathcal{D}q \exp\left(-\frac{1}{\hbar}S_E[q]\right) \tag{3.46}$$

Let us note, at this point that, usually, potentials in quantum mechanical problems satisfy the following scaling property

$$V(q,g) = \frac{1}{g}V(\sqrt{q}q, 1) \tag{3.47}$$

with g the coupling constant. For example, one has for the anharmonic potential

$$V(q,g) = \frac{m\omega^2}{2}q^2 + gq^4 = \frac{1}{g}\left(\frac{m\omega^2}{2}(\sqrt{g}q)^2 + (\sqrt{g}q)^4\right) \tag{3.48}$$

We then see that if we explicitly write the dependence of the Action on the coupling constant, $S_E = S_E[q,g]$ we can rewrite the path-integral (3.46) in the form

$$Z_E = \int \mathcal{D}q \exp\left(-\frac{1}{\hbar}S_E[q,g]\right) = \int \mathcal{D}q \exp\left(-\frac{1}{g\hbar}S_E[\sqrt{g}q, 1]\right) \tag{3.49}$$

or, after making the change of variables $\sqrt{g}q \to q$,

$$Z_E = \int \mathcal{D}q \exp\left(-\frac{1}{g\hbar}S_E[q]\right) \tag{3.50}$$

(We have absorbed the change of the measure into the overall multiplicative constant that, as always, we do not write explicitly).

Eq.(3.50) shows us that g plays at the quantum level the same rôle as $\hbar$. Then, a semiclassical limit , which corresponds to the $\hbar \to 0$ limit, also corresponds to the weak-coupling limit $g \to 0$. To see this correspondence more in detail, consider the Euclidean classical equation of motion,

$$\ddot{q}_{cl} = \frac{dV}{dq_{cl}}[q,g] \tag{3.51}$$

which, after the scaling $q \to x = \sqrt{g}q$ becomes

$$\ddot{x}_{cl} = \frac{dV}{dx_{cl}}[x, 1] \tag{3.52}$$

Then, x_{cl} does not depend on g and one has

$$q_{cl}(t, g) = \frac{1}{\sqrt{g}} x_{cl}(t) \tag{3.53}$$

We then see that the change of variables that we did for the harmonic oscillator,

$$q = q_{cl} + \sqrt{\hbar}x(t) \tag{3.54}$$

is in general a reasonable one. Indeed, we see that it takes the form

$$q = \frac{1}{\sqrt{g}} x_{cl} + \sqrt{\hbar}x(t) \tag{3.55}$$

so that in the semiclassical limit ($\hbar \to 0$ or $g \to 0$) the classical configuration dominates.

Bad news: Finite action configurations (as for example the classical solution) have zero measure in the path integral.

Consider for example Z for the harmonic oscillator which we have computed to be (now in Euclidean time)

$$Z = \mathcal{N} \lim_{M\to\infty} \int dc_1 dc_2 \dots dc_M \exp\left(-\frac{m}{2\hbar}\sum_n \lambda_n c_n^2\right) \tag{3.56}$$

(here $\mathcal{N}$ is some constant normalizing Z). Now, after the change of variables $x_i = c_i\sqrt{m\lambda_i/\hbar}$, (3.56) can be rewritten in the form

$$Z = \bar{\mathcal{N}} \lim_{M\to\infty} \int dx_1 dx_2 \dots dx_M \exp\left(-\frac{1}{2}\sum_n x_n^2\right) \tag{3.57}$$

or

$$Z = \bar{\mathcal{N}} \lim_{M\to\infty} (2\pi)^{M/2} \tag{3.58}$$

Let us then define the normalization constant $\bar{\mathcal{N}}$ so that Z is normalized to 1,

$$Z = \lim_{M\to\infty} \frac{1}{(2\pi)^{\frac{M}{2}}} \int dx_1 dx_2 \dots dx_M \exp\left(-\frac{1}{2}\sum_n x_n^2\right) \tag{3.59}$$

Suppose that we don't know the value of Z (which we have just accomodated to be 1) and try to approximate it by an approximation of the kind we have been applying before. A "finite action" in this example means

$$\sum_i x_i^2 = R^2 \leq K \tag{3.60}$$

Passing to spherical coordinates and integrating over angles gives for (3.57)

$$Z = \lim_{M\to\infty} \frac{1}{(2\pi)^{\frac{M}{2}}} \frac{2\pi^{\frac{M}{2}}}{\Gamma(\frac{M}{2})} \int_0^\infty R^{M-1} \exp\left(-\frac{R^2}{2}\right) dR \tag{3.61}$$

If we call Z_M^K the contribution, at M fixed, of finite action configurations, we have

$$Z_M^K = \frac{2^{1-\frac{M}{2}}}{\Gamma(\frac{M}{2})} \int_0^K dR R^{M-1} \exp\left(-\frac{R^2}{2}\right) \leq \frac{2^{1-\frac{M}{2}}}{\Gamma(\frac{M}{2})} \int_0^K R^{M-1} dR \tag{3.62}$$

or

$$Z_M^K \leq \frac{2^{1-\frac{M}{2}}}{M\Gamma(\frac{M}{2})} K^M = \left(\frac{K^2}{2}\right)^{\frac{M}{2}} \frac{1}{\Gamma(\frac{M}{2}+1)} \tag{3.63}$$

But then, for $M \to \infty$ we have, using the Sterling formula,

$$\Gamma\left(\frac{M}{2}+1\right) \sim \sqrt{\pi M}\left(\frac{M}{2}\right)^{M/2} \exp\left(\frac{M}{2}\right) \quad \text{for } M \to \infty \tag{3.64}$$

we see that for the finite action configuration contributions one gets

$$Z^K = \lim_{M\to\infty} Z_M^K \leq 0 \tag{3.65}$$

or, since the integral is semipositive definite,

$$Z^K = 0 \tag{3.66}$$

There is a way out of this problem of zero measure for finite action configurations contribution. If instead of just taking the action we consider also the contribution of the Jacobian coming from spherical coordinates,

$$S_{eff} = \frac{R^2}{2} - \frac{M-1}{2} \log R^2 \tag{3.67}$$

then the extremum condition

$$\left.\frac{\delta S_{eff}}{\delta R}\right|_{R_{cl}} = 0 \tag{3.68}$$

gives

$$R_{cl} = \sqrt{M-1} \tag{3.69}$$

At the extremum the Action takes the value

$$S_{eff} = \frac{M}{2}(1 - \log M) \tag{3.70}$$

This is, the relevant configurations contributing to the path-integral have infinite (effective) action!

More on the Euclidean action

Barrier penetration for a quantum mechanical has to be studied with care. Consider for example the potential in figure 1.

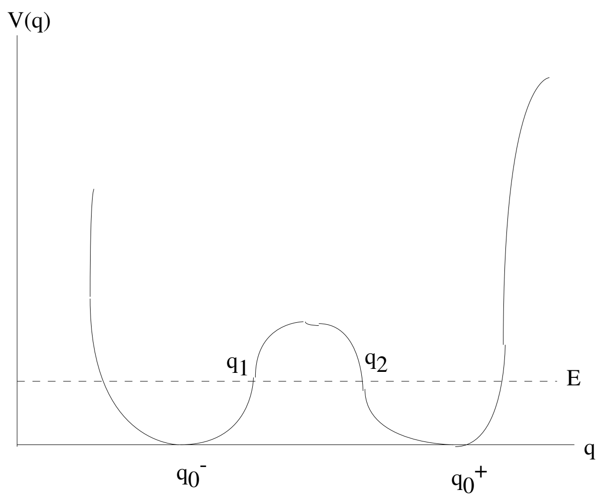


Figure 1

The transition amplitude for a particle in such a potential can be computed using the BKW method. The transmission coefficient for a particle of energy E tunneling from one side of the barrier to the other side is given by (see for example [7])

$$|T(E)| \sim \exp(\Big(-\frac{2}{\hbar}\int_{q_1}^{q_2}\sqrt{2(V(q)-E)}dq\Big)\,(1+O(\hbar)) \tag{3.71}$$

(Here q_1 and q_2 are the classical turning points at energy E).

Evidently, $T(E)$ vanishes, for $\hbar \to 0$ more rapidly than any power of $\hbar$. Now, we have seen that $\hbar \to 0$ is equivalent to $g \to 0$ (g the coupling constant of the problem). It is for this reason that one can not obtain (3.71) by doing ordinary perturbation theory and one has to develop a semiclassical approximation like BKW.

How can we attack this problem using a semiclassical technique within the path-integral approach? We have seen that the equivalent to a semiclassical approach starts, in the path-integral, by a change of variables of the form

$$q(t) = q_{cl}(t) + \sqrt{\hbar}x(t) \tag{3.72}$$

But now we face a real problem:

<u>There is no classical trajectory connecting q_1 with q_2.</u>

Real problems are faced by great men and women with **imagination**. In this case, passing to **imaginary** time. Indeed, we have seen that in the Euclidean Action, the potential appears with an opposite sign,

$$S_E = \int(T+V)dt = \int(T-(-V))dt \tag{3.73}$$

Then, the Euclidean potential V_E is not that given in figure 1 but that in figure 2:

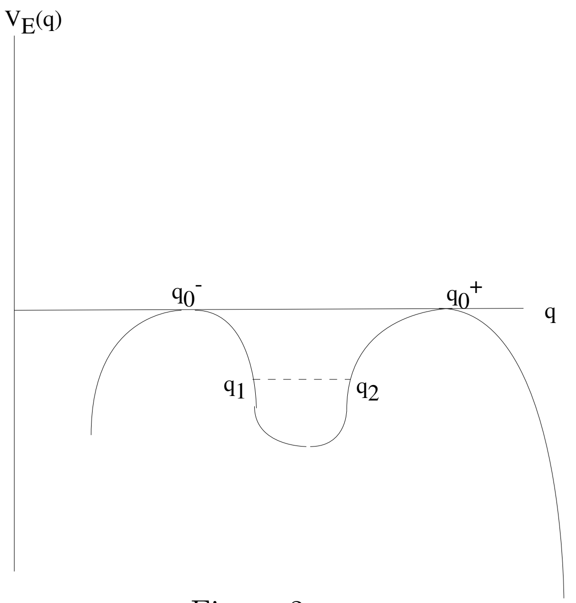


Figure 2

There is now a trajectory that can be joined classically from q_1 to q_2 and this is true even for $E \to 0$, that is, when the particle passes from q_0^- to q_0^+. The application of path-integral techniques in Euclidean time to this kind of problems was initiated, as we already mentionned, in [6]. The method was afterwards elaborated for different quantum mechanical problems in [8].

Clase 4

We shall develop here a semiclassical treatment of quantum mechanical problems that are usually solved by employing the BKW method. The advantage of our path-integral approach lies in the fact that, contrary to BKW, it can be easily extended to quantum systems with an infinite number of degrees of freedom, i.e. *field theories*.

But before doing this, let us recall the method that Rieman invented for dealing with asymptotic aproximations of hypergeometric functions in a posthumouse paper [10] and Debye afterwards adapted to the evaluation of asymptotic expansions of Bessel functions [11]. Following the beautiful book by Copson [12], we shall call the former *the saddle point method* and the latter *the steepest descent method*.

Debye's method of steepest descent

Suppose we want to evaluate an integral of the form

$$\mathcal{I} = \int dz I(z) \tag{4.1}$$

with its integrand $I(z)$ given by

$$I(z) = f(z)\exp(\nu w(z)) \tag{4.2}$$

for large and positive values of ν. Here $z = x + iy$ is a complex variable and both $f(z)$ and $w(z) = u(x, y) + iv(x, y)$ are analytic functions of z, independent of ν and regular in the domain that contains the path of integration, which is an arc or closed curve in the complex plane.

Recalling the kind of path-integrals in which we are interested,

$$Z_E = \int \mathcal{D}q \exp\left(-\frac{1}{\hbar} S_E[q]\right) \tag{4.3}$$

we see that if we identify z with q, ν with $1/\hbar$, $w(z)$ with $-S_E[q]$ and put $f=1$, the semiclassical $\hbar \to 0$ limit corresponds to large values of ν.

The idea of the steepest descent method is to deform the path of integration so as to satisfy the following conditions:

1. the path passes through a zero $z_0 = x_0 + iy_0$ of $w'(z) = dw/dz$.

   This corresponds to an extrema of the action (a classical trajectory).

2. the imaginary part of $w(z)$ is constant on the path.

On such a path I(z) becomes

$$I(z) = \exp(i\nu v(x_0, y_0)) f(x+iy) \exp(\nu u(x,y)) \tag{4.4}$$

Consider $u = u(x,y)$ as a surface in orthogonal coordinates (x,y), with u as the vertical axis of a 3-axis system. Being $w(z)$ an analytic function, its derivative $w'(z) = \lim_{\Delta z \to 0} \Delta w/\Delta z$ can be calculated choosing $\Delta z = \Delta x + i0$

$$w'(z) = \frac{dw}{dz} = \frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x} \tag{4.5}$$

Then, using the Cauchy-Riemann condition

$$\frac{dv}{dx} = -\frac{du}{dy} \tag{4.6}$$

one gets

$$w'(z) = \frac{\partial u}{\partial x} - i\frac{\partial u}{\partial y} \tag{4.7}$$

Now, if as stated z_0 is a zero of $w'(z)$, one has

$$\left.\frac{\partial u}{\partial x}\right|_{x_0,y_0} = 0$$
$$\left.\frac{\partial u}{\partial y}\right|_{x_0,y_0} = 0 \tag{4.8}$$

so that the tangent plane to the surface at (x_0, y_0, u_0) is horizontal. But we have also

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \tag{4.9}$$

so that the point z_0 has to be a saddle point, not minimum or maximum.

The curves chosen by Debye when deforming the contour, which correspond to $v =$ constant are orthogonal to the level curves $u =$constant and then corresponds to the maps on the (x, y)-plane of the steepest curves on the surface. The Debye curves are then the steepest curves passing through the saddle point.

On the steepest path through the saddle point z_0 we have

$$w(z) = w(z_0) - \tau \tag{4.10}$$

with τ real, parametrizing the steepest curve (remember that the imaginary part v of w is constant so that in fact eq.(4.10) reduces to $u(z) = u(z_0) - \tau$).

If we call s the arc of the path we have

$$\left(\frac{d\tau}{ds}\right)^2 = \left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial u}{\partial y}\right)^2 = |w'(z)|^2 \tag{4.11}$$

so that $d\tau/ds = \pm|w'(z)|$ and it can only change of sign if it passes through another saddle point (or singularity, but this is rare to happen). The variable τ is usually monotonic on a steepest path from a saddle point and either increases to $+\infty$ pr decreases to $-\infty$ but this last would lead to a divergent integral.

Then, we have to chose paths on which τ is positive, these are the paths of the steepest descent from the saddle point. Whenever this is possible, in order to evaluate integral I_ν it only remains to consider the asymptotic behavior of

$$I_\nu = \exp\left(\nu v(z_0)\right) \int_0^\infty f \exp(-\nu\tau) \frac{dz}{d\tau} d\tau \tag{4.12}$$

where ν is large and positive.

Now, in general it is rather difficult to get, starting from (4.10), the parametric expression for the path. That is, expressing z as a function of τ in order to calculate $dz/d\tau$ is cumbersome. Let us see how does one proceeds with the example of the asymptotic expansion of the inverse of the Γ-function. Hankel's integral giving $1/\Gamma(\nu)$ can be written, for ν real and positive, in the form

$$\frac{1}{\Gamma(\nu)} = \frac{1}{2\pi i \nu^{\nu-1}} \int_C \exp(\nu(z - \log z)) dz \tag{4.13}$$

where C is a curve that starts at $\infty \exp(-\pi i)$, goes round the origin and ends at $\infty \exp(\pi i)$.

With the previous notation we have

$$f(z) = 1 \,, \quad w(z) = z - \log z \tag{4.14}$$

Here we have just one saddle point, $z_0 = 1$ since

$$w'(z) = 0 \Rightarrow 1 - \frac{1}{z} = 0 \tag{4.15}$$

so that (4.10) reads

$$z - \log z = 1 - \tau \tag{4.16}$$

Near the saddle point, τ is small and if we expand the left hand side around $z = 1$ we get

$$1 + \frac{1}{2}(z-1)^2 \approx 1 - \tau \tag{4.17}$$

or

$$z \approx 1 \pm i\sqrt{2\tau} \tag{4.18}$$

There are then two directions of steepest descent from the saddle point in which the phase of $z - 1$ is

$$\text{phase}(z-1) = \pm\pi/2 \tag{4.19}$$

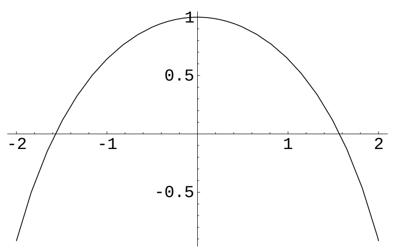
If we consider the imaginary part of eq.(4.16) we get that the equation for the saddle point becomes

$$\mathcal{I}m(z - \log z) = 0 \Rightarrow y - \arctan\frac{y}{x} = 0 \tag{4.20}$$

This equation is satisfied by $y = 0$ which gives two directions of steepest ascent from the saddle point and by

$$x = y \cot y \tag{4.21}$$

The upper and lower halves of this curve, which is symmetrical about the real axis, are the two paths of steepest descent



Since

$$\frac{dx}{dy} = \frac{\sin 2y - 2y}{2\sin^2 y} \tag{4.22}$$

x decreases steadily as y increases from 0 to π and tends to $-\infty$ and the same for y decreasing from 0 to $-\pi$. The curve $x = y\cot y$ is thus asymptoric at $x = -\infty$ to $\pm\pi$. Let us change variables in (4.16) according to

$$z = 1 + Z\,, \qquad \tau = -\frac{t^2}{2} \tag{4.23}$$

getting

$$Z - \log(1 + Z) = \frac{t^2}{2} \tag{4.24}$$

On can solve this equation, with Z multivalued as a function of t, One obtains two solutions vanishing at $t = 0$ in terms of t

$$Z_1(t) = \sum_1^\infty a_n t^n\,, \qquad Z_2(t) = \sum_1^\infty a_n(-t)^n \tag{4.25}$$

The series are convergent for $|t| < 2\sqrt{\pi}$. The first 5 coefficients a_n are

$$a_1 = 1\,,\ \ a_2 = \frac{1}{3}\,,\ \ a_3 = \frac{1}{36}\,,\ \ a_4 - \frac{1}{270}\,,\ \ a_5 = \frac{1}{4320} \tag{4.26}$$

Replacing Z by the original variable $Z = z - 1 \approx \pm i\sqrt{2\tau}$ we have

$$\begin{aligned} z_1 &= 1 + \sum_1^\infty a_n i^n (2\tau)^{n/2}\,, & (4.27) \\ z_2 &= 1 + \sum_1^\infty a_n (-i)^n (2\tau)^{n/2} & (4.28) \end{aligned}$$

with (4.27) giving the upper path of steepest descent and (4.28) the lower one. Hence

$$\frac{1}{\Gamma(\nu)} = \frac{\exp(\nu)}{2\pi i \nu^{\nu-1}} \int_0^\infty \exp(-\nu\tau) \left(\frac{dz_1}{d\tau} - \frac{dz_2}{d\tau} \right) \tag{4.29}$$

Now, since

$$\frac{dz}{d\tau} = \frac{z}{1-z} \tag{4.30}$$

one easily finds that

$$\frac{dz_1}{d\tau} - \frac{dz_2}{d\tau} = \frac{1}{z_2 - 1} - \frac{1}{z_1 - 1} \tag{4.31}$$

After inspection of the radius of convergence of the series, we can substitute the power series in the integrand and integrate it. The answer is

$$\frac{1}{\Gamma(\nu)} = \exp(\nu)\nu^{-\nu}\sqrt{\frac{\nu}{2\pi}} \left(1 - \frac{1}{12\nu} + \frac{1}{288\nu^2} - \ldots \right) \tag{4.32}$$

The anharmonic oscillator

Coming back to path-integral quantization, let us consider the anharmonic oscillator in one dimension, with potential

$$V = \frac{\lambda}{4}(q^2 - q_0^2)^2 \tag{4.33}$$

There are two constant classical solutions corresponding to this potential

$$q_{cl} = \pm q_0 \tag{4.34}$$

Within the (Minkowski space) path-integral framework, they provide two natural candidates for a path-integral change of variables,

$$q(t) = q_0 + \sqrt{\hbar} x \tag{4.35}$$

or

$$q(t) = -q_0 + \sqrt{\hbar} x \tag{4.36}$$

However, both changes of variables are useless for describing the well-known quantum tunnel effect that takes place for this kind of potentials.

Indeed, let us call ψ_r the gaussian wave packet solution of a problem with just the well on the right, with E_0 as lowest energy eigenvalue and normalized so that the integral of $|\psi_r|^2$ on the right is 1. Then a solution of the complete problem can be written as the superposition of two gaussians, one $\psi_r(q)$ centered in q_0 the other one $\psi_l(q)$ centered in $-q_0$, $\psi_l(q) = \psi_r(-q)$

$$\psi(q) = \alpha\psi_r(q) + \beta\psi_r(-q) \tag{4.37}$$

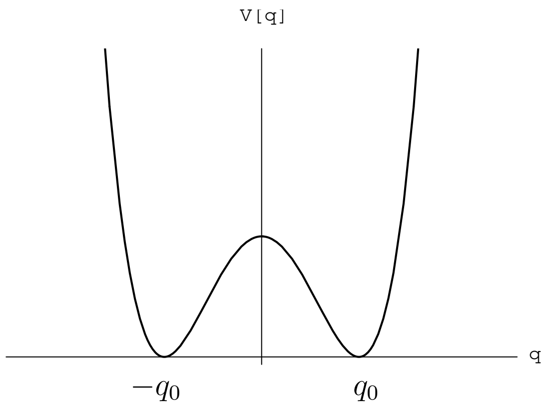
Being the problem parity invariant, the parity operator $\mathcal{P}$,

$$\mathcal{P}\psi(q) = \psi(-q) \tag{4.38}$$

commutes with the Hamiltonian, $[\hat{H}, \mathcal{P}] = 0$ and then states will simultaneously be eigenfunctions of $\hat{H}$ and $\mathcal{P}$ so that (4.37) becomes

$$\psi_\pm(q) = \frac{1}{\sqrt{2}}(\psi_r(q) \pm \psi_r(-q)) \tag{4.39}$$

Being ψ_r gaussian functions, the ψ_+ solution has no nodes and the ψ_- one just one node so that the former corresponds to the ground state and the later to the first excited state



That is, the ground state is given by,

$$\psi_0 = \frac{1}{\sqrt{2}}(\psi_r(q) + \psi_r(-q)) \tag{4.40}$$

while the antisymmetric combination describes the first excited state.

$$\psi_1 = \frac{1}{\sqrt{2}}(\psi_r(q)) - \psi_r(-q)) \tag{4.41}$$

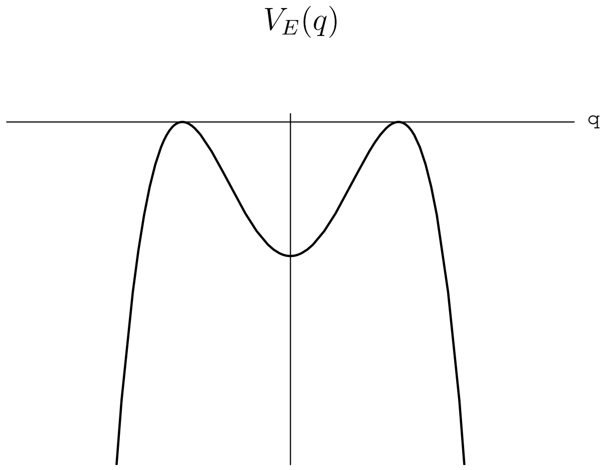
The form of these wave functions makes evident the tunnel effect.

What about a path-integral approach to this problem? On the one hand, we know that if we make a smart choice for q_{cl} for the shift of path-integral variables, this q_{cl} will dominate the quantum result and we will be able to control quantum fluctuations non perturbatively. To achieve this, such classical solution should have, in itself, some information of the tunneling but this in turn becomes an oxymoron: tunneling is a phenomenon that does not exist classically.

In other words, one needs a a classical solution interpolating between both minima in order to capture in the (fixed) classical term of the change of variables the leading contribution to the path-integral. But there is no classical solution in Minkowski space to take into account this. And, if one insists in performing a change of variables like (4.35), one will be just studying small fluctuations around the minimum at q_0. If, instead, one makes the change of variables (4.36), one will be describing small fluctuations around the minimum at $-q_0$. But mall fluctuation cannot lead to tunneling.

Notice also that using q_0 or $-q_0$ to center the fluctuation analysis implies breaking the original symmetry under parity ($q \to -q$) of the problem. In order to describe a true ground state reflecting the symmetry of the problem one needs a classical trajectory joining both minima. But there is no such trajectory in Minkowski space.

We have seen however that in Euclidean space the potential reverse its sign in the Euclidean action appearing in the exponential, so that it looks as in the following figure,



Then, for such an Euclidean potential, we can expect to have classical solutions interpolating between the two extrema. Such trajectories will in some sense "restore" the symmetry under parity transformations implied by a nos symmetric choice.

Now, such a kind of solution of the equation of motion in Euclidean space does exist

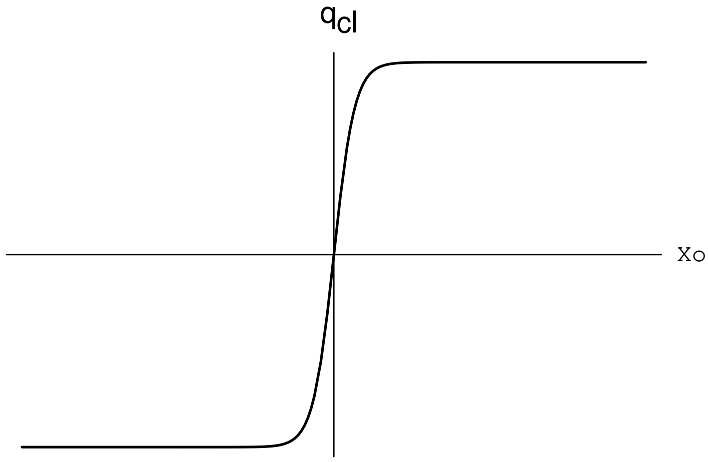
$$\ddot{q} = \lambda(q^2 - q_0^2)q \tag{4.42}$$

$$q = q_{cl}(x_0) = q_0 \tanh\left(\sqrt{\frac{\lambda}{2}} q_0 x_0\right) \tag{4.43}$$

and one easily sees that it interpolates between both minima. Remarkably, the Euclidean energy associated to this solution is zero (the kinetic energy T is positive definite and the Euclidean potential energy negative) and also $|T| = |V|$,

$$E = T_E + V_E = q_0^4 \cosh^{-4}\left(\sqrt{\frac{\lambda}{2}} q_0 x_0\right) - q_0^4 \cosh^{-4}\left(\sqrt{\frac{\lambda}{2}} q_0 x_0\right) = 0 \tag{4.44}$$

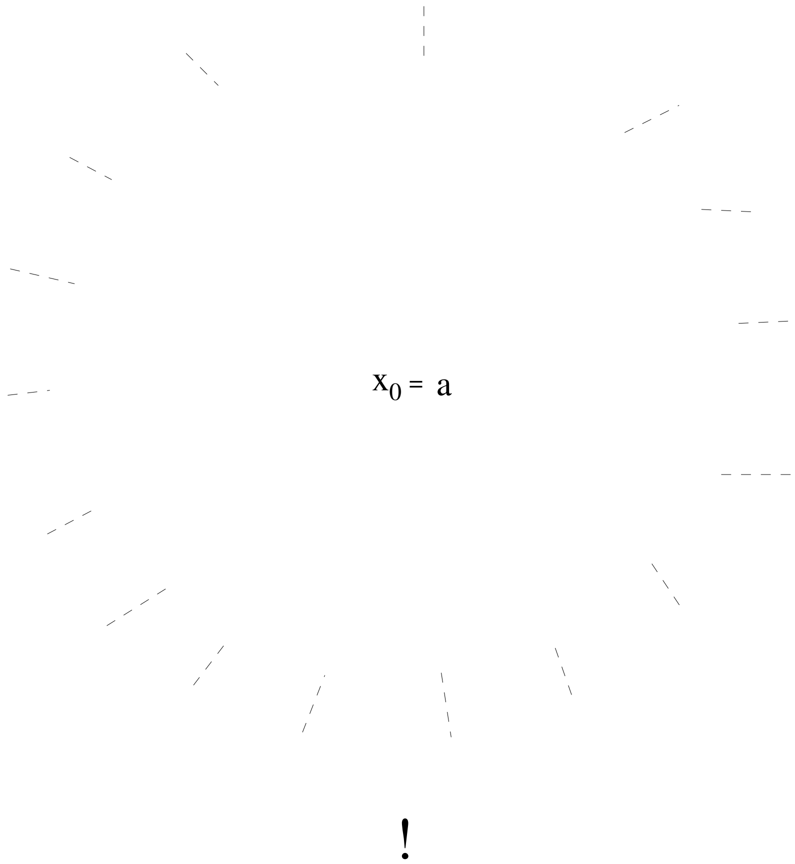
The shape of this solutions is shown in the figure below,



Now, since in the horizontal axis we are representing (Euclidean) time and the solution differs from a constant just in a neighborhood of 0, we see that the solution is localized around Euclidean time 0. Of course, because of translation (in time) invariance, there is an infinite family of solutions centered at an arbitrary <u>instant</u> a,

$$q_{cl}(x_0; a) = q_0 \tanh\left(\sqrt{\frac{\lambda}{2}} q_0 (x_0 - a)\right) \tag{4.45}$$

We can understand why 't Hooft called this solution an *instanton* and represented it as in the next figure



A "snapshot" of an instanton

Instanton: a classical solution of the Euclidean equations of motion with finite (localized) action and zero energy.

The instanton provides a trajectory joining two minima so that the path-integral calculation "centered" around the instanton restores, as we shall see, the symmetry broken by the existence of trivial minima.

Consider the transition amplitude that we have already defined,

$$Z_{i\to f} = \langle q_f | \exp\left(-\frac{HT}{\hbar}\right) |q_i\rangle \tag{4.46}$$

with, as usual, $T = t_f - t_i$. We have learnt that

$$Z_{i\to f} = \mathcal{N}^{-1} \int \mathcal{D}q \exp\left(-\frac{1}{\hbar} S[q]\right) \tag{4.47}$$

To see how physical quantities can be calculated from $Z_{i\to f}$ using instantons, let us observe that the ground state energy can be derived in quantum mechanics as follows: one considers

$$H|n\rangle = E_n|n\rangle \tag{4.48}$$

and then, one can write for the partition function

$$Z_{i\to f} = \sum_n \langle q_f|n\rangle\langle n|q_i\rangle \exp\left(-\frac{E_n T}{\hbar}\right) \tag{4.49}$$

Now, in the $T\to\infty$ limit, it is the ground state energy the one which makes the most important contribution to the sum

$$\lim_{T\to\infty} Z_{i\to f} \sim \langle q_f|0\rangle\langle 0|q_i\rangle \exp\left(-\frac{E_0 T}{\hbar}\right) \tag{4.50}$$

Then, from the exponential damping of $Z_{i\to f}$ when $T\to\infty$ we can read the ground state energy

$$\lim_{T\to\infty} Z_{i\to f} = \lim_{T\to\infty}\int \mathcal{D}q \exp\left(-\frac{1}{\hbar}S[q]\right) \sim \lim_{T\to\infty}\exp\left(-\frac{E_0 T}{\hbar}\right) \tag{4.51}$$

This will be our task in what follows. Let us then try to compute the path-integral in the l.h.s. by making the change of variables

$$q(x_0) = q_{cl}(x_0) + \sqrt{\hbar}\, x(x_0) \tag{4.52}$$

with q_{cl} the instanton given by eq.(4.43). However, we have seen that, due to translation (in time) invariance, (4.45) is a more general solution depending on a, the "center" of the instanton. Then, we can think in a change of variables of the form

$$q(x_0) = q_{cl}(x_0; a) + \sqrt{\hbar}\, x(x_0) \tag{4.53}$$

But now a could be any real number and we have an infinite set of possible changes of variables. Even worse,

Lemma: If $q_{cl}(x_0; a)$ is a solution of the Euler-Lagrange equations of motion for any value of the real parameter a, i.e. if

$$\left.\frac{\delta S}{\delta q(x_0)}\right|_{q=q_{cl}} = 0 \tag{4.54}$$

then the quadratic form

$$S^{(2)} = \int dy_0 \left. \frac{\delta^2 S}{\delta q(x_0) \delta q(y_0)} \right|_{q=q_{cl}} \tag{4.55}$$

has a *zero-mode* of the form

$$x^{(0)} = A \frac{dq_{cl}(x_0; a)}{da} \tag{4.56}$$

with A a normalization constant

(Trivial) **Proof**: In view of (4.54), we have

$$\frac{d}{da} \left. \frac{\delta S}{\delta q(x_0)} \right|_{q=q_{cl}} = \frac{d}{da} \left(\frac{\delta S}{\delta q_{cl}(x_0, a)} \right) = 0 \tag{4.57}$$

or

$$\int dy_0 \left. \frac{\delta^2 S}{\delta q(x_0) \delta q(y_0)} \right|_{q=q_{cl}} \frac{dq_{cl}(y_0; a)}{da} = 0 \tag{4.58}$$

which can be symbolically written as

$$\hat{S}^{(2)} x^{(0)} = 0 \tag{4.59}$$

Why is this result a problem? Remember that, when studying the harmonic oscillator, the strategy was to make an expansion of the path-integral variable $x(x_0)$ precisely in terms of the eigenfunctions of $S^{(2)}$. With this, when integrating over the coefficients of the associated Bessel-Fourier expansion, we faced the evaluation of gaussian integrals. Now, in the expansion *all* eigenfunctions have to be taken into account, including possible zero-mode (whenever they are normalizable). The gaussian integral over each coefficient c_i lead to the inverser of the corresponding eigenvalue squared root. Integral over all coefficients then lead to the inverse of the square root of a determinant. But in the present case, since there is a zero eigenvalue, the determinant vanishes! In other words, the integral of the coefficient associated to the zero mode is no more gaussian.

Let us see this phenomenon in detail. Concerning, the classical action, if we consider the change of variables (4.53) and perform an expansion, we have

$$S[q] = S[q_{cl}] + \frac{\hbar}{2}\int dx_0 dy_0\, x(y_0)\, \frac{\delta^2 S}{\delta q(x_0)\delta q(y_0)}\bigg|_{q=q_{cl}} x(x_0) + O(\hbar^{3/2}) \quad (4.60)$$

(There is no linear term since q_{cl} is a solution of the eqs. of motion). The Euclidean action $S[q_{cl}]$, it takes the form

$$S[q_{cl}] = \int_{-\infty}^{\infty}(\frac{1}{2}\dot{q}^2 + V(q_{cl}))dx_0 \quad (4.61)$$

with V the fumction defined in (4.33)

$$V = \frac{\lambda}{4}(q^2 - q_0^2)^2 \quad (4.62)$$

Now, since the Euclidean energy for the instanton is zero, we have

$$\int_{-\infty}^{\infty}\frac{1}{2}\dot{q}^2 dx_0 = \int_{-\infty}^{\infty} V(q_{cl})dx_0 \quad (4.63)$$

or

$$S[q_{cl}] = \int_{-\infty}^{\infty}\dot{q}^2 dx_0 = -\int_{-\infty}^{\infty}\dot{q}_{cl}dq_{cl} = \int_{-\infty}^{\infty}\sqrt{2V(q_{cl})}dx_0 \quad (4.64)$$

(We have again used that the energy is zero in writing the last identity).

Using the explicit form of the potential, given by eq.(4.33), we finally have

$$S[q_{cl}] = \frac{2\sqrt{2}}{3}\frac{\mu^3}{\lambda} \quad (4.65)$$

where we have introduced a parameter μ by writing

$$\frac{\mu^2}{\lambda} = q_0^2 \quad (4.66)$$

so that the potential can be expressed as

$$V[q] = -\frac{\mu^2}{2}q^2 + \frac{\lambda}{4}q^4 + \frac{\mu^4}{4\lambda} \quad (4.67)$$

It is important to notice that the classical action for the instanton (4.65) is larger than the action corresponding to the trivial minima of the potential, $q = \pm q_0$ which of course corresponds to $S[\pm q_0] = 0$. That is, the contribution to the path-integral of the instanton solution will be suppressed by a factor

$$\exp\left(-\frac{2\sqrt{2}\mu^3}{3\hbar\lambda}\right) \tag{4.68}$$

with respect to that coming from the trivial solution to the equations of motion.

Let us ignore for the moment the existence of an a-dependence in the classical solution and proceed to the change of variables,

$$q(x_0) = q_{cl}(x_0) + \sqrt{\hbar}\, x(x_0) \tag{4.69}$$

Up to order $\hbar$, we have for the action,

$$S[q] = S[q_{cl}] + \frac{\hbar}{2}\int dx_0 x(x_0) S^{(2)}\left[q_{cl}\right] x(x_0) \tag{4.70}$$

with

$$S^{(2)}\left[q_{cl}\right] = -\frac{d^2}{dx_0^2} - V''[q_{cl}] \tag{4.71}$$

and the second derivative of the potencial given by

$$V''[q_{cl}] = \mu^2 - 3\lambda q_{cl}^2 \tag{4.72}$$

Then, up to and including terms of order $\hbar$, we have for $Z_{i\to f}$ once the change of variables (4.69) is performed,

$$Z_{i\to f} = \exp\left(-\frac{1}{\hbar}S[q_{cl}]\right)\int \mathcal{D}x \exp\left(-\frac{1}{2}\int_{-\infty}^{\infty} dx_0\, x(x_0)\hat{S}^{(2)}\left[q_{cl}\right]x(x_0)\right) \tag{4.73}$$

Now we consider, as in the harmonic oscillator case, a Bessel-Fourier expansion,

$$x(t) = \sum_n c_n x^{(n)} \tag{4.74}$$

where the basis is provided by the eigenfunctions of $S^{(2)}$,

$$\hat{S}^{(2)}\left[q_{cl}\right] x^{(n)} = \lambda_n x^{(n)} \tag{4.75}$$

It is at this point that we will be in trouble, since this basis includes the zero mode (4.56). Indeed, if we naively proceed as before, by writing for the path-integral measure

$$\mathcal{D}x = \prod_n dc_n \tag{4.76}$$

and use the expansion (4.74) in (4.73), we end with

$$Z_{i\to f} = \exp\left(-\frac{1}{\hbar}S[q_{cl}]\right)\prod_n \int_{-\infty}^{\infty} dc_n \exp\left(-\sum_m \lambda_m c_m^2\right) \tag{4.77}$$

or

$$Z_{i\to f} = \exp\left(-\frac{1}{\hbar}S[q_{cl}]\right)\prod_n \lambda_n^{-1/2} \tag{4.78}$$

Now, since we know that there is a zero eigenvalue the product in (4.78) explodes.

It is important to stress that the zero-mode (4.56) is normalizable (otherwise, it would not have been included in the Bessel-Fourier basis and the problem would not have existed). That is, one has

$$x^{(0)} = \left|\frac{dq_{cl}}{da}\right|^{-1/2}\frac{dq_{cl}}{da} \tag{4.79}$$

Now, since $q_{cl} = q_{cl}(x_0 - a)$, one has

$$\frac{d}{dx_0} = -\frac{d}{da} \tag{4.80}$$

We can then write

$$x^{(0)} = \left|\frac{dq_{cl}}{dx_0}\right|^{-1/2}\frac{dq_{cl}}{dx_0} \tag{4.81}$$

Moreover, since we have

$$S[q_{cl}] = 2T = \int_{-\infty}^{\infty} dx_0 \dot{q}_{cl}^2 = \int_{-\infty}^{\infty} dx_0 \left(\frac{dq_{cl}}{da}\right)^2 = \left|\frac{dq_{cl}}{da}\right|^2 \tag{4.82}$$

we can write

$$x^{(0)} = \frac{1}{\sqrt{S[q_{cl}]}}\frac{dq_{cl}}{dx_0} \tag{4.83}$$

We have to do something with this zero-mode. The method we shall apply was originally developped by Bogoliubov and Tyablikov[13] long ago and specialized to the case of field theories by Faddeev and Popov [14]. We shall particularly follow this last paper, except that the symmetry here is a time-translation symmetry while that treated by Faddeev and Popov was a gauge symmetry.

The Faddeev-Popov idea can be understood as follows; in performing the change of variables, one possibility is to write

$$q(x_0) = q_{cl}(x_0; a = 0) + \sqrt{\hbar} \sum_n c_n x^{(n)} \tag{4.84}$$

We have explicitly indicated that we have no parameter a in the classical solution. (Note that a is present in the zero-mode eigenfunction in the sum). Moreover, the zero eigenfunction introduces an infinite when performing the gaussian integrations. What if we exclude the zero mode from the sum? We can think, correctly, that we shall not be describing all trajectories. That is, integration over the Bessel-Fourier coefficents with exception of c_0 would not exhaust integration over all trajectories. However, if instead of (4.84), we write

$$q(x_0) = q_{cl}(x_0; a) + \sqrt{\hbar} \sum_{n \neq 0} \tilde{c}_n x^{(n)}(x_0) \tag{4.85}$$

one can show that what has disappeared from the complete sum when not including the zero-mode reappears through the a-parameter in the classical solution. Indeed, consider the expansion

$$q_{cl}(x; a) = q_{cl}(x) + a \frac{dq_{cl}(x_0)}{dx_0} + \ldots = q_{cl}(x) + a \sqrt{S[q_{cl}]}\, x^{(0)} + \ldots \tag{4.86}$$

We see that the term lacking from the sum in (4.84) reappears in the change of variables provided we identify the Bessel-Fourier coefficient that should accompany $x^{(0)}$ with a. That is, alternatives (4.84) and (4.85) corresponds respectively to integrate over all $c_n' s$ or over all but c_0 which is replaced by a:

$$\text{either} \quad \prod_n dc_n \qquad \text{or} \quad da \prod_{n \neq 0} d\tilde{c}_n \tag{4.87}$$

Of course if we use the measure in the right hand side but with the expansion (4.86) interrupted to order a, nothing has changed. The idea is to employ the exact change of variables (4.85) with that measure:

$$\begin{aligned} q(x_0) &= q_{cl}(x;a) + \sqrt{\hbar}\sum_{n\neq 0}\tilde{c}_n x^{(n)}(x_0) \\ \mathcal{D}q &= da\prod_{n\neq 0} d\tilde{c}_n \end{aligned} \tag{4.88}$$

If we manage to implement this, the integral over c_0, which arised from an approximation (the expansion of the action up to order $\hbar$) and, being not gaussian (the corresponding $S^{(2)}$ eigenvalue was zero) created a problem, now is replaced by an integration over a which will be done exactly.

If one recalls that a represented the "center" of the instanton one can understood why this method is also known as the *collective coordinates* method: a represents a collective coordinate associated with time translation invariance. This collective coordinate should be carefully separated from the rest of the modes (exactly as one separates the motion of the center of mass in ordinary mechanical problems). This is the sense of the Faddev-Popov procedure.

We have stated when studying the harmonic oscillator that, appart from an unimportant normalization constant, we shall identify measures as follows

$$\mathcal{D}x \equiv \prod_n dc_n \tag{4.89}$$

Now we are proposing to use a different measure and then we cannot ignore a possible Jacobian Δ_{FP}:

$$\prod_n dc_n = \Delta_{FP}\, da\prod_{n\neq 0} d\tilde{c}_n \tag{4.90}$$

This Jacobian, usually called the *Faddeev-Popov determinant*, is then defined as

$$\Delta_{FP} = \frac{\partial(c_0, c_{n\neq 0})}{\partial(a, \tilde{c}_{n\neq 0})} \tag{4.91}$$

Clase 5

The Faddeev-Popov determinant

Let us consider the identity

$$1 = \Delta \int_{\infty}^{\infty} \delta \left[F[a] \right] da \tag{5.1}$$

where Δ is such that the equality holds. We shall see that such a formal identity can be used to implement the change of variables (4.58) provided one choses an adequate function $F[a]$ and then $\Delta = \Delta_{FP}$ will act as the Jacobian we were looking for when trying a change of variables in order to handle the zero-mode problem.

We shall assume that F is such that the equation

$$F[a] = 0 \tag{5.2}$$

has just one solution (When this does not happen one faces a *Gribov problem*).

Given Z in the form

$$Z_{i\to f} = \int \mathcal{D}q \exp\left(-\frac{1}{\hbar}S[q]\right) \tag{5.3}$$

we insert identity (5.1) (some people calls this, with some bad faith, the "Faddeev-Popov trick")

$$Z_{i\to f} = \int \mathcal{D}q \int_{-\infty}^{\infty} da \exp\left(-\frac{1}{\hbar}S[q]\right)\Delta\delta \left[F[a] \right] \tag{5.4}$$

We are at the point in which if we chose $F[a]$ adequately, the so-called trick will effectively implement the change of variables that is needed in order to overcome the zero-mode problem. Indeed, consider the following choice for functional $F[a]$

$$F[a] = \int_{-\infty}^{\infty} q(x_0) \frac{1}{\sqrt{S[q_{cl}]}} \dot{q}_{cl}(x_0 - a) dx_0 \tag{5.5}$$

We can easily see that the condition $F[a] = 0$ corresponds, for the choice (5.5), to the enforcement of the orthogonality between the zero mode and $q(x_0)$. Indeed, if we insert expansion (4.54),

$$q(x_0) = q_{cl}(x_0 - a) + \sqrt{\hbar} \sum_{n \neq 0} \tilde{c}_n x^{(n)}(x_0) \tag{5.6}$$

in (5.5), we get

$$\begin{aligned} F[a] &= \int_{-\infty}^{\infty} q_{cl}(x_0 - a) \frac{1}{S[q_{cl}]} \dot{q}_{cl}(x_0 - a) dx_0 + \\ &\sqrt{\hbar} \sum_{n \neq 0} \tilde{c}_n \int_{-\infty}^{\infty} x^{(n)}(x_0) \frac{1}{S[q_{cl}]} \dot{q}_{cl}(x_0 - a) dx_0 \end{aligned} \tag{5.7}$$

Now, the first term in (5.7) can be integrated exactly and vanishes in view of the boundary conditions for q_{cl} (Remember that $S[q_{cl}]$ is a constant, eq.(4.65)). Recalling that $\dot{q}_{cl}$ in the second line of (5.7) can be identified with the zero mode, one ends with

$$F[a] = \frac{\sqrt{\hbar}}{\sqrt{S[q_{cl}]}} \sum_{n \neq 0} \tilde{c}_n \int_{-\infty}^{\infty} x^{(n)}(x_0) x^{(0)}(x_0) dx_0 \tag{5.8}$$

and hence we see that condition (5.2) does enforce orthogonality between non-zero and zero modes.

So we have

$$\begin{aligned} Z_{i \to f} &= \int \mathcal{D}q \int_{-\infty}^{\infty} da \exp\left(-\frac{1}{\hbar} S[q]\right) \Delta \times \\ &\delta\left[\int_{-\infty}^{\infty} q(x_0) \frac{1}{S[q_{cl}]} \dot{q}_{cl}(x_0 - a) dx_0\right] \end{aligned} \tag{5.9}$$

Let us now perform the change of variables

$$\begin{aligned} q(x_0) &\to q(x_0 - a) \\ \mathcal{D}q &\to \mathcal{D}q \end{aligned} \tag{5.10}$$

so that we can write

$$Z_{i\to f} \;=\; \int \mathcal{D}q \int_{-\infty}^{\infty} da \exp\Big(-\frac{1}{\hbar}S[q]\Big)\Delta \times$$
$$\delta\left[\int_{-\infty}^{\infty} q(\tau)\frac{1}{S[q_{cl}]}\dot{q}_{cl}(\tau)d\tau\right] \tag{5.11}$$

where we have called $\tau = x_0 - a$. If we expand q in the "old" basis,

$$q(\tau) = q_{cl}(\tau) + \sqrt{\hbar}\sum_n c_n x^{(n)}(\tau) \tag{5.12}$$

we have

$$Z_{i\to f} = \exp\Big(-\frac{1}{\hbar}S[q_{cl}]\Big)\int_{-\infty}^{\infty} dc_o \prod_{n\neq 0} dc_n da \exp\Big(-\frac{1}{\hbar}S^{(2)}[c_n]\Big)\Delta \times$$
$$\delta\left[\int_{-\infty}^{\infty}\left(q_{cl}(\tau) + \sqrt{\hbar}c_0 x^{(0)}(\tau) + \sqrt{\hbar}\sum_{n\neq 0} c_n x^{(n)}(\tau)\right)x^{(0)}(\tau)d\tau\right] \tag{5.13}$$

One can easily see that the delta function reduces to

$$\delta\left[\int_{-\infty}^{\infty}(q_{cl}(\tau) + \sqrt{\hbar}c_0 x^{(0)}(\tau) + \sqrt{\hbar}\sum_{n\neq 0} c_n x^{(n)}(\tau)x^{(0)}(\tau)d\tau = \right.$$
$$\left. = \delta[\sqrt{\hbar}\int_{-\infty}^{\infty} c_0 {x^{(0)}}^2 d\tau]\right] = \delta[c_0] \tag{5.14}$$

where we have used orthogonality between non-zero and zero modes and the fact that

$$\int_{-\infty}^{\infty} d\tau q_{cl}(\tau)x^{(0)}(\tau) = \frac{1}{\sqrt{S[q_{cl}]}}\int_{-\infty}^{\infty} d\tau q_{cl}(\tau)\dot{q}_{cl}(\tau) = 0 \tag{5.15}$$

and disregarded also a $\sqrt{\hbar}$ factor in the last equality in (5.14). Then, we see that the integral over c_0 can be trivially performed so that (5.13) becomes

$$Z_{i\to f} = \exp\Big(-\frac{1}{\hbar}S[q_{cl}]\Big)\int_{-\infty}^{\infty}\prod_{n\neq 0} dc_n da \exp\Big(-\frac{1}{\hbar}S^{(2)}[c_n]\Big)\Delta \tag{5.16}$$

We see that the c_0 integration has been traded by an a integration so that Δ is in fact the Faddeev-Popov Jacobian, $\Delta = \Delta_{FP}$, that we were looking after. The zero-mode that put us in trouble was associated to translation (in time) invariance but now the Faddeev-Popov method eliminated the problem since the Bessel-Fourier coefficient accompanying this zero mode is no more present in the infinite product of integrations. Instead, we have an integration over the collective coordinate associated with the "center" a of the instanton. We can then integrate safely the non-zero modes getting the product of non-zero eigenvalues finally obtaining

$$\begin{aligned} Z_{i\to f} &= \exp\left(-\frac{1}{\hbar}S[q_{cl}]\right)\int_{-\infty}^{\infty}\Delta_{FP}\prod_{n\neq 0}(\lambda_n)^{-1/2}da \\ &= \exp\left(-\frac{1}{\hbar}S[q_{cl}]\right)\int_{-\infty}^{\infty}\Delta_{FP}\left(\det{}'\hat{S}^{(2)}\right)^{-1/2}da \end{aligned} \tag{5.17}$$

where we indicate with $\det'$ that the zero eigenvalue has been already extracted from the eigenvalue product.

Explicit calculation of the Fadeev-Popov determinant

Let us come back to the identity (5.1),

$$1 = \Delta_{FP}\int_{\infty}^{\infty}\delta\left[F[a]\right]da$$

Our choice for $F[a]$, called the "natural" choice since it implies orthogonality between zero and non-zero modes was (5.5)

$$F[a] = \int_{-\infty}^{\infty} q(x_0)\frac{1}{\sqrt{S[q_{cl}]}}\dot{q}_{cl}(x_0-a)dx_0$$

Now, we know that

$$\delta[F[a]\,] = \left|\frac{dF}{da}\right|^{-1}_{a=a_0}\delta(a-a_0) \tag{5.18}$$

with a_0 the (unique by assumption) zero of F,

$$F[a_0] = 0 \tag{5.19}$$

Inserting (5.19) in the identity (5.1) we have

$$1 = \Delta_{FP} \left| \frac{dF}{da} \right|^{-1}_{a=a_0} \int_{\infty}^{\infty} \delta[a - a_o] da \tag{5.20}$$

or

$$\Delta_{FP} = \left| \frac{dF}{da} \right|_{a=a_0} \tag{5.21}$$

So, Δ_{FP} is the determinant of the operator resulting from differentiation of the "gauge fixing" function $F[a]$ with respect to the parameter associated with the invariance (in this case under translations in time).

We are now ready to explicitly compute Δ_{FP},

$$\Delta_{FP} = \left| \frac{d}{da} \int_{-\infty}^{\infty} q(x_0) \frac{1}{\sqrt{S[q_{cl}]}} \dot{q}_{cl}(x_0 - a) dx_0 \right|_{a=a_0} \tag{5.22}$$

Note that the action at the classical solution is independent of a. Moreover, we have seen that

$$\frac{dq_{cl}(x_0 - a)}{da} = -\dot{q}_{cl}(x_0 - a) \tag{5.23}$$

so that (5.22) can be rewritten as

$$\Delta_{FP} = \frac{1}{\sqrt{S[q_{cl}]}} \left| \int_{-\infty}^{\infty} q(x_0) \ddot{q}_{cl}(x_0 - a) dx_0 \right|_{a=a_0} \tag{5.24}$$

We now expand as in (5.6)

$$\Delta_{FP} = \frac{1}{\sqrt{S[q_{cl}]}} \left| \int_{-\infty}^{\infty} q_{cl}(x_0 - a) \ddot{q}_{cl}(x_0 - a) dx_0 \right|_{a=a_0} + O(\hbar^{1/2}) \tag{5.25}$$

or

$$\Delta_{FP} = \frac{1}{\sqrt{S[q_{cl}]}} \left| \int_{-\infty}^{\infty} \dot{q}_{cl}^2(x_0 - a) dx_0 \right|_{a=a_0} + O(\hbar^{1/2}) = \sqrt{S[q_{cl}]} + O(\hbar^{1/2}) \tag{5.26}$$

At the order we are working, $O(\hbar^{1/2})$ terms can can be disregarded in the Faddeev-Popov Jacobian (5.26), so that we can write for Z,

$$Z_{i\to f} = \exp\left(-\frac{1}{\hbar}S[q_{cl}]\right)\sqrt{S[q_{cl}]}\int_{-\infty}^{\infty}\left(\det{}'S^{(2)}\right)^{-1/2} da \tag{5.27}$$

Gaussian integrations

We have assumed that integrations over the $c_n{}'s$ ($n \neq 0$) were all gaussian and written the product as $\det{}'S^{(2)}$. We have however to show this and, moreover, compute this determinant. Let us remember that the operator $S^{(2)}$ was given by

$$S^{(2)} = -\frac{d^2}{dx_0^2} + V''[q_{cl}] \tag{5.28}$$

with

$$V''[q_{cl}] = -\mu^2 + 3\lambda q_{cl}^2 \tag{5.29}$$

Now, the corresponding eigenvalue problem,

$$S^{(2)}x^{(n)} = \lambda_n x^{(n)} \tag{5.30}$$

with

$$x^{(n)}(t_i) = x^{(n)}(t_f) = 0 \tag{5.31}$$

can be thought as a one-dimensional stationary quantum mechanical problem. Indeed, taking x_0 as a space coordinate, eq.(5.30) can be identified with the time-independent Schrödinger equation for a particle in a potential V''. We know that the lowest eigenfunction for such a problem [7] has no nodes, the next to the lowest one node and so on. We already know an eigenfunction, the zero-mode,

$$x^{(0)} = \frac{1}{\sqrt{S[q_{cl}]}}\dot{q}_{cl}(x_0) \tag{5.32}$$

But, since q_{cl} is a monotonic function, the zero-mode (5.32) has no nodes and this is then the eigenfunction corresponding to the lowest eigenvalue. Then, all other eigenvalues are positive definite and the path-integrals over the $c_n{}'s$ are all gaussians.

It remains now to compute the determinant of $S^{(2)}$.

How to compute determinants

The analysis that follows is closely related to the so called "Node's theorem" in Quantum mechanics[2]. Consider the equation [15]

$$D\psi = \lambda\psi \tag{5.33}$$

where

$$D = [-\partial_t^2 + \omega(t)] \tag{5.34}$$

with $\omega(t)$ some bounded function of t. Let us call $\psi_\lambda(t)$ the solution of equation (5.33) obeying the initial conditions

$$\begin{aligned} \psi_\lambda(-T/2) &= 0 \\ \partial_t\psi_\lambda(-T/2) &= 1 \end{aligned} \tag{5.35}$$

Now, when the operator D acts on functions that vanish at $\pm T/2$, it has an eigenvalue λ_n if and only if $\psi_{\lambda_n}(+T/2) = 0$.

Indeed if, for a given $\bar{\lambda}$, $\psi_{\bar{\lambda}}$ not only satisfies condition (5.35) but also vanishes at $t = T/2$, then it is one of the eigenfunctions with eigenvalue $\lambda_n = \bar{\lambda}$. Conversely, we shall consider that any eigenfunction vanishing at $t = \pm T/2$ has a positive derivative at $t = -T/2$ (the negative case can be treated in the same way).

In summary, taking λ as a parameter, one has a large number of values of λ for which there exists solutions of (5.33) satisfying conditions (5.35). When one adds the condition that the solution should also vanish at $t = T/2$, the space of solutions is smaller: the possible values of the corresponding $\lambda's$ becomes in fact discrete and the smaller eigenvalue will be called λ_0 ($\lambda_0 > 0$).

Now, consider two functions $\omega^{(1)}(t)$ and $\omega^{(2)}(t)$ and let us call $\psi_\lambda^{(1)}$ and $\psi_\lambda^{(2)}$ the corresponding solutions of eq.(5.33). One can prove that [16]

$$\frac{\det[-\partial_t^2 + \omega^{(1)}(t) - \lambda]}{\det[-\partial_t^2 + \omega^{(2)}(t) - \lambda]} = \frac{\psi_\lambda^{(1)}(T/2)}{\psi_\lambda^{(2)}(T/2)} \tag{5.36}$$

[2] The Node's theorem is valid for the discrete spectrum of quantum systems in $d = 1$ dimensional space. It states that if the potential is bounded, the $n + 1$ eigenfunction has n zeroes.

Proof : The l.h.s. is a meromorphic function of λ with a simple zero at each $\lambda_n^{(1)}$ and a simple pole at each $\lambda_n^{(2)}$. According to the analytic Fredholm theorem it goes to 1 for $\lambda \to \infty$ in any direction (except along the real positive axis). The r.h.s. is a meromorphic function with exactly the same zeros and poles and it also goes to 1 for $\lambda \to \infty$, except along the real positive axis, as can be seen from elementary differential equations theory. Thus the ratio of the two sides is an analytic function of λ that goes to 1 for $\lambda \to \infty$ (except along the real positive axis). That is to say, it is 1. Q.E.D.

Let us define a quantity N by the equation

$$\frac{\det[-\partial_t^2 + \omega(t)]}{\psi_0(T/2)} = \pi\hbar N^2 \tag{5.37}$$

In view of eq.(5.36), such quantity N has to be independent of ω. It then takes the same value for constant ω (an harmonic oscillator) or for any other ω. But for constant ω one can easily find a solution of eq.(5.33) with $\lambda = 0$ and satisfying boundary conditions (5.34),

$$\psi_0^{osc} = \frac{\sinh(\sqrt{\omega}(t + T/2))}{\sqrt{\omega}} \tag{5.38}$$

(This solution is obtained by Wick rotation of the "normal one".) Then, for the harmonic oscillator we can write eq.(5.37) in the form

$$N^{-1}\det{}^{1/2}\left[-\partial_t^2 + \omega\right] = \left(\frac{\pi\hbar \sinh(\sqrt{\omega}(t + T/2))}{\sqrt{\omega}}\right)^{1/2} \tag{5.39}$$

The idea is now to profit from the knowledge of a zero eigenvalue ψ_0^{ins} in the instanton case, that is when the "potential" is V'' , to write a formula analogous to (5.39). Then, since N is independent of the potential, we will be able to express the determinant in the instanton case in terms of known functions and the determinant for the harmonic oscillator just by making the ration between eq.(5.39) and the analogous for the instanton case.

The zero-mode we know in the instanton case is:

$$x^{(0)} = \frac{1}{\sqrt{S_{cl}}}\frac{dq_{cl}}{dt} \tag{5.40}$$

The interval for this solution is $(-\infty, +\infty)$ but we shall consider the interval $(-T/2, +T/2)$. When the interval is finite zero modes do not exist (invariance under translations in time does no more hold in finite intervals). There will exist, instead, a lowest eigenvalue λ_0. Now, if we consider the complete product of eigenvalues, divide it by the lowest one and then take the limit $T \to \infty$ we shall be calculating the det$'$ determinant we were looking for. The behavior of the zero mode (5.40) is

$$\lim_{t\to\pm\infty} x^{(0)} = A\exp(-2q_o|t|) \tag{5.41}$$

One knows that there exist a second linear independent solution $\bar{x}^{(0)}$. One easily finds (for example by integrating the expression for the wronskian that has to be a constant)

$$\lim_{t\to\pm\infty} \bar{x}^{(0)} = \pm A\exp(2q_0|t|) \tag{5.42}$$

Combining $x^{(0)}$ and $\bar{x}^{(0)}$ we can get a solution satisfying conditions (5.35). The answer is, for large T

$$\psi_0^{ins}(t) \approx \frac{1}{2A}\left(\exp(q_0T)x^{(0)} + \exp(-q_oT)\bar{x}^{(0)}\right) \quad \text{for large } T \tag{5.43}$$

We need now the lowest eigenvalue of the problem in the finite interval. That is, we have to analyse the case of small λ for which an iterative process in the equivalent integral equation problem works. The answer for the solution is

$$\begin{aligned}\psi_\lambda(t) &= \psi_0^{ins} - \lambda(2A^2)^{-1}\int_{-T/2}^{t} dt'[\bar{x}^{(0)}(t)x^{(0)}(t') - \\ &\qquad \bar{x}^{(0)}(t')x^{(0)}(t)]dt' + O(\lambda^2)\end{aligned} \tag{5.44}$$

Then, at $t = T/2$ we can write, in view of eq.(5.43),

$$\begin{aligned}\psi_\lambda(t) &= 1 - \lambda(4A^2)^{-1}\int_{-T/2}^{T/2} dt'[\exp(q_0T)x^{(0)2}(t') - \\ &\qquad \exp(-q_0T)\bar{x}^{(0)2}(t')]dt' + O(\lambda^2)\end{aligned} \tag{5.45}$$

For large T it is the first term in the integral that dominates and one has

$$\psi_\lambda(T/2) = 1 - \lambda(4A^2)^{-1}\exp(q_0T) \tag{5.46}$$

The only way in which this is zero at $\lambda = \lambda_0$ is if

$$\lambda_0 = 4A^2 \exp(-q_0 T) \tag{5.47}$$

So, finally, we can make the ratio of determinants, divide by the lowest eigenvalue and take the limit of $T \to \infty$. We get

$$\frac{\det'[-\partial_t^2 + V'']}{\det[-\partial_t^2 + \omega^{(2)}]} = \frac{\psi_0^{ins}(T/2)}{\lambda_0 \exp(q_0 T)} = \frac{1}{2A^2} \tag{5.48}$$

Let us remember that the constant A can be determined by looking at the classical action (see eq.(5.40)) and the definition of A given through eq.(5.41). One finds

$$A = \frac{1}{4q_0} \tag{5.49}$$

Clase 6

Multi-instantons

We have been able to write the one instanton contribution to the transition amplitude $Z_{i\to f}$ (eq.(5.27) in the form

$$Z^{(1)}_{i\to f} = \exp\left(-\frac{1}{\hbar}S[q_{cl}]\right)\sqrt{S[q_{cl}]}\int_{-\infty}^{\infty}\left(\det{}'S^{(2)}\right)^{-1/2} da \tag{6.1}$$

This time, we have added a superscript (1) to indicate explicitly that formula (6.1) is giving the one-instanton contribution to Z.

Now, suppose that instead of $Z_{i\to f}$, one needs the $\langle q(T)q(0)\rangle$ defined as

$$\langle q(T)q(0)\rangle = \frac{1}{Z}\int \mathcal{D}q \exp\left(-\frac{1}{\hbar}S[q]\right) q(T)q(0) \tag{6.2}$$

Here, for notation simplicity, we omit the subscript $i \to f$ in the partition function which is just written as Z,

$$Z = \int \mathcal{D}q \exp\left(-\frac{1}{\hbar}S[q]\right) \tag{6.3}$$

In principle, the path-integrals one has to perform in QFT include contributions from the "0-instanton" sector, the "1-instanton" sector, the "2-instanton" sector, etc, whenever these solutions exist. This means, in our quantum mechanics example, that we should include such configurations in (6.2), both in the numerator and the denominator. One possibility is to consider Z as s sum over different instanton sectors,

$$Z = \sum_{n=0,\pm1,\ldots} Z^{(n)} \tag{6.4}$$

Equation (6.1) just represents a given 1-instanton ($n = 1$) contribution to this sum.

It is convenient to stop at this point in order to comment the sense of what we are calling $n = 0, \pm 1, \ldots$ instanton sectors. Consider the non trivial solution that we called in the precedent section the instanton. It corresponds to a solution which interpolates smoothly between $-q_0$ at $x_0 = -\infty$ and $+q_0$ at $x_0 = +\infty$. We can make manifest the dependence of such a solution on its border properties by defining the following local quantity associated to any configuration $q(x_0)$ (not necessarily a solution)

$$\rho(x_0) = \frac{1}{2q_0} \frac{dq}{dx_0} \tag{6.5}$$

Although this quantity can depend on the local properties of the configuration q, the following integral just depends on its border values

$$Q[q] = \int_{-\infty}^{+\infty} dx_0 \rho(x_0) = \frac{1}{2q_0} (q(+\infty) - q(-\infty)) \tag{6.6}$$

Any configuration satisfying the same boundary conditions as the instanton solution we discussed will have the same value for Q which we have normalized to 1

$$Q[\text{instanton solution}] = 1 \tag{6.7}$$

This value does not depend on local details but on the global properties of the configuration. It can be seen as an example of a topological quantity. A smooth local deformation of the instanton solution will be of course no more an exact solution of the eq. of motion but will still have the same topological charge.

Now, suppose we want to classify solutions of the classical eqs. of motion of the anharmonic oscillator according to the value that Q takes for such solutions. Now, solutions, depend on the imposed boundary conditions which precisely determine the value of Q. If, as we did in the previous discussion, we seek for solutions which, at $T \to \pm\infty$ correspond to a minimum of the potential, the boundary conditions are

$$\begin{aligned} \lim_{T \to -\infty} q(T) &= \pm q_0 \\ \lim_{T \to +\infty} q(T) &= \pm q_0 \end{aligned} \tag{6.8}$$

Then, there are just three possible values of Q. Either $Q = 0$ (for $q(-\infty) = q(+\infty)$) or $Q = \pm 1$ (for $q(-\infty) = -q(+\infty)$). Hence, in the present quantum mechanical example, with the specified boundary conditions, we have just three "topologically distinct" sectors.

Let us now compute the 0-instanton sector contribution to Z. To this end, we chose as classical representative of such a sector the two trivial solutions $q(t) = \pm q_0$ and make them enter into play through a change of path-integral variables in (6.3) of the form

$$q(x_0) = \pm q_0 + \sqrt{\hbar} y(x_0) = \pm q_0 + \sqrt{\hbar} \sum_n c_n^0 y_n^0 \tag{6.9}$$

Here y_n^0 are, for convenience, eigenfunctions of the quadratic form resulting from the expansion around the trivial solution. Writing $q(x_0)$ in this form, we are sure that the integral we shall compute, up to quadratic terms in y, is restricted to the 0-instanton sector since y represent small fluctations that cannot change the topological $Q[\pm q_0] = 0$ charge of the classical part.

Once the shift (6.9) is performed in (6.3) one ends with

$$Z^{(0)} = \int \prod_n dc_n^0 \exp\left(- \sum_n {c_n^0}^2 \lambda_n^{(0)} \right) = \det{}^{-1/2} S''[\pm q_0] \equiv \frac{A}{2} \tag{6.10}$$

A 1-instanton contribution to Z is that given by eq.(6.1),

$$\begin{aligned} Z^{(1)} &= \exp\left(-\frac{1}{\hbar} S[q_{cl}]\right) \sqrt{S[q_{cl}]} \int_{-\infty}^{\infty} \left(\det{}' S^{(2)}\right)^{-1/2} da \\ &\equiv \exp\left(-\frac{1}{\hbar} S[q_{cl}]\right) \sqrt{S[q_{cl}]} B \int_{-\infty}^{\infty} da \end{aligned} \tag{6.11}$$

(We assumed that the 1-instanton determinant is a independent, as it can be proved, see below).

We can then write

$$Z = A + B \exp\left(-\frac{1}{\hbar} S[q_{cl}]\right) \sqrt{S[q_{cl}]} \int_{-\infty}^{\infty} da + \ldots \tag{6.12}$$

Concerning the numerator in (6.2), we can write analogously

$$\mathrm{Num} = \mathrm{q_0^2 A} + \mathrm{B} \exp\left(-\frac{1}{\hbar} \mathrm{S[q_{cl}]}\right) \sqrt{\mathrm{S[q_{cl}]}} \int_{-\infty}^{\infty} \mathrm{q_{cl}(0,a) q_{cl}(T,a) da} + \ldots \tag{6.13}$$

Then, calling

$$R = \frac{B}{A} \exp\left(-\frac{1}{\hbar} S[q_{cl}]\right) \sqrt{S[q_{cl}]} \tag{6.14}$$

we have for $\langle q(T)q(0)\rangle$,

$$\langle q(T)q(0)\rangle = \frac{q_0^2 + R\int_{-\infty}^{\infty} q_{cl}(0,a)q_{cl}(T,a)da + \ldots}{1 + R\int_{-\infty}^{\infty} da + \ldots} \tag{6.15}$$

In the semiclassical limit we can write

$$\langle q(T)q(0)\rangle \approx (q_0^2 + R\int_{-\infty}^{\infty} q_{cl}(0,a)q_{cl}(T,a)da + \ldots)(1 - R\int_{-\infty}^{\infty} da + \ldots) \tag{6.16}$$

or

$$\langle q(T)q(0)\rangle \approx q_0^2 \left(1 - R\int_{-\infty}^{\infty} da(1 - \frac{q_{cl}(0,a)}{q_0}\frac{q_{cl}(T,a)}{q_0}\right) \tag{6.17}$$

Using the explicit form for the 1-instanton solution we can write or

$$\langle q(T)q(0)\rangle \approx q_0^2 \left(1 - RT\int_{-\infty}^{\infty} dx(1 - \tanh\left(\frac{\mu x}{\sqrt{2}}\right)\tanh\left(\frac{\mu(x-1)}{\sqrt{2}}\right)\right) \tag{6.18}$$

(We have made $a/T = x$).

In order to write this result more compactly let us define

$$D = R\exp\left(\frac{S[q_{cl}]}{\hbar}\right)\int_{-\infty}^{\infty} \ldots dx \tag{6.19}$$

so that (6.18) becomes

$$\begin{aligned} \langle q(T)q(0)\rangle &\approx q_0^2 \left(1 - DT\exp\left(-\frac{S[q_{cl}]}{\hbar}\right)\right) \\ &= q_0^2 \left(1 - DT\exp\left(-\frac{2\sqrt{2}\mu^3}{\hbar\lambda}\right)\right) \end{aligned} \tag{6.20}$$

We can see already at this point that, although $S[q_0] < S[q_{cl}]$, as $T \to \infty$ the 1-instanton contribution grows bigger and bigger and at a certain time T_{inst}, it becomes dominant. This happens when

$$1 - DT_{inst}\exp\left(-\frac{2\sqrt{2}\mu^3}{\hbar\lambda}\right) \approx 0 \tag{6.21}$$

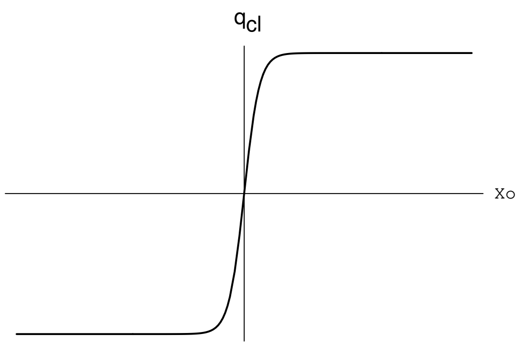
or

$$T_{inst} \approx \frac{1}{D} \exp\left(-\frac{2\sqrt{2}\mu^3}{\hbar\lambda}\right) \tag{6.22}$$

Let us now try to include multi-instanton contributions. This means, contribution coming from configurations which can be thought as superposition of instanton ($Q = +1$) and anti-instanton ($Q = -1$) solutions.

Visualizing multi-instantons

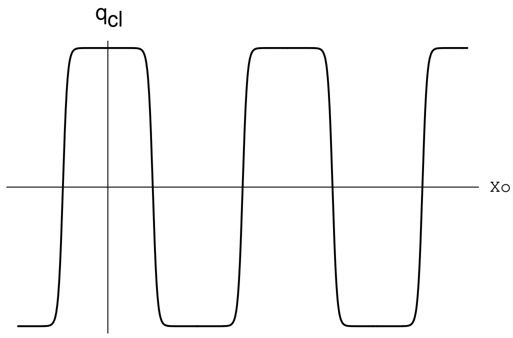
The shape of the 1-instanton solution we were considering up to now is



The exact 1-instanton solution

It is clear that the $Q = 1$-instanton solution is "localized" (in time) over an extension $\Delta = 1/\omega$, where $\omega = \sqrt{2\mu}$ is the frequency associated with the anharmonic oscillator ($\omega^2 = V''[q_0]$). It is just in this region that the solution differs from the trivial constant one.

Now, we can imagine that other $Q = 1$-instantons should exist. Even without knowing the "multi-instanton" solution exactly, we can guess that its shape should be that shown in next page:



An approximate multi-instanton solution

Note that the topological charge of the multi-instanton represented in the figure is still $Q = +1$. It results from the sum of 3 instantons and 2 anti-instantons and its charge is computed as $Q = 1-1+1-1+1 = +1$. Thats is the reason why it is called a “multi” instanton.

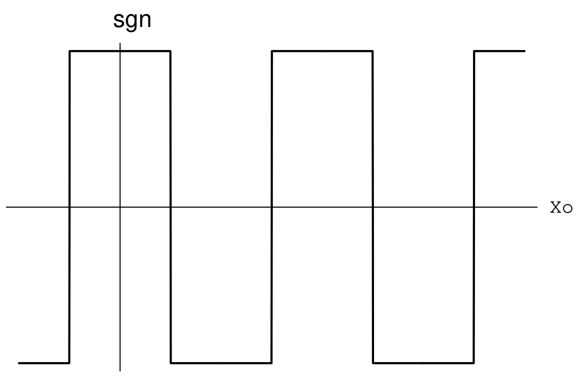
We shall see at the end of this discussion that, even when one knows the exact multi-instanton solution, within the order one is working, the approximate solution is enough. That is the reason why we shall still simplify our calculation by replacing hyperbolic tangents by sign functions

$$\tanh\left(\frac{\mu}{\sqrt{2}}(x_0 - a)\right) \approx \mathrm{sgn(x_0 - a)} \tag{6.23}$$

With this approximation, the multi-instanton solution can be written as

$$q_{cl}^{(n)} \approx q_0 \prod_{j=1}^{2k+1=n} \mathrm{sgn}(x_0 - a_j) \tag{6.24}$$

and it looks like



A superposition of sign functions

One can easily compute the action for such an approximate multi-instanton solution

$$S[q_{cl}^n] = nS[q_{cl}^1] + O\left(\exp(-\mu(a_i - a_j)\right) \tag{6.25}$$

where $a_i - a_j$ represents the "distance" between the two instanton centers a_i and a_j.

Then, if we want to compute the contribution of <u>all</u> multi-instanton solutions to the amplitude for a transition between the initial state $|q_i = -q_o\rangle$ and the final state $|q_f = +q_0\rangle$ we have to consider the following sum of path-integrals

$$\langle q_f = q_0|\exp\left(-\frac{HT}{\hbar}\right)|q_i = -q_o\rangle = \mathcal{N}\sum_{n\ \mathrm{odd}} \exp\left(-\frac{n}{\hbar}S[q_{cl}^1]\right. \\ \left.\int da_1 da_2 \dots da_n \det{}'^{-1/2} S''[q_{cl}^n]\right) \tag{6.26}$$

Were the determinant absent, calculation of the $a's$ integrals is trivial

$$\begin{aligned}
I_0 &= \int_{t_i}^{t_f} da_1 \int_{t_i}^{a_1} da_2 \int_{t_i}^{a_2} da_3 \dots \int_{t_i}^{a_{n-2}} da_{n-1} \int_{t_i}^{a_{n-1}} da_n \\
&= \int_{t_i}^{t_f} da_1 \int_{t_i}^{a_1} da_2 \int_{t_i}^{a_2} da_3 \dots \int_{t_i}^{a_{n-2}} da_{n-1}(a_{n-1} - t_i) \\
&= \int_{t_i}^{t_f} da_1 \int_{t_i}^{a_1} da_2 \int_{t_i}^{a_2} da_3 \dots \frac{1}{2}(a_{n-2} - t_i)^2 \\
&= \dots \\
&= \int_{t_i}^{t_f} \frac{1}{(n-1)!} da_1 (a_1 - t_i)^{n-1} = \frac{1}{n!}(t_f - t_i)^n
\end{aligned} \tag{6.27}$$

Now we have to take into account the determinant of the operator S'',

$$S'' = -\frac{d^2}{dx_0^2} + V''[q_{cl}^n] \tag{6.28}$$

Except for the regions where the function q_{cl} changes from $\pm q_0$ to $\mp q_0$, $V''[q_{cl}^n]$ just coincides with the corresponding value for the harmonic oscillator, namely $V'' = \omega^2$. We have already calculated the determinant for this last potential finding, in Minkowski space (see eq.(3.24)

$$F[T = t_f - t_i] \equiv \det{}^{-1/2} S''[V_{har}] = \left(\frac{m\omega}{2\pi i\hbar \sin(\omega T)}\right)^{\frac{1}{2}} \tag{6.29}$$

so that rotating to Euclidean space and taking the $T \to \infty$ limit

$$F[T] = \sqrt{\frac{m\omega}{\pi\hbar}} \exp(-\frac{\omega T}{2}) \tag{6.30}$$

We shall compute as a problem the corresponding expression for the actual problem, i.e., when the multi-instanton determinant is included. Here, we shall make an approximation which starts by writing

$$\det{}'^{-1/2} S''[q_{cl}^m] = \left(\frac{m\omega}{\pi\hbar}\right)^{1/2} \exp\left(\frac{-\omega T}{2}\right) K^n \tag{6.31}$$

where K^n is a factor connecting the harmonic oscillator result with that corresponding to the instanton result. We have then

$$\begin{aligned} \langle -q_0 | \exp\left(-\frac{HT}{\hbar}\right| + q_o\rangle &= \sum_{n \text{ odd}} \exp\left(-\frac{n}{\hbar} S[q_{cl}^1]\right) \frac{T^n}{n!} \left(\frac{m\omega}{\pi\hbar}\right)^{1/2} K^n \\ & \exp\left(-\frac{\omega T}{2}\right) \end{aligned} \tag{6.32}$$

Analogously, one finds for the transition amplitude from $|q_i = +q_0\rangle$ to $|q_f = -q_0\rangle$ the same value

$$\begin{aligned} \langle +q_0 | \exp\left(-\frac{HT}{\hbar}\right| - q_o\rangle &= \sum_{n \text{ odd}} \exp\left(-\frac{n}{\hbar} S[q_{cl}^1]\right) \frac{T^n}{n!} \left(\frac{m\omega}{\pi\hbar}\right)^{1/2} K^n \\ & \exp\left(-\frac{\omega T}{2}\right) \end{aligned} \tag{6.33}$$

In contrast, when the transition corresponds to $|q_i = \pm q_0\rangle$ going to $|q_f = \pm q_0\rangle$ one gets a sum over even $n's$,

$$\langle \pm q_0| \exp\left(-\frac{HT}{\hbar}\right| \pm q_o\rangle = \sum_{n \text{ even}} \exp\left(-\frac{n}{\hbar}S[q_{cl}^1]\right)\frac{T^n}{n!}\left(\frac{m\omega}{\pi\hbar}\right)^{1/2} K^n \exp\left(-\frac{\omega T}{2}\right) \tag{6.34}$$

We know that the ground state has to be symmetric under parity. Then, it has to be constructed from a combination of the non-symmetric states $|\pm q_0\rangle$ (the same is true for the antisymmetric first excited state)

$$|\pm\rangle = \frac{1}{\sqrt{2}}(|+q_0\rangle \pm |-q_0\rangle) \tag{6.35}$$

One easily finds, for the ground-state to ground-state amplitude

$$\langle +| \exp\left(-\frac{HT}{\hbar}\right|+\rangle = \left(\frac{2m\omega}{\pi\hbar}\right)^{1/2} \exp\left(-\left(\frac{\omega}{2} - \exp(-\frac{S[q_{cl}]}{\hbar})K\right)T\right) \tag{6.36}$$

In the limit $T \to \infty$ one can read from the coefficient in front of T in the argument of the exponential, the lowest energy of the system i.e. the ground state energy E_0

$$\frac{E_0}{\hbar} = \frac{\omega}{2} - K\exp\left(-\frac{S[q_{cl}]}{\hbar}\right) \tag{6.37}$$

For the first excited state one has to evaluate $\langle -| \exp\left(-\frac{HT}{\hbar}\right| -\rangle$ getting for the corresponding energy E_1

$$\frac{E_1}{\hbar} = \frac{\omega}{2} + K\exp\left(-\frac{S[q_{cl}]}{\hbar}\right) \tag{6.38}$$

Then, the energy gap $\Delta E = E_1 - E_0$ is given by

$$\Delta E = 2\hbar K \exp\left(-\frac{S[q_{cl}]}{\hbar}\right) = 2\hbar K \exp\left(-\frac{2\sqrt{2}\mu^3}{3\lambda\hbar}\right) \tag{6.39}$$

Note that this is a non-perturbative result (the coupling constant λ appears as $1/\lambda$ in the exponential argument). Concerning K, there are two ways for computing it. The longer one is just to evaluate the determinant for operator S''. This is left as a problem. A shorter way is based in the following trick: we have seen tha in computing transition amplitudes we were summing over 1-instanton contribution, 3-instanton contribution, etc,

$$\langle +q_0 | \exp\left(-\frac{HT}{\hbar}\right| - q_o\rangle = 1\!-\!\text{inst} + 3\!-\!\text{inst} + \ldots \tag{6.40}$$

Looking at the first term in the series, we see that

$$1\!-\!\text{inst} = \left(\frac{m\omega}{\pi\hbar}\right)^{1/2} KT \exp\left(-\left(\frac{\omega T}{2} + \frac{S[q_{cl}]}{\hbar}\right)\right) \tag{6.41}$$

On the other hand, the 1-instanton contribution includes the determinant in a 1-instanton background whose calculation was described in the precedent section. The asnwer for the determinant was:

$$\frac{\det'(S''[q^1_{cl}])}{\det(-\partial_t^2 + \omega^2)} = \frac{8\mu^2}{\lambda} \tag{6.42}$$

From eqs.(6.41)-(6.42) we then have

$$K = \frac{1}{4}\sqrt{\frac{2\sqrt{2}\mu}{3\pi}} \tag{6.43}$$

and

$$\Delta E = \sqrt{\frac{\hbar\mu}{\sqrt{2}\pi}} \exp\left(-\frac{2\sqrt{2}\mu^3}{3\lambda\hbar}\right) \tag{6.44}$$

We have promised to discuss why is it enough to consider approximate multi-instanton solutions (and not exact ones). In our calculation instantons and anti-instantons were widely separated and hence superposed approximate 1-instanton solutions were employed. This corresponds to a dilute gas approximation. Of course, for close configurations non-linearities become important. Now, one can prove *a posteriori* that such an approximation is justified. Indeed, consider the amplitude

$$\langle -q_0 | \exp(-HT/\hbar| + q_0\rangle = \mathcal{A} \sum_{n \text{ odd}} \left(KT \exp\left(-S[q_{cl}]/\hbar\right)\right)^n (1 + O(\hbar)) \tag{6.45}$$

$$\mathcal{A} = \left(\frac{m\omega}{\pi\hbar}\right)^{1/2} \exp(-\omega T/2) \tag{6.46}$$

Then, calling

$$\kappa = KT \exp\left(-S[q_{cl}]/\hbar\right) \tag{6.47}$$

we have a series of the form

$$S = \sum \frac{\kappa^n}{n!} \tag{6.48}$$

Now, the terms in the sum grow with n until $n \approx \kappa$; after this point, they begin to decrease rapidly. That is, relevant terms correspond to $n \leq \kappa$ or

$$\text{density} \equiv \frac{n}{T} \leq K\left(-S[q_{cl}]/\hbar\right) \tag{6.49}$$

Then, for small $\hbar$ the density of instantons is exponentially small or, what is the same, their separation is large.

Clase 7

Path-integral formulation for field theories

We have seen that the path-integral defining Z, the partition function that describes the behavior of a a quantum mechanical system with one degree of freedom was given by (see eq.(1.47))

$$Z_{i\to f} = \langle q_f t_f | q_i t_i \rangle = \int \mathcal{D}p\mathcal{D}q \exp\left(\frac{i}{\hbar}\int_{t_i}^{t_f} (p\dot{q} - h(p,q))dt\right) \qquad (7.1)$$

where the path-integral measure was defined as

$$\mathcal{D}p\mathcal{D}q = \lim_{N\to\infty}\lim_{\epsilon\to 0}\left(\prod_{i=1}^{N-1} dq_i \frac{dp_i}{2\pi\hbar}\right)\frac{dp_N}{2\pi\hbar} \qquad (7.2)$$

with the exponential in the integrand interpreted as

$$S = S_{N,N-1} + S_{N-1,N-2} + \ldots \qquad (7.3)$$

$$S_{j,j-1} = \epsilon\left(p_j \frac{q_j - q_{j-1}}{\epsilon} - h(p_j, q_j)\right) \qquad (7.4)$$

If the system has n degrees of freedom, labeled by an index α ($\alpha = 1, 2, \ldots, n$), one has, instead of eqs.(7.1)-(7.4),

$$\begin{aligned} Z_{i\to f} &= \lim_{N\to\infty}\lim_{\epsilon\to 0}\prod_{\alpha=1}^{n}\prod_{i=1}^{N-1}\left(\int dq_i^\alpha \frac{dp_i^\alpha}{2\pi\hbar}\right)\frac{dp_N^\alpha}{2\pi\hbar} \\ &\quad \exp\left(\frac{i}{\hbar}\sum_{(j,\alpha)}^{(N,n)}(p_j^\alpha(q_j^\alpha - q_{j-1}^\alpha) - \epsilon h(p_j^\alpha, q_j^\alpha))\right) \end{aligned} \qquad (7.5)$$

or, compactly

$$Z_{i\to f} = \langle q_f t_f | q_i t_i \rangle = \int \mathcal{D}p^\alpha \mathcal{D}q^\alpha \exp\left(\frac{i}{\hbar}\int_{t_i}^{t_f} (p^\alpha \dot{q}^\alpha - h(p^\alpha, q^\alpha))dt\right) \tag{7.6}$$

which is nothing but expression (7.1) but extended to a system with n degrees of freedom.

What about a system with infinite many degrees of freedom, like a field theory? For simplicity, we start with a scalar theory with Lagrangian density $\mathcal{L}$ defined as

$$\mathcal{L} = \mathcal{L}(\phi, \partial_\mu \phi) \tag{7.7}$$

with $\partial_\mu = \partial/\partial x^\mu$ and $(x^\mu) = (t, \vec{x})$ $(\mu = 0, 1, 2, 3))$.

We shall call Lagrangian the integral of the Lagrangian density over space (which we took for the moment to be R^3)

$$L \equiv \int d^3x \mathcal{L}(\phi, \partial_\mu \phi) \tag{7.8}$$

so that the action is

$$S = \int_{t_i}^{t_f} dt L = \int_{t_i}^{t_f} dt \int d^3x \mathcal{L}(\phi, \partial_\mu \phi) \tag{7.9}$$

As a shorthand we shall write

$$S = \int_M d^4x \mathcal{L}(\phi, \partial_\mu \phi) \tag{7.10}$$

remembering that the four dimensional integral corresponds to a manifold $M = T \times R^3$ $(T = (t_i, t_f))$. Eventually, we shall make $t_i = -\infty$ and $t_f = \infty$ and then the manifold will be just the usual 4-dimensional Minkowski space.

The Euler-Lagrange equations of motion read

$$\partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)}\right) = \frac{\partial \mathcal{L}}{\partial \phi} \tag{7.11}$$

Concerning the conjugate momentum Π and the Hamiltonian density H one has

$$\Pi = \frac{\partial \mathcal{L}}{\partial \dot{\phi}} \tag{7.12}$$

and

$$H = \Pi\dot{\phi} - \mathcal{L} \tag{7.13}$$

We can at this point apply what we have learnt for the quantum-mechanical problem with partition function (7.6). To this end, we consider in space R^3, cubical cells of volume ϵ^3. Then, we define

$$q^\alpha(t) \equiv \frac{1}{\epsilon^3}\int_{V_\alpha} d^3x\phi(\vec{x}, t) \tag{7.14}$$

where V_α represents the manifold defined by the α^{th}-cell. Each q^α can be seen as the "mean-value" of ϕ over V_α. We then consider a time partition also with size ϵ so that space-time is now divided in N^4 cubical cells. Then, Lagrangian L is written as

$$L = \int d^3x\mathcal{L} = \sum_\alpha \epsilon^3\mathcal{L}^\alpha(q^\alpha, q^{\alpha+\vec{d}}, \dot{q}^\alpha) \tag{7.15}$$

The "conjugate momentum" to q^α is then

$$p^\alpha(t) = \frac{\partial L}{\partial \dot{q}^\alpha} = \epsilon^3\frac{\partial \mathcal{L}^\alpha}{\partial \dot{q}^\alpha} = \epsilon^3\Pi^\alpha(t) \tag{7.16}$$

The Hamiltonian density is then given by

$$H = \sum_\alpha \epsilon^3(\Pi^\alpha\dot{q}^\alpha - \mathcal{L}^\alpha) = \sum_\alpha \epsilon^3 H^\alpha \tag{7.17}$$

Now equation (7.6) can be used just by identifying Π^α with p^α, noting that the sum over cells, $\sum_\alpha$, becomes in the limit $\epsilon \to 0$ an integral over the space manifold, $\int d^3x$. Then, instead of formula (7.6) we have, for the partition function of the scalar field theory

$$Z = \int \mathcal{D}\Pi\mathcal{D}\phi \exp\left(\frac{i}{\hbar}\int_M d^4x(\Pi(x)\dot{\phi}(x) - H(\Pi, \phi))\right) \tag{7.18}$$

(Here $(x) = (t, \vec{x})$).

We can take as initial state at $t_i = -\infty$ the state of least energy for the system, i.e. the "vacuum state". Again at $t_f = \infty$ we shall consider that the system is in its vacuum state (unique, for the moment). Then Z can be interpreted as the vacuum to vacuum amplitude for the system with Lagrangian L and Hamiltonian H. Adding a source J one defines the generating functional $Z[j]$,

$$\begin{aligned} Z[j] &= \int \mathcal{D}\Pi\mathcal{D}\phi \exp\left(\frac{i}{\hbar}\int_M d^4x(\Pi(x)\dot{\phi}(x) - H(\Pi,\phi))\right) \\ & \quad \exp\left(\frac{i}{\hbar}\int_M d^4x j(x)\phi(x)\right) \end{aligned} \tag{7.19}$$

After differentiation with respect to $j(x)$ one can compute vacuum expectation values of products of fields and, from these, physical quantities like scattering amplitudes.

Let us now copy the steps that led us in the case of just one degree of freedom, from the Hamiltonian formulation of Z to the corresponding Lagriangian formulation. Suppose the Hamiltonian for the scalar field theory is quadratic in the momenta

$$H = \frac{1}{2}\Pi^2 + \frac{1}{2}(\partial_i\phi)^2 + V[\phi] \tag{7.20}$$

with $\vec{x} = (x^i)$, $(i = 1, 2, 3)$. Then, as in the quantum mechanical case, we can factor out a gaussian integral over momenta and end with a path integral over "coordinates", in this case <u>over fields</u>. Indeed, the integral over momenta reads

$$\int \mathcal{D}\Pi \exp\left(-\frac{i}{2\hbar}\int d^4x(\Pi^2 - 2\Pi\dot{\phi})\right) = \exp\left(\frac{i}{2\hbar}\int d^4x\dot{\phi}^2\right)\mathcal{N} \tag{7.21}$$

with

$$\mathcal{N} = \int \mathcal{D}\Pi \exp\left(-\frac{i}{2\hbar}\int d^4x(\Pi - \dot{\phi})^2\right) = \int \mathcal{D}\Pi \exp\left(-\frac{i}{2\hbar}\int d^4x\Pi^2\right) \tag{7.22}$$

With this, $Z[j]$ reads

$$Z[j] = \int \mathcal{D}\phi \exp\left(\frac{i}{\hbar}\int_M d^4x(\frac{1}{2}\partial_\mu\phi\partial^\mu\phi - V[\phi] + j\phi)\right) \tag{7.23}$$

or

$$Z[j] = \int \mathcal{D}\phi \exp\left(\frac{i}{\hbar}\int_M d^4x(\mathcal{L} + j\phi)\right) \tag{7.24}$$

This expression is perfectly covariant. Of course the integration measure $\mathcal{D}\phi = \prod_x d\phi(x)$ has no completely rigorous definition but, as we shall see, one can give it a precise meaning by comparing results with usual perturbation theory and many people take the resulting identity as a definition of the measure. To start with, let us point that the quantity

$$G^{(n)}(x_1, x_2, \ldots, x_n) \equiv \frac{1}{Z[0]} \left(\frac{\hbar}{i}\right)^n \left.\frac{\delta^n Z}{\delta j(x_1)\delta j(x_2)\ldots\delta j(x_n)}\right|_{j=0} \tag{7.25}$$

is nothing but the n-point Green function for the quantum field theory. That this is the case can be seen by comparing, pertubatively, the result obtained from the path-integral expression (7.24) with that arising in the canonical approach.

A simple example

Let us consider a "$\lambda\phi^3$ theory, i.e. one with potential

$$V[\phi] = \frac{\lambda}{3!}\phi^3 + \frac{1}{2}m^2\phi^2 \tag{7.26}$$

(m will be identified with the mass of the scalar particle associated to the field $\phi(x)$). Signs are chosen so as to guarantee the stability of the solution $\phi = 0$ of the classical equations of motion.

From here on we shall work in Euclidean space where $Z[j]$ reads (after integrating by parts the kinetic energy term for ϕ and neglecting surface terms)

$$Z[j] = \int \mathcal{D}\phi \exp\left(-\frac{1}{\hbar}\int_M d^4x(\phi\frac{(-\Box + m^2)}{2}\phi + \frac{\lambda\phi^3}{3!} + j\phi)\right) \tag{7.27}$$

Let us call A the differential operator appearing in the quadratic part of the Lagrangian,

$$A = -\Box + m^2 \tag{7.28}$$

Consider now the case in which the "coupling constant" λ vanishes, that is, the "free" (i.e. non self-interacting) case. Calling $Z_0[j]$ the corresponding generating functional, we have

$$Z_0[j] = \int \mathcal{D}\phi \exp\left(-\frac{1}{\hbar}\int_M d^4x(\phi\frac{A}{2}\phi + j\phi)\right) \tag{7.29}$$

Now, for the $j = 0$ case the path-integral can be done following a line similar to that of the quantum mechanical example in which we expanded the path-integral variable in Bessel-Fourier series. Indeed, we can consider the eigenvalue problem

$$A\varphi_n = \lambda_n \varphi_n \tag{7.30}$$

and then define the path-integral measure in the form

$$\mathcal{D}\phi = \prod_n dc_n \tag{7.31}$$

where c_n are the Bessel-Fourier coefficients in the expansion of ϕ in terms of φ_n. We then have (ignoring $\hbar$ factors, π factors that when comupting v.e.v.'s play no role)

$$Z_0[j=0] = \prod_n \int dc_n \exp\left(-\sum_n \lambda_n c_n^2\right) = \left(\prod_n \lambda_n\right)^{-1/2} \equiv \det{}^{-1/2} A \tag{7.32}$$

Now we can also compute $Z_0[j \neq 0]$ just as we did for the quantum mechanical case (i.e., by completing squares). We write

$$Z_0[j] = \int \mathcal{D}\phi \exp\left(-\frac{1}{2\hbar}\int d^4x(\phi - \bar{\phi})A(\phi - \bar{\phi}) - \frac{1}{2\hbar}\int d^4x C\right) \tag{7.33}$$

One easily finds that (7.33) coincides with (7.29) whenever

$$C = -\int d^4x' j(x) A^{-1}(x,x') j(x') \tag{7.34}$$

and

$$\bar{\phi} = -\int d^4x' A^{-1}(x,x') j(x') \tag{7.35}$$

with $A^{-1}(x,x')$ the Green function for the operator A.

Then, one has

$$\begin{aligned} Z_0[j] &= \exp\left(\frac{1}{2\hbar}\int d^4x d^4x' j(x) A^{-1}(x,x') j(x')\right) \times \\ &\quad \int \mathcal{D}\phi \exp\left(-\frac{1}{2\hbar}\int d^4x(\phi - \bar{\phi})A(\phi - \bar{\phi})\right) \end{aligned} \tag{7.36}$$

or

$$Z_0[j] = \exp\left(\frac{1}{2\hbar}\int d^4x d^4x' j(x) A^{-1}(x,x') j(x')\right) Z_0[j=0] \qquad (7.37)$$

We can use formula (7.37) in order to compute the vaccum expectation value of the product of two fields at different points in the free theory

$$\langle\phi(x_1)\phi(x_2)\rangle_0 = \frac{\hbar^2}{Z_0[0]}\frac{\delta^2 Z_0[j]}{\delta j(x_1)\delta j(x_2)}\bigg|_{j=0} \qquad (7.38)$$

One finds

$$\langle\phi(x_1)\phi(x_2)\rangle_0 = \hbar A^{-1}(x_1,x_2) \qquad (7.39)$$

which is nothing (appart from a $\hbar$ factor) but the (two-points) Green function $G_0^{(2)}(x_1,x_2)$ for the free theory

$$AG_0^{(2)}(x,y) = (-\Box_x + m^2)G_0^{(2)}(x,y) = \delta^{(4)}(x-y) \qquad (7.40)$$

$$G_0^{(2)}(x,y) = G_0^{(2)}(x-y) = \int \frac{d^4k}{(2\pi)^4}\exp\left(ik(x-y)\right)\frac{1}{k^2+m^2} \qquad (7.41)$$

We can then write the generating functional $Z[j]$ in terms of the partition function $Z_0[j=0]$ in the form

$$Z_0[j] = \exp\left(-\frac{1}{2}\int d^4x d^4x' j(x) G_0^{(2)}(x,x') j(x')\right) Z_0[j=0] \qquad (7.42)$$

The Wick theorem

This "theorem" due a work of to Gian Carlo Wick [19] (and also Freeman Dyson [20]), allows to evaluate vacuum expectation values of product of n fields (v.e.v.'s) at different points x_i $(i=1,2,\ldots,n)$ in terms of the vacuum expectation values of product of just two fields.

From the partition function for the free theory defined by eq.(7.42) one easily finds the formula that proves the theorem. For example, for the 4-points function one gets, after differentiation

$$\begin{aligned}\langle\phi(x_1)\phi(x_2)\phi(x_3)\phi(x_4)\rangle_0 \;=\;& G_0^{(2)}(x_1,x_2)G_0^{(2)}(x_3,x_4) + \\ & G_0^{(2)}(x_1,x_3)G_0^{(2)}(x_2,x_4) + \\ & G_0^{(2)}(x_1,x_4)G_0^{(2)}(x_2,x_3)\,. \qquad (7.43)\end{aligned}$$

It is evident that in the present free case, all v.e.v.'s with an odd number of points vanish. Concerning the even case with $n > 2$, it just gives analogous results as that for the case $n = 4$. The next lecture will be devoted to the evaluation of v.e.v.'s in the (self-)interacting theory. We shall show how one can develop a perturbative scheme so that finally any Green function can be expressed in terms of free two-point Green functions thanks to Wick theorem. But before let us study some formal properties of $Z[j]$.

Integration by parts and equations of motion

One easily convince onself that

Lemma 1: The path-integral of a total functional derivative is zero. In particular

$$\int \mathcal{D}\phi \frac{\delta}{\delta\phi(y)} \exp\left(-S[\phi] + \int d^4x j\phi\right) = 0 \tag{7.44}$$

For a "proof" of this result, one goes back to the definition of the path-integral measure in terms of cells of volume ϵ^4 and uses the equivalent result for normal integrals provided boundary terms vanish.

Now, from eq.(7.44) one has,

$$\int \mathcal{D}\phi \left(j(y) - \frac{\delta S}{\delta\phi(y)}\right) \exp\left(-S[\phi] + \int d^4x j\phi\right) = 0 \tag{7.45}$$

or, writing (7.45) in terms of j,

$$\int \mathcal{D}\phi \left(j(y) - \frac{\delta S}{\delta\phi(y)}\left[\frac{\delta}{\delta j}\right]\right) \exp\left(-S[\phi] + \int d^4x j\phi\right) = 0 \tag{7.46}$$

where by $\frac{\delta S}{\delta\phi(y)}\left[\frac{\delta}{\delta j}\right]$ we understand computing the derivative with respect to ϕ and then replacing ϕ by $\delta/\delta j$ in the resulting expression. But then we can rewrite (7.46) in the form

$$\left(j(y) - \frac{\delta S}{\delta\phi(y)}\left[\frac{\delta}{\delta j}\right]\right) Z[j] = 0 \tag{7.47}$$

This equation is known as the Schwinger-Dyson equation of motion, the quantum version of the classical equation of motion in the presence of a source. One can derive an infinite set of equations using this master equation. For example , writing for S

$$S[\phi] = \frac{1}{2}\int d^4x\phi A\phi + S_{int}[\phi] \tag{7.48}$$

one has

$$\frac{\delta S}{\delta\phi} = A\phi + \frac{\delta S_{int}}{\delta\phi} \tag{7.49}$$

In the particular case of the $\lambda\phi^3$ theory,

$$\frac{\delta S}{\delta\phi} = A\phi + \frac{1}{2}\lambda\phi^2 \tag{7.50}$$

Then, the Schwinger-Dyson equation reads

$$\left(A\frac{\delta}{\delta j(y)} + \frac{\lambda}{2}\left(\frac{\delta}{\delta j(y)}\right)^2 - j(y)\right) Z[j] = 0 \tag{7.51}$$

Now, differentating with respect of $j(z)$ and then putting $j = 0$ one gets, after some algebra

$$A\,\frac{\delta^2}{\delta j(y)\delta j(z)}Z[j]\bigg|_{j=0} - \delta^4(y-z)Z[0] + \frac{\lambda}{2}\frac{\delta}{\delta j(z)}\left(\frac{\delta}{\delta j(y)}\right)^2 Z\bigg|_{j=0} = 0 \tag{7.52}$$

Dividing by $Z[0]$ and rearranging we finally have

$$(-\Box + m^2)\langle\phi(y)\phi(z)\rangle + \frac{\lambda}{2}\langle\phi^2(y)\phi(z)\rangle = \delta^4(y-z) \tag{7.53}$$

to be compared with the classical equation of motion

$$(-\Box + m^2)\phi(y) + \frac{\lambda}{2}\phi^2(y) = 0 \tag{7.54}$$

Conversely, one can show that the solution of the Schwinger-Dyson equation is precisely the generating functional $Z[j]$. In fact, writing

$$Z[j] = \exp\left(-W[j]\right) \tag{7.55}$$

one can think of W as a sort of Lapalce transform of S and then if one essays for the solutions $Z_{sol}[j]$ of Schwinger-Dyson equation a Laplace transform one finds that the argument in the exponential should be in fact the action S.

Delta functionals

One can see that a sensible path-integral representation for the "*delta-*functional" $\delta[F[\phi]]$ is

$$\delta[F[\phi]] = \int \mathcal{D}\lambda \exp\left(-\int d^4xF[\phi]\lambda(x)\right) \tag{7.56}$$

the natural extension of the definition of the δ-function in terms of Fourier transforms. To see that (1.49) gives a sound result let us consider the following (restraint) path-integral

$$Z_r[j] = \int \mathcal{D}\phi\delta[F[\phi]] \exp\left(-S + \int d^4xj\phi\right) \tag{7.57}$$

Using the representation (7.56) we have

$$Z_r[j] = \int \mathcal{D}\phi\mathcal{D}\lambda \exp\left(-S + \int d^4x(j\phi - \lambda(x)F[\phi])\right) \tag{7.58}$$

But in view of our Lemma 1, we have

$$\int \mathcal{D}\phi\mathcal{D}\lambda\frac{\delta}{\delta\lambda} \exp\left(-S + \int d^4x(j\phi - \lambda(x)F[\phi])\right) = 0 \tag{7.59}$$

or

$$\int \mathcal{D}\phi F[\phi] \exp\left(-S + \int d^4x(j\phi - \lambda(x)F[\phi])\right) = 0 \tag{7.60}$$

Which can be written compactly as

$$F\left[\frac{\delta}{\delta J}\right] Z_r[j] = 0 \tag{7.61}$$

this showing that indeed Z_r corresponds to a restraint path-integral, restrained in the sense that it only takes into account configurations for which $F = 0$.

Problemas series 2,3,4

1. Using the change of variables

$$q(t) = q_{cl}(t) + x(t)$$

   show that for va free particle one has

$$Z_{i\to f} = F(t_f, t_i) \exp\left(\frac{im}{2\hbar}\frac{(q_f - q_i)^2}{t_f - t_i}\right)$$

   Find F and show that Z satisfies the superposiction principle.

2. Solvce problem 3, section 50 of Landau and Lifshitz *Non-relativistic Quantum Mechanics* book, where the gap between the fundamental abd the first excited states for the anharmonic oscillator is discussed using the WKB method.

3. Consider the N-dimensional integral

$$I = \int d^N q \exp\left(\vec{q}^{\,2} - g(\vec{q}^{\,2})^2\right)$$

   with $\vec{q} = (q_1, q_2, \ldots, q_N)$ and g a coupling constant. Calling $\vec{q}_{cl}$ the extremum of the "action" S, $S = \vec{q}^{\,2} - g(\vec{q}^{\,2})^2)$ find

   (a) q_{cl} (there are N solutions!)

   (b) Consider the change of variables $q = q_{cl} + x$ taking into account that when one chooses a given solution, the integral diverges in the remaining $N-1$ directions (a zero-mode problem)

   (c) Handle properly the zero-mode problem which is related to the $O(N)$ invariance of the action. To this end, integrate out angles, exponentiate the spherical coordinate Jacobian and then use the steepest descent method for the resulting "complete action" which includes the Jacobian effect.

4. Consider an N-dimensional Euclidean space-time. Calling $\vec{z} = (z_i) = (x_p, y_q)$ (with $p = 1, \ldots, k$ and $q = k+1, \ldots, N$), analyze the integral

$$I = \int_{-\infty}^{\infty} d^N z \exp\left((iS(z)\right)$$

Suppose that

$$\frac{\partial S}{\partial y} = 0$$

Use the Fadeev-Popov method to solve the problem of divergences in the integral. Consider a “gauge condition” $y = f(x)$ and show the factorization of the infinity. Compare with the method we employed in QFT.

5. Study the Euclidean and Minkowski Green function $G_0^{(2)}$ for a scalat theory in $d = 4$ and $d = 2$ space-time dimensions.

Clase 8

We are now ready to study the Green functions of the scalar theory when the interaction is present. Let us begin by evaluating the two-point function $G^{(2)}$,

$$G^{(2)}(u,v) = \frac{1}{Z}\int \mathcal{D}\phi \exp\left(-S_0 - \frac{\lambda}{3!}\int d^4x\phi^3(x)\right)\phi(u)\phi(v) \equiv \frac{\text{num}}{Z} \tag{8.1}$$

with Z defined as

$$Z = \int \mathcal{D}\phi \exp\left(-S_0 - \frac{\lambda}{3!}\int d^4x\phi^3(x)\right) \tag{8.2}$$

and S_0 the free (quadratic) action.

Consider first the denominator Z. Let us expand the interaction term in powers of λ,

$$Z = \sum_{n=0}^{\infty}(-1)^n\frac{\lambda^n}{n!(3!)^n}\int dx_1^4 d^4x_2\dots d^4x_n\langle\phi^3(x_1)\phi^3(x_2)\dots\phi^3(x_n)\rangle_0 \times Z_0 \tag{8.3}$$

The v.e.v. in the r.h.s. corresponds to a free theory. As we have seen, only if it corresponds to an even number of fields it will be different from zero. We can then write

$$Z = \sum_{k=0}^{\infty}\frac{\lambda^{2k}}{(2k)!(3!)^{2k}}\int dx_1^4 d^4x_2\dots d^4x_{2k}\langle\phi^3(x_1)\phi^3(x_2)\dots\phi^3(x_{2k})\rangle_0 \times Z_0 \tag{8.4}$$

Consider the first non trivial term, namely $k=1$

$$I^2 = \frac{\lambda^2}{(3!)^2 2!}\int dx_1^4 d^4x_2\langle\phi(x_1)\phi(x_1)\phi(x_1)\phi(x_2)\phi(x_2)\phi(x_2)\rangle_0 \times Z_0 \tag{8.5}$$

Since the v.e.v. corresponds to a free theory, we can apply the Wick theorem to decompose it into products of two-point Green functions for the free theory, which are known. Consider first the case in which the three "contractions" corresponds to fields ϕ at different points x_1 and x_2. The resulting product gives

$$p_{1,2} = G_0^{(2)}(x_1 - x_2)^3 \tag{8.6}$$

How many of these cubic terms are there? Consider the first $\phi(x_1)$. It can be contracted with any of the three ϕ at points x_2. Once a given contraction is done, the second $\phi(x_1)$ has just two possible $\phi(x_2)$ to contract with. The remaining last $\phi(x_1)$ has just no option an only one possible contraction. So the number of terms identical to $p_{1,2}$ is 3.2. We then have

$$P_{1,2} = 3.2 p_{1,2} = 3! G_0^{(2)}(x_1 - x_2)^3 \tag{8.7}$$

But there is also the possibility that two of the $\phi(x_1)$ points contract between themselves. There is for the remaining $\phi(x_1)$ no other possibility than contracting with one of the three $\phi(x_2)$. And the last two $\phi(x_2)$ have to contract among themselves. Let us call $p_{1,1}$ this kind of terms. They take the form

$$p_{1,1} = G_0^{(2)}(0) G_0^{(2)}(x_1 - x_2) G_0^{(2)}(0) \tag{8.8}$$

Concerning the number of times this term appears, the first $\phi(x_1)$ has 2 possible choices to contract with another $\phi(x_1)$. The third $\phi(x_1)$ can choose among 3 $\phi(x_2)$'s and then the other two $\phi(x_2)$ have just one possibility. So the factor is again $3.2 = 3!$. Using the same notation we have

$$P_{1,1} = 3! G_0^{(2)}(0) G_0^{(2)}(x_1 - x_2) G_0^{(2)}(0) \tag{8.9}$$

We then have

$$\begin{aligned}\langle \phi(x_1)\phi(x_1)\phi(x_1)\phi(x_2)\phi(x_2)\phi(x_2)\rangle_0 &= P_{1,2} + P_{1,1} \\ &= 3! G_0^{(2)}(x_1 - x_2)\left(G_0^{(2)}(x_1 - x_2)^2 + G_0^{(2)}(0)^2\right)\end{aligned} \tag{8.10}$$

We then have

$$Z = \left(1 + \frac{\lambda^2}{2.3!}\int d^4x_1 d^4x_2 G_0^{(2)}(x_1 - x_2)\left(G_0^{(2)}(x_1 - x_2)^2 + G_0^{(2)}(0)^2\right) + \ldots\right) Z_0 \tag{8.11}$$

It will be convenient at this point to write the free two-point Green function in terms of its Fourier transform (7.41)

$$G_0^{(2)}(z) = \int \frac{d^4k}{(2\pi)^4} \exp{(ikz)} \frac{1}{k^2+m^2} \tag{8.12}$$

so that calling $z = x_1 - x_2$, we have for the term with $G_0^{(2)}(x_1 - x_2)^3$

$$\begin{aligned} I_3 &= \frac{\lambda^2}{2.3!} \int d^4x_1 d^4x_2 \int \frac{d^4k_1}{(2\pi)^4} \int \frac{d^4k_2}{(2\pi)^4} \int \frac{d^4k_3}{(2\pi)^4} \\ & \quad \exp{(ik_1z)} \exp{(ik_2z)} \exp{(ik_3z)} \frac{1}{k_1^2+m^2} \frac{1}{k_2^2+m^2} \frac{1}{k_3^2+m^2} \end{aligned} \tag{8.13}$$

Integrating out x_1 we get a delta-function,

$$\int d^4x_1 \exp{(i(k_1+k_2+k_3)x_1)} = (2\pi)^4 \delta^{(4)}(k_1+k_2+k_3) \tag{8.14}$$

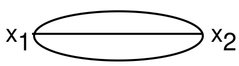
Integrating out x_2 we get the same result so that we have

$$\begin{aligned} I_3 &= \frac{1}{(2\pi)^4} \int d^4k_1 d^4k_2 d^4k_3 \delta^{(4)}(k_1+k_2+k_3) \delta^{(4)}(k_1+k_2+k_3) \\ & \quad \frac{1}{k_1^2+m^2} \frac{1}{k_2^2+m^2} \frac{1}{k_3^2+m^2} \end{aligned} \tag{8.15}$$

Finally, integrating over k_3,

$$I_3 = \frac{\delta^{(4)}(0)}{(2\pi)^4} \int d^4k_1 d^4k_2 \frac{1}{k_1^2+m^2} \frac{1}{k_2^2+m^2} \frac{1}{(k_1+k_2)^2+m^2} \tag{8.16}$$

We can represent this result by the following (Feynman) diagram:

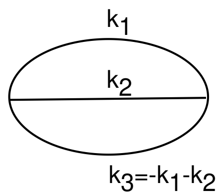


In this diagram

1. Each line represents a propagator . That is, a two-point free Green function in coordinate space, $G_0^{(2)}(x_1 - x_2)$ or its Fourier transform in momentum space, $1/(k^2+m^2)$.

2. Each <u>independent</u> loop represents an integral. (In the diagram above one considers as independent any pair of loops among the 3 possible loops of two lines since always the third one can be expressed in terms of the other two).

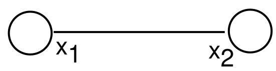
We can be more specific and draw the Feynman diagram, in momentum space indicating the momentum associated with each line:



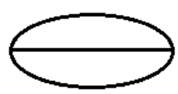
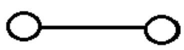
Concerning $G_0^{(2)}(0)^2 G_0^{(2)}(x_1 - x_2)$, which can be more instructively written as

$$G_0^{(2)}(0)G_0^{(2)}(x_1 - x_2)G_0^{(2)}(0) = G_0^{(2)}(x_1 - x_1)G_0^{(2)}(x_1 - x_2)G_0^{(2)}(x_2 - x_2) \tag{8.17}$$

its Feynman diagram can be drawn as follows:



Then, in coordinate space, we can represent diagramaticalle Z in the form

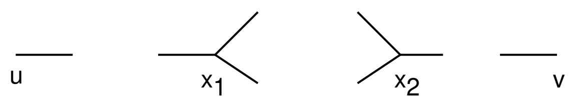
1 + λ^2( + ) + . . .

What about the numerator in (8.1)? Expanding as we did for Z, we have

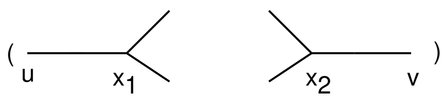
$$\text{num} \;=\; \Big(\langle\phi(u)\phi(v)\rangle_0 + \frac{\lambda^2}{2!(3!)^2}\int d^3x_1 d^3x_2 \langle\phi(u)\phi^3(x_1)\phi^3(x_2)\phi(v)\rangle_0 + \ldots\Big) \times Z_0 \tag{8.18}$$

Since Z_0 appears as an overall factor both in the numerator and the denominator, we shall omit it from here on.

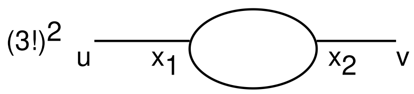
In order to analyse the number of ways we can do contractions when applying the Wick theorem in the second term in the r.h.s. of (8.18), it will be useful to draw the following picture in which each field $\phi(z)$ is represented by a short line with a z in one of its end,



To form a diagram there is first the possibility of joining the line that represents $\phi(u)$ with any of the three lines at x_1 representing the thre fields at the vertex x_1, or with any of the three lines at x_2. We then count a factor 2 and draw one of the two equivalent diagrams, say joining $\phi(u)$ with one $\phi(x_1)$ and $\phi(v)$ with one $\phi(x_2)$ But in fact we have three choices for joining $\phi(u)$ with one leg in x_1 and three other choices for joining $\phi(u)$ with x_2. We then have a factor 2.3.3:

2.3.3 

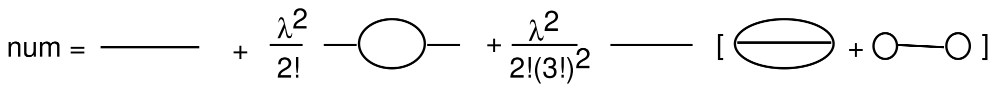
Now, for joining each one of the two remaining legs in x_1 with one in x_2 we have two choices. The overall factor is then 2.3.3.2 or $3!^2$ and the diagram (including the factor counting the number of times it appears) will be then drawn as follows:



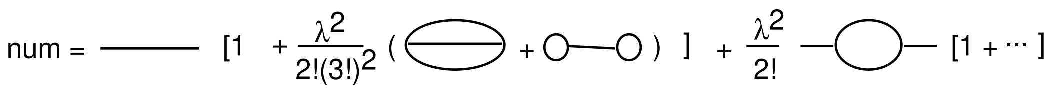
There are other two possible diagrams appearing in the numerator:

Diagrams like the two above are called *disconnected* diagrams in contrast with the one in the top of this page which is a connected one.

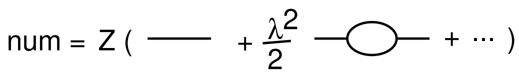
Graphically, we can then represent the numerator in the form



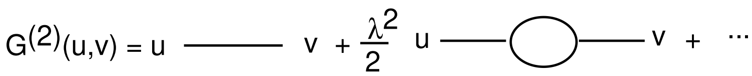
We can graphically factorize a propagator in the first and third terms in the r.h.s. and the same can be done with the second and the following term which is not represented in the figure above. Then, the numerator can be rearranged as



We then see that each bracket is nothing but Z so that we can write:



The factor Z in the numerator is then cancelled out by the denominator so that we finally have for $G^{(2)}(u,v)$

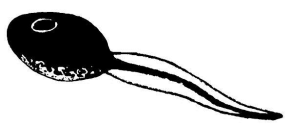


We see that only connected diagrams contribute to $G^{(2)}(u,v)$. This is true in general and this is the reason why one calls $Z[j]$ the generating functional for <u>connected</u> diagrams.

<u>Some comments</u>

I- Diagrams like

are called <u>tadpole</u> diagrams.



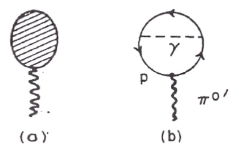
The following article in the Webster 's New Collegiate Dictionary clarifies the figure and the name:
tad.pole 'tad-,pol/n
[ME taddepol, fr. tode toad + polle head]
(15c):

a larval amphibian; specif: a frog or toad larva that has a rounded body with a long tail bordered by fins and external gills soon replaced by internal gills and that undergoes a metamorphosis to the adult.

It was Sidney Coleman who called such a diagram a "tadpole diagram" in one of his papers in collaboration with Sheldon Glashow [17]. They write:

If the scalar meson octet does exist, there is the possibility of a class of Feynman diagrams that vanish for every other kind of particle. These are the scalar tadpoles, diagrams with only one external line. In the limit of exact unitary symmetry, all scalar tadpoles vanish; but as we turn on the various symmetry-breaking interactions, the scalar tadpoles acquire nonzero values... Figure 1 shows a typical electromagnetic contribution to the π_0' *tadpole.*

FIG. 1. (a) The general scalar meson tadpole diagram. (b) A typical $\pi^{0\prime}$ tadpole, due to electromagnetism.



The editor of the journal was not satisfied, but he finally accepted since Coleman proposed spermion instead[3].

Consider for example the tadpole diagram on page 103, with one loop in each extreme. This diagram is problematic: the integral over each loop, which corresponds to $G_0^{(2)}(0)$, is divergent since, according to our Feynman recipe

$$G_0^{(2)}(0) = \int \frac{d^4k}{(2\pi)^2} \frac{1}{k^2+m^2} \propto \int_0^\infty \frac{k^3 dk}{k^2+m^2} \tag{8.19}$$

[3] As Coleman and Glashow indicate in footnote 11 of their work, the contribution of tadpole diagrams was first suggested by Julian Schwinger in his renowned paper *A Theory of the Fundamental Interactions* [18].

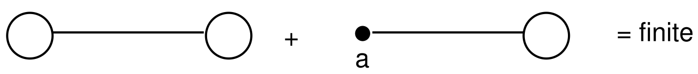
Now, this divergence can be easily eliminated by a procedure that is standard in Quantum Field Theory. One adds to the $\lambda\phi^3$ Lagrangian L a "counterterm" which in the present case should be linear in ϕ, so that the resulting Lagrangian $\bar{L}$ reads

$$\bar{L} = L + a\lambda\phi \tag{8.20}$$

with a some constant to be chosen The presence of this new term in the action generates new diagrams. For example, when computing Z there will be a new diagram of order λ^2 coming from the product of a linear $\lambda\phi^3$ and a linear λa factor in the expansion of the λ terms of the action: Calling T this new contribution

$$T = a\lambda^2\int d^4x_1 d^4x_2 \langle\phi(x_1)\phi^3(x_2)\rangle_0 = a\lambda^2 G_0^{(0)}(0)\int d^4x_1 d^4x_2 G_0^{(2)}(x_1 - x_2) \tag{8.21}$$

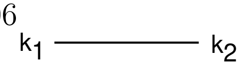
The new diagram is, then, also divergent. An appropriate choice of the constant a allows to cancel this divergent contribution with that arising from the tadpole. Diagramatically one has



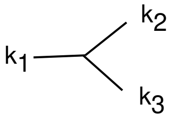
One can arrange a so that the new term just cancels the divergency without adding any finite contribution.

In this way one can eliminate any divergency coming from tadpole diagrams. The procedure is the path-integral version of normal ordering in the operator approach to QFT. The rationale of the approach can be schematized as follows: unless one communicates with God to know wheter L or $\bar{L}$ describes the actual physical system, both Lagrangian are candidates on an equal foot.

Let us summarize the recipe for constructing any Feynman diagram in the $\lambda\phi^3$ theory:



$$= \delta^{(4)}\ (k_1+k_2)\frac{1}{k_1{}^2+m^2}$$

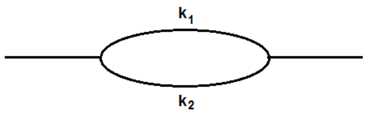


$$= \lambda\ \delta^{(4)}\ (k_1+k_2+k_3)$$

It remains to calculate the combinatorial factor with which each diagram enters. The analysis goes as follows: the n^{th} term in the expansion has an overall factor $(-\lambda)^n/(n!(3!)^n)$. If we call N the number of times a given diagram appears in such a term, the factor f we are looking for is

$$f = N\frac{(-\lambda)^n}{n!(3!)^n} \tag{8.22}$$

Given a diagram, there is a number called its <u>*order*</u> (order of the symmetry group) r. For example, the simple loop diagram has an order $r = 2$:



This number r corresponds to the number of diagrams that are identical when one interchanges lines in a symmetry operation. In the example above, there are 2 identical diagrams, one with the 'k_1' line above the 'k_2' line and one with the 'k_1' line below the' k_2'one. Then, $r = 2$. Now, one can proove that for a $\lambda\phi^3/3!$ interaction term N is related to r as follows

$$N = \frac{n!(3!)^n}{r} \tag{8.23}$$

One then has

$$f = \frac{(-\lambda)^n}{r} \tag{8.24}$$

One has to add some factors if one is working in momentum space. First, there is a $1/(2\pi)^4$ for each propagator. But each loop integral leads to a $2\pi^4\delta(k_1+k_2+\ldots)$ at each vertex where k_1, k_2, … join. Finally, there is an overall $2\pi^4\delta(k_1+k_2+\ldots)$ which guarantees conservation of four momenta. Then, if we call l the number of lines and V the number of vertices, we have a factor F of the form

$$F = \left(\frac{1}{(2\pi)^4}\right)^l \left((2\pi)^4\right)^V (2\pi)^4 = \left(\frac{1}{(2\pi)^4}\right)^{l-V-1} \tag{8.25}$$

The quantity $l-V-1$ is just the number L of independent loops of the diagram. Then, in momentum space the total factor $\mathcal{F}$ in front of a given diagram is

$$\mathcal{F} = \frac{(-\lambda)^n}{r}\left(\frac{1}{(2\pi)^4}\right)^L \tag{8.26}$$

II- Usually, the denominator (Z) in v.e.v.'s is written as

$$\text{den} = \exp\left(\sum_i D_i(\text{vac})\right) \tag{8.27}$$

where $D_i(\text{vac})$ is the contribution of the ith connected vacuum diagram (i.e. a diagram without external legs). The exponential appears because one has a symmetry factor of the form

$$\prod_i \frac{1}{n_i! r_i^{n_i}}$$

for any diagram with n_1 components of the "type" 1, n_2 of the type 2, …. The numerator is just the denominator times the sum of connected components without vacuum components,

$$\text{num} = \exp\left(\sum_i D_i(\text{vac})\right) \times \sum \text{Diag}(\textit{connected}, \textit{external legs}) \tag{8.28}$$

or. compactly

$$\text{num} = \exp\left(\sum_i D_i(\text{vac})\right) \times \sum \text{D}(c,e) \tag{8.29}$$

so that

$$\frac{\text{num}}{\text{den}} = \sum \mathrm{D}(c, e) \tag{8.30}$$

III- We have seen that the generating functional for the free theory can be computed to be

$$Z_0[j] = Z_0[0] \exp\left(\frac{1}{2\hbar} \int d^4x_1 d^4x_2 j(x_1) G_0^{(2)}(x_1 - x_2) j(x_2)\right) \tag{8.31}$$

with $G_0^{(2)}(x_1 - x_2)$ defined through

$$(-\Box + m^2) G_0^{(2)}(x_1 - x_2) = \delta^{(4)}(x_1 - x_2) \tag{8.32}$$

Now, when an interaction L_{int} is added, the generating functional cannot be computed exactly,

$$Z[j] = \int \mathcal{D}\phi \exp(-\frac{1}{\hbar} S_0) \exp\left(-\frac{1}{\hbar} \int d^4x L_{int}[\phi]\right) \exp\left(\int d^4 j(x)\phi(x)\right) \tag{8.33}$$

The interaction Lagrangian can be written in the form

$$L_{int} = exp\Big(-\int d^4z j(z)\phi(z)\Big) \, L_{int}\left[\frac{\delta}{\delta j(x)}\right] \exp\Big(\int d^4z j(z)\phi(z)\Big) \tag{8.34}$$

and the same for any functional of L_{int}. In particular

$$\begin{aligned} \exp\left(-\frac{1}{\hbar} \int d^4x L_{int}[\phi]\right) \quad &= \quad \exp\Big(-\int d^4z j(z)\phi(z)\Big) \\ &\times \exp\left(-\frac{1}{\hbar} \int d^4x L_{int}\left[\frac{\delta}{\delta j}\right]\right) \exp\Big(\int d^4z j(z)\phi(z)\Big) \end{aligned} \tag{8.35}$$

and then

$$Z[j] = \exp\left(-\frac{1}{\hbar} \int d^4x L_{int}\left[\frac{\delta}{\delta j}\right]\right) Z_0[j] \tag{8.36}$$

this formula can be taken as the perturbative (in the sense the exponential has to be expanded in powers of the coupling constant appearing in L_{int}) definition of the path-integral.

Clase 9

Quantization of Gauge Theories

The Abelian case

Electrodynamics, a paradigm of gauge field theories, admits a formulation in which the basic objects are *gauge fields* $A_\mu(x)$ ($\mu = 0, 1, 2, \ldots d$ with $d+1$ the dimensionality of space-time). In terms of these gauge fields (*connections* in the mathematical language of fiber bundles) one constructs the electromagnetic field tensor $F^{\mu\nu}(x)$ (the *curvature*),

$$F^{\mu\nu} = \partial^\mu A^\nu - \partial^\nu A^\nu \tag{9.1}$$

The Maxwell equations (in the absence of sources) are compactly written as

$$\partial_\mu F^{\mu\nu} = 0 \tag{9.2}$$

$$\partial_\mu {}^*F^{\mu\nu} = 0 \tag{9.3}$$

where we have defined the *dual* of $F^{\mu\nu}$ as

$${}^*F^{\mu\nu} = \frac{1}{2}\varepsilon^{\mu\nu\alpha\beta}F_{\alpha\beta} \tag{9.4}$$

In three dimensional space the observable in the theory, the electric E^i and magnetic B^i fields ($i = 1, 2, 3$) are defined as

$$E^i \equiv F^{0i} \qquad\qquad B^i \equiv^* F^{0i} \tag{9.5}$$

Eq.(1.1) can be derived, *à la* Euler-Lagrange, from the Maxwell action,

$$S = \frac{1}{4}\int d^4x F^{\mu\nu}F_{\mu\nu} \tag{9.6}$$

while eq.(1.6) is just a mathematical property (the Bianchi identity) of the curvature defined in (9.1).

One calls the theory a gauge theory because of their invariance under what is known as a gauge transformation

$$A_\mu(x) \to A_\mu(x)^\Lambda = A_\mu(x) + \partial_\mu \Lambda(x) \tag{9.7}$$

with Λ an arbitrary real function satisfying $\partial_i \partial_j \Lambda = \partial_j \partial_i \Lambda$. It is easy to see that under such a transformation observables remain unchanged

$$E^{\Lambda i}(x) = E^i(x) \qquad\qquad B^{\Lambda i}(x) = B^i(x) \tag{9.8}$$

We can identify Λ with an algebra valued function in the $U(1)$ algebra. Indeed, a group element of $U(1)$ can be always written as

$$g(x) = \exp(i\Lambda(x)) \tag{9.9}$$

and in that case the gauge transformation (9.7) takes a form which looks more complicated but which can be more easily generalized to other groups

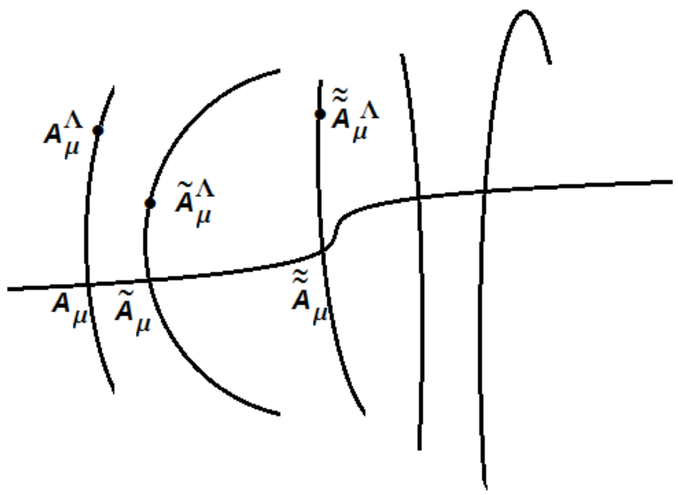
$$A_\mu(x) \to A_\mu(x)^\Lambda = A_\mu(x) + \frac{1}{i} g^{-1} \partial_\mu g \tag{9.10}$$

Note that at each point x (9.9) represents an element of $U(1)$ an this can be seen as a local $U(1)$ gauge group.

In order to quantize this theory starting from the classical action (9.6) one has to identify the basic field variable to integrate out in a path-integral formulation which we learnt is an *integral over all fields.* For the moment we shall consider the Lagrangian formulation and leave the Hamiltonian one for a more detailed analysis in a future chapter. In the Lagrangian formulation, it seems natural to take A_μ as the variables to integrate to define a path-integral partition function..

One faces however a problem: the gauge field has d components (d=4 in Minkowski $3+1$ space-time) but due to gauge-invariance not all of them are independent in the sense that several choices of A describe the same physical situation, at last at the classical level. If we were to represent graphically the would be domain of path-integration, it should look as in the next figure.

In the figure below the vertical lines are the orbits of the gauge group. One moves along an orbit by gauge transforming a connection.



An infinitesimal gauge transformation corresponds to a movement tangent to the orbit. That is, given an infinitesimal parameter ϵ, one has for the gauge transformation,

$$A_\mu^\epsilon(x) = A_\mu(x) + \partial_\mu \epsilon \tag{9.11}$$

and then $\partial_\mu \epsilon$ represent a vector tangent to the orbit. A finite gauge transformation Λ moves A_μ along the orbit to another value A_μ^Λ associated to the same curvature $F_{\mu\nu}$ i.e, the observable electric and magnetic fields take the same value. $A_\mu, \tilde{A}_\mu, \tilde{\tilde{A}}_\mu, \ldots$ represent gauge fields giving different $F_{\mu\nu}$, i.e, different values of observables.

The fact that the space over which one integrates has this strucure poses a problem which can be seen as one of zero-modes, which we learnt to handle in the quantum mechanical case of an anharmonic oscillator. Here it is $\partial_\mu \epsilon$ who plays the role that the center a of the instanton played. As in that case, we shall overcome the problem, in the Lagrangian formulation, by restricting the integration to one representative of the gauge fields for each orbit. The line crossing each orbit in points $A_\mu, \tilde{A}_\mu, \tilde{\tilde{A}}_\mu, \ldots$ represents such choice.

That is, one takes the whole family of gauge-transformed of a given configuration as one point in the integration domain. Different "points" then corresponds to inequivalent gauge fields (i.e., gauge fields that are not the gauge transformed one of the other).

The process of selecting one representative is what was called in the classical electrodynamics course gauge fixing. Indeed, even at the classical level, when studying Maxwell equations it was necessary to fix the gauge. That is, to chose a gauge condition (for example the Lorentz condition, de Coulomb condition, etc).

A gauge condition can be written as an equation of the form

$$F[A_\mu] = 0 \tag{9.12}$$

which is the equation for the curve crossing orbits in the figure.

Suppose we adopt the usual definition of the Lagrangian version of the partition function

$$Z = \int \mathcal{D}A_\mu \exp\left(-\frac{1}{\hbar}S[A]\right) \tag{9.13}$$

and we consider a clasical configuration (like the instanton configuration in the case of the anharmonic oscillator) which we cal $A^{cl}_\mu(x)$ to write

$$A_\mu(x) = A^{cl}_\mu(x) + \sqrt{\hbar}a_\mu(x) \tag{9.14}$$

Then, in terms of the new variable a_μ one has

$$Z = \exp\left(-\frac{1}{\hbar}S[A^{cl}_\mu]\right)\int \mathcal{D}a_\mu \exp\left(-\int d^4x d^4z a_\mu(x) S^{(2)}_{\mu\nu}(x,z) a_\nu(z)\right) \tag{9.15}$$

When writing the action in terms of the new variable a_μ, we have supposed that there are no linear terms (i.e., A^{cl}_μ is a solution of the eqs. of motion. The simplest one that we can use is just the trivial solution $A^{cl}_\mu = 0$.

Denoting by Λ a gauge transformation parameter we write the gauge transformed of A^{cl}_μ as

$$A^{cl}_\mu(x,\Lambda) = A^{cl}_\mu(x) + \partial_\mu \Lambda(x) \tag{9.16}$$

Then, if

$$\frac{\delta S}{\delta A_\mu^{cl}(x,\Lambda)} = 0 \tag{9.17}$$

$$\frac{\delta}{\delta\Lambda(z)}\frac{\delta S}{\delta A_\mu^{cl}(x,\Lambda)} = \int d^4y \frac{\delta^2 S}{\delta A_\mu^{cl}(x,\Lambda)\delta A_\nu^{cl}(y,\Lambda)}\frac{\delta A_\nu^{cl}(y,\Lambda)}{\delta\Lambda(z)} = 0 \tag{9.18}$$

or

$$\int d^4y S_{\mu\nu}^{(2)}\frac{\delta A_\nu^{cl}(y,\Lambda)}{\delta\Lambda(z)} = 0 \tag{9.19}$$

this meaning that the quadratic form $S_{\mu\nu}^{(2)}$ has a zero mode $a_\mu^{(0)}$ of the form

$$a_\mu^{(0)}(y) = \frac{\delta A_\nu^{cl}(y,\Lambda)}{\delta\Lambda(z)} \tag{9.20}$$

with all the problems it implies for the path-integration. Now, in view of (9.16), we see that the explicit form of this zero mode is

$$a_\mu^{(0)} = \partial_\mu\delta^{(4)}(y-z) \tag{9.21}$$

We want to proceed exactly as we did for the anharmonic oscillator zero-mode, except that now we have a zero mode related to invariance under gauge transformations instead of that for the oscillator, related to translation invariance. Orthogonality of zero and non zero modes $a_\mu^{(n)}$ now imply

$$\int d^4z\partial_\mu\delta^{(4)}(x-z)a_\mu^{(n)} = -\partial_\mu a_\mu^{(n)}(x) = 0 \tag{9.22}$$

this suggests that the "natural" gauge condition to impose on the gauge field A_μ (and consecuently on a_μ and also on the eigenfunctions $a_\mu^{(n)}$ is in this Abelian case the Lorentz gauge. Indeed, if we impose

$$\partial_\mu a_\mu(x) = 0 \tag{9.23}$$

we can expect to eliminate the zero mode from the quadratic integration and trade it for an integration over the gauge parameter Λ. (Evidently, we can chose the *background* configuration A_μ^{cl} to also satisfy the Lorentz condition).

Then exactly as we did in the case of the anharmonic oscillator, we start by a natural gauge condition which in this case reads

$$F[A] = -\int d^4y A_\mu(y) a_\mu^{(0)}(y;x) = \partial_\mu A_\mu(x) \tag{9.24}$$

where the second identity follows from eq.(9.22). We then write a resolution of the identity in the form

$$1 = \Delta_{FP}[A] \int \delta\Big(F[A^\Lambda]\Big)\,\mathcal{D}\Lambda \tag{9.25}$$

We have called $\Delta_{FP}[A]$ (the Faddeev-Popov determinant) a functional of the gauge fields to be adjusted so that the r.h.s. is 1. We introduce this resolution in the partition function (9.13)

$$Z = \int \mathcal{D}A_\mu \mathcal{D}\Lambda \Delta_{FP}[A] \exp\left(-\frac{1}{\hbar}S[A]\right)\delta\Big(F[A^\Lambda]\Big) \tag{9.26}$$

and make a shift in the path-integral variable A_μ, in the form

$$A_\mu(x) = A_\mu^{cl}(x) + \sqrt{\hbar}a_\mu = A_\mu^{cl}(x) + \sqrt{\hbar}\sum_n c_n a_\mu^{(n)}(x) \tag{9.27}$$

with c_n the Bessel-Fourier coefficients of the expansion of a_μ in terms of the basis $\{a_\mu^{(n)}\}$ of the quadratic operator $S_{\mu\nu}^{(2)}$,

$$\int dy S_{\mu\nu}^{(2)}(x,y) a_\mu^{(n)}(y) = \lambda_n a_\mu^{(n)}(x) \tag{9.28}$$

In this form, we pass from the integration measure $\mathcal{D}A_\mu\mathcal{D}\Lambda$ to $\prod_n dc_n \mathcal{D}\Lambda$ and we have, instead of (9.15),

$$\begin{aligned} Z &= \exp\left(-\frac{1}{\hbar}S[A_\mu^{cl}]\right)\int \mathcal{D}\Lambda \prod_n dc_n \Delta_{FP}[A^{cl}, \sqrt{\hbar}c_n] \exp\Big(-{\sum}' c_n^2\lambda_n\Big)\,\delta[c_0] \\ &= \exp\Big(-\frac{1}{\hbar}S[A_\mu^{cl}]\Big)\,\Delta_{FP}\left(\det{}' S^{(2)}\right)^{-1/2}\int D\Lambda \end{aligned} \tag{9.29}$$

We have kept the term of orden 0 in $\hbar$ in the Faddeev-Popov determinant since the rest of it (which depends on the $c_n's$) gives higher order corrections to our calculation.

Let us now evaluate Δ_{FP}. Starting from (9.25) we expand up to first order in Λ since the δ function constraints the possible values of Λ to $\Lambda = 0$ (see discussion below).

$$1 = \Delta_{FP}[A] \int \delta\left(F[A] + \frac{\delta F}{\delta A_\mu}\partial_\mu \Lambda\right) \mathcal{D}\Lambda \tag{9.30}$$

or, using explicitly that F corresponds to the Lorentz condition

$$1 = \Delta_{FP}[A] \int \delta(\partial_\mu A_\mu + \partial_\mu \partial_\mu \Lambda]) \, \mathcal{D}\Lambda = \Delta_{FP}[A] \frac{1}{\det \Box} \tag{9.31}$$

We then have, in the Lorentz gauge

$$\Delta_{FP}[A] = \det \Box \tag{9.32}$$

and then the final form of Z is

$$Z = (\det \Box) \exp\left(-\frac{1}{\hbar} S[A_\mu^{cl}]\right) \left(\det{}' S^{(2)}\right)^{-1/2} \int D\Lambda \tag{9.33}$$

Note that given the Lorentz condition, we have

$$\partial_\mu A_\mu^\lambda = \partial_\mu A_\mu + \Box \Lambda \tag{9.34}$$

Supose that A_μ satisfies the Lorentz condition. Is it possible that other $\Lambda \neq 0$ exists so that also A_μ^Λ also satisfies this condition? From (9.34) we see that this implies

$$\Box \Lambda = 0 \tag{9.35}$$

This equation has no well behaved solutions other than the trivial one. But for the case of non-Abelian gauge theories, there always exist non-trivial solutions and this poses a problem called the Gribov problem: one cannot fix univoquely the gauge and there are always *Gribov copies*: there are gauge equivalent configurations all satisfying a given gauge condition. One can see one of the problems implied by this by noting that in (1.33) we were able to extract the $1/\Box$ precisely because eq.(1.39) has just the trivial solution $\Lambda = 0$.

For a generic gauge condition $F[a] = 0$, the Faddeev Popov determinant is in the Abelian case,

$$\Delta_{FP} = \det\left(\frac{\delta F}{\delta A_\mu}\partial_\mu\right) \tag{9.36}$$

The Non-Abelian case

Consider a gauge group G with a Lie Algebra $\mathcal{G}$ generated by the set of generators t^a,

$$[t^a, t^b] = f^{abc} t^c \quad a = 1, 2, \dots \dim \mathcal{G} \tag{9.37}$$

$$\operatorname{tr} t^a t^b = \frac{1}{2} \delta^{ab} \tag{9.38}$$

$$f^{abc} f^{pbc} = \delta^{ap} C[G] \tag{9.39}$$

with $C[G]$ the quadratic Casimir of G.

For non-Abelian gauge groups, the gauge field takes values in the Lie Algebra $\mathcal{G}$,

$$A_\mu(x) = A_\mu^a(x) t^a \tag{9.40}$$

A gauge transformation is associated now with an element g of the gauge group G,

$$A_\mu(x) \to A_\mu^g = g^{-1} A_\mu g + \frac{1}{i} g^{-1} \partial_\mu g \tag{9.41}$$

with

$$g = \exp\left(i t^a \Lambda^a(x)\right) \tag{9.42}$$

For an infinitesimal gauge transformation one has

$$g = \exp\left(i\Lambda(x)\right) \approx 1 + i\Lambda(x) + O(\Lambda^2) \tag{9.43}$$

where we have writen $\Lambda = \Lambda^a t^a$. In this case one has

$$A_\mu^g \approx (1 - i\Lambda) A_\mu (1 + i\Lambda) + \partial\mu\Lambda + O(\Lambda^2) \tag{9.44}$$

or

$$\delta A_\mu \equiv A_\mu^g - A_\mu = \partial_\mu \Lambda + i[A_\mu, \Lambda] \tag{9.45}$$

We write compactly eq.(9.45) in the form

$$\delta A_\mu = D_\mu[A] \Lambda \tag{9.46}$$

where the covariant derivative $D_\mu[A]$ is defined as

$$D_\mu[A] = \partial\mu + i[A_\mu, \ ] \tag{9.47}$$

In gauge theories the covariant derivative should be defined from the ordinary one in such a way as to become a true vector operator, so that equations written using the covariant derivative preserve their physical properties under gauge transformations. One can easily see that this is the case for $D_\mu[A]$ since under gauge transformations, the covariant derivative changes according to

$$D_\mu[A] \to D_\mu[A^g] = g^{-1}D[A_\mu]g \tag{9.48}$$

Such transformation is a covariant transformation law. The covariant derivative generates *infinitesimal gauge transformations* tangent to the orbits of the gauge group.

We are now at the point where the curvature $F_{\mu\nu}$ can be defined as an object will also belongs to the Lie Algebra $\mathcal{G}$,

$$F_{\mu\nu} = F^a_{\mu\nu} t^a \tag{9.49}$$

To see its origin, let us calculate the commutator of two covariant derivatives acting over some Lie Algebra valued function. The idea is to make infinitesimal tangent displacement along two different directions and calculate the change when the order of differentiation is changed,

$$[D_\mu, D_\nu]f = i[\partial_\mu A_\nu - \partial_\nu A_\mu + i[A_\mu, A_\nu], f] \equiv i[F_{\mu\nu}, f] \tag{9.50}$$

The curvature or non-Abelian field strength is then defined as

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + i[A_\mu, A_\nu] \tag{9.51}$$

or, in components,

$$F^a_{\mu\nu} = \partial_\mu A^a_\nu - \partial_\nu A^a_\mu + i f^{abc} A^b_\mu A^c_\nu \tag{9.52}$$

One easily sees that under a gauge transformation it does not remain invariant (as it happens in the Abelian case) but it changes as the covarian derivative,

$$F_{\mu\nu}(x) \to F^g_{\mu\nu}(x) = g^{-1} F_{\mu\nu}(x) g \tag{9.53}$$

Let us now write the non-Abelian version of Maxwell Lagrangian. This Lagrangian was written for the first time by Yang and Mills [21].

It corresponds to the "natural" extension of the Maxwell Lagrangian[4],

$$L = \frac{a}{2}\mathrm{tr} F_{\mu\nu}F^{\mu\nu} \tag{9.54}$$

with a a constant to be adjusted later. Using (9.53) and thanks to the trace operation "tr" over gauge group elements one proves that the Yang-Mills Lagrangian is gauge invariant,

$$L[A^g] = L[A] \tag{9.55}$$

The equations of motion are

$$D_\mu F^{\mu\nu} = \partial_\mu F^{\mu\nu} + i[A_\mu F^{\mu\nu}] = 0 \tag{9.56}$$

Concerning the Bianchi identity, in the non-Abelian case it takes the form

$$D_\mu {}^*F^{\mu\nu} = 0 \tag{9.57}$$

Again the dual of $F_{\mu\nu}$ is defined as

$${}^*F_{\mu\nu} = \frac{1}{2}\varepsilon_{\mu\nu\alpha\beta}F^{\alpha\beta} = \varepsilon_{\mu\nu\alpha\beta}(\partial^\alpha A^\beta + A^\alpha A^\beta) \tag{9.58}$$

Coupling constants

In the $U(1)$ case (i.e., electrodynamics) there is no coupling constant in the gauge-field Lagrangian. Abelian gauge fields do not interact among themselves. In electrodynamics, an Abelian gauge theroy, the coupling constant appears when we couple gauge fields (photons) to matter fields (electrons). Only the matter fields are charged while photons are neutral.

In the non-Abelian case, also the gauge field is charged. This can be clearly seen if we make the shift

$$A_\mu \to eA_\mu \tag{9.59}$$

[4]Perhaps this naturalness explains why C.N. Yang and his student R.L. Mills haven't received a Nobel prize for proposing a so natural theory that is now at the basis of standard model which unifies electromagnetic, weak and strong interactions.

The curvature changes as

$$\partial_\mu A_\nu - \partial_\nu A_\mu + i[A_\mu, A_\nu] \to e(\partial_\mu A_\nu - \partial_\nu A_\mu + ie[A_\mu, A_\nu]) \equiv eF_{\mu\nu} \quad (9.60)$$

We also call $F_{\mu\nu}$ the curvature in which the (coupling) constant e appears multiplying the commutator.

If we make this replacement in the Yang-Mills Lagrangian (9.54) we have

$$L = \frac{e^2 a}{2} \mathrm{tr} F_{\mu\nu} F^{\mu\nu} \quad (9.61)$$

Hence, if we chose $a = 1/e^2$ we finally have

$$L = \frac{1}{2} \mathrm{tr} F_{\mu\nu} F^{\mu\nu} \quad (9.62)$$

with

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + ie[A_\mu, A_\nu] \quad (9.63)$$

and in this form it is clear that in the non-Abelian case, the gauge field is self-interacting with a coupling constant e: the cubic and quartic self-interacting terms in the Lagrangian are

$$L^{cubic} = -2ie^3 \mathrm{tr} \partial_\mu A_\nu [A_\mu, A_\nu] \quad (9.64)$$

$$L^{quartic} = \frac{e^4}{2} \mathrm{tr} [A_\mu, A_\nu][A_\mu, A_\nu] \quad (9.65)$$

So, e is really the gauge-field charge.

Clase 10

Quantization of non-Abelian gauge theories

Except for the fact one has to care of the non-commutative character of objects, quantization of non-Abelian gauge theories poses the same problems as those of electrodynamics and hence their solution follows the same Faddeev-Popov recipe. One starts from the partition function

$$Z = \int \mathcal{D}A_\mu \exp\left(-\frac{1}{\hbar}S[A]\right) \tag{10.1}$$

where now

$$\mathcal{D}A_\mu = \prod_x \prod_{\mu=0}^{d} \prod_{a=1}^{dim\ \mathcal{G}} dA_\mu^a(x) \tag{10.2}$$

and the Yang-Mills action is given by

$$S = \frac{\mathrm{tr}}{2}\int d^dx F_{\mu\nu}F^{\mu\nu} = \frac{1}{4}\int d^dx F_{\mu\nu}^a F^{a\mu\nu} \tag{10.3}$$

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + ie[A_\mu, A_\nu] \tag{10.4}$$

Evidently, the path-integral measure should be defined in a gauge-invariant way,

$$\mathcal{D}A_\mu = \mathcal{D}A_\mu^g \tag{10.5}$$

where

$$A_\mu^g = g^{-1}A_\mu g - \frac{i}{e}g^{-1}\partial_\mu g \tag{10.6}$$

Again, one has to avoid the problem of overcounting gauge configuration which are gauge copies. To this end, one inserts a resolution of the identity of the form

$$1 = \Delta_{FP}[A] \int \mathcal{D}g\delta(F[A^g]) \tag{10.7}$$

where $\mathcal{D}g$ is a Haar measure

$$\mathcal{D}g = \prod_x dg(x) \tag{10.8}$$

so that at each point x, $dg(x)$ is a measure over the group volume. One then gets for Z ,

$$Z = \int \mathcal{D}A_\mu \mathcal{D}g \Delta_{FP}[A] \exp\left(-\frac{1}{\hbar}S[A]\right) \delta(F[A^g]) \tag{10.9}$$

One now proceeds exactly as in the Abelian case, either by expanding up to quadratic terms and showing that the zero mode is eliminated or one can follow the more heuristic approach that will be adopted in what follows: one proves that $\Delta_{FP}[A] = \Delta_{FP}[A^g]$ and since $S[A] = S[A^g]$ and $\mathcal{D}A_\mu = \mathcal{D}A^g_\mu$ one writes, instead of (10.9),

$$Z = \int \mathcal{D}A_\mu \mathcal{D}g \Delta_{FP}[A^g] \exp\left(-\frac{1}{\hbar}S[A^g]\right) \delta(F[A^g]) \tag{10.10}$$

or, renaming $A^g_\mu \to A_\mu$,

$$\begin{aligned} Z &= \int \mathcal{D}g \int \mathcal{D}A_\mu \Delta_{FP}[A] \exp\left(-\frac{1}{\hbar}S[A]\right) \delta(F[A]) \\ &= V \int \mathcal{D}A_\mu \Delta_{FP}[A] \exp\left(-\frac{1}{\hbar}S[A]\right) \delta(F[A]) \end{aligned} \tag{10.11}$$

where V is the (infinite) gauge group volume which has been factored out leaving the path-integral over a representative of each physical configuration, selected by the gauge condition $F[A] = 0$ which is imposed by the delta function.

Concerning the Faddeev-Popov determinant, it can be computed as in the Abelian case, by noting that, if there were no Gribov copies, the condition $F[A^g] = 0$ has just a solution, say that for $g = I$ (one can always rename gauge group elements so that this is true). Since the path-integral where the FP determinant has to be employed is constrained by $\delta(F[A^g])$, this means that the integral over g is constrained to gauge group elements close to the identity. One can then consider in the integrand in the r.h.s. of eq.(10.7)

$$g \approx 1 + i\epsilon + O(\epsilon^2) \ , \qquad \mathcal{D}g = \mathcal{D}\epsilon \tag{10.12}$$

so that

$$1 = \Delta_{FP}[A] \int \mathcal{D}\epsilon \delta\left(F[A] + \frac{\delta F}{\delta A_\mu} D_\mu \epsilon \right) = \frac{\Delta_{FP}}{\det\left(\frac{\delta F}{\delta A_\mu} D_\mu\right)} \tag{10.13}$$

and hence

$$\Delta_{FP} = \det\left(\frac{\delta F}{\delta A_\mu} D_\mu\right) \tag{10.14}$$

We see that even in the Lorentz gauge, which in this case implies dim$\mathcal{G}$ conditions,

$$\partial_\mu A^a_\mu = 0 \tag{10.15}$$

the Faddeev-Popov determinant depends on the gauge field,

$$\Delta_{FP} = \det(\partial_\mu D_\mu[A]) = \det(\Box + ie\partial_\mu[A_\mu, \ ]) = \det(\Box + ie[A_\mu, \partial_\mu]) \tag{10.16}$$

One can see that the Lorentz gauge is not the natural gauge in the non-Abelian case. Indeed, extrema of the Yang-Mills action satisfy

$$\frac{\delta S}{\delta A^{cl}_\mu(x, \Lambda)} = 0 \tag{10.17}$$

where we have explicitly indicated that $A^{cl}_\mu(x, \Lambda)$, solution of (10.17), depends, before gauge-fixing, of a gauge parameter Λ. Then, as usual, one has

$$\frac{\delta}{\delta\Lambda}\left(\frac{\delta S}{\delta A^{cl}_\mu(x, \Lambda)}\right)\delta\Lambda = \int d^4y \frac{\delta^2 S}{\delta A^{cl}_\mu(x, \Lambda)\delta A^{cl}_\nu(y, \Lambda)} D_\nu[A^{cl}]\Lambda = 0 \tag{10.18}$$

this showing that $D_\mu[A^{cl}]\Lambda$ is a zero mode.

Then, when expanding in eigenfunctions of the quadratic operator appearing in (10.18), orthogonality between zero and non-zero modes implies

$$\int d^4x a_\nu^{(n)} D_\nu[A^{cl}]\Lambda = 0 \tag{10.19}$$

this in term giving

$$D_\nu[A^{cl}]a_\nu^{(n)}(x) = 0 \tag{10.20}$$

Then, in order to eliminate the zero mode from the path-integration, it is natural to select, as gauge condition,

$$D_\mu[A^{cl}]A_\mu(x) = 0 \tag{10.21}$$

The corresponding Faddeev-Popov determinant reads,

$$\Delta_{FP} = \det\Big(D_\mu[A^{cl}]D_\mu[A]\Big) \tag{10.22}$$

The generating functional

Let us now add a source and construct the generating functional for the Yang-Mills theory,

$$Z[j] = \int \mathcal{D}g \int \mathcal{D}A_\mu \Delta_{FP}[A] \exp\Big(-\frac{1}{\hbar}S[A] - \frac{1}{\hbar}\int d^4x j_\mu(x)A_\mu(x)\Big)\delta(F[A]) \tag{10.23}$$

We want to mantain gauge-invariance at this level but then, since the source term is not in general invariant,

$$\delta_\epsilon \int d^4x j_\mu(x)A_\mu(x) = \int d^4x j_\mu(x)\delta_\epsilon A_\mu(x) = \int d^4x j_\mu(x) D_\mu\epsilon \tag{10.24}$$

we have to impose the condition

$$D_\mu[A]j_\mu(x) = 0 \tag{10.25}$$

so as to ensure that, by integration by parts, the r.h.s. in (10.25) vanishes. Apart from the fact that the Yang-Mills action is not quadratic and hence we cannot perform the path-inegral exactly, we have the problem of the Faddeev-Popov determinant which in principle depends on A_μ and also the delta function. We postpone the problem of Δ_{FP} and attack the problem of the delta function.

A possible functional representation for the delta function is

$$\delta(F[A]) = \lim_{\alpha\to 0} \frac{1}{\sqrt{2\pi\hbar\alpha}} \exp\left(-\frac{1}{\hbar\alpha}\mathrm{tr}\int d^4x F[A]^2\right) \tag{10.26}$$

Hence, one can work at fixed α using this representation and then, at the end of the calculations, take the limit of vanishing α to effectively restrict the path-integral to the gauge condition $F[A] = 0$. In the case $F[A] = \partial_\mu A_\mu$, the quadratic part of the effective Lagrangian is

$$L^{(2)} = \frac{1}{4}(\partial_\mu A^a_\nu - \partial_\nu A^a_\mu)^2 + \frac{1}{2\alpha}\partial_\mu A^a_\mu \partial_\nu A^a_\nu \tag{10.27}$$

In the corresponding quadratic action we can integrate by parts so that the quadratic Lagrangian can be written as

$$L^{(2)} = \frac{1}{2}A^a_\mu \left(-\Box\delta_{\mu\nu} + \left(1 - \frac{1}{\alpha}\right)\partial_\mu\partial_\nu\right) A^a_\nu \tag{10.28}$$

Concerning the cubic and quartic terms (1.65) and (1.66), i.e., the self-interacting terms, they can be written as

$$L^{int} = ef^{abc}(\partial_\mu A^a_\nu)A^b_\mu A^c_\nu + e^2(A^a_\mu A^a_\mu A^b_\mu A^b_\mu - A^a_\mu A^a_\nu A^b_\mu A^b_\nu) \tag{10.29}$$

Having in mind a perturbative expansion, we start by writing $Z_0[j]$, the generating functional without including interaction terms,

$$Z_o[j] = \int \mathcal{D}A_\mu \Delta_{FP} \exp\left(-\frac{1}{2\hbar}\int d^4x(A^a_\mu M^{ab}_{\mu\nu}A^b_\nu + j^a_\mu A^a_\mu)\right) \tag{10.30}$$

where

$$M^{ab}_{\mu\nu} = \left(-\Box\delta_{\mu\nu} + \left(1 - \frac{1}{\alpha}\right)\partial_\mu\partial_\nu\right)\delta^{ab} \tag{10.31}$$

Let us suppose for the moment that the Faddeev-Popov determinant does not contribute to the quadratic part of the theory so that it can be ignored. Then, the path-integral can be performed and the answer is

$$Z_0[j] = (\det M)^{-1/2} \exp\left(\frac{1}{4\hbar}\int d^4x d^4y j^a_\mu(x)(M^{-1})^{ab}_{\mu\nu}(x,y)j^b_\nu(y)\right) \tag{10.32}$$

(Note that since we have included just the inverse of a square root of $\det \mathcal{M}$ coming from the A integration, we have to remember that in fact there are $N = 4 \times \dim \mathcal{G}$ integrations over the gauge field components so that the determinant should include this N appart from being an operator determinant with index x, y). Concerning with $(M^{-1})^{ab}_{\mu\nu}$, it is the Green function of the operator $M^{ab}_{\mu\nu}$,

$$M^{ab}_{\mu\nu}\,(M^{-1})^{bc}_{\nu\rho}(x,y) = \delta^{(4)}(x-y)\delta_{\mu\rho}\delta^{ac} \tag{10.33}$$

Defining the Fourier transform of $(M^{-1})^{ab}_{\mu\nu}$ through

$$(M^{-1})^{ab}_{\mu\nu}(x) = \int \frac{d^4k}{(2\pi)^4}(\bar{M}^{-1})^{ab}_{\mu\nu}(k)\exp(ik_\mu x_\mu) \tag{10.34}$$

one easily finds

$$\left(-k^2\delta_{\mu\nu} + \left(1 - \frac{1}{\alpha}\right)k_\mu k_\nu\right)(\bar{M}^{-1})^{ac}_{\nu\rho}(k) = \delta_{\mu\rho}\delta^{ac} \tag{10.35}$$

If one now proposes

$$(\bar{M}^{-1})^{ac}_{\nu\rho}(k) = -\delta^{ac}\left(A(k^2)\delta_{\nu\rho} + B(k^2)k_\nu k_\rho\right) \tag{10.36}$$

one finds

$$A(k^2) = \frac{1}{k^2}\;, \qquad B(k^2) = \frac{\alpha - 1}{k^4} \tag{10.37}$$

so that the gauge field propagator reads

$$(\bar{M}^{-1})^{bc}_{\nu\rho}(k) = -\frac{1}{k^2}\delta^{bc}\left(\delta_{\nu\rho} - (1-\alpha)\frac{k_\nu k_\rho}{k^2}\right) \tag{10.38}$$

We can take now the limit $\alpha \to 0$ which in fact corresponds to the Lorentz gauge. The corresponding propagator is

$$(\bar{M}^{-1})^{ac}_{\nu\rho}(k) = -\frac{1}{k^2}\delta^{ac}\left(\delta_{\nu\rho} - \frac{k_\nu k_\rho}{k^2}\right) \tag{10.39}$$

One can easily verify the dentity

$$k_\nu(\bar{M}^{-1})^{ac}_{\nu\rho}(k) = 0 \tag{10.40}$$

which is nothing but the Lorentz gauge condition for the propagator. (Remember that the propagator is the Fourier transform of the v.e.v. $\langle A^a_\nu(x)A^c_\rho(y)\rangle_0$).

It is interesting to note that in the absence of a gauge fixing one should have from the begining $\alpha = \infty$. In that case eq.(10.35) would read

$$\left(-k^2\delta_{\mu\nu} + k_\mu k_\nu\right)(\bar{M}^{-1})^{ac}_{\nu\rho}(k) = \delta_{\mu\rho}\delta^{ac} \tag{10.41}$$

One easily convinces oneself by repeating the steps (10.38)-10.37) that, in that case, $(\bar{M}^{-1})^{bc}_{\nu\rho}(k)$ does not exists. This is another manifestation of the necessity of fixing the gauge.

There is a family of gauge fixing which is very usefull for perturbative calculations and which can be implemented as follows. Since the generating function is gauge invariant, it cannot depend on the gauge condition one choses. Supose then that, instead of considering the condition $F[A] = 0$ one choses

$$F[A] = c(x) \tag{10.42}$$

then, Z_0 should read

$$Z_0[j] = V\int \mathcal{D}A_\mu \Delta_{FP}[A] \exp\left(-\frac{1}{\hbar}S[A] - \int d^4x j_\mu(x)A_\mu(x)\right)\delta(F[A]-c) \tag{10.43}$$

Since Z_0 cannot depend on c, we can integrate trivially over c both sides of this identity with an arbitrary weight that we choose to be

$$\int \mathcal{D}c \exp\left(-\frac{1}{\hbar\alpha}\int d^4x c(x)^2\right) \tag{10.44}$$

with α now arbitrary. In the l.h.s. of (10.44) this corresponds to a constant factor that we absorb in the definition. In the r.h.s., the delta function allows to integrate out c so that finally one gets

$$\begin{aligned} Z_0[j] &= V\int \mathcal{D}A_\mu \Delta_{FP}[A] \exp\left(-\frac{1}{\hbar}S[A] - \int d^4x j_\mu(x)A_\mu(x)\right) \\ &\quad \exp\left(-\frac{1}{\alpha}\int d^4x F[A]^2\right) \end{aligned} \tag{10.45}$$

Comparing (10.45) with the result we derived for the Lorentz gauge, we see that all the derivation goes the same except that now α is arbitrary compared to the $\alpha = 0$ case for the Lorentz gauge. Then, the propagator is still given by (10.38) but with α any positive real number. In particular, the choice $\alpha = 1$ greatly simplifies its form when $F[A] = \partial_\mu A_\mu$,

$$(\bar{M}^{-1})^{bc}_{\nu\rho}(k) = -\frac{1}{k^2}\delta^{bc}\delta_{\nu\rho} \tag{10.46}$$

This gauge is known as the Feynman gauge.

Clase 11

Path-integrals for fermionic fields

We have developed, through the path-integral approach, an alternative quantization procedure for scalar fields, gauge fields and, in general, bosonic fields. That is, fields that obey Bose-Einstein statistics with canonical <u>commutation</u> relations. Indeed, in the canonical approach to a Quantum Field Theory, given the Lagrangian say for a scalar field,

$$\mathcal{L} = \int d^3x L[\phi, \partial_\mu \phi] \tag{11.1}$$

and the canonical conjugate momentum defined as

$$\pi(\vec{x}, t) = \frac{\delta L}{\delta\left(\partial_0 \phi(\vec{x}, t)\right)} \tag{11.2}$$

one imposes canonical commutation relations *at equal times*,

$$[\phi(\vec{x}, t), \pi(\vec{y}, t)] = \hbar i \delta^{(3)}(\vec{x} - \vec{y}) \tag{11.3}$$

$$[\phi(\vec{x}, t), \phi(\vec{y}, t)] = [\pi(\vec{x}, t), \pi(\vec{y}, t)] = 0 \tag{11.4}$$

then inverts (11.2) to obtain $\partial_0 \phi$ in terms of π and ϕ and in this way one constructs the Hamiltonian,

$$H = \int d^3x (\pi \partial_0 \phi - L) \tag{11.5}$$

One can then follow the standard quantum analysis of the system.

What about fermions? In the canonical approach, one modifies the prescription above in order to take into account Pauli principle and the consequent Fermi-Dirac statistics. There is in fact a previous problem related to the fact that the Dirac Lagrangian for a (massive) fermion is first order,

$$L = \frac{i}{2}(\bar{\psi} i\partial\!\!\!/ \psi - (\partial_\mu \bar{\psi})\gamma^\mu \psi) - m\bar{\psi}\psi \tag{11.6}$$

with $\bar{\psi} = \psi^\dagger \gamma^0$. Now, being first order, the canonical momentum is related to (and not independent from) ψ and $\psi^\dagger$ and the Hamiltonian cannot be constructed following the steps described above for scalars. One can however extract the Hamiltonian from the energy-momentum tensor,

$$H = \int d^3x T_{00} \tag{11.7}$$

and define equal-time *anti-commutation* relations which are consistent with the Fermi-Dirac statistic

$$\{\psi_\alpha(\vec{x},t)\psi^\dagger_\beta(\vec{y},t)\} = \delta^{(3)}(\vec{x}-\vec{y})\gamma^0_{\alpha\beta} \tag{11.8}$$

$$\{\psi_\alpha(\vec{x},t)\psi_\beta(\vec{y},t)\} = \{\psi^\dagger_\alpha(\vec{x},t)\psi^\dagger_\beta(\vec{y},t)\} = 0 \tag{11.9}$$

As in the bosonic case, a Fock space can be constructed for fermions with creation and annhilation operators satisfying anticommutation relations,

$$\{c_n, c^\dagger_m\} = \delta_{nm} \tag{11.10}$$

$$\{c_n, c_m\} = \{c^\dagger_n, c^\dagger_m\} = 0 \tag{11.11}$$

The quantum theory for fermionic fields can be then developed without major problems.

Concerning the path-integral approach, we have seen that all the quantum aspects of the scalar theory (or in general a bosonic theory) was concentrated in the path-integral measure $\mathcal{D}\phi$,

$$\mathcal{D}\phi = \prod_n dc_n \tag{11.12}$$

with the $c'_n s$ the Bessel-Fourier coefficients of an expansion of the classical scalar field. These c-number coefficients were reminiscent of the creation and annhilation operators in the canonical approach. If we think in something similar for fermion fields, it is evident that if we propose such a Bessel-Fourier expansion, we shall end with a quantum theory corresponding to commuting fields, since we are doing nothing different from what we did in the bosonic case. Only if we try for the Bessel-Fourier expansion *not c-numbers but anticommuting numbers* we have a chance of ending with the correct quantum theory.

Let us see this in detail by starting, for simplicity with two elements a and $\bar{a}$ satisfying a Grassmann algebra

$$\{a, \bar{a}\} = a\bar{a} + \bar{a}a = 0 \tag{11.13}$$

$$\{a, a\} = 2a^2 = 0 \ , \quad \{\bar{a}, \bar{a}\} = 2\bar{a}^2 = 0 \tag{11.14}$$

(Cf. these equations with eqs.(11.10)-(11.11)). In this two-dimensional case, any polynomial of a and $\bar{a}$ can have at most $2^2 = 4$ terms,

$$P[a, \bar{a}] = f_0 + f_1 a + f_2 \bar{a} + f_3 a\bar{a} \tag{11.15}$$

with the coefficients f_i being c-numbers. We shall call a and $\bar{a}$ the generators of the two-dimensional Grassmann algebra.

Consider the simple case in which the polynomial takes the form

$$f(a) = f_0 + f_1 a \tag{11.16}$$

We can define two operators $\hat{a}$ and $\hat{a}^*$ acting on $f(a)$,

$$\hat{a} f(a) = a f(a) \tag{11.17}$$

$$\hat{a}^* f(a) = \frac{d}{da} f(a) \tag{11.18}$$

From (11.16)-(11.18) we have

$$\hat{a} f(a) = a f_0 \tag{11.19}$$

$$\hat{a}^* f(a) = f_1 \tag{11.20}$$

One easily sees that

$$\hat{a}^2 f(a) = a^2 f_0 = 0 \tag{11.21}$$

$$\hat{a}^{*2} f(a) = a^* f_1 = 0 \tag{11.22}$$

$$(\hat{a}\hat{a}^* + \hat{a}^*\hat{a}) f(a) = f(a) \tag{11.23}$$

Comparing this algebra with that defined in eqs.(11.10)-(11.11) for fermionic creation and annihilation operators we see that they coincide. We have then been able to obtain a simple representation for an algebra of the kind Fermi-Dirac creation and annihilation operators obey. One should just act over functions of the anticommuting Grassmann algebra above defined. It is then reasonable to infer that a and $\bar{a}$ are the reasonable subyacent (anticommuting) numbers to try a Bessel-Fourier expansion for classical fermionic fields and then to integrate in order to define a path-integration over fermions.

But we have first to define integration over anticommuting numbers, a task that, according to Coleman [22], make strong men quail. We want to know, for example, the value of the following integral

$$I = \int f(a) da \tag{11.24}$$

with $f(a)$ defined as in (11.16). In fact, we need just to define the following two integrals

$$\int da \quad \text{and} \quad \int a da \tag{11.25}$$

To find a sensible definition, we demand the usual properties

$$\begin{aligned} \int da[f(a) + g(a)] &= \int da f(a) + \int da g(a) && (11.26) \\ \int da f(a+b) &= \int da f(a) && (11.27) \end{aligned}$$

Now, we have

$$\int da f(a+b) = \int da[f_0 + f_1(a+b)] = f_0 \int da + f_1 \int da(a+b) \tag{11.28}$$

or

$$\int da f(a+b) = f_0 \int da + f_1 \int da a + f_1 (\int da) b \tag{11.29}$$

From eq.(11.27) the r.h.s. cannot depend on b. This means that the integral in the parenthesis in the third term should vanish and then

$$\int da = 0 \tag{11.30}$$

But then, eq.(11.29) redues to

$$\int da f(a+b) = f_1 \int daa \tag{11.31}$$

We have from eq.(11.27) that the r.h.s. in (11.31) should be just the integral of $f(a)$. Then

$$f_1 \int daa = f_0 \int da + f_1 \int daa \tag{11.32}$$

This equation reduces to an identity independently of the value of $\int daa$. We can then fix this value as a normalization

$$\int daa = 1 \tag{11.33}$$

We summarize in the next formula these results emphazing the coincidence between integrals and what one should expect for derivatives of anticommuting objects

$$\begin{array}{ll} \int da = 0\ , & (d/da)1 = 0 \\ \int daa = 1\ , & (d/da)a = 1 \end{array} \tag{11.34}$$

In some sense, integrating and differentiating anticommuting objects coincide: the integral and the derivative of a constant vanishes and the integral or derivative of the variable gives 1. We impose that differentials and variables anticommute

$$\{a, da\} = 0 \tag{11.35}$$

so that

$$\int ada = -\int daa = -1 \tag{11.36}$$

All these properties are also demanded to $\bar{a}$,

$$\int d\bar{a} = 0\ , \quad \int d\bar{a}\bar{a} = 1\ , \quad \{\bar{a}, d\bar{a}\} = 0 \tag{11.37}$$

We also impose

$$\{da, d\bar{a}\} = \{a, d\bar{a}\} = \{\bar{a}, da\} = 0 \tag{11.38}$$

Consider now as an example the following integral

$$I = \int da d\bar{a} \exp(-\lambda \bar{a} a) \tag{11.39}$$

Only the first two term survive in an expansion in powers of λ so that one has

$$\begin{aligned} \int d\bar{a} da \exp(-\lambda \bar{a} a) &= \int d\bar{a} da (1 - \lambda \bar{a} a) = -\lambda \int d\bar{a} da \bar{a} a \\ &= \lambda \int da d\bar{a} \bar{a} a = \lambda \int da a = \lambda \end{aligned} \tag{11.40}$$

This result should be contrasted with that obtained with an integral over (commuting) real numbers,

$$\int_{-\infty}^{\infty} dx \exp(-\lambda x^2) = \frac{\sqrt{\pi}}{\sqrt{\lambda}} \tag{11.41}$$

Remember that representing a gaussian path-integral for bosonic fields as an infinite product of gaussian integrals over real numbers we got the result of the inverse of the square root of the product of $\lambda' s$. In the present case, one can then advance that an infinite product of integrals of the kind (11.39) will lead to the product of $\lambda' s$ without a square root and appearing in the numerator. To see this in detail, let us extend the number of anticommuting generators of the Grassman algebra to $2N$, satisfying the anticommutation relations

$$\{a_n, a_m\} = \{\bar{a}_n, \bar{a}_m\} = \{a_n, \bar{a}_m\} = 0 \tag{11.42}$$

Concerning integrals, the rules extend to

$$\int \prod_{n,m=1}^{N} d\bar{a}_m da_n = \left\{ \begin{array}{cccccccc} \bar{a}_1 & \bar{a}_2 & \bar{a}_3 & \cdots & \bar{a}_N & a_1 & a_2 & \cdots & a_N \\ & \bar{a}_2 & \bar{a}_3 & \cdots & \bar{a}_N & a_1 & a_2 & \cdots & a_N \\ & & \bar{a}_3 & \cdots & \bar{a}_N & a_1 & a_2 & \cdots & a_N \\ & & & & . & & & & \\ & & & & . & & & & \\ & & & & . & & & & \\ & & & & & a_1 & a_2 & \cdots & a_N \\ & & & & & & a_2 & \cdots & a_N \\ & & & & . & & & & \\ & & & & . & & & & \\ & & & & . & & & & \\ & & & & & & & & a_N \end{array} \right\} = \left\{ \begin{array}{c} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{array} \right\} \tag{11.43}$$

Consider now the integral

$$I_N = \int \prod_{n,m=1}^{N} d\bar{a}_m da_n \exp\left(-\lambda_1 \bar{a}_1 a_1 - \lambda_2 \bar{a}_2 a_2 - \ldots - \lambda_N \bar{a}_N a_N\right) \tag{11.44}$$

If one now expands the exponential, according to eq.(**??**) only the term containing the $N's$ $\bar{a}'s$ and the $N's$ $a's$ will give a nonvanishing result and then we have

$$I_N = \prod_n (-1)^N \lambda_n \tag{11.45}$$

From this result, we infer the following one

$$\int \prod_{n,m} d\bar{a}_m da_n \exp\left(-\begin{pmatrix} \bar{a}_1 & \bar{a}_2 \ldots \bar{a}_N \end{pmatrix} A \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_N \end{pmatrix} \right) = \det(A) \tag{11.46}$$

for A some diagonalizable matrix. As advanced, we got a determinant in the numerator (instead of square root of a determinant in the denominator). Also note that under the change

$$a_j \to \mu a_j \tag{11.47}$$

with μ some real number, the differential changes as

$$da_j \to \frac{1}{\mu} da_j \tag{11.48}$$

This follows from the fact that $\int daa = 1$. It should not be a suprise in view of the connection already signaled between the integral and the derivative for anticommuting objects.

Since we are not cowards as quails, we are ready to define a path integral for fermions. Consider

$$Z = \int \mathcal{D}\bar{\psi}\mathcal{D}\psi \exp\Big(S[\bar{\psi}, \psi]\Big) \tag{11.49}$$

with S the Dirac action,

$$S = \int d^4x L \tag{11.50}$$

and L the Lagrangian defined in (11.6) which we can compactly write

$$L = \bar{\psi}(i\partial\!\!\!/ + m)\psi \tag{11.51}$$

We now consider now the eigenvalue problem for this free Dirac operator

$$(i\partial\!\!\!/ + m)\varphi_n(x) = \lambda_n \varphi_n(x) \tag{11.52}$$

$$\int d^4x \varphi_n^\dagger \varphi_m = \delta_{nm} \tag{11.53}$$

and expand the fermionic variables in (11.49) in terms of the basis provided by this operator,

$$\psi(x) = \sum_n^\infty a_n \varphi_n(x) \ , \qquad \bar{\psi}(x) = \sum_n^\infty \bar{a}_n \varphi^\dagger(x) \tag{11.54}$$

where the Bessel-Fourier coefficients are **anticommuting Grassmann objects**. Note that we are taking $\bar{\psi}$ and ψ as independent path-integral variables and not as related through the usual operation, $\bar{\psi} = \psi^\dagger\gamma^0$.

We then define the path-integral over all fermion fields as an integral over the Grassmann coefficients,

$$\mathcal{D}\bar{\psi} \equiv \prod_n^\infty d\bar{a}_n \ , \quad \mathcal{D}\psi \equiv \prod_n^\infty da_n \tag{11.55}$$

and then the partition function for free fermions can be written in the form,

$$\begin{aligned} Z_0 &= \prod_n^\infty \int d\bar{a}_n da_n \exp\left(-\int d^4x \sum_n \bar{a}_n \varphi_n^\dagger (i\partial\!\!\!/ + m) \sum_m a_m \varphi_m\right) \\ &= \prod_n^\infty \int d\bar{a}_n da_n \exp\left(-\sum_n \lambda_n \bar{a}_n a_n\right) = \mathcal{N} \prod_n \lambda_n \\ &\equiv \det(i\partial\!\!\!/ + m) \end{aligned} \tag{11.56}$$

where we have absorbed a normalization constant $\mathcal{N}$ and defined the determinant of the Dirac operator as the (possibly unbounded) product of its eigenvalues. One should compare this result for free fermions,

$$Z_0^{fermions} = \det(i\partial\!\!\!/ + m) \tag{11.57}$$

with that arising in the case of a free scalar field

$$Z_0^{bosons} = \det{}^{-1/2}(\Box + m) \tag{11.58}$$

If instead of considering free fermions, we study the case in which they are subjected to an external electromagnetic field with potential A_μ, so that the resulting Lagrangian is

$$L = \bar{\psi}(i\partial\!\!\!/ + m + eA\!\!\!/)\psi \tag{11.59}$$

the path-integral quantization proceeds in the same way. One ends with

$$Z[A] = \det(i\partial\!\!\!/ + m + eA\!\!\!/) \tag{11.60}$$

We can now use the identity, valid for matrices but also for operators,

$$\log \det A = \operatorname{tr} \log A \tag{11.61}$$

in order to write

$$Z[A] = \exp\left(\operatorname{tr} \log(i\partial\!\!\!/ + m + eA\!\!\!/)\right) \tag{11.62}$$

Consider now

$$P[A] = \log\left(\frac{i\partial\!\!\!/ + m + eA\!\!\!/}{i\partial\!\!\!/ + m}\right) = \log(1 + eG_0 A\!\!\!/) \tag{11.63}$$

with G_0 the free fermion Green function,

$$(i\partial\!\!\!/ + m)G_0(x - y) = \delta^{(4)}(x - y) \tag{11.64}$$

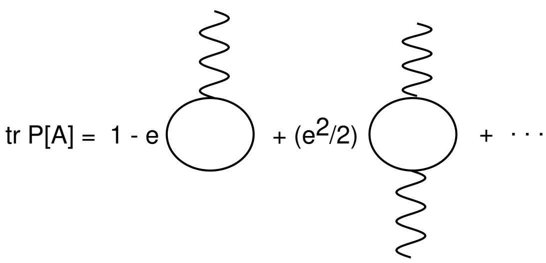
Expanding $P[A]$ one has

$$P[A] = \sum_n \frac{1}{n!}(-e)^n \left(G_0 A\!\!\!/\right)^n \tag{11.65}$$

Then, the normalized version of $Z[A]$ can be identified with

$$Z[A] = \exp(\mathrm{tr}P[A]) \tag{11.66}$$

which gives nothing but the vacuum to vacuum amplitude for fermions in an external fields, as can be seen by comparing with the result obtained via canonical quantization,



This result was proved for the first time by Matthews and Salam [23].

BRST Symmetry

BRST à la ancienne

In 1975 Becchi, Rouet and Stora [37] and, independently Tyutin [38], discovered that although gauge symmetry has been fixed, Z as defined in (16.24) has still a symmetry related to gauge invariance but with an anticommuting parameter. This symmetry showed its power when it made possible to prove, in a clear and complete form, the renormalizability of non-Abelian gauge theories. In particular, the proof helped in taking seriously these theories as candidates to describe electroweak and strong interactions.

The way in which BRST invariance was discovered is related to the Faddeev-Popov method but it can be seen that a BRST principle can replace and generalize this method. We shall discuss both approaches, starting by the historical one.

Let us start from formula (10.11) for the Euclidean partition function of a non-Abelian gauge theory and, for simplicity, put $\hbar = 1$

$$Z = \mathcal{N} \int \mathcal{D}A_\mu \Delta_{FP}[A] \exp\left(-S[A]\right) \delta(F[A]) \tag{11.67}$$

We then introduce a path-integral representation for the delta function enforcing the gauge condition.

$$Z = \int \mathcal{D}A_\mu \mathcal{D}b \exp\left(-\frac{1}{4}\mathrm{tr}\int d^4x F_{\mu\nu}^2\right) \Delta_{FP}[A] \exp\left(-\mathrm{tr}\int d^4x F[A] b(x)\right) \tag{11.68}$$

Here $b(x) = b^a(x)t^a$ is a Lie-algebra valued auxilliary scalar. Concerning the Faddeev-Popov determinant, it is given by

$$\det \Delta_{FP}[A] = \det D_\mu^{ab}[A] \frac{\delta F^c[A]}{\delta A_\mu^b} = \det \mathcal{M}^{ac}[A] \tag{11.69}$$

As we have seen, for the Lorentz and Coulomb gauge one has

$$\begin{aligned} \det \mathcal{M}^{ac}[A] &= \det D_\mu^{ac}\partial_\mu \quad \text{Lorentz gauge} \\ \det \mathcal{M}^{ac}[A] &= \det D_i^{ac}\partial_i \quad \text{Coulomb gauge} \end{aligned} \tag{11.70}$$

We have lernt that determinants of differential operators like $\mathcal{M}$ can be represented as path integrals over anticommuting variables,

$$\det \mathcal{M} = \int \mathcal{D}\bar{c}^a \mathcal{D}c^a \exp\left(-\int d^4x \bar{c}^a \mathcal{M}^{ab} c^b\right) \tag{11.71}$$

where $\bar{c}(x) = \bar{c}^a(x)t^a$ and $c(x) = c^a(x)t^a$ are anticommuting Lie algebra valued scalars. Although integration over anticommuting variables was originally introduced to describe fermions, it should be stressed here that $\bar{c}$ and c are a pair of anticommuting <u>scalars</u> (not fermions transforming like 1/2 spinors). Concerning gauge transformations, remember that the Faddeev-Popov determinant is gauge invariant and this, in turn, is a result of the fact that $\mathcal{M}$ transforms covariantly. Then, if one is going to use fields $\bar{c}$ and c to exponentiate $\mathcal{M}$, these fields have to transform in the adjoint representation of the gauge group to get a scalar effective action,

$$\begin{aligned} \bar{c} &\rightarrow \bar{c}^g = g^{-1}(x)\bar{c}g(x) \\ c &\rightarrow c^g = g^{-1}(x)cg(x) \end{aligned} \tag{11.72}$$

The fact that $\bar{c}$ and c are not physical fields, in the sense that they will not appear as physical *in* or *out* states when computing v.e.v.'s makes plausible the name of *ghosts* that they have received.

We can then write the path-integral Z in the compact form

$$Z = \int \mathcal{D}A_\mu \mathcal{D}b\mathcal{D}\bar{c}\mathcal{D}c \exp\left(-\int d^4x L_{eff}\right) \tag{11.73}$$

where the effective Lagrangian L_{eff} is defined as

$$L_{eff} = \mathrm{tr}\left(\frac{1}{4}F_{\mu\nu}^2 + bF[A] + \bar{c}\mathcal{M}c\right) \tag{11.74}$$

Being the gauge fixed to the condition $F[A] = 0$, this effective Lagrangian is not gauge invariant due to the presence of the term containing, precisely, the gauge fixing function $F[A]$. One can see however that L_{eff} has still a remanent invariance, associated to gauge transformations, parametrized by an anticommuting global quantity instead of the commuting local parameter $\epsilon(x)$ associated to general gauge transformations.

The transformation laws that define this new invariance (called BRST invariance) are inferred from the gauge transformations laws of the physical fields, in this case the gauge field. Under an infinitesimal gauge transformation the gauge fields transforms as

$$\delta_{gauge}A_\mu^a = D_\mu^{ab}[A]\epsilon^b(x) \tag{11.75}$$

We then define a BRST transformation δ_{BRST} as if it were a gauge transformation but with a parameter $\epsilon^a(x) = c^a(x)\lambda$, with λ an anticommuting constant parameter,

$$\delta_{BRST}A_\mu^a = D_\mu^{ab}[A]c^b(x)\lambda \tag{11.76}$$

Since c^a is an anticommuting ghost, the product with *lambda* gives a commuting result for function $\epsilon(x)$.

It is not difficult to define BRST transformation laws for b, $\bar{c}$ and c ensuring that L_{eff} is BRST invariant under the complete set of transformations. Indeed, one can see that if one imposes

$$\begin{aligned} \delta_{BRST}\bar{c}^a &= -b^a\lambda \\ \delta_{BRST}c^a &= -\frac{1}{2}f^{abc}c^bc^c\lambda \\ \delta_{BRST}b^a &= 0 \end{aligned} \tag{11.77}$$

one has

$$\delta_{BRST}L_{eff} = 0 \tag{11.78}$$

This can be proved by direct calculation. Instead, we shall use an important property associated to δ_{BRST} namely that it is nilpotent

$$\delta^2_{BRST} = 0 \tag{11.79}$$

It is important to stress that this property holds without use of the fact $\lambda^2 = 0$. In fact, from here on, we shall omit parameter λ in the definition of δ_{BRST}. This means that δ_{BRST} so defined is now an anticommuting object that takes a commuting field and transforms it into some anticommuting object.

To prove (16.37) one has to see that it is satisfied for each one of the fields. For example, one can easily see that acting on $\bar{c}$ it is nilpotent,

$$\delta^2_{BRST}c = \delta_{BRST}b = 0 \tag{11.80}$$

It is a little more involved to consider the case in which δ_{BRST} acts on A_μ,

$$\begin{aligned} \delta^2_{BRST}A_\mu &= \delta_{BRST}\left(\partial_\mu c + g[A_\mu, c]\right) \qquad (11.81) \\ &= -\frac{1}{2}\partial_\mu[c,c] + [\partial_\mu c, c] + [[A_\mu, c], c] - \frac{1}{2}[A_\mu, [c,c]] \end{aligned}$$

Now, the two first terms in the second line of (16.39) cancel each other. Concerning the remaining two, they are zero due to a Jacobi identity for anticommuting objects. We shall prove this just for the $SU(2)$ case but the proof can be extended to a general gauge group. One has, in components

$$\begin{aligned}\varepsilon^{abc}A_\mu^b c^c \varepsilon^{apq}c^p - \frac{1}{2}\varepsilon^{abc}A_\mu^a \varepsilon^{bpq}c^p c^q \quad &= \quad A_\mu^b c^b c^c - A_\mu^c c^b c^b - \frac{1}{2}A_\mu^b c^b c^c + \\ &\quad \frac{1}{2}A_\mu^b c^c c^b \end{aligned} \tag{11.82}$$

Because c^a is anticommuting, the second term in (16.40) is zero while the third and forth terms add to cancel the first one. One then has proved that $\delta^2_{BRST}A_\mu = 0$. We leave as a problem to show that $\delta^2_{BRST}\bar{c} = \delta^2_{BRST}c = 0$.

Once nilpotency is proved, it is easy to see that L_{eff} is BRST invariant. Of course the Yang-Mills term in L_{eff} remains invariant under BRST transformation, which is a particular case of a gauge transformation,

$$\delta_{BRST}\mathrm{tr}F^2_{\mu\nu} = 0 \tag{11.83}$$

Concerning the remaining two terms in (16.32), let us note that they can be written as the BRST transformed of a given functional. Indeed, consider

$$\delta_{BRST}(\bar{c}F[A]) = (\delta_{BRST}c)F[A] - \bar{c}\frac{\delta F[A]}{\delta A_\mu}\delta_{BRST}A_\mu \tag{11.84}$$

(Here we have taken into account that $\bar{c}$ and δ_{BRST} anticommute). Now, using (16.35) we have,

$$-\delta_{BRST}(\bar{c}F[A]) = bF[A] + \bar{c}\frac{\delta F[A]}{\delta A}D_\mu c \tag{11.85}$$

We can then write

$$L_{eff} = \frac{1}{4}\mathrm{tr}F^2_{\mu\nu} - \mathrm{tr}\delta_{BRST}(\bar{c}F[A]) \tag{11.86}$$

and then

$$\delta_{BRST}L_{eff} = \frac{1}{4}\mathrm{tr}\delta_{BRST}F^2_{\mu\nu} - \mathrm{tr}\delta^2_{BRST}(\bar{c}F[A]) = 0 \tag{11.87}$$

Clase 12

The Quantum Noether theorem

We shall discuss here the issue of symmetries at the quantum level within the path-integral framework. That is, we shall investigate how the symmetries of a classical theory are affected by quantization. This means, in the path-integral approach, that we should investigate how the path-integral measure (the place where the quantum aspects are taken into account) reacts to a symmetry transformation.

The Noether theorem at the classical level

Let us briefly review the Noether theorem for a classical system. For simplicity we consider a scalar field with Lagrangian

$$L = L[\phi, \partial_\mu \phi] \tag{12.1}$$

and a certain field transformation S

$$\phi(x) \xrightarrow{S} \phi'(x) = \phi(x) + \delta_S \phi(x) \tag{12.2}$$

We shall call S a symmetry of the theory if the Euler-Lagrange equations (that is, the equations that govern the dynamics of the system) are not affected by its action. In other words, the equations of motion must remain unaltered. Mathematically, the condition is that the only possible change of the Lagrangian under S can be a total derivative which in the covariant field theory case corresponds to the divergence of a 4-vector,

$$\delta_S L = \partial_\mu \Lambda^\mu[\phi, \partial_\mu \phi] \tag{12.3}$$

Eq.(12.3) should hold independently of the eqs. of motion. Indeed, if this is the case, the action change can be written in the form

$$S' = S + \delta_S S \tag{12.4}$$

$$\delta_S S[\phi, \partial_\mu \phi] = \int_M d^4x \partial_\mu \Lambda^\mu[\phi, \partial_\mu \phi] = \int_{\partial M} \Lambda^\mu dS_\mu \tag{12.5}$$

where ∂M is the boundary of the manifold M on which dynamics is defined. Since Euler-Lagrange equations are derived by varying fields and their derivatives under the condition

$$\delta\phi\,|_{\partial M} = 0 \tag{12.6}$$

the eqs. of motion for S' in eq.(12.4) will not differ from those for S since, in view of (12.6), the change of the action on the boundary vanish.

The Noether therem [24] then states that

For each symmetry S of the theory there exists a locally conserved current j^μ, that is, a current satisfying

$$\partial_\mu j^\mu(x) = 0 \tag{12.7}$$

Equivalently, there is a charge Q,

$$Q = \int d^3x j^0(x) \tag{12.8}$$

which is conserved,

$$\frac{dQ}{dt} = 0 \tag{12.9}$$

The proof of this theorem is constructive. One considers the change in the Lagrangian,

$$\delta_S L = \frac{\partial L}{\partial \phi} \delta_S \phi(x) + \frac{\partial L}{\partial(\partial_\mu \phi)} \delta_S \partial_\mu \phi + O(\phi^2) \tag{12.10}$$

or, using (12.3) in the l.h.s.

$$\partial_\mu \Lambda^\mu = \frac{\partial L}{\partial \phi} \delta_S \eta(x) + \frac{\partial L}{\partial(\partial_\mu \phi)} \partial_\mu \delta_S \phi + O(\eta^2) \tag{12.11}$$

Now, one uses the Euler-Lagrange equations

$$\frac{\delta L}{\delta \phi} = \partial_\mu \frac{\partial L}{\partial(\partial_\mu \phi)} \tag{12.12}$$

so that one has

$$\partial_\mu \frac{\partial L}{\partial(\partial_\mu \phi)} \delta_S \phi(x) + \frac{\partial L}{\partial(\partial_\mu \phi)} \partial_\mu \delta_S \phi(x) = \partial_\mu \Lambda^\mu \tag{12.13}$$

or

$$\partial_\mu \left(\frac{\partial L}{\partial(\partial_\mu \phi)} \delta_S \phi(x) \right) = \partial_\mu \Lambda^\mu \tag{12.14}$$

We have then found the locally conserved current j^μ associated with the symmetry S

$$j^\mu = \Lambda^\mu - \frac{\partial L}{\partial(\partial_\mu \phi)} \delta_S \phi \tag{12.15}$$

$$\partial_\mu j^\mu = 0 \tag{12.16}$$

Nota bene: Usually, when $\Lambda \neq 0$ the symmetry is a space-time symmetry of the physical system (like Lorentz symmetry, diffeomorphism symmetry, etc). Internal symmetries (for example gauge symmetry) correspond in general to $\Lambda = 0$ although there are some exceptions (for example, in the case of Chern-Simons theories for which $\Lambda \neq 0$ under gauge transformations).

The Noether theorem at the quantum level

Let us consider the case of a global symmetry, that is a symmetry for which the parameter of the transformation should be constant. For example in the case of a complex scalar field $\phi = \phi_1 + i\phi_2$ with the usual quartic potential,

$$L = \frac{1}{2} \partial_\mu \phi^* \partial^\mu \phi - V[\phi^* \phi] \tag{12.17}$$

there is a symmetry under constant phase transformations of the form

$$\phi \to \exp(i\epsilon)\phi \tag{12.18}$$

with ϵ constant. This is a global $U(1)$ symmetry.

Lagrangian (12.17) is strictly invariant under (12.18) since $\delta_\epsilon L = 0$, so that we can compute the Noether current from (12.15) putting $\Lambda = 0$. Taking ϕ^* and ϕ as the two independent components of the complex field one gets.

$$j^\mu_\epsilon = -\frac{1}{2}\partial_\mu\phi\delta\phi^* - \frac{1}{2}\partial_\mu\phi^*\delta\phi \tag{12.19}$$

or, using

$$\delta\phi = i\epsilon\phi\,, \qquad \delta\phi^* = -i\epsilon\phi^* \tag{12.20}$$

$$j^\mu = \frac{i}{2}\left(\phi^*\partial^\mu\phi - \phi\partial^\mu\phi^*\right) \tag{12.21}$$

(Here we have written $j^\mu_\epsilon = \epsilon j^\mu$, factorizing the constant phase which plays no role in the definition of the conserved charge).

This is a classical current which is conserved at the classical level,

$$\partial_\mu j^\mu = 0 \tag{12.22}$$

How can we test whether (12.22) remains valid at the quantum level? Within the path integral approach, if the symmetry is still present, we can expect to have,

$$\langle\partial_\mu j^\mu\rangle = 0 \tag{12.23}$$

A simple method to compute the l.h.s. in (12.23) and see whether it vanish is the following (it is called the Noether method, although the path-integral was not invented yet in the times of Emmy Noether). One starts from generating functional for the theory

$$Z = \int \mathcal{D}\phi^*\mathcal{D}\phi\exp(-\int d^4xL) \tag{12.24}$$

one promotes the global infinitesimal symmetry transformation (12.18), (12.20) into a local infinitesimal change of variables,

$$\phi \to \phi' = \phi + i\epsilon(x)\phi\,, \qquad \phi^* \to \phi^{*\prime} = \phi^* - i\epsilon(x)\phi^* \tag{12.25}$$

Obviously, the Lagrangian is no more invariant under this transformation. Evidently, the change in the Lagrangian should come from terms containing derivatives of $\epsilon(x)$ (which are absent in the global case). One easily sees that

$$
\begin{aligned}
\delta_{\partial_\mu \epsilon(x)} L &= i\partial_\mu \left(\frac{\partial L}{\partial(\partial_\mu \phi)} \phi \right) \epsilon(x) - i\partial_\mu \left(\frac{\partial L}{\partial(\partial_\mu \phi^*)} \phi^* \right) \epsilon(x) \\
&+ i\frac{\partial L}{\partial(\partial_\mu \phi)} (\partial_\mu \epsilon) \phi - i\frac{\partial L}{\partial(\partial_\mu \phi^*)} (\partial_\mu \epsilon) \phi^* \\
&= (\partial_\mu j^\mu(x))\, \epsilon(x) + j^\mu(x) \partial_\mu \epsilon(x) \qquad (12.26) \\
&= j^\mu(x) \partial_\mu \epsilon(x) \qquad (12.27)
\end{aligned}
$$

Note that we have used the expression for the global current j^μ in which the global parameter can be factorized out of the divergence. As it was to be expected, the change produced by trhe x dependence of the transformation parameter $\epsilon(x)$ gives a term proportional to its derivative.

Another source of change in Z may come from the path integral measure. In principle we cannot be sure that transformation (12.25) leaves invariant the path integral measure. We call $J[\epsilon]$ the corresponding Jacobian,

$$\mathcal{D}\phi^{*\prime} \mathcal{D}\phi' = J[\epsilon] \mathcal{D}\phi^* \mathcal{D}\phi \qquad (12.28)$$

and then write the path-integral defining Z in terms of the new variables

$$Z = \int \mathcal{D}\phi^{*\prime} \mathcal{D}\phi' J[\epsilon] \exp\Big(-\int d^4x L - \int d^4x (\partial_\mu \epsilon(x)) j^\mu(x) \Big) \qquad (12.29)$$

The path-integral in the r.h.s. cannot depend on the arbitrary parameter $\epsilon(x)$ introduced in the change of variables. Or, in other words, Z cannot depend on $\epsilon(x)$. We can then write, after differentiating with respect to $\epsilon(x)$

$$0 = \frac{1}{Z} \frac{\delta Z}{\delta \epsilon(x)} = \langle \partial_\mu j^\mu(x) \rangle + \frac{\delta \log J[\epsilon]}{\delta \epsilon(x)} \qquad (12.30)$$

That is

$$\langle \partial_\mu j^\mu(x) \rangle = -\frac{\delta \log J[\epsilon]}{\delta \epsilon(x)} \qquad (12.31)$$

Eq.(12.31) clearly indicates the condition under which a classical global symmetry survives quantization: the Jacobian associated to the local change of variables associated to the classical symmetry transformation should be trivial, $J[\epsilon] = 1$, so that

$$\langle \partial_\mu j^\mu(x) \rangle = 0 \tag{12.32}$$

This is completely consistent with the path-integral approach to quantization: the path integral measure is the place where quantum aspects are taken into account and then a quantum violation of a classical symmetry, an anomaly[5], should manifest as a non-trivial Jacobian associated with the symmetry transformation.

What about the case of symmetry transformations which are already local. A local change of variables will leave the lagrangian invariant and hence no current term is generated in the transformed action so as to follow the Noether method as above. There is a way out to this which we shall exemplify with the scalar field example. In order to have a complex scalar field theory with a local (gauge) invariance one should replace the derivatives in (12.17) by covariant derivatives including a gauge field connection (one then adds also an invartiant kinetic Lagrangian for the gauge field). It is only when both fields are transformed that the Lagrangian remains invariant. The trick to obtain a formula like (12.31) or (12.32) is to proceed within the path integral approach making a change of variables only for the matter fields (i.e, changing ϕ^* and ϕ but not the gauge field connection). This will produce a change in the Lagrangian which is not compensated by any change in the gauge field part and the change will be precisely of the form calculated above. One then proceeds as before, getting an answer for the v.e.v. of the matter field current divergence.

Non-Abelian gauge symmetry

Consider a gauge group G with generators

$$\begin{aligned} [t^a, t^b] &= i f^{abc} t^c\,, \quad a,b,c = 1,2,3,\ldots,\mathrm{dim}G \\ \mathrm{tr}(t^a t^b) &= \frac{1}{2}\delta^{ab} \end{aligned} \tag{12.33}$$

[5]**anomalous**: deviating from what is standard, normal, or expected: *there are a number of anomalies in the present system.*
abnormal: deviating from what is normal or usual, typically in a way that is undesirable or worrying: *the illness is recognizable from the patients abnormal behaviour*

and the gauge field connection

$$A_\mu(x) = A_\mu^a(x) t^a \tag{12.34}$$

with curvature

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + e[A_\mu, A_\nu] \tag{12.35}$$

Under a gauge transformation $g(x)$,

$$g(x) = \exp\left(i\theta^a(x) t^a\right) \ , \quad g(x) : M \to G \tag{12.36}$$

the gauge field and curvature change as

$$\begin{aligned} A_\mu^g &= g^{-1} A_\mu g - \frac{i}{e} g^{-1} \partial_\mu g \\ F_{\mu\nu}^g &= g^{-1} F_{\mu\nu}\, g \end{aligned} \tag{12.37}$$

but the Yang-Mills Lagrangian

$$L_{YM} = \frac{1}{2} \mathrm{tr} F_{\mu\nu} F^{\mu\nu} \tag{12.38}$$

remains invariant

$$\delta_g L_{YM} = 0 \tag{12.39}$$

According to the Noether theorem, there is a conserved current associated with this symmetry. Under an infinitesimal gauge transformation the gauge field transforms according to

$$\delta_\epsilon A_\mu = \frac{1}{e}\left(\partial_\mu \epsilon + ie[A_\mu, \epsilon]\right) = \frac{1}{e} D_\mu[A]\epsilon \tag{12.40}$$

where ϵ is defined through

$$g(x) = \exp(i\epsilon^a t^a) \approx I + i\epsilon^a t^a = I + i\epsilon \tag{12.41}$$

Then, using eq.(12.15) with $\Lambda = 0$ we have for the conserved current

$$\tilde{j}_g^\mu = \frac{\partial L}{\partial(\partial_\mu A_\nu^a)} \delta_\epsilon A_\nu^a = \frac{1}{e} F^{\mu\nu\, a} D_\nu \epsilon^a \tag{12.42}$$

Let us first consider a <u>global</u> gauge transformation. Then $\epsilon(x) = \epsilon$ with $\partial_\mu \epsilon = 0$ and the current takes the form

$$\tilde{j}_g^\mu = f^{abc} F^{\mu\nu\, a} A_\nu^b \epsilon^c \tag{12.43}$$

Extracting the infinitesimal parameter ϵ from $\tilde{j}^\mu$ we can define a conserved current taking values in the Lie algebra of the gauge group,

$$j_{gl}^{a\,\mu} = f^{abc} F^{\mu\nu\,b} A_\nu^c \tag{12.44}$$

or

$$j_g^\mu = -i[F^{\mu\nu}, A_\nu] \tag{12.45}$$

The global charge is then defined as

$$Q_g = -i \int d^3x [F^{0i}, A_i] \,, \qquad \frac{dQ_g}{dt} = 0 \tag{12.46}$$

or, in components

$$Q_g^a = f^{abc} \int d^3x F^{0i\,b} A_i^c \,, \qquad \frac{dQ_g^a}{dt} = 0 \,, \quad a = 1, 2, \ldots \dim G \tag{12.47}$$

One can also write this charge in the form

$$Q_g^a = e f^{abc} \int d^3x E^{i\,b} A_i^c \tag{12.48}$$

and, from this, compute, using the canonical Poisson brackets for A_i and E^i , the algebra satisfied by Q_g^a which is of course nothing but the gauge algebra,

$$\{Q_g^a, Q_g^b\} = ie f^{abc} Q_g^c \tag{12.49}$$

From the equations of motion,

$$D_\mu F^{\mu\nu} = 0 \tag{12.50}$$

we have

$$D_i F^{i0} = 0 \tag{12.51}$$

or

$$\partial_i E^i = -ie[A^i, F^{0i}] = e j_g^0 \tag{12.52}$$

Then, we can write the conserved charge as a surface integral,

$$Q_g = \frac{1}{e} \int d^3x \partial_i E^i = \frac{1}{e} \int_{\partial M} dS_i E^i \tag{12.53}$$

this showing that charge conservation is related to the behavior of the theory at the boundary. Finally, one easily checks that the charge generators transform covariantly under gauge transformations. Indeed, from the results above one finds, for a gauge transformation element $h(x)$

$$Q_g \stackrel{h(x)}{\longrightarrow} Q'_g = h(x)^{-1} Q_g h(x) \tag{12.54}$$

Have we lost any information in considering global and not local gauge transformations? The point in the derivation above where we used the fact that ϵ was constant was when we considered eq.(12.42)

$$\tilde{j}^{\mu}_g = \frac{\partial L}{\partial(\partial_\mu A^a_\nu)} \delta_\epsilon A^a_\nu = \frac{1}{e} F^{\mu\nu\, a} D_\nu \epsilon^a \tag{12.55}$$

and put $\partial_\mu \epsilon = 0$. Let us then keep this term in the covariant derivative and write for the charge, which we call $\tilde{Q}_{local}$

$$\tilde{Q}_{local} = \frac{1}{e} \int d^3x F^{0i,a} D^{ab}_i \epsilon^b \tag{12.56}$$

We can rewrite this formula in the form

$$\begin{aligned}
\tilde{Q}_{local} &= \frac{1}{e} \int d^3x F^{0i,a} \partial_i \epsilon^a(x) + i f^{abc} \int d^3x F^{0i,a} A^b_i \epsilon^c(x) \\
&= \frac{1}{e} \int d^3x \partial_i (F^{0i,a} \epsilon^a(x)) - \frac{1}{e} \int d^3x (\partial_i F^{0i,a}) \epsilon^a(x) + \\
&\quad i f^{abc} \int d^3x F^{0i,a} A^b_i \epsilon^c(x)
\end{aligned} \tag{12.57}$$

Now, the second and third terms cancel because of the equations of motion while the first one can be integrated using Gauss-Stokes theorem to give,

$$\tilde{Q}_{local} = \frac{1}{e} \int_{\partial M} DS_i\, F^{0i,a} \epsilon^a(x) \tag{12.58}$$

Normally, one considers gauge transformations that go the the identity on the boundary. This means

$$\lim_{|x|\to\infty} g(x) = I \qquad \text{or} \qquad \epsilon(x)|_{\partial M} = 0 \tag{12.59}$$

We then have $\tilde{Q}_{local} = 0$ and there is no new charge! What about the case in which $\epsilon(x)|_{\partial M} = \epsilon_0 \neq 0$? In that case one has

$$\tilde{Q}_{local} = \frac{1}{e}\epsilon_0^a \int_{\partial_M} DS_i \, F^{0i,a} = \epsilon_0^a Q_g^a \tag{12.60}$$

and one recovers the same charge found when considering global transformations.

We conclude that there is no use in distinguishing global and local gauge transformations: at the level of conserved charges, one obtains the same answer whether one considers local gauge transformations or its particular case of local ones.

Adding matter

Let us add to the Yang-Mills Lagrangian (12.38)

$$L_{YM} = \frac{1}{2}\mathrm{tr} F_{\mu\nu}F^{\mu\nu} \tag{12.61}$$

matter fields which we choose to be described by fermions in some representation R of the gauge group G. Except indication, we shall take for simplicity R as the fundamental representation. The minimal gauge invariant coupling corresponds to the Dirac Lagrangian

$$L_F = \bar{\psi}(i\partial\!\!\!/ + qA\!\!\!/)\psi = \bar{\psi}^r \left(i\partial\!\!\!/ \delta^{rs} + qA_\mu^a \left(T^a\right)_{rs} \gamma^\mu \right) \psi^s \tag{12.62}$$

where T^a_{rs} are the gauge group generators in the fundamental representation of G, $a = 1, 2, \ldots, \mathrm{dim}G$; $r, s = 1, 2, \ldots, \mathrm{dim}R$. Concerning q, it represents the charge of ψ with respect to its coupling to A_μ. It may or not coincide with the coupling e appearing in the curvature. We shall then write $q = re$ with r a number depending on the representation for matter fields.

Under gauge transformations

$$\begin{aligned} \psi \to \psi^g &= g^{-1}\psi \\ \bar{\psi} \to \bar{\psi}^g &= \bar{\psi} g \\ A_\mu \to A_\mu^g &= g^{-1}A_\mu g - \frac{i}{e} g^{-1}\partial_\mu g \end{aligned} \tag{12.63}$$

the total Lagrangian

$$L = L_{YM} + L_F \tag{12.64}$$

remains invariant , $\delta_g L = 0$. Note that acting on fermions g should be written in the form

$$g = \exp\left(irT^a \epsilon^a(x)\right) \tag{12.65}$$

whith r depending on the representation.

As in the pure gauge case, global and local symmetry lead to the same conserved Noether charge. We shall then consider the global case. The Noether current picks now two contributions, one coming from the Yang-Mills Lagrangian, which we have already computed, the new one coming from the fermion Lagrangian,

$$j^\mu_\epsilon = \frac{\partial L}{\partial(\partial_\mu A^a_\nu)}\delta A^a_\nu + \frac{\partial L}{\partial_\mu \psi^r}\delta\psi^r - \delta\bar{\psi}^r \frac{\partial L}{\partial_\mu \psi^r} \tag{12.66}$$

with

$$\begin{aligned} \delta A_\mu &\approx \frac{1}{e} D_\mu[A]\epsilon = i[A_\mu, \epsilon] \\ \delta\psi &= \psi^\epsilon - \psi \approx -ir\epsilon\psi\ , \quad \delta\bar{\psi} = \bar{\psi}^\epsilon - \bar{\psi} \approx +ir\bar{\psi}\epsilon \end{aligned} \tag{12.67}$$

Defining $j^\mu_\epsilon = j^{\mu a}\epsilon^a$ we have

$$j^{\mu a} = \overbrace{f^{abc}F^{\mu\nu b}A^c_\nu}^{\text{gauge current}} + \overbrace{i2r\bar{\psi}\gamma^\mu T^a\psi}^{\text{matter current}} \equiv j^{\mu a}_{\text{gauge}} + j^{\mu a}_{\text{matter}} \tag{12.68}$$

It is important to note that the equations of motion for the gauge field take the form

$$D_\nu F^{\mu\nu} = \frac{e}{2} j^{\mu a}_{\text{matter}} \tag{12.69}$$

so that we see that the matter part of the Noether current acts as the source for the gauge field strength. Neither the gauge current nor the matter current are conserved by separate,

$$\partial_\mu j^{\mu a}_{\text{gauge}} \neq 0\ , \qquad \partial_\mu j^{\mu a}_{\text{matter}} \neq 0 \tag{12.70}$$

but of course their sum is

$$\partial_\mu \left(j^{\mu a}_{\text{gauge}} + j^{\mu a}_{\text{matter}}\right) = 0 \tag{12.71}$$

One can see however that the matter field is *covariantly conserved*. Indeed, from eq.(12.69) one has

$$D_\mu D_\nu F^{\mu\nu} = e D_\mu j^{\mu a}_{\text{matter}} \tag{12.72}$$

which, in view of the antisymmetry of the field strength can be written as

$$\frac{1}{2}[D_\mu, D_\nu] F^{\mu\nu} = e D_\mu j^{\mu a}_{\text{matter}} \tag{12.73}$$

Now, using

$$[D_\mu, D_\nu]\Phi = ie[F_{\mu\nu}, \Phi] \tag{12.74}$$

one can see that the left hand side of (12.73) is zero,

$$\frac{1}{2}[D_\mu, D_\nu] F^{\mu\nu} = \frac{ie}{2}[F_{\mu\nu}, F^{\mu\nu}] = 0 \tag{12.75}$$

so that one has

$$0 = D_\mu j^{\mu a}_{\text{matter}} \tag{12.76}$$

Of course in the Abelian case, since $D_\mu = \partial_\mu$ one has matter current conservation,

$$\partial_\mu j^\mu_{\text{matter}} = \partial_\mu \left(\bar\psi \gamma^\mu \psi\right) = 0 \qquad \text{U(1) symmetry} \tag{12.77}$$

but this can be viewed just as a mathematicl result since it is a consequence of antisymmetry of the field strength,

$$\partial_\mu \partial_\nu F^{\mu\nu} = \partial_\mu j^\mu_{\text{matter}} = 0 \tag{12.78}$$

Other global symmetries of the matter Lagrangian

We start, for simplicty, with the $U(1)$ case. One can easily see that the (Euclidean) Dirac Lagrangian

$$L_F = \bar\psi(i\partial\!\!\!/ + eA\!\!\!/)\psi = \bar\psi D\!\!\!\!/[A]\psi \tag{12.79}$$

is invariant under global chiral transformations. Define the γ_5 matrix

$$\gamma_5 = i\gamma_0\gamma_1\gamma_2\gamma_3 \tag{12.80}$$

satisfying the very important property

$$\gamma_\mu \gamma_5 = -\gamma_5 \gamma_\mu \tag{12.81}$$

which guarantees that

$$\gamma_5 \not{D}[A] = -\not{D}[A]\gamma_5 \tag{12.82}$$

Then, under the global chiral (or γ_5) transformation with parameter α

$$\begin{aligned} \psi &\to \psi' = \exp(\gamma_5 \alpha)\psi \\ \bar{\psi} &\to \bar{\psi}' = \bar{\psi}\exp(\gamma_5 \alpha) \end{aligned} \tag{12.83}$$

the Lagrangian remains invariant,

$$\begin{aligned} L \to L' &= \bar{\psi}\exp(\gamma_5\alpha)\not{D}[A]\exp(\gamma_5\alpha)\psi \\ &= \bar{\psi}\exp(\gamma_5\alpha)\exp(-\gamma_5\alpha)\not{D}[A]\psi = L \end{aligned} \tag{12.84}$$

Here we have used, extending (12.30) and using that α is constant

$$\not{D}[A]\exp(\gamma_5\alpha) = \exp(-\gamma_5\alpha)\not{D}[A] \tag{12.85}$$

We have then

$$\delta_5 L = 0 \tag{12.86}$$

so that we know there should be a Noether conserved current

$$j_5^\mu = \frac{\partial L}{\partial(\partial_\mu \psi)}\delta_5\psi \tag{12.87}$$

with

$$\delta_5\psi = \psi' - \psi \approx \alpha\gamma_5\psi \tag{12.88}$$

so that one gets

$$j_5^\mu = i\bar{\psi}\gamma^\mu\gamma_5\psi \tag{12.89}$$

which transforms as an axial quadrivector under Lorentz transformations.

We then have, together with the gauge current, an axial current which is classically conserved

$$\partial_\mu j^\mu = 0\,, \quad j^\mu = i\bar{\psi}\gamma^\mu\psi \tag{12.90}$$

$$\partial_\mu j_5^\mu = 0\,, \quad j_5^\mu = i\bar{\psi}\gamma^\mu\gamma_5\psi \tag{12.91}$$

Together with the electric charge, we then have an axial charge which is conserved,

$$\frac{dQ_5}{dt} = 0 \,, \quad Q_5 = \int d^3x j_5^0(t,x) \tag{12.92}$$

If we write a four component spinor in the form

$$\psi = \begin{pmatrix} \psi^1 \\ \psi^2 \\ \psi^3 \\ \psi^4 \end{pmatrix} \equiv \begin{pmatrix} \psi^+ \\ \psi^- \end{pmatrix} \tag{12.93}$$

and use the chiral representation for Dirac matrices which corresponds to a γ_5 matrix in the form

$$\gamma_5 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \equiv \begin{pmatrix} I_{2\times 2} & 0 \\ 0 & -I_{2\times 2} \end{pmatrix} \tag{12.94}$$

we see that the chiral charge can be written as

$$Q_5 = \int d^3x \left(|\psi^+|^2 - |\psi^-|^2 \right) \tag{12.95}$$

to be compared with the electric charge

$$Q = \int d^3x \left(|\psi^+|^2 + |\psi^-|^2 \right) \tag{12.96}$$

Of course, a mass term for the fermion breaks chiral invariance. Indeed, consider a mass term of the form

$$L_{mass} = im\bar{\psi}\psi \tag{12.97}$$

(The "i" in front of the mass arises in Euclidean space so as to ensure a correct pole in the propagator). Then, under an infinitesimal chiral change this term changes as

$$\delta_5 L_{mass} = im\alpha\bar{\psi}\gamma_5\psi \tag{12.98}$$

We can still define the axial current in the form $j_5^\mu = i\bar{\psi}\gamma^\mu\gamma_5\psi$ but, instead of eq.(12.91), the following equation for the chiral current holds,

$$\partial_\mu j_5^\mu = 2im\bar{\psi}\gamma_5\psi \tag{12.99}$$

If $m \approx 0$, one can consider that the axial current is a *partially conserved* current.

The axial symmetry, as the gauge symmetry, is a $U(1)$ symmetry. In fact, if we look for its action on each Weyl component of the 4 component spinor one has (in Minkowski space for convenience of the argument)

$$\begin{aligned} \psi^+ &\stackrel{chiral\ t.}{\longrightarrow} \exp(i\alpha)\psi^+ \\ \psi^- &\stackrel{chiral\ t.}{\longrightarrow} \exp(-i\alpha)\psi^- \end{aligned} \tag{12.100}$$

to be compared with gauge transformations

$$\begin{aligned} \psi^+ &\stackrel{gauge\ t.}{\longrightarrow} \exp(i\alpha)\psi^+ \\ \psi^- &\stackrel{gauge\ t.}{\longrightarrow} \exp(i\alpha)\psi^- \end{aligned} \tag{12.101}$$

It is then natural to define two charges $Q_\pm$,

$$Q_\pm = \frac{1}{2}(Q \pm Q_5) \tag{12.102}$$

so that Q_+ changes the right and Q_- the left parts of the fermion spinor. In this context, one says that the massless fermion Lagrangian has a $U(1)_L \times U(1)_R$ symmetry.

Noether symmetries at the quantum level

We define the generating functional $Z[A]$ for fermions in a gauge field background as

$$Z[A] = \int \mathcal{D}\bar{\psi}\mathcal{D}\psi \exp\left(\int d^4x\bar{\psi}(i\partial\!\!\!/ + qA\!\!\!/)\psi\right) = \det D\!\!\!\!/[A] \tag{12.103}$$

The gauge field background can be thought of as an external source to obtain current correlation function through differentiation (and eventually putting the background to zero). For example

$$\begin{aligned} \langle j^\mu(x)\rangle &= \frac{1}{Z}\frac{\delta Z}{\delta A_\mu(x)}\bigg|_{A=0} \\ \langle j^\mu(x)j^\nu(y)\rangle &= \frac{1}{Z}\frac{\delta^2 Z}{\delta A_\mu(x)\delta A_\mu(y)}\bigg|_{A=0} \end{aligned} \tag{12.104}$$

Evidently, the Lagrangian in (12.103) is invariant under $U(1)$ gauge transformations,

$$\begin{aligned} \psi \to \psi' &= \exp(i\alpha)\psi \\ \bar{\psi} \to \bar{\psi}' &= \bar{\psi}\exp(-i\alpha) \end{aligned} \tag{12.105}$$

so that at the classical level we have

$$\partial_\mu j^\mu = 0 \,, \qquad j_\mu = i\bar{\psi}\gamma^\mu\psi \tag{12.106}$$

In order to study the symmetry at the quantum level we promote, following the Noether method, the symmetries to local change of variables. We first consider gauge transformations and then define the change of fermion variables

$$\begin{aligned} \psi \to \psi' &= \exp(i\alpha(x))\psi \\ \bar{\psi} \to \bar{\psi}' &= \bar{\psi}\exp(-i\alpha(x)) \end{aligned} \tag{12.107}$$

(Note that the background field is not affected by the transformation: it is not a "complete" gauge transformation but just a change of fermion variables. This change produces a change in the Lagrangian, which is proportional to $\partial_\mu\alpha$,

$$\delta_{\alpha(x)}L = i\bar{\psi}\gamma^\mu\psi\partial_\mu\alpha = j^\mu\partial_\mu\alpha \tag{12.108}$$

At the path integral level, the change should be also implemented in the path integral measure and can in principle be accompanied by a Jacobian

$$\mathcal{D}\bar{\psi}\mathcal{D}\psi \to \mathcal{D}\bar{\psi}'\mathcal{D}\psi' = J^{-1}[\alpha]\mathcal{D}\bar{\psi}\mathcal{D}\psi \tag{12.109}$$

Then, when we perform the change of variables in Z we end with

$$\begin{aligned} Z[A] &= J[\alpha]\int \mathcal{D}\bar{\psi}'\mathcal{D}\psi' \exp\left(-\int d^4x\left(\bar{\psi}'(i\partial\!\!\!/ + eA\!\!\!/) + j^\mu\partial_\mu\alpha(x)\right)\psi'\right) \\ &= J[\alpha]\int \mathcal{D}\bar{\psi}'\mathcal{D}\psi' \exp\left(-\int d^4x\left(\bar{\psi}'(i\partial\!\!\!/ + eA\!\!\!/) - \alpha(x)\partial_\mu j^\mu\right)\psi'\right) \end{aligned} \tag{12.110}$$

As usual, we now use the fact that Z is independent of the arbitrary function $\alpha(x)$ so that, differentiating $\log Z$ with respect to α and then putting $\alpha = 0$ we obtain

$$\langle \partial_\mu j^\mu \rangle = - \left. \frac{\delta \log J[\alpha]}{\delta \alpha} \right|_{\alpha=0} \tag{12.111}$$

Now, $Z[A]$ in (12.110) can also be written, after the change of variables as

$$\begin{aligned} Z[A] &= J[\alpha] \int \mathcal{D}\bar{\psi}'\mathcal{D}\psi' \exp\left(- \int d^4x \bar{\psi}'(i\not{\partial} + e\not{A} + \gamma^\mu \partial_\mu \alpha)\psi' \right) \\ &= J[\alpha] \int \mathcal{D}\bar{\psi}'\mathcal{D}\psi' \exp\left(- \int d^4x \bar{\psi}'(i\not{\partial} + e\not{A}^\alpha)\psi' \right) \\ &= J[\alpha] \det(i\not{\partial} + e\not{A}^{\alpha(x)}) \end{aligned} \tag{12.112}$$

where we have written

$$A_\mu^{\alpha(x)} = A_\mu + \frac{1}{e}\partial_\mu \alpha(x) \tag{12.113}$$

But we have also $Z[A] = \det(i\not{\partial} + e\not{A})$ and then we can write

$$\det(i\not{\partial} + e\not{A}) = J[\alpha] \det(i\not{\partial} + e\not{A}^{\alpha(x)}) \tag{12.114}$$

so that the Jacobian can be seen as the ratio of two determinants: that of the Dirac operator in a background and the one for the gauge transformed background.

Naively, one would say that since

$$i\not{\partial} + e\not{A}^{\alpha(x)} = \exp(-i\alpha)(i\not{\partial} + e\not{A}) \exp(i\alpha) \tag{12.115}$$

and determinants are endowed with the cyclic property, one has

$$\begin{aligned} \det\left(i\not{\partial} + e\not{A}^{\alpha(x)}\right) &= \det\left(\exp(-i\alpha)(i\not{\partial} + e\not{A}) \exp(i\alpha)\right) \\ &= \det\left(\exp(i\alpha) \exp(-i\alpha)(i\not{\partial} + e\not{A})\right) \\ &= \det(i\not{\partial} + e\not{A}) \end{aligned} \tag{12.116}$$

this giving

$$J[\alpha] = 1 \tag{12.117}$$

But the point is the following: the Dirac operator being unbounded, neither $Z[A]$ nor the identity (12.114) are well-defined. A regularization prescription for computing the determinant should be adopted. It may happen that a given regularization can not preserve the cyclic property. This would mean that gauge invariance is spoiled. Only when one finds a gauge-invariant regularization one is sure that $J[\alpha] = 1$ or, equivalently, that the path-inegral measure can be regularized in a gauge invariant way. As we shall see, when gauge fields are coupled to Dirac fermions this is always possible (In contrast, for Weyl fermions this is not, in general, the case).

Now, suppose that a gauge invariant regularization is found and used to regularize the Jacobian in such a way that $J = 1$. Consider then a chiral transformation instead of a gauge one. Repeating all the precedent steps, one starts from a chiral change of variables

$$\begin{aligned} \psi \to \psi' &= \exp(\gamma_5\alpha(x))\psi \\ \bar\psi \to \bar\psi' &= \bar\psi \exp(\gamma_5\alpha(x)) \end{aligned} \tag{12.118}$$

This change produces a change in the Lagrangian, which again is proportional to $\partial_\mu\alpha$ (the Lagrangian is invariant under chiral *global* transformations,

$$\delta_{\gamma_5\alpha(x)} L = \bar\psi\gamma^\mu\gamma_5\psi\partial_\mu\alpha = j_5^\mu\partial_\mu\alpha \tag{12.119}$$

At the path integral level, we define the Jacobian

$$\mathcal{D}\bar\psi\mathcal{D}\psi \to \mathcal{D}\bar\psi'\mathcal{D}\psi' = J_5^{-1}[\alpha]\mathcal{D}\bar\psi\mathcal{D}\psi \tag{12.120}$$

so that when we perform the change of variables in Z we end with

$$Z[A] = J_5[\alpha]\int \mathcal{D}\bar\psi'\mathcal{D}\psi' \exp\left(-\int d^4x \left(\bar\psi'(i\partial\!\!\!/ + eA\!\!\!/) + j_5^\mu\partial_\mu\alpha(x)\right)\psi'\right) \tag{12.121}$$

Again, we use the fact that Z is independent of the arbitrary function $\alpha(x)$ so that, differentiating $\log Z$ with respect to α and then putting $\alpha = 0$ we obtain

$$\langle\partial_\mu j_5^\mu\rangle = -\left.\frac{\delta \log J_5[\alpha]}{\delta\alpha}\right|_{\alpha=0} \tag{12.122}$$

An analogous treatment as that followed for gauge transformations leads for $J_5[\alpha]$ to

$$\det(i\partial\!\!\!/ + eA\!\!\!/) = J[\alpha]\det\Big(\exp\left(\gamma_5\alpha(x)\right)(i\partial\!\!\!/ + eA\!\!\!/)\exp\left(\gamma_5\alpha(x)\right)\Big) \tag{12.123}$$

or,

$$J_5[\alpha] = \frac{\det(i\partial\!\!\!/ + eA\!\!\!/)}{\det\Big(\exp\left(\gamma_5\alpha(x)\right)(i\partial\!\!\!/ + eA\!\!\!/)\exp\left(\gamma_5\alpha(x)\right)\Big)} \tag{12.124}$$

One again needs to regularize determinants. If one uses (as one should if gauge invariance is to be respected) the regularization leading to $J = 1$, then, in principle, there is no reason to expect that also $J_5 = 1$. We shall show below how one effectively computes this Jacobian, finding that it is not trivial.

One sees that in the denominator, one has something similar to the previous case when gauge transformations were consider, except that the exponentials have the same sign. Even if the cyclic property of the determinant is respected, these exponentials do not cancel but add. A curious fact takes place for the case in which $\alpha(x) = \alpha =$ constant. Since γ_5 anticommutes with the Dirac operator, one can pass one of the exponentials through it by changing its sign and cancel the other one. Then, one would obtain $J_5 = 1$ in this way while the cyclic property would give in principle a different result since, when it is used, the arguments in the exponentials add. The answer to this apparent contradiction will be clear after computation of the chiral Jacobian.

Clase 13

The chiral Jacobian à la Fujikawa

We have seen through the use of the Noether method how to to compute the vacuum expectation value of the divergence of the chiral current. The result showed that, at the quantum level, there is a possible source of non conservation arising from a chiral Jacobian,

$$\langle \partial_\mu j_5^\mu \rangle = - \left. \frac{\delta \log J_5[\alpha]}{\delta \alpha} \right|_{\alpha=0} \tag{13.1}$$

Here, the chiral current j_5^μ is defined as

$$j_5^\mu = i\bar{\psi}\gamma^\mu\gamma_5\psi \tag{13.2}$$

and the chiral Jacobian J_5 was defined either in terms of change of the path-integral measure under a chiral local change of fermionic variables,

$$\begin{array}{rcl} \psi \to \psi' & = & \exp(\gamma_5\delta\alpha(x))\psi \\ \bar{\psi} \to \bar{\psi}' & = & \bar{\psi}\exp(\gamma_5\delta\alpha(x)) \end{array} \tag{13.3}$$

$$\mathcal{D}\bar{\psi}\mathcal{D}\psi \to \mathcal{D}\bar{\psi}'\mathcal{D}\psi' = J_5^{-1}[\delta\alpha]\mathcal{D}\bar{\psi}\mathcal{D}\psi \tag{13.4}$$

or in terms of a ratio of determinants (the ratio of partition functions before and after the chiral change),

$$J_5[\alpha] = \frac{\det(i\partial\!\!\!/ + eA\!\!\!/)}{\det\Big(\exp\left(-\gamma_5\alpha(x)\right)\left(i\partial\!\!\!/ + eA\!\!\!/\right)\exp\left(-\gamma_5\alpha(x)\right)\Big)} \tag{13.5}$$

Fujikawa original derivation [25] of the chiral Jacobian was based in (13.4) for an infinitesimal change of variables. We shall describe in the first part of this class the way he followed. Then, we shall give the alternative derivation, based on definition (13.5) which is more appropriate for computing Jacobians (not necessarily chiral ones) for finite changes of variables [26].

At the point, the important point to stress in (13.5) is that each determinant in the ratio is in principle ill-defined: The Dirac operator is unbounded in the sense that its eigenvalues grow without bound so, in computing the Jacobian, we shall need to handle this problem as we had already done when computing the determinant for the harmonic oscillator.

Fujikawa took seriously the definition of the path integral measure in terms of the (anti-commuting) Bessel-Fourier coefficients in the expansion of fermions,

$$\begin{aligned}\psi(x) &= \sum_n c_n \varphi_n(x) \\ \bar{\psi}(x) &= \sum_n \bar{c}_n \varphi_n^\dagger(x)\end{aligned} \tag{13.6}$$

Here c_n and $\bar{c}_n$ are Grassman valued objects and $\{\varphi_n(x)\}$ an adequate basis of Dirac spinors,

$$\begin{aligned}&\int \varphi_n^\dagger(x)\varphi_m(x)d^4x = \delta_{nm} \\ &\sum_n \varphi_n(x)\otimes\varphi_n^\dagger(y) = I_{4\times4}\delta^{(4)}(x-y)\end{aligned} \tag{13.7}$$

Note that in the expansion of $\bar{\psi}$ we use $\varphi^\dagger$ instead of $\bar{\varphi}$. We shall come to this election later.

We have seen that a possibility for putting to work the Feynman recipe of *integrating over all fields* was to define the path-integral measure as

$$\mathcal{D}\bar{\psi}\mathcal{D}\psi = \prod_n d\bar{c}_n dc_n \tag{13.8}$$

Then, in order to compute the Jacobian in (13.4), we have to consider a Bessel-Fourier expansion for the chirally transformed fermion fields,

$$\begin{aligned}\psi'(x) &= \sum_n c'_n \varphi_n(x) \\ \bar{\psi}'(x) &= \sum_n \bar{c}'_n \varphi_n^\dagger(x)\end{aligned} \tag{13.9}$$

That is, the coefficients $\bar{c}'_n, c'_n$ are different from those in (13.6) but spinorial basis is of course the same. The Jacobian is then defined in terms of the expansion coefficients that make the measure change from $\mathcal{D}\bar{\psi}\mathcal{D}\psi$ to $\mathcal{D}\bar{\psi}'\mathcal{D}\psi'$,

$$\prod_n d\bar{c}_n dc_n = J_5[\delta\alpha] \prod_n d\bar{c}'_n dc'_n \tag{13.10}$$

Now, from (13.3) we have

$$\psi'(x) = \exp(\gamma_5 \delta\alpha(x))\psi(x) \tag{13.11}$$

so that we can easily connect c'_n with c_n,

$$\sum_n c'_n \varphi_n(x) = \exp(\gamma_5 \delta\alpha(x)) \sum_n c_n \varphi_n(x) \tag{13.12}$$

Multiplying by $\varphi^\dagger_m(x)$ and integrating one gets, after using ortonormality in the l.h.s.

$$c'_m = \int d^4x \varphi^\dagger_m(x) \exp(\gamma_5 \delta\alpha(x)) \sum_n c_n \varphi_n(x) = \sum_n C_{mn} c_n \tag{13.13}$$

where

$$C_{mn} = \int d^4x \varphi^\dagger_m(x) \exp(\gamma_5 \delta\alpha(x)) \varphi_n(x) \tag{13.14}$$

The same result is obtained for $\bar{\psi}$,

$$\bar{c}'_m = \sum_n C_{mn} \bar{c}_n \tag{13.15}$$

so that one can write for the Jacobian,

$$J_5^{-1}[\delta\alpha] = \det{}^{-2} C_{mn} \tag{13.16}$$

where the negative power in the l.h.s. is due to the definition (13.4) while that in the r.h.s. arises because c'_n and $\bar{c}'_n$ are Grassman objects and, as we have seen, Jacobians in that case appear with negative powers in comparison with the commuting case.

Now we shall calculate C_{mn}. To this end, we use the fact that we are considering an infinitesimal transformation and write (13.14) in the form

$$C_{mn} \approx \int d^4x \varphi^\dagger_m(x)(1 + \gamma_5 \delta\alpha(x))\varphi_n(x) = \delta_{mn} + \int d^4x \varphi^\dagger_m(x) \gamma_5 \varphi_n(x) \delta\alpha(x) \tag{13.17}$$

Then, we can write the C_{mn} determinant in the form

$$\det C = \begin{vmatrix} 1+\int d^4x a_{11}\delta\alpha(x) & \int d^4x a_{12}\delta\alpha(x) & \dots & \int d^4x a_{1n}\delta\alpha(x) \\ \int d^4x a_{21}\delta\alpha(x) & 1+\int d^4x a_{22}\delta\alpha(x) & \dots & \int d^4x a_{2n}\delta\alpha(x) \\ \dots & \dots & \dots & \dots \\ \int d^4x a_{n1}\delta\alpha(x) & \int d^4x a_{n2}\delta\alpha(x) & \dots & 1+\int d^4x a_{nn}\delta\alpha(x) \end{vmatrix} \tag{13.18}$$

where

$$a_{nm} = \varphi_n^\dagger(x)\gamma_5\varphi_m(x) \tag{13.19}$$

Then, to first order in $\delta\alpha$ the diagonal factors contribute and one ends with

$$\det C = 1 + \sum_n \int d^4x a_{nn}\delta\alpha(x) = \exp\left(\sum_n \int d^4x a_{nn}\delta\alpha(x)\right) \tag{13.20}$$

or

$$J_5[\delta\alpha] = \exp\left(2\int d^4x \sum_n \varphi_n^\dagger(x)\gamma_5\varphi_n(x)\delta\alpha(x)\right) \tag{13.21}$$

Defining

$$\mathcal{A} = 2\sum_n \varphi_n^\dagger(x)\gamma_5\varphi_n(x) \tag{13.22}$$

and using eq.(13.1), we finally have for the anomaly equation

$$\langle \partial_\mu j_5^\mu \rangle = -\mathcal{A} \tag{13.23}$$

Our next task is to compute $\mathcal{A}$ to determine whether the Jacobian is trivial ($\mathcal{A} = 0$) or not. A first naive manipulation shows that $\mathcal{A}$ is ill-defined. Indeed, one has, after using properties of Dirac matrices

$$\begin{aligned} \mathcal{A} &= 2\sum_n \varphi_n^\dagger(x)\gamma_5\varphi_n(x) = 2\mathrm{tr}\left(\gamma_5 \sum_n \varphi_n(x)\otimes\varphi_n^\dagger(x)\right) \\ &= 2\mathrm{tr}\left(\gamma_5\delta^{(4)}(0)\right) = 0\times\infty \quad !!! \end{aligned} \tag{13.24}$$

We then see that $\mathcal{A}$ needs some regularization and the question concerning the triviality of the Jacobian changes now to that concerning the search of a consistent regularization for $\mathcal{A}$ (which, after rendered finite, could vanish or not).

As we mentioned above, the ambiguity manifested in (13.24) is related to the fact that the Dirac operator eigenvalues grow with no bound leading to a divergent result for determinants so that a regularization procedure is necessary. Following Fujikawa, let us then introduce a regularization which consists in a large eigenvalue cut off.

To this end, we shal use the following trivial identity

$$1 = \lim_{M^2\to\infty} \exp\left(-\frac{\lambda_n^2}{M^2}\right) \tag{13.25}$$

Here λ_n are real eigenvalues which grow with increasing n as rapidly as those associated with the Dirac operator $\not{D}[A]$ (which are real since $\not{D}$ is hermitian) . But, in principle, they can be different from $\not{D}[A]$ eigenvalues (for example, one can chose them as the eigenvalues of the free Dirac operator, $i\not{\partial}$).

Insertingt identity (13.25) in the expression for $\mathcal{A}$, we have

$$\mathcal{A} = 2\sum_n \varphi_n^\dagger(x)\gamma_5 \left(\lim_{M^2\to\infty} \exp\left(-\frac{\lambda_n^2}{M^2}\right)\right)\varphi_n(x) \tag{13.26}$$

We still call $\mathcal{A}$ the l.h.s. in (13.26) because we have just inserted the identity and nothing has changed regarding the convergence. The next is the crucial step: we define the regularized anomaly $\mathcal{A}_{reg}$ through the interchange between the sum (responsible for the infinite in (13.24)) and the limit:

$$\mathcal{A}_{reg} = 2\lim_{M^2\to\infty} \sum_n \varphi_n^\dagger(x)\gamma_5 \exp\left(-\frac{\lambda_n^2}{M^2}\right)\varphi_n(x) \tag{13.27}$$

Moreover, in order to be able to manipulate naively the basis spinors, we can make a point-splitting so that

$$\mathcal{A}_{reg} = 2\lim_{y\to x}\lim_{M^2\to\infty} \sum_n \varphi_n^\dagger(y)\gamma_5 \exp\left(-\frac{\lambda_n^2}{M^2}\right)\varphi_n(x) \tag{13.28}$$

Operations that were ill defined when performed in (13.24) are now safe,

$$\mathcal{A}_{reg} \;\; = \;\; 2\lim_{y\to x}\lim_{M^2\to\infty} \sum_n \varphi_n^\dagger(y)\gamma_5 \exp\left(-\frac{\lambda_n^2}{M^2}\right)\varphi_n(x)$$

$$= 2\lim_{y\to x}\lim_{M^2\to\infty} \mathrm{tr}\left(\gamma_5 \sum_n \exp\left(-\frac{\lambda_n^2}{M^2}\right)\varphi_n(x)\otimes\varphi_n^\dagger(y)\right) \tag{13.29}$$

Now, a second choice: evidently, the results should be independent of the choice of basis $\{\varphi_n\}$ and eigenvalues λ_n (provided they grow with n as rapidly as those of $\not{D}$. Then, we can exploit this freedom to simplify calculations. Indeed, if we chose the basis as that provided by $\not{D}$ and the eigenvalues λ_n as those precisely associated to $\not{D}$,

$$\not{D}[A]\varphi_n(x) = \lambda_n\varphi_n(x) \tag{13.30}$$

It is important to notice that, being the Dirac operator a gauge covariant one, its eigenvalues are gauge independent an so will be the resulting regularization procedure. We can then write

$$\begin{aligned}
\mathcal{A}_{reg} &= 2\lim_{y\to x}\lim_{M^2\to\infty} \mathrm{tr}\left(\gamma_5 \sum_n \exp\left(-\frac{\not{D}_x^2[A]}{M^2}\right)\varphi_n(x)\otimes\varphi_n^\dagger(y)\right) \\
&= 2\lim_{y\to x}\lim_{M^2\to\infty} \mathrm{tr}\left(\gamma_5 \exp\left(-\frac{\not{D}_x^2[A]}{M^2}\right)\sum_n \varphi_n(x)\otimes\varphi_n^\dagger(y)\right) \\
&= 2\lim_{y\to x}\lim_{M^2\to\infty} \mathrm{tr}\left(\gamma_5 \exp\left(-\frac{\not{D}_x^2[A]}{M^2}\right)\delta^{(4)}(x-y)\right)
\end{aligned} \tag{13.31}$$

At this point, we can use the Fourier transform representation for the delta function,

$$\mathcal{A}_{reg} = 2\lim_{y\to x}\lim_{M^2\to\infty} \mathrm{tr}\left(\gamma_5 \exp\left(-\frac{\not{D}_x^2[A]}{M^2}\right)\frac{1}{(2\pi)^4}\int d^4k \exp(ik(x-y))\right) \tag{13.32}$$

Acting with the Dirac operator one can easily reduce (13.32) to

$$\begin{aligned}
\mathcal{A}_{reg} &= 2\lim_{y\to x}\lim_{M^2\to\infty} \mathrm{tr}\left(\gamma_5 \frac{1}{(2\pi)^4}\int d^4k \exp(ik(x-y))\right. \\
&\qquad \left.\times \exp\left(-\frac{1}{M^2}(\not{D}_x[A]-\not{k})^2\right)\right)
\end{aligned} \tag{13.33}$$

One can safely take at this stage the limit $y \to x$,

$$\mathcal{A}_{reg} = 2\lim_{M^2\to\infty} \mathrm{tr}\left(\gamma_5 \frac{1}{(2\pi)^4}\int d^4k \exp\left(-\frac{1}{M^2}(\not{D}_x[A]-\not{k})^2\right)\right) \tag{13.34}$$

It is convenient to change the integration variables

$$p^\mu = \frac{k^\mu}{M} \tag{13.35}$$

and work a little the Dirac matrices in the exponential so as to obtain

$$\mathcal{A}_{reg} = 2 \lim_{M^2\to\infty} \mathrm{tr}\left(\gamma_5 \frac{M^4}{(2\pi)^4} \int d^4p \exp\left(-\frac{\not{D}_x}{M^2} + \frac{\not{D}\not{p} + \not{p}\not{D}}{M} - \not{p}^2\right)\right) \tag{13.36}$$

or

$$\begin{aligned} \mathcal{A}_{reg} &= 2 \lim_{M^2\to\infty} \mathrm{tr}\left(\gamma_5 \frac{M^4}{(2\pi)^4} \exp\left(-\frac{\not{D}_x}{M^2}\right) \int d^4p \exp\left(-p^2\right)\right. \\ &\quad \times \left.\exp\left(\frac{\not{D}\not{p} + \not{p}\not{D}}{M}\right)\right) \end{aligned} \tag{13.37}$$

We can expand the exponentials in $1/M$ powers and just keep, in view of the M^4 factor in front of the integral, terms up to $1/M^4$. Moreover, those terms in the expansion containing odd powers of p vanish after integration because of parity properties. Concerning even powers, after multiplying by γ_5 and computing the Dirac trace terms containg γ_5 and less than four γ matrices vanish when the Dirac trace is evaluated. The answer is then

$$\mathcal{A}_{reg} = -\frac{e^2}{2} F_{\mu\nu} F_{\alpha\beta} \lim_{M^2\to\infty} \left(\mathrm{tr}\gamma_5 [\gamma^\mu, \gamma^\nu][\gamma^\alpha, \gamma^\beta]\right) \int d^4p \exp\left(-p^2\right) \tag{13.38}$$

where we have used

$$\begin{aligned} \not{D}^2 &= D_\mu D_\nu \gamma^\mu \gamma^\nu \\ &= D_\mu D_\nu \left(\frac{\gamma^\mu\gamma^\nu + \gamma^\nu\gamma^\mu}{2} + \frac{\gamma^\mu\gamma^\nu - \gamma^\nu\gamma^\mu}{2}\right) \\ &= D_\mu D_\nu \left(g^{\mu\nu} + [\gamma^\mu, \gamma^\nu]\right) = D_\mu^2 + \frac{ie}{2} F_{\mu\nu} [\gamma^\mu, \gamma^\nu] \end{aligned} \tag{13.39}$$

The product of commutators in (13.38) is proportional to $\varepsilon^{\mu\nu\alpha\beta}\gamma_5$ so that after computing the trace and the gaussian integral one ends with

$$\mathcal{A}_{reg} = -\frac{e^2}{32\pi^2} \varepsilon^{\mu\nu\alpha\beta} F_{\mu\nu} F_{\alpha\beta} \tag{13.40}$$

and the anomaly equation finally reads

$$\langle \partial_\mu j_5^\mu \rangle = \frac{e^2}{16\pi^2} \varepsilon^{\mu\nu\alpha\beta} F_{\mu\nu} F_{\alpha\beta} \tag{13.41}$$

<u>Note 1</u>: If instead of using $\exp(-\not{D}^2/M^2)$ as regulator one uses $\exp(-\not{\partial}^2/M^2)$ the regulation also works. In fact, the derivation proceeds exactly as before since $\not{\partial}^2 = \not{D}[A=0]$. This means that instead of the result (13.40) for $\mathcal{A}_{reg}$, one would get $\bar{\mathcal{A}}_{reg} = 0$. Of course, this would mean that the basis and eigenvalues one should use are those of the free Dirac operator. Evidently these eigenvalues are not gauge invariant so there is no guarantee that this alternative regularization preserves gauge-invariance. And, in fact, if following the Noether method one controls conservation of the gauge current, one finds an anomaly which, appart from a sign, coincides precisely with the one found for the chiral current with the gauge invariant regularization. Mathematically, both regularizations are mathematically correct. It is on physical grounds that one should decide which regularization should be selected. If gauge invariance is to be maintained at any cost, a gauge invariant regularization must be adopted and a chiral anomaly will be the prize to pay.

<u>Note 2</u>: In theories with Weyl fermions one is led to choose as the Dirac operator one of the form

$$D[A] = i\not{\partial} + (1-\gamma_5)\not{A} \tag{13.42}$$

This is not an hermitian operator,

$$D^\dagger[A] = i\not{\partial} + \not{A}(1-\gamma_5) = i\not{\partial} + (1+\gamma_5)\not{A} \tag{13.43}$$

Then, one cannot ensure that eigenvalues of $D^2[A]$ are positive-definite and this operator cannot be used as regulator in the same form as done above. One can solve this problem in two ways: one possibility is to analytically continue the operator

$$D^s[A] = i\not{\partial} + (1-s\gamma_5)\not{A}) \tag{13.44}$$

with s a complex variable such that $s = -s^*$ so that $D^{s\dagger}[A] = D^s[A]$. One then works with this hermitian operator and at the end of the calculations one puts $s = 1$. A second possibility is to work with a regulator R defined as

$$R = \frac{1}{2}(D[A]D^\dagger[A] + D^\dagger[A]D[A]) \tag{13.45}$$

Interestingly enough, the answer obtained for the anomaly in the two cases is different. In the first case one obtains what is known as the covariant anomaly, the name referring to the fact the resulting $\mathcal{A}$ is gauge covariant (in fact, it is proportional to F^*F) as in the Dirac fermion case. Using R as regulator, one finds that $\mathcal{A}$ is not gauge covariant. However, this last anomaly gives a partition function $Z[A]$ that behaves correctly under gauge transformations for A. Moreover, the difference in applicating two consecutive gauge transformations to $Z[A]$ in different order close the correct gauge group algebra (Wess-Zumino consistency equations). One calls this anomaly the consistent anomaly. The physical situation will determine which regularization one should adopt.

Jacobians as ratio of determinants

We have seen (eq.(13.5) that the chiral Jacobian can also be defined as a ratio of determinants

$$J_5[\delta\alpha] = \frac{\det(i\not\partial + e\not A)}{\det\Big(\exp\left(-\gamma_5\delta\alpha(x)\right)(i\not\partial + e\not A)\exp\left(-\gamma_5\delta\alpha(x)\right)\Big)} \tag{13.46}$$

so that, taking logarithms one has

$$\log\left(J_5[\delta\alpha]\right) = \log\det\not D[A] - \log\det\left(\exp\left(-\gamma_5\delta\alpha\right)\not D[A]\exp\left(-\gamma_5\delta\alpha\right)\right) \tag{13.47}$$

or, using the formula relating $\log\det$ with $\mathrm{Tr}\log$ (with Tr both a functional and Dirac matrices trace)

$$\log\left(J_5[\alpha]\right) = \mathrm{Tr}\log\not D[A] - \mathrm{Tr}\log\left(\exp\left(-\gamma_5\delta\alpha\right)\not D[A]\exp\left(-\gamma_5\delta\alpha\right)\right) \tag{13.48}$$

Expanding the exponentials and keeping up to order $\delta\alpha$,

$$\begin{aligned} -\log\left(J_5[\alpha]\right) &= -\mathrm{Tr}\log\not D[A] + \mathrm{Tr}\log\left((1-\gamma_5\delta\alpha)\not D[A]\,(1-\gamma_5\delta\alpha)\right) \\ &= -\mathrm{Tr}\log\not D[A] + \mathrm{Tr}\log\left(\not D[A] - \gamma_5\delta\alpha\not D[A] - \not D[A]\gamma_5\delta\alpha\right) \\ &= \mathrm{Tr}\log\not D^{-1}[A]\left(\not D[A] - \gamma_5\delta\alpha\not D[A] - \not D[A]\gamma_5\delta\alpha\right) \end{aligned} \tag{13.49}$$

Calling

$$G[A] = \not{D}^{-1}[A] \tag{13.50}$$

so that

$$G[A]\not{D}^{-1}[A] = \not{D}^{-1}[A]\, G[A] = I \tag{13.51}$$

we can rewrite (13.49) in the form

$$-\log\left(J_5[\alpha]\right) = \mathrm{Tr}\log\left(1 - G[A]\gamma_5\delta\alpha\not{D}[A] - \gamma_5\delta\alpha\right) \tag{13.52}$$

Now we expand the logarithm keeping again terms up to $\delta\alpha$,

$$\log\left(J_5[\alpha]\right) = -\mathrm{Tr}\left(G[A]\gamma_5\delta\alpha\not{D}[A] - \gamma_5\delta\alpha\right) \tag{13.53}$$

Ignoring at the moment possible ambiguities introduced by divergencies, one can at this point use in (13.53) the cyclic property of the trace so as to make the Green function act on the Dirac operator, this giving the identity

$$\log\left(J_5[\alpha]\right) = -2\mathrm{Tr}\gamma_5\delta\alpha \tag{13.54}$$

and then

$$-\left.\frac{dJ_5[\alpha]}{d\alpha}\right|_{\delta\alpha=0} = 2\mathrm{Tr}\gamma_5 = 2\mathrm{tr}\sum_n \varphi_n^\dagger(x)\gamma_5\varphi_n(x) \tag{13.55}$$

where we have explicitly written the functional trace in the last line using the basis $\{\varphi_n\}$ so that one can see that this method leads to the same answer as that obtained following Fujikawa's one, eqs.(13.22)-(13.23).

Problems

Problem 1 : Prove the identity

$$\sum_n \varphi_n^\dagger(x)\gamma_5\varphi_n(y) = \mathrm{tr}\left(\gamma_5 \sum_n \varphi_n(y) \otimes \varphi_n^\dagger(x)\right) = \mathcal{A}$$

Problem 2: Give the details for passing from (13.37) to (13.40).

Problem 2: Show that in two dimensions, the chiral anomaly for massless Dirac fermions is

$$\mathcal{A}^{d=2} = \frac{e}{2\pi}\varepsilon^{\mu\nu}F_{\mu\nu}$$

Clase 14

Zero-modes; the Index Theorem

We have discussed the necessity of regularizating Jacobians and fermion determinants originated in the fact the Dirac operator eigenvalues grow without bound. But there is a second very important problem with physical consequences namely when the Dirac operator has zero-modes, i.e., eigenfunctions with vanishing eigenvalues. One can prove that this indeed happens under certain circumstances which are precisely determined by the so called Index theorem [27].

Consider then the case in which the Dirac operator has N normalizable zero modes,

$$\not{D}\varphi_n^0 = 0\,, \quad n = 1, 2, \ldots, N \tag{14.1}$$

(As we have already mentioned, non-normalizable eigenvalues do not pose a problem since they should not be included in the functional basis used to define the path-integral measure). Then, once one has found a consistent regularization prescription for curtting-off the divergent contribution of large eigenvalues, one is left with a product of eigenvalues which includes a zero of order N and hence the natural finite result for the Dirac operator determinant would be zero!

We shall discuss now how a non-trivial result can be derived when zero-modes exist and, at the same time, we shall discover the conditions under which zero-modes of the Dirac operator do exist.

A natural procedure to postpone the zero-mode problem when the Dirac operator has zero modes is to add a mass term to the Lagrangian, so that the generating functional (which we now call Z_m) reads

$$Z_m[A] = \int \mathcal{D}\bar{\psi}\mathcal{D}\psi \exp\Big(-\int d^4x\bar{\psi}(i\partial\!\!\!/ + eA\!\!\!/ + m)\psi\Big) \tag{14.2}$$

Let us the introduce an operator $D_m[A]$ defined as

$$D_m[A] = D\!\!\!\!/[A] + m \tag{14.3}$$

When the diagonal mass term is included, one can show that the resulting operator has no zero modes. Now, the zero-modes of the original operator $D\!\!\!\!/$ continue to be eigenmodes of D_m but in this case with non-zero eigenvalue m,

$$D_m[A]\,\varphi_n^0 = m\varphi_n^0 \tag{14.4}$$

Of curse, if one naively takes the $m \to 0$ limit after eliminating the large eigenvalue contribution one should obtain a trivial answer, since one can write for D_m,

$$\det D_m[A] = m^N \prod_n \lambda'_n \tag{14.5}$$

where we indicate with λ'_n the remaining eigenvalues of D_m which, in the $m \to 0$ limit do not become zero modes of $D\!\!\!\!/$. A natural question to pose is whether the operator D_m has no new zero eigenvalues even when it contains a diagonal term as the mass term produces. The answer is no: zero modes, in order to be included in the basis used to define the path-integral should be normalizable. Otherwise, they can be excluded and they do not appear in the product of eigenvalues defining the Dirac operator determinant. But, when this operator has diagonal elements (as it is the case with D_m), one can prove that no normalizable zero modes exist (We refer the interested persons to [27] and [28] for a thorough discussion of this point).

Consider then a Jacobian arising from gauge transformations of the fermionic variables in $Z_m[A]$. We have seen that it can be written as a ratio of determinants in the form

$$J_m[\delta\alpha] = \frac{\det D_m[A]}{\det D_m[A^{\delta\alpha}]} \tag{14.6}$$

where we have used the fact that the infinitesimal gauge transformation of fermions can be rewritten in terms of the gauge transformed gauge filed $A_\mu^{\delta\alpha}$ Now, we can use (14.5) in each determinant in the ratio (assuming some regularization for the product of non-zero eigenvalues),

$$J_m[\delta\alpha] = \left.\frac{m^N \prod_n \lambda'_n}{m^N \prod_n \lambda'_n}\right|_{reg'} \tag{14.7}$$

Here we have used that the non-zero eigenvalues λ'_n are gauge invariant and indicated with reg' the fact that the product of non-zero eigenvalues should be consistently regularized. Evidently, the numerator cancels the denominator so that finally we have $J_m = 1$. Then, taking the $m \to 0$ limit one can write

$$J \equiv \lim_{m\to 0} J_m = 1 \tag{14.8}$$

where J represents the gauge Jacobian for the *originally massless* theory.

This procedure will be the natural one each one one faces the problem of Dirac operator zero modes: one adds a "small" mass, perform all calculations and at the end one takes the $m \to 0$ limit. Hence, it will be convenient to directly define a "primed" determinant just extracting from the beginning the zero eigenvalues

$$\det{}' \not{D} \equiv \prod_n \lambda'_n \tag{14.9}$$

That this procedure is mathematically well-defined is proved in [28].

What about J_5? As we have seen, a mass term breaks global chiral invariance so that, when following Noether method one gets, after the (local) chiral change,

$$\begin{aligned} Z_m[A] \;&=\; J_{5m}[\delta\alpha] \exp\left(-\int d^4x\bar\psi D_m[A]\psi\right) \\ &\quad \exp(-2m\int d^4x\bar\psi\gamma_5\psi\delta\alpha(x)) \exp(\int d^4x\delta\alpha(x)\partial_\mu j_5^\mu) \end{aligned} \tag{14.10}$$

We can then write

$$0 = \left.\frac{1}{Z}\frac{\delta Z_m}{\delta\alpha(x)}\right|_{\alpha=0} = \left.\frac{\delta J_{5m}}{\delta\alpha(x)}\right|_{\alpha=0} - 2m\langle\bar\psi\gamma_5\psi\rangle + \langle\partial_\mu j_5^\mu\rangle \tag{14.11}$$

At this point we can take the $m \to 0$ limit. First, we note that the Jacobian does not depend on m. One can convince oneself of this by rememebring that the non trivial contribution to chiral Jacobians comes from the trace of a term containing γ_5 times the product of as many Dirac operators as the dimension of space-time is. Now, the mass term is proportional to the identity so that it reduces the number of γ_μ matrices in the product leading to a vanishing trace (see Problem 1). We then have

$$\langle \partial_\mu j_5^\mu \rangle = - \left. \frac{\delta \log J_5}{\delta \alpha(x)} \right|_{\alpha=0} + \lim_{m\to 0} 2m \langle \bar{\psi}\gamma_5 \psi \rangle \tag{14.12}$$

or, using the result for the Fujikawa Jacobian in $3+1$ dimensions,

$$\langle \partial_\mu j_5^\mu \rangle = -\frac{e^2}{16\pi^2} \mathrm{tr}^{\mathrm{c}}({}^*F_{\mu\nu}F^{\mu\nu}) + \lim_{m\to 0} 2m \langle \bar{\psi}\gamma_5 \psi \rangle \tag{14.13}$$

(We have considered the general case of a non-Abelian gauge theory; the only change in the Fujikawa Jacobian is the appearence of the trace over group indices, tr^c).
We are left with the computation of $\langle \bar{\psi}\gamma_5\psi \rangle$,

$$\langle \bar{\psi}\gamma_5 \psi \rangle = \frac{1}{Z} \int \mathcal{D}\bar{\psi}\mathcal{D}\psi \exp(-\int d^4x \bar{\psi} D_m[A]\psi)\bar{\psi}\gamma_5\psi \tag{14.14}$$

Concerning the normalization Z, one has, after expanding in Bessel-Fourier series in terms of the $\not{D}[A]$ basis,

$$Z = \int \prod_n d\bar{c}_n dc_n \exp(-\sum_k \bar{c}_k c_k(\lambda_k + m)) = \prod_k (\lambda_k + m) \tag{14.15}$$

Concerning the numerator, which we call $\mathcal{N}$,

$$\mathcal{N} = \int \prod_n d\bar{c}_n dc_n \exp(-\sum_k \bar{c}_k c_k(\lambda_k + m)) \sum_{pq} \bar{c}_p c_q \varphi_p^\dagger(x)\gamma_5\varphi_q(x) \tag{14.16}$$

In view of anticommuting integration rules, only those terms in the sum with $p = q$ will contribute

$$\mathcal{N} = \sum_p \varphi_p^\dagger(x)\gamma_5\varphi_p(x) \int \prod_n d\bar{c}_n dc_n \exp(-\sum_k \bar{c}_k c_k(\lambda_k + m))\bar{c}_p c_p \tag{14.17}$$

When computing Z , one expands the exponential and only the term containing once each eigenvalue contributes. Here, it is the term containing each but one, since there is also the product $\bar{c}_p c_p$ to integrate. And this for $p = 1, 2, \ldots$. The answer is then

$$\mathcal{N} = \sum_p \varphi_p^\dagger(x)\gamma_5\varphi_p(x)(\lambda_1 + m)(\lambda_2 + m)\ldots(\lambda_{p-1} + m)(\lambda_{p+1} + m)\ldots \tag{14.18}$$

Then, from (14.15) and (14.18) we have

$$\langle\bar{\psi}\gamma_5\psi\rangle = \sum_p \varphi_p^\dagger(x)\gamma_5\varphi_p(x)\frac{1}{\lambda_p + m} \tag{14.19}$$

Then, we have

$$\lim_{m\to 0} 2m\langle\bar{\psi}\gamma_5\psi\rangle = \lim_{m\to 0} 2m\sum_p \varphi_p^\dagger(x)\gamma_5\varphi_p(x)\frac{1}{\lambda_p + m} \tag{14.20}$$

Once the limit is taken, the m factor in the numerator eliminates all contributions from the sum except those corresponding to $\lambda_p = 0$ since in that case the m factor in the numerator is cancelled by that in the denominator. That is, only zero modes give a non-vanishing contribution and one ends with

$$\lim_{m\to 0} 2m\langle\bar{\psi}\gamma_5\psi\rangle = 2\sum_{p=1}^{N} \varphi_p^{0\dagger}(x)\gamma_5\varphi_p^0(x) \tag{14.21}$$

We can then write (14.13) in the form

$$\langle\partial_\mu j_5^\mu\rangle = -\frac{e^2}{16\pi^2}\mathrm{tr}^{\mathrm{c}}({}^*F_{\mu\nu}F^{\mu\nu}) + 2\sum_{p=1}^{N} \varphi_p^{0\dagger}(x)\gamma_5\varphi_p^0(x) \tag{14.22}$$

Let us integrate this identity over all space. The l.h.s. can be seen to vanish using Stokes theorem,

$$\int d^4x\langle\partial_\mu j_5^\mu\rangle = \langle\int dS_\mu j_5^\mu\rangle = 0 \tag{14.23}$$

Here we interchanged the expectation value with integration and also assumed that fields go to zero on the boundary. So we have

$$2\sum_{p=1}^{N}\int d^4x\varphi_p^{0\dagger}(x)\gamma_5\varphi_p^0(x) = \frac{e^2}{16\pi^2}\mathrm{tr}^{\mathrm{c}}\int d^4x({}^*F_{\mu\nu}F^{\mu\nu}) \tag{14.24}$$

The l.h.s. can be computed after the following observation. In view of the property $\gamma_5 \not{D} = -\not{D}\gamma_5$ one has that if χ^0 is a a zero mode, $\gamma_5\chi^0$ also is,

$$\not{D}\chi^0 = 0 \Rightarrow \gamma_5\not{D}\chi^0 = -\not{D}\gamma_5\chi^0 = 0 \qquad (14.25)$$

Then, we can combine zero-modes so that they are also eigenfunctions of γ_5 with eigenvalue ± 1,

$$\varphi^{0\pm} = \frac{1 \pm \gamma_5}{2}\chi^0 \, , \qquad \gamma_5\varphi^{0\pm} = \pm\varphi^{0\pm} \qquad (14.26)$$

Then, the sum in (14.24) becomes

$$\sum_{p=1}^{N} \int d^4x \varphi_p^{0\dagger}(x)\gamma_5\varphi_p^0(x) = n_+ - n_- \qquad (14.27)$$

so that we end with the formula

$$\frac{e^2}{32\pi^2}\mathrm{tr}^{\mathrm{c}} \int d^4x ({}^*F_{\mu\nu}F^{\mu\nu}) = n_+ - n_- \qquad (14.28)$$

In physical problems, one demands that $F_{\mu\nu}$ vanish at infinity. This condition is equivalent to demand that A_μ is a pure gauge at infinity, which can be identified with a very large sphere S^3,

$$A_\mu|_{S^3} \approx \frac{1}{e} g^{-1}\partial_\mu g(x) \qquad (14.29)$$

with g and element of the gauge group G. Then, every gauge field A_μ produces, through its relation with some g at infinity, a certain mapping of the sphere S^3 onto the gauge group G,

$$g(x) : S^3 \to G \qquad (14.30)$$

Now, it is known that there exists an infinite number of different classes of such mappings of $S^3 \to G$ if G is a non-Abelian simple Lie group, each (homotopy) class characterized by an integer q. This number is called the Pontryagin index and, given a gauge field A_μ, it can be written as

$$q = \frac{e^2}{32\pi^2}\mathrm{tr}^{\mathrm{c}} \int d^4x ({}^*F_{\mu\nu}F^{\mu\nu}) \, , \quad q \in Z \qquad (14.31)$$

It is clear that if two such mappings belong to different homotopy classes with index q and $\bar{q}$, the corresponding fields $A_\mu^{(q)}$ and $A_\mu^{(\bar{q})}$ cannot be continuously deformed one into another.

We can then rewrite (14.28) in the form

$$q = n_+ - n_- \tag{14.32}$$

which is the integral form of the Atiyah-Singer theorem that establishes a relation between the Pontryagin index (which is a topological object) for a given gauge field and the difference between the number of zero modes with opposite chirality of the Dirac operator in such a gauge field background.

An application of the Index Theorem: The $U(1)_A$ problem

Consider the QCD Lagrangian

$$L_{QCD} = \frac{1}{2}\mathrm{tr}^{\mathrm{c}} F_{\mu\nu}F^{\mu\nu} + \theta\frac{g^2}{16\pi^2}\mathrm{tr}^{\mathrm{c}}\tilde{F}_{\mu\nu}F^{\mu\nu} + \bar{q}_i^f(i\partial\!\!\!/\delta^{ij} + A\!\!\!/^a T^{a\,ij})q_j^f + \bar{q}Mq \tag{14.33}$$

Here, concerning the color gauge symmetry $SU(3)_c$, we have

$$a = 1, 2, \ldots, 8, \qquad i = 1, 2, 3 \tag{14.34}$$

The quark fields q are then in the fundamental representation of $SU(3)_c$ and we call T^a the generators in that representation. There is also a flavor index f which corresponds to exact (or approximate) symmetries (charge, isospin, ...). The associated global symmetry group is taken as $SU(N_f)$. If, for example, $N_f = 3$, the corresponding quark fields are

$$\begin{aligned} q_j^1 &: \text{ up} \\ q_j^2 &: \text{ down} \\ q_j^3 &: \text{ strange} \end{aligned} \tag{14.35}$$

If $N_f = 4$ there is a forth "charmed" quark,

$$q_j^4 \; : \; \text{charm} \tag{14.36}$$

Appart from the Yang-Mills Lagrangian for the gauge field A_μ taking values in the Lie algebra of $SU(3)_c$, there is the CP violating θ term with a coefficient θ small enough as to account for the small measured CP violation. Being a total derivative, this term does not change the Yang-Mills-matter equations of motion,

$$\tilde{F}^a_{\mu\nu} F^{a\,\mu\nu} = \partial_\mu K^\mu \tag{14.37}$$

with

$$K^\mu = 2\mathrm{tr}^{\mathrm{c}} \varepsilon^{\mu\nu\alpha\beta} \left(A_\nu \partial_\alpha A_\beta + \frac{2}{3} A_\nu A_\alpha A_\beta \right) \tag{14.38}$$

However, this term is crucial due to its global (topological) properties. Indeed, we have already seen that in Euclidean space, one has

$$\frac{g^2}{16\pi^2} \int d^4x \tilde{F}^a_{\mu\nu} F^{a\,\mu\nu} = n \tag{14.39}$$

with n the integer labeling the Chern class to which A_μ belongs.

Concerning the matter Lagrangian, note that the covariant derivative is diagonal in flavor indices and that there is a mass term which should eventually be put to zero. In that case, there is a *non-Abelian global chiral symmetry*: the Lagrangian is invariant under the change

$$q^f \to \exp \left(\gamma_5 \delta\alpha^A \left(T^A \right)^{ff'} \right) q^{f'} \tag{14.40}$$

with T^A the flavor generators. One can combine this symmetry with the flavor symmetry by writing the complete symmetry group as the product $SU(N_f)_L \otimes SU(N_f)_R$. This symmetry is not violated at the quuantum level. Indeed, when one computes the Fujikawa Jacobian associated with the transformation (14.40), one gets

$$\log J_5[\delta\alpha] = \frac{g^2}{16\pi^2} \mathrm{tr}^{\mathrm{c}}(t^a t^b) \mathrm{tr}^{\mathrm{f}} T^A \int \tilde{F}^a_{\mu\nu} F^{b\mu\nu} \delta\alpha^A = 0 \tag{14.41}$$

a vanishing result since $\mathrm{tr}^{\mathrm{f}} T^A$ vanishes. In order to take profit of this symmetry in connection with the observed particle spectrum, one proceeds as follows: one breaks the chiral non-Abelian (say $SU(3)$) symmetry spontaneously so that, according to Goldstone theorem, 8 massless bosons arise. One can associate this eight Goldstone bosons with the known mesons $\pi^+, \pi^-, \pi^0, K^0, \bar{K}^0, K^+, K^-, \eta$. We know that this mesons are not massless so that they cannot be represented by massless Goldstone bosons. It is to solve this problem that one introduces the mass term (with a small mass M) in the QCD Lagrangian so that the original symmetry becomes approximate and then mesons can be interpreted as quasi Goldston bosons having a small (but non zero) mass.

Now, for massless quarks, there was one more quiral (axial) symmetry that we have not yet taken into account. Indeed, under the $U(1)$ global chiral transformation,

$$q^f \to \exp(i\alpha\gamma_5)q^f \tag{14.42}$$

the QCD Lagrangian is invariant. The associated conserved current is the so called $U_A(1)$ current

$$j_5^\mu = q_i^f \gamma^\mu \gamma_5 q_i^f \,, \qquad \partial_\mu j_5^\mu = 0 \tag{14.43}$$

Then, instead of a $SU(N_f)_L \otimes SU(N_f)_R$ symmetry, one in fact has, in the massless case, a $U(N_f)_L \otimes U(N_f)_R$.

Now, repeating the arguments above (i.e, introducing spontaneous breaking ob the symmetry and a small quark mass) one ends with 9 (and not 8) quasi Goldston bosons! Eight of them were already identified. Concerning the new (ninth) one, one can search in a Table of particles and discover that there is in fact a meson, called η' which can be considered as a candidate. However, its mass is too big in comparison to the masses of the other 8 mesons,

$$m_{\eta'} = 1\mathrm{GeV}\,, \quad m_{\pi^0} \approx 135\mathrm{MeV} \tag{14.44}$$

One cannot use the same mechanism to arrive to such different masses. Then, η' cannot be the ninth Goldston boson This is the so called $U(1)_A$ Problem: where is the ninth Goldston boson?

To answer this question, let us first note that the $U_A(1)$ current is anomalous: in contrast with the non-Abelian case, one has now a non-trivial Fujikawa Jacobian,

$$\log J_5^{U_A(1)}[\delta\alpha] = \frac{g^2 N_f}{32\pi^2}\int \tilde{F}^a_{\mu\nu} F^{a\mu\nu}\delta\alpha(x) \tag{14.45}$$

where the factor N_f comes from the fact each flavor contributes with one Jacobian. Then, we have for the current anomaly,

$$\langle \partial_\mu j_5^\mu \rangle = \frac{g^2 N_f}{32\pi^2} \tilde{F}^a_{\mu\nu} F^{a\mu\nu} = \frac{g^2 N_f}{32\pi^2} \partial_\mu K^\mu \tag{14.46}$$

Of course, a conserved current can be constructed by substracting an adequate term,

$$\tilde{j}_5^\mu = j_5^\mu - \frac{g^2 N_f}{32\pi^2} \partial_\mu K^\mu \tag{14.47}$$

but since K_μ is not gauge-covariant, the corresponding charges, although conserved, are not gauge-covariant and hence they are not related to measurable quantities. At this point one could think that the problem is solved: since there is an anomaly, the $U_A(1)$ symmetry is dead at the quantum level. Then, there is no sense in searching for a ninth Goldston boson since there is no $U_A(1)$ symmetry to spontaneously break. To be convinced that the $U_A(1)$ symmetry is no more present, let us compute $\langle \bar{q}(x) q(x) \rangle$. This is the v.e.v. of an object which is not invariant under $U_A(1)$ transformations. If we find a non zero result, this would definitively mean that the anomaly has broken the invariance and that the $U_A(1)$ problem is solved. So, we start by computing

$$N[A^{(1)}; x] = \int \mathcal{D}\bar{q} \mathcal{D}q \exp(-S_{QCD}) \bar{q}(x) q(x) \tag{14.48}$$

in the 1-instanton sector; that is, for gauge fields belonging to the first Pontryagin class. We know, from the Atiyah-Singer theorem that, for $n = 1$, there is at least one zero mode. In fact 't Hooft has shown [29] that in this sector there is just one zero mode which can be exactly constructed,

$$\varphi_0(x) = \frac{\rho}{(\rho^2 + (x - x^0)^2)^2} u_0 \,, \quad \not{D} \varphi_0(x) = 0 \tag{14.49}$$

where ρ and x^0 are 5 parameters appearing in the one instanton solution,

$$A_\mu^{(1)} = \eta^a_{\mu\nu} \frac{x_\nu - x^0_\nu}{(\rho^2 + (x - x^0)^2)^2} \tag{14.50}$$

and associated with scale and translation invariance. Concerning u_0, it is a constant spinor of positive chirality.

Then, when expanding in the $\not{D}$ basis (14.48) becomes

$$
\begin{aligned}
N[A^{(1)};x] &= \int \prod_n d\bar{c}_n dc_n \exp\left(-\sum_{n\neq 0}\lambda_n \bar{c}_n c_n\right) \\
&\quad \times \left(\sum_{p\neq 0}\bar{c}_p c_p \varphi_p^\dagger \varphi_p + \bar{c}_0 c_0 \varphi_0^\dagger \varphi_0\right)
\end{aligned} \tag{14.51}
$$

One easily computes (see problem 2) this integral obtaining

$$
N[A^{(1)};x] = \prod_{n\neq 0}\lambda_n \left(\varphi_0^\dagger(x)\varphi_0(x)\right) \tag{14.52}
$$

Now, when computing $\langle \bar{q}(x)q(x)\rangle$, one should also integrate over $A'_\mu s$ and take into account zero modes. Associated to translation invariance, to avoid the zero-mode, one shoul integrate, as we have seen, over the collective coordinate x_0. But then one will have to compute (in the $n = 1$ sector)

$$
\langle \bar{q}(x)q(x)\rangle \sim \int dx_0 N[A^{(1)};x] \int dx_0 \sim \int dx_0 \varphi_0^\dagger(x-x_0)\varphi_0(x-x_0) = 1 \tag{14.53}
$$

We thus see that the $U_A(1)$ symmetry is effectively anomalous and there is no possibility of generating a ninth quasi Goldstone boson. This, we have seen for the $n = 1$ sector but can be extended to multi-instanton configurations (within the dilute gas approximation).

Let us note that a global chiral transformation with angle α,

$$
q \to \exp(\gamma_5 \alpha) q \tag{14.54}
$$

leads, as we have seen, to a Jacobian

$$
J_5 = \exp\left(-\frac{g^2 N_f \alpha}{16\pi^2}\right) \mathrm{tr}^{\mathrm{c}} \int d^4x \tilde{F}_{\mu\nu} F^{\mu\nu} \tag{14.55}
$$

So, the theta term in the QCD Lagrangian can be cancelled out by performing a chiral change in the fermion fields such that $\alpha = \theta/N_f$. In fact, we can rotate the fermion fields just one half of this value and obtain a QCD Lagrangian with half a theta angle and so on. Hence, we have a whole family of partition functions Z_θ with $dZ_\theta/d\theta = 0$. This is not possible for pure Yang-Mills theory since there are no quarks to rotate,

$$Z_\theta^{YM} \neq Z_0^{YM} \tag{14.56}$$

Let us then define

$$E_{YM}(\theta) = \frac{1}{Z_\theta^{YM}} \frac{d^2 Z_\theta^{YM}}{d\theta^2}\bigg|_{\theta=0} \tag{14.57}$$

This gives

$$\begin{aligned} E_{YM}(\theta) &= \left(\frac{g^2}{64\pi^2}\right)^2 \int d^4x d^4y \tilde{F}^a_{\mu\nu}(x) F^{a\,\mu\nu}(x) \tilde{F}^b_{\alpha\beta}(x) F^{b\,\alpha\beta}(y) \\ &= \left(\frac{g^2}{64\pi^2}\right)^2 U(0) \end{aligned} \tag{14.58}$$

Since when quarks are added one can show that Z does not depend on θ, the quark contribution to E should be such that it precisely cancels (14.58),

$$E_{quarks}(\theta) = -\left(\frac{g^2}{64\pi^2}\right)^2 U(0) \tag{14.59}$$

Writing $g^2 = e^2/N_c$ and working up to $1/N_f$, Witten [30] has proven that the Fourier transformed of the total U,

$$U_{tot}(k) = \int d^4k \exp(ik(x-y)) \langle \tilde{F}^a_{\mu\nu}(x) F^{a\,\mu\nu}(x) \tilde{F}^b_{\alpha\beta}(x) F^{b\,\alpha\beta}(y) \rangle \tag{14.60}$$

takes the form

$$U_{tot}(k) = \sum_n \frac{N_f^2 a_n}{k^2 - M_n^2} + \sum_n \frac{N_f c_n}{k^2 - m_n^2} \tag{14.61}$$

where the first sum extends over gluons with mass M_n and the second over mesons, with mass m_n and the coefficients are defined as

$$\begin{aligned} a_n &= \frac{1}{N_c} \langle 0 | \tilde{F}^b_{\alpha\beta} F^{b\,\alpha\beta} | \text{gluon ``n''} \rangle \\ c_n &= \frac{1}{\sqrt{N_c}} \langle 0 | \tilde{F}^b_{\alpha\beta} F^{b\,\alpha\beta} | \text{meson ``n''} \rangle \end{aligned} \tag{14.62}$$

Since $U_{tot}(0)$ should vanish, one has

$$0 = \sum_n \frac{N_f^2 a_n}{M_n^2} + \sum_n \frac{N_f c_n}{m_n^2} \tag{14.63}$$

showing that at least one meson should have a mass such that

$$\frac{m^2}{M^2} \sim \frac{1}{N_f} = \frac{1}{N_f = 3} = 0.333 \tag{14.64}$$

Let us call that meson η' and try to compute $c_{\eta'}$,

$$\sqrt{N_c} c_{\eta'} = \langle 0|\tilde{F}F|\eta'\rangle \tag{14.65}$$

or, using the anomaly equation

$$\sqrt{N_c} c_{\eta'} = \frac{32\pi^2 N_c}{e^2 N_f}\langle 0|\partial_\mu j_5^\mu|\eta'\rangle = p_\mu \frac{32\pi^2 N_c}{e^2 N_f}\langle 0|j_5^\mu|\eta'\rangle = p_\mu p^\mu \frac{32\pi^2 N_c}{e^2 N_f} f_{\eta'} \tag{14.66}$$

Now, $p^\mu p_\mu = m_{\eta'}^2$ so that one has

$$c_{\eta'} = \frac{32\pi^2 N_c}{e^2 N_f} m_{\eta'}^2 f_{\eta'} \tag{14.67}$$

or

$$m_{\eta'}^2 = c_{\eta'} \frac{e^2 N_f}{32\pi^2 \sqrt{N_c} f_{\eta'}} \tag{14.68}$$

This is the so called Veneziano-Witten formula. Using results from the lattice one finally ends with $m_{\eta'} \sim 1$ GeV, which is OK.

Problems

Problem 1 : Compute that the Jacobian for chiral transformations of massive fermions showing that it coincides with the one obtained in the massless case.

Problem 2 : Obtain, from (14.51), the result (14.52).

SECOND PART

Clase 15

Hamiltonian Quantization of Gauge Theories

Statement of the problem

We have seen that the path-integral formulation of Quantum Mechanics for a system with Hamiltonian $H(p, q)$ is governed by the path-integral

$$Z = \mathcal{N} \int \mathcal{D}p\mathcal{D}q \exp\left(\frac{i}{\hbar}\int dt(p(t)\dot{q}(t) - H(p,q))\right) \tag{15.1}$$

The extension to the case of a scalar field theory (i.e., a system with an infinite number of degrees of freedom) was straightforward. Instead of (15.1) one has

$$Z = \mathcal{N} \int \mathcal{D}\pi\mathcal{D}\varphi \exp\left(\frac{i}{\hbar}\int d^4x(\pi(x)\dot{\varphi}(x) - H(\pi,\varphi))\right) \tag{15.2}$$

with $\varphi(x)$ the scalar field and $\pi(x)$ its canonical conjugate momentum.

In cases in which the Hamiltonian is quadratic in the momenta, one can trivially integrate ovre π and end with (for the case of the scalar field)

$$Z = \bar{\mathcal{N}} \int \mathcal{D}\varphi \exp\left(\frac{i}{\hbar}\int d^4x L(\varphi, \partial_\mu\varphi)\right) \tag{15.3}$$

where $L(\varphi, \partial_\mu\varphi)$ is the Lagrangian for the system.

The rationale of this approach is then the following: one can prove that the Hamiltonian path integral (15.2) gives the same answer that one obtains by quantizing the scalar theory within the operator approach. Now, since this Hamiltonian path-integral is equivalent to the

Lagrangian version (15.3), one concludes that one can quantize a field theory using the Lagrangian approach represented by (15.3).

Assuming this connection to be correct, we quantized gauge theories in the first part of this course directly attacking the problem within the Lagrangian formulations. That is, we assumed that the passage from a Hamiltonian formulation to a Lagrangian one proceeds, for gauge theories, as it does in the case of scalar theories. But then we faced difficulties connected to the gauge invariance of the gauge field Lagrangian. These difficulties were solved somehow heuristically by following the Faddeev-Popov recipe, which was precisely invented (in 1967) by working directly in the Lagrangian formulation of Z [14].

It is natural to suspect that these difficulties reflect those that would appear if one tries to start by establishing the Hamiltonian formulation of a theory endowed with gauge invariance. If these difficulties could be overcome within the Hamiltonian formulation *without appealing to any heuristic recipe*, then it should be possible to pass to a Lagrangian formulation in which the Faddeev-Popov "trick" is no more a trick but the result of a correct treatment at the Hamiltonian level. This was in fact the way followed by Faddeev [31] in 1969 to "convince himself" [32] that the Lagrangian Faddeev-Popov method was indeed correct.

Let us then start this program by writing the Hamiltonian for a non-Abelian gauge theory. To this end, we have as a first step to identify canonical variables, i.e., coordinates and conjugate momenta in this system (with an infinite number of degrees of freedom). We know the Lagrangian of such a theory, the well-honored Yang-Mills Lagrangian,

$$L = -\frac{1}{4}\mathrm{tr}F^{\mu\nu}F_{\mu\nu} \tag{15.4}$$

with $A_\mu = A^a_\mu t^a$ (t_a being the generators of the Lie algebra associated to the gauge group),

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + e[A_\mu, A_\nu] \tag{15.5}$$

and classical equations of motion

$$D_\mu F^{\mu\nu} = 0 \tag{15.6}$$

$$D_\mu = \partial_\mu + e[A_\mu,\ ] \tag{15.7}$$

Now, to identify canonical coordinates and momenta it is convenient to pass to the so called first order formulation of gauge theories. This consist in writing, instead of Lagrangian (15.4),

$$L = -\frac{1}{2}\mathrm{tr}F^{\mu\nu}\left(\partial_\mu A_\nu - \partial_\nu A_\mu + e[A_\mu, A_\nu] - \frac{1}{2}F_{\mu\nu}\right) \tag{15.8}$$

$$S = \int d^4x L \tag{15.9}$$

but here both A_μ and $F_{\mu\nu}$ are taken as independent variables. Now, the Euler-Lagrange equation associated to the variation of $F_{\mu\nu}$ is trivially derived,

$$\frac{\delta S}{\delta F^a_{\mu\nu}} = 0 \longrightarrow F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + e[A_\mu, A_\nu] \tag{15.10}$$

That is, the correct relation between the curvature and the gauge potential is not defined but obtained as the result of the equation of motion.

To obtain the equation of motion associated to the variation of A_μ it is convenient to integrate by parts in action (15.9)

$$\begin{aligned} S &= \frac{1}{4}\mathrm{tr}\int d^4x F^{\mu\nu}F_{\mu\nu} - \mathrm{tr}\int d^4x F^{\mu\nu}(\partial_\mu A_\nu + eA_\mu A_\nu) \\ &= \int d^4x \frac{1}{4}\mathrm{tr}F^{\mu\nu}F_{\mu\nu} + \mathrm{tr}\int d^4x\,(\partial_\mu F^{\mu\nu} - eF^{\mu\nu}A_\mu)\,A_\nu \end{aligned} \tag{15.11}$$

It is now easy to see that variation with respect to A_μ leads to the Yang-Mills equation of motion (15.6),

$$\frac{\delta S}{\delta A^a_\mu} = 0 \longrightarrow D^\mu F_{\mu\nu} = 0 \tag{15.12}$$

Once we have established the first order formulation, we can to define the the Yang-Mills Hamiltonian. Inspired in the notation of Maxwell theory, let us write

$$E_k = F_{0k}\,, \qquad B_k = \frac{1}{2}\varepsilon_{kpq}F_{pq}\,, \quad k,p,q = 1,2,3 \tag{15.13}$$

One easily finds for the first order Lagrangian

$$L = \mathrm{tr}\left(E_k\partial_0 A_k - \frac{1}{2}(E_k^2 + B_k^2) + A_0(\partial_k E_k + e[A_k, E_k])\right) \qquad (15.14)$$

It is now clear that if we consider the A_k's as playing the rol of coordinates, the E_k's as those of momenta, the B_k's as functionals of the A_k's ($B_k = B_k[A]$), the first two terms can be viewed as the canonical way of writing a Lagrangian in terms of the Hamiltonian, $L = p\dot{q} - H$. This, provided we identify

$$\begin{aligned} p_k &\rightarrow E_k \\ q_k &\rightarrow A_k \\ H &= \frac{1}{2}\mathrm{tr}\left(E_k^2 + B_k^2[A]\right) \end{aligned} \qquad (15.15)$$

Concerning the A_0 component, since by construction there is no possible $\dot{A}_0$ term in the Lagrangian, it cannot be taken as a coordinate (it would represent "half" degree of freedom since the conjugate momentum would be lacking) but as a Lagrange multiplier enforcing a given constraint

$$C = C[A, E] \equiv \partial_k E_k + e[A_k, E_k] = D_k[A]E_k = 0 \qquad (15.16)$$

which is nothing but the non-Abelian version of the Gauss law in the absence of external sources, $\partial_k E_k = 0$.

Given A_k and E_k as canonical coordinates and momenta, we can easily define the Poisson brackets which, for a mechanical system are gicen by

$$\{p_i, q_j\} = \delta_{ij}\,, \qquad \{f, g\} = \frac{\partial f}{\partial p_k}\frac{\partial g}{\partial q_k} - \frac{\partial f}{\partial q_k}\frac{\partial g}{\partial p_k} \qquad (15.17)$$

The natural definition of equal time Poisson bracket for a system which, like Yang-Mills, has an infinite number of degrees of freedom is then

$$\left\{E_i^a(\vec{x}, x_0), A_j^b(\vec{y}, x_0)\right\} = \delta_{ij}\delta^{ab}\delta^{(3)}(\vec{x} - \vec{y}) \qquad (15.18)$$

Concerning A_0, we have to learn how to treat in the Hamiltonian formalism a constraint like (15.16). Let us first note that the constraint $C = C^a t^a$,

$$C^a = (D_k E_k)^a = D_k^{ab} E_k^b \qquad (15.19)$$

satisfies the equation

$$\left\{C^a(\vec{x},x_0),C^b(\vec{y},x_0)\right\}=ef^{abc}C^c(\vec{x},x_0)\delta^{(3)}(\vec{x}-\vec{y}) \tag{15.20}$$

which is nothing but a realization, in terms of fields, of the gauge Lie algebra which for generators t^a is defined by

$$[t^a,t^b]=if^{abc}t^c \tag{15.21}$$

Moreover, if we define

$$\mathcal{H}=\int d^3xH=\int d^3x\frac{1}{2}(E_k^2+B_k^2) \tag{15.22}$$

it is also easy to see that

$$\{\mathcal{H},C^a(x)\}=0 \tag{15.23}$$

A system satisfying eqs.(15.20) and (15.23) is said to obey the **generalized Hamiltonian dynamics** developed by Dirac and has to be treated accordingly [1].

Before explaining Dirac method, let us describe another property of the gauge theory constraints that can be easily established. Consider a time independent infinitesimal gauge transformation g,

$$g=\exp\left(i\alpha(\vec{x})\right)\approx I+i\alpha(\vec{x}) \tag{15.24}$$

$$\alpha(\vec{x})=\alpha^a(\vec{x})t^a \tag{15.25}$$

Under such a transformation E_k transforms covariantly,

$$E_k\to E_k'=g^{-1}E_kg\approx\left(I-i\alpha(\vec{x})\right)E_k\left(I+i\alpha(\vec{x})\right) \tag{15.26}$$

so that

$$\delta E_k\equiv E_k'-E_k=i[E_k,\alpha] \tag{15.27}$$

or

$$\delta E_k^a\equiv E_k'^a-E_k^a=f^{abc}\alpha^bE_k^c \tag{15.28}$$

this formula then gives the infinitesimal transformation law for the "chromoelectric" field. Now, it is easy to see that

$$\left\{E_k^a(\vec{x},t),C^b(\vec{y})\right\}=-\delta^{(3)}(\vec{x}-\vec{y})f^{abc}E_k^c(\vec{x},t) \tag{15.29}$$

so that

$$\int d^3y \left\{ E_k^a(\vec{x},t), C^b(\vec{y}) \right\} \alpha^b(\vec{y}) \;=\; -\int d^3y \delta^{(3)}(\vec{x}-\vec{y}) f^{abc} E_k^c(\vec{x},t)\alpha^b(\vec{y})$$
$$= f^{abc}\alpha^b E_k^c(\vec{x},t) = \delta E_k^a \qquad (15.30)$$

We conclude that the C^a's give a realization of the generators of infinitesimal gauge transformations for E_k^a since its Poisson bracket with E_k^a precisely gives δE_k^a . Analogously one can prove (see Problem 2) that

$$\delta A_k^a(\vec{x},t) = \int d^3y \left\{ A_k^a(\vec{x},t), C^b(\vec{y})\alpha^b(\vec{y}) \right\} \qquad (15.31)$$

Dirac generalized Hamiltonian dynamics

Let us consider a mechanical system with n degrees of freedom. Let p_i, q_i , $i = 1, 2, \ldots, n$, be the canonical variables spanning the $2n$ dimensional phase space Γ^{2n}. Consider the action

$$S = \int dt \left(\sum_{i=1}^{n} p_i \dot{q}_i - H(p,q) - \sum_{a=1}^{m} \lambda^a \varphi^a(p,q) \right) \qquad (15.32)$$

where λ^a $(a = 1, 2, \ldots, m < n)$ are Lagrange multipliers enforcing the <u>constraints</u>

$$\varphi^a(p,q) = 0\,, \qquad a = 1, 2, \ldots m \qquad (15.33)$$

Such an action defines a generalized Hamiltonian system whenever the following conditions are satisfied

$$\{H(pq), \varphi^a(p,q)\} = \sum_b c^{ab}(p,q)\varphi^b(p,q) \qquad (15.34)$$

$$\{\varphi^a(p,q), \varphi^b(p,q)\} = \sum_c c^{abc}(p,q)\varphi^c(p,q) \qquad (15.35)$$

The first equation guarantees that no new constraints are generated while the system evolves in time. The second one ensures that constraints close a certain algebra with respect to Poisson brackets and are, in some sense, linearly independent: no new constraint arises by operating with the original ones.

In the case of a gauge theory, we have seen that (15.34) is satisfied with $c^{ab}(p,q) = 0$; also (15.35) holds if we put $c^{abc}(p,q) = f^{abc}$, the structure constants of the gauge algebra.

This generalized Hamiltonian system defined in a $2n$ dimensional phase space should be equivalent to a usual Hamiltonian system but in a reduced (due to the existence of constraints) phase space which should have $n-m$ degrees of freedom and which will be called $\Gamma^{*2(n-m)}$.

How to connect Γ^{2n} with $\Gamma^{*2(n-m)}$? Paradoxically, in order to reduce the original phase space, one strats, according to Dirac method, by adding m subsidiary conditions

$$\chi^a(p,q) = 0\,, \qquad a = 1,2,\ldots m \tag{15.36}$$

Functions χ^a and φ^b should satisfy

$$\det\left|\{\varphi^a,\chi^b\}\right| \neq 0 \tag{15.37}$$

$$\{\chi^a,\chi^b\} = 0 \tag{15.38}$$

then, the subspace Γ^* of Γ^{2n} defined by

$$\begin{aligned} \varphi^a(p,q) &= 0 \\ \chi^a(p,q) &= 0 \end{aligned} \tag{15.39}$$

is the reduced phase-space $\Gamma^{*2(n-m)}$ we were looking for. To see this, let us first note that, owing to conditions (15.38), it is possible to identifiy m coordinates with the subsidiary functions χ^a. One then choses

$$q = (\chi^a, q_l^*)\,, \qquad l = m+1, m+2, \ldots n \tag{15.40}$$

Accordingly, one writes

$$p = (p^a, p_l^*) \tag{15.41}$$

At this point, one can see the reason for condition (15.37). Indeed, because of the choice of subsidiary functions one has

$$\det\left|\{\varphi^a,\chi^b\}\right| = \det\left|\{\varphi^a,q^b\}\right| = \det\left|\frac{\partial\varphi^a}{\partial p^b}\right| \neq 0 \tag{15.42}$$

Now, this last condition can be interpreted as the necessary condition to solve the homogeneous system

$$\varphi^a(p,q) = 0 \tag{15.43}$$

in order to obtain the "redundant" momenta p^a, conjugate to the "redundant" coordinates that we have identified with the subsidiary functions. Once this is achieved, one explicitly knows how to write the p^a's in terms of the q^l's and the p^l's, so that one has effectively reduced the phase space from Γ^{2n} to $\Gamma^{*2(n-m)}$. In formulae, this means that one inverts (15.43) to write

$$p^a = p^a(q^l, p^l)\,, \qquad l = m+1, m+2, \ldots n \tag{15.44}$$

That is, $\Gamma^{*2(n-m)}$ is constructed from Γ^{2n} by making

1. $\chi^a = q^a = 0\,, \qquad a = 1, 2, \ldots, m$
2. $p^a = p^a(q^l, p^l)\,, \quad l = m+1, m+2, \ldots n$

We shall call generically q^* the coordinates q^l $(l = m+1, m+2, \ldots n)$ and, analogously, p^* the momenta p^l. These are identified as coordinates and conjugate momenta of the $\Gamma^{*2(n-m)}$ space.

Then, the Hamiltonian of the reduced system can be written from the original one by identifying

$$H^* = H^*(p^*, q^*) = \left. H(p,q)\right|_{\varphi=\chi=0} \tag{15.45}$$

We can at this point ask the following question: In what sense are the original constrained system with phase space Γ^{2n} and the reduced one with phase space $\Gamma^{*2(n-m)}$ equivalent? To answer it, let us write the equations of motion in Γ^{2n} space,

$$\dot{p}^i + \frac{\partial H}{\partial q^i} + \sum_a \lambda^a \frac{\partial \varphi^a}{\partial q^i} = 0\,, \quad i = 1, 2 \ldots n \tag{15.46}$$

$$\dot{q}^i - \frac{\partial H}{\partial p^i} - \sum_a \lambda^a \frac{\partial \varphi^a}{\partial p^i} = 0\,, \quad i = 1, 2 \ldots n \tag{15.47}$$

$$\varphi^a = 0\,, \qquad a = 1, 2 \ldots m \tag{15.48}$$

Note that here the λ^a's are arbitrary and then the solutions to (15.46)-(15.48) will exhibit this arbitrariness until we eliminate it appropriately. Precisely the subsidiary conditions $\chi^a = 0$ do this work. As a result, one ends with equations of motion in terms of variables p^* and q^* which are the Hamiltonian equations for our system. Indeed, since we have chosen $\chi^a = q^a = 0$ we have

$$\dot{q}^a = 0\,, \quad a = 1, 2, \ldots, m \tag{15.49}$$

and then, from eq.(15.47) we can write

$$\frac{\partial H}{\partial p^b} + \sum_a \lambda^a \frac{\partial \varphi^a}{\partial p^b} = 0\,, \quad a, b = 1, 2, \ldots, m \tag{15.50}$$

On the other hand, the "true" coordinates q^* obey, again from (15.47),

$$\dot{q}^l = \frac{\partial H}{\partial p^l} + \sum_a \lambda^a \frac{\partial \varphi^a}{\partial p^l}\,, \quad l = l+1, l+2, \ldots, n \tag{15.51}$$

Now, Hamiltonian H^* should lead to the simple Hamilton eqs.

$$\dot{q}^l = \frac{\partial H^*}{\partial p^l} \tag{15.52}$$

Then, from (15.51) and (15.52) we can establish the identity

$$\frac{\partial H^*}{\partial p^l} = \frac{\partial H}{\partial p^l} + \sum_a \lambda^a \frac{\partial \varphi^a}{\partial p^l} \tag{15.53}$$

Now, writing

$$\frac{\partial H^*}{\partial p^l} = \frac{\partial H}{\partial p^l} + \sum_a \frac{\partial H^*}{\partial p^a} \frac{\partial p^a}{\partial p^l} \tag{15.54}$$

we conclude that

$$\sum_a \lambda^a \frac{\partial \varphi^a}{\partial p^l} = \sum_a \frac{\partial H^*}{\partial p^a} \frac{\partial p^a}{\partial p^l} \tag{15.55}$$

This, together with eq.(15.50) leads to

$$\sum_b \lambda^b \left(\frac{\partial \varphi^b}{\partial p^a} \frac{\partial p^a}{\partial p^l} + \frac{\partial \varphi^b}{\partial p^l} \right) = 0 \tag{15.56}$$

Now, this identity is guaranteed by the fact that $\varphi^a = 0$ and hence we see that the simple Hamilton equations in the reduced phase space are consistent:

$$\dot{q}^l = \frac{\partial H^*}{\partial p^l} \ , \qquad \dot{p}^l = -\frac{\partial H^*}{\partial q^l} \tag{15.57}$$

and equivalent to the equations of motion of the original constrained Hamiltonian.

The path-integral for generalized Hamiltonian systems

Being $H^*(p^*, q^*)$ a "normal" Hamiltonian with $n-m$ coordinates, $n-m$ momenta and no constraints, we can just apply our recipe to define the quantum theory using the usual partition functional Z defined as

$$\begin{aligned} Z^{\Gamma^{*2(n-m)}} &= \int \prod_{l=m+1}^{n} \mathcal{D}p^l \mathcal{D}q^l \exp\left(\frac{i}{\hbar}\int dt (\sum_l p_l \dot{q}_l - H^*(p_l, q_l)\right) \\ &= \int \mathcal{D}p^* \mathcal{D}q^* \exp\left(\frac{i}{\hbar}\int dt (p^*\dot{q}^* - H^*(p^*, q^*)\right) \end{aligned} \tag{15.58}$$

Now, suppose one wants to define the path integral Z over the whole Γ^{2n} space. It is easy to prove that the following path-integral over Γ^{2n} gives the correct answer

$$\begin{aligned} Z^{\Gamma^{2n}} = \int \prod_{i=1}^{n} \mathcal{D}p^i Dq^i \prod_{a=1}^{m} \mathcal{D}\lambda^a \delta(\chi^a) \det |\{\varphi^b, \chi^c\}| \\ \exp\left(\frac{i}{\hbar}\int dt (\sum_i p_i \dot{q}_i - H(p_i, q_i) - \sum_a \lambda^a \varphi^a(p, q)\right) \end{aligned} \tag{15.59}$$

Let us see that this integral coincides with $Z^{\Gamma^{*2(n-m)}}$. First one integrates over the λ^a's,

$$\begin{aligned} Z^{\Gamma^{2n}} = \int \prod_{i=1}^{n} \mathcal{D}p^i Dq^i \prod_{a=1}^{m} \delta(\chi^a)\delta(\varphi^a) \det |\{\varphi^b, \chi^c\}| \\ \exp\left(\frac{i}{\hbar}\int dt (\sum_i p_i \dot{q}_i - H(p_i, q_i)\right) \end{aligned} \tag{15.60}$$

The rôle of the determinant in definition (15.59) becomes now clear. Since $\chi^a = q_a$, one has

$$\{\varphi^b, \chi^c\} = \{\varphi^b, q^c\} = \frac{\partial \varphi^b}{\partial p_c} \tag{15.61}$$

Then, the product over a in the first line of (15.60) can be written as

$$\prod_{a=1}^{m} \delta(\chi^a)\delta(\varphi^a) \det |\{\varphi^b, \chi^c\}| = \prod_{a=1}^{m} \delta(q_a)\delta(\varphi^a) \det |\frac{\partial \varphi^b}{\partial p_c}| \tag{15.62}$$

Now, using the identity

$$\delta(f(x)) = \frac{1}{|f'(x_o)|}\delta(x - x_0) \tag{15.63}$$

we can write

$$\delta(\varphi^a) \det |\frac{\partial \varphi^b}{\partial p_c}| = \delta(p_a - p_a(p^*, q^*)) \tag{15.64}$$

where we have identified the constraints in (15.64) with $f(x)$ and the solutions to the constraint equations with the zeros x_0. We can then write

$$\begin{aligned} Z^{\Gamma^{2n}} = \int \prod_{i=1}^{n} \mathcal{D}p^i Dq^i \prod_{a=1}^{m} \delta(q_a)\delta(p_a - p_a(p^*, q^*)) \\ \exp\left(\frac{i}{\hbar}\int dt (\sum_i p_i \dot{q}_i - H(p_i, q_i)\right) \end{aligned} \tag{15.65}$$

Now, the delta functions $\delta(q_a)$ eliminate, after integration, the dependence on the first m coordinates. Concerning the delta functions over momenta, integration over p_a allow to replace these momenta in terms of the p^*'s and the q^*'s. But then, one is left with the path integral (15.58) that defines $Z^{\Gamma^{*2(n-m)}}$ and hence we can write

$$Z^{\Gamma^{*2(n-m)}} = Z^{\Gamma^{2n}} \tag{15.66}$$

We shall see in what follows that when this method is applied to the case of a gauge theory, $Z^{\Gamma^{2n}}$ will precisely lead (once momenta E_k are integrated out) to the Lagrangian Faddeev-Popov proposal which then turns out to be a result and not just a recipe for quantizing theories with a gauge invariance. Identity (15.66) guarantees that this is indeed the correct answer since it is equivalent to work in the appropriately reduced phase space according to the generalized Hamiltonian dynamics.

Problem 1: Prove that

(i)

$$\left\{C_i^a(\vec{x},x_0),C_j^b(\vec{y},x_0)\right\}=ef^{abc}C^c(\vec{x},x_0)\delta^{(3)}(\vec{x}-\vec{y})$$

(ii)

$$\left\{\int d^3xH(x),C^a\right\}=0$$

Problem 2: Prove that

$$\delta A_k^a(\vec{x},t)=\int d^3y\left\{A_k^a(\vec{x},t),C^b(\vec{y})\alpha^b(\vec{y})\right\}$$

Clase 16

Gauge theories as generalized Hamiltonian systems

We have seen that one can define the path integral Z over the whole Γ^{2n} space of a Hamiltonian system with n degrees of freedom and m constraint and showed that it coincides with the results obtained by working directly in the "normal" $\Gamma^{*2(n-m)}$ Hamiltonian system, where the Feynman recipe can be applied naively. The form that Z takes in terms of Γ^{2n} variables is

$$Z^{\Gamma^{2n}} = \int \prod_{i=1}^{n} \mathcal{D}p^i Dq^i \prod_{a=1}^{m} \mathcal{D}\lambda^a \delta(\chi^a) \det |\{\varphi^b, \chi^c\}|$$
$$\exp\left(\frac{i}{\hbar}\int dt(\sum_i p_i\dot{q}_i - H(p_i,q_i) - \sum_a \lambda^a \varphi^a(p,q)\right) \quad (16.1)$$

where, appart from constraints φ^a ($a = 1, 2, \ldots, m$), we have introduced subsidiary conditions $\chi^a = 0$ which have to satisfy

$$\det \left|\{\varphi^a, \chi^b\}\right| \neq 0\ , \quad a, b = 1, 2, \ldots, m \quad (16.2)$$

$$\{\chi^a, \chi^b\} = 0 \quad (16.3)$$

It remains then to relate each quantity appearing in (16.1) with the appropriate gauge field theory object. We have seen that coordinates and momenta are simply given by A_i and $E_i = F_{0i}$. A_0 plays the rôle of a (functional) Lagrange multiplier and, accordingly, the Gauss law acts as a contraint,

$$q \to A_i^a(x)\ , \quad p \to E_i^a(x)\ , \quad \lambda \to A_0^a(X)\ , \quad a = 1, 2, \ldots, \mathrm{dim}G$$

$$\varphi^a \to D_k^{ab}[A_k]E_k^b(x) = \partial_k E_k^a(x) + igf^{acb}A_k^c(x)E_k^b(x) \tag{16.4}$$

Concerning Poisson brackets, we have

$$\begin{aligned} \{E_i^a(\vec{x},t), A_k^b(\vec{y},t)\} &= \delta_{ik}\delta^{ab}\delta^{(3)}(\vec{x}-\vec{y}) \\ \{A_i^a(\vec{x},t), A_k^b(\vec{y},t)\} = 0 \ , & \qquad \{E_i^a(\vec{x},t), E_k^b(\vec{y},t)\} = 0 \end{aligned} \tag{16.5}$$

Finally, the Hamiltonian is given by

$$H = \frac{1}{2}(E_k^2 + B_k^2[A]) \tag{16.6}$$

where

$$B_i[A] = \varepsilon_{ijk}\left(\partial_j A_k + A_j A_k\right) \tag{16.7}$$

We have now to decide what subsidiary conditions we chose. In the mechanical problem, the simplest choice was just to identify χ^a with "the first" m (m being the number of constraints) coordinates q^a. Here we have dim$G$ constraints and hence we have to chose dim$G$ subsidiary conditions. Now, at each space-time point we have $4\times$dim$G$ coordinates and then, a natural choice is to chose as subsidiary conditions (i.e. as a gauge condition, as we can already suspect)

$$\chi^a = \partial_k A_k^a \ , \quad a = 1, 2, \ldots, \mathrm{dim}G \tag{16.8}$$

One easily sees that this choice satisfies conditions (16.2)-(16.3). Indeed, from (16.5) one trivialy has

$$\{\partial_i A_i^a(\vec{x},t), \partial_k A_k^b(\vec{y},t)\} = 0 \tag{16.9}$$

Concerning (16.2), one finds that (see Problem 3)

$$\{D_k E_k^a, \partial_i A_i^b\} = \partial_i D_i^{ab}\delta^{(3)}(\vec{x}-\vec{y}) \tag{16.10}$$

so that in fact one has just to be sure that

$$\det \partial_i D_i^{ab}[A] \neq 0 \tag{16.11}$$

in order the approach to be valid. In fact one can prove that the operator $\partial_i D_i^{ab}$ has, for non-Abelian gauge groups, zero-modes (the Gribov problem [35]) and hence the Coulomb gauge (which corresponds to the subsidiary condition we are aconsidering) is not an appropriate gauge condition condition. Now, Singer has proven [36] that a very general class of gauge conditions (including the Lorentz gauge condition) has also this problem. One can see however that zero-modes appear for "large" (in a certain -Sobolev norm- sense) values of the gauge field and hence the problem is absent if one works perturbatively (this regime corresponding to "small" values of the gauge field).

We are ready to write Z in the complete phase space Γ^{2n}. At the path-integral level, we have seen that a delta function imposes the subsidiary or gauge condition

$$\chi^a[A] = \partial_i A_i^a = 0 \tag{16.12}$$

It is then natural to separate "coordinates" A_i^a into longitudinal and transverse components, so that the first ones are trivially zero when (16.12) is imposed. We then write (for notational simplicity we do not explicitly write the internal index a),

$$A_k = A_k^T + A_k^L \tag{16.13}$$

where, by definition,

$$\partial_i A_i^T = 0 \tag{16.14}$$

Concerning the longitudinal part, it can be written as a gradient,

$$A_i^L = \partial_i \Lambda \tag{16.15}$$

Then, given a general gauge field A_k, not necessarily in the Coulomb gauge, Λ is adjusted as follows. From (16.13) one has

$$\partial_k A_k = \nabla \Lambda \tag{16.16}$$

This equation can be inverted using the Laplacian Green function to get

$$\Lambda(x) = \frac{1}{4\pi} \int d^3x \frac{1}{|\vec{x} - \vec{y}|} \partial_i A_i \tag{16.17}$$

And this gives the required value for Λ.

From this discussion it is clear that what we called q^* in the mechanical formulation of the Faddeev method corresponds to the transverse components of the gauge field. Concerning the p^* momenta, they will just be the transverse components of the electric field,

$$q^* \to A_k^T \ , \quad p^* \to E_k^T \tag{16.18}$$

What about the longitudinal part E_i^l of the electric field? Precisely the constraint equations fix this longitudinal part through the Gauss law as can be seen by writing

$$E_k^L = \partial_k \Pi \tag{16.19}$$

and solving

$$D_k[A]E_k = \nabla^2 \Pi - g[A_k, \partial_k \Pi] - g[A_k, E_k^T] = 0 \tag{16.20}$$

That is, solving (16.20) one gets $\Pi = \Pi[A^t, E^t]$ or $E_k^L = E_k^L[A^t, E^t]$ so that everything is expressed under the path-integral, in terms of the reduced set of coordinates and momenta once constraints and subsidiary conditions are imposed. Note that, at a given time, at each point $\vec{x}$ of space $A_k^T(\vec{x}, t)$ is defined by two functions (each one taking values in the Lie algebra of the gauge group). this means that the gauge field has two polarizations.

In summary, we can be confident when writing Z in the "whole" phase space Γ^{2n},

$$Z = \int \mathcal{D}E_k \mathcal{D}A_k DA_0 \prod_a \delta(\partial_k A_k^a) |\det \partial_k D_k| \times$$
$$\exp\left(\frac{i}{4\hbar} \mathrm{tr} \int d^4x \left(2E_k \dot{A}_k - E_k^2 - B_k^2 + 2A_0 D_k E_k\right)\right) \tag{16.21}$$

One now integrates by parts the last term in the argument of the exponential so as to obtain

$$Z = \int \mathcal{D}E_k \mathcal{D}A_k DA_0 \prod_a \delta(\partial_k A_k^a) |\det \partial_k D_k| \times$$
$$\exp\left(\frac{i}{4\hbar} \mathrm{tr} \int d^4x \left(2E_k \dot{A}_k - E_k^2 - B_k^2 - 2E_k D_k A_0\right)\right) \tag{16.22}$$

The integral over E_k has just a quadratic and a linear term and then it can be easily integrated out. Noting that the coefficient in the linear term can be accomodated as $2(\partial_0 A_k - \partial_k A_0 + g[A_0, A_k]) = 2F_{0k}$, the result is

$$Z = \int \mathcal{D}A_k \mathcal{D}A_0 \exp\left(\frac{i}{4\hbar}\mathrm{tr}\!\int\! d^4x \left(F_{ij}^2 - 2F_{0i}^2\right)\right) \prod_a \delta(\partial_k A_k^a) |\det \partial_k D_k| \tag{16.23}$$

or,

$$Z = \int \mathcal{D}A_\mu \exp\left(\frac{i}{\hbar}\mathrm{tr}\!\int\! d^4x \frac{1}{4} F_{\mu\nu}F^{\mu\nu}\right) \prod_a \delta(\partial_k A_k^a) \Delta_{FP} \tag{16.24}$$

where we have written

$$\Delta_{FP} = |\det \partial_k D_k| \tag{16.25}$$

in order to make contact with the Faddeev-Popov prescription in the Lagrangian formalism. Being derived in the correct (generalized) Hamiltonian approach, the prescription is now justified.

BRST Symmetry

BRST à la ancienne

In 1975 Becchi, Rouet and Stora [37] and, independently Tyutin [38], discovered that although gauge symmetry has been fixed, Z as defined in (16.24) has still a symmetry related to gauge invariance but with an anticommuting parameter. This symmetry showed its power when it made possible to prove, in a clear and complete form, the renormalizability of non-Abelian gauge theories. In particular, the proof helped in taking seriously these theories as candidates to describe electroweak and strong interactions.

The way in which BRST invariance was discovered is related to the Faddeev-Popov method but it can be seen that a BRST principle can replace and generalize this method. We shall discuss both approaches, starting by the historical one.

Let us begin by rewriting (16.24) for a general gauge condition $F[A] = 0$ and introduce a path-integral representation for the delta function enforcing these condition. For simplicity we put $\hbar = 1$ and rotate to Euclidean space,

$$Z = \int \mathcal{D}A_\mu \mathcal{D}b \exp\left(-\frac{1}{4}\mathrm{tr}\!\int\! d^4x F_{\mu\nu}^2\right) \Delta_{FP}[A] \exp\left(-\mathrm{tr}\int d^4x F[A] b(x)\right) \tag{16.26}$$

Here $b(x) = b^a(x)t^a$ is a Lie-algebra valued auxilliary scalar. Concerning the Faddeev-Popov determinant, it is given by

$$\det \Delta_{FP}[A] = \det D_\mu^{ab}[A] \frac{\delta F^c[A]}{\delta A_\mu^b} = \det \mathcal{M}^{ac}[A] \tag{16.27}$$

As we have seen, for the Lorentz and Coulomb gauge one has

$$\begin{aligned} \det \mathcal{M}^{ac}[A] &= \det D_\mu^{ac}\partial_\mu \quad \text{Lorentz gauge} \\ \det \mathcal{M}^{ac}[A] &= \det D_i^{ac}\partial_i \quad \text{Coulomb gauge} \end{aligned} \tag{16.28}$$

We have lernt that determinants of differential operators like $\mathcal{M}$ can be represented as path integrals over anticommuting variables,

$$\det \mathcal{M} = \int \mathcal{D}\bar{c}^a \mathcal{D}c^a \exp\left(-\int d^4x \bar{c}^a \mathcal{M}^{ab} c^b\right) \tag{16.29}$$

where $\bar{c}(x) = \bar{c}^a(x)t^a$ and $c(x) = c^a(x)t^a$ are anticommuting Lie algebra valued scalars. Although integration over anticommuting variables was originally introduced to describe fermions, it should be stressed here that $\bar{c}$ and c are a pair of anticommuting scalars (not fermions transforming like 1/2 spinors). Concerning gauge transformations, remember that the Faddeev-Popov determinant is gauge invariant and this, in turn, is a result of the fact that $\mathcal{M}$ transforms covariantly. Then, if one is going to use fields $\bar{c}$ and c to exponentiate $\mathcal{M}$, these fields have to transform in the adjoint representation of the gauge group to get a scalar effective action,

$$\begin{aligned} \bar{c} &\rightarrow \bar{c}^g = g^{-1}(x)\bar{c}g(x) \\ c &\rightarrow c^g = g^{-1}(x)cg(x) \end{aligned} \tag{16.30}$$

The fact that $\bar{c}$ and c are not physical fields, in the sense that they will not appear as physical *in* or *out* states when computing v.e.v.'s makes plausible the name of *ghosts* that they have received.

We can then write the path-integral Z in the compact form

$$Z = \int \mathcal{D}A_\mu \mathcal{D}b\mathcal{D}\bar{c}\mathcal{D}c \exp\left(-\int d^4x L_{eff}\right) \tag{16.31}$$

where the effective Lagrangian L_{eff} is defined as

$$L_{eff} = \mathrm{tr}\left(\frac{1}{4}F_{\mu\nu}^2 + bF[A] + \bar{c}\mathcal{M}c\right) \tag{16.32}$$

Being the gauge invariance fixed, this effective Lagrangian is not gauge invariant due to the presence of the term containing, precisely, the gauge condition $F[A]$. However, one can see that L_{eff} has a more restricted invariance, associated to gauge transformations parametrized by an anticommuting global quantity instead of the commuting local parameter $\epsilon(x)$ associated to general gauge transformations. The transformation laws that define this new invariance (BRST invariance) are inferred from the gauge transformations laws of the physical fields, in this case the gauge field. Under an infinitesimal gauge transformation the gauge fields transforms as

$$\delta_{gauge}A_\mu^a = D_\mu^{ab}[A]\epsilon^b(x) \tag{16.33}$$

We then define a BRST transformation δ_{BRST} as if it were a gauge transformation but with a parameter $\epsilon^a(x) = c^a(x)\lambda$, with λ an anticommuting constant parameter,

$$\delta_{BRST} A^a_\mu = D^{ab}_\mu[A] c^b(x)\lambda \tag{16.34}$$

It is not difficult to define BRST transformation laws for b, $\bar{c}$ and c ensuring that L_{eff} is BRST invariant under the complete set of transformations. Indeed, one can see that if one imposes

$$\begin{aligned} \delta_{BRST}\bar{c}^a &= -b^a\lambda \\ \delta_{BRST}c^a &= -\frac{1}{2}f^{abc}c^b c^c\lambda \\ \delta_{BRST}b^a &= 0 \end{aligned} \tag{16.35}$$

one has

$$\delta_{BRST} L_{eff} = 0 \tag{16.36}$$

This can be proved by direct calculation. Instead, we shall use an important property associated to δ_{BRST} namely that it is <u>nilpotent</u>

$$\delta^2_{BRST} = 0 \tag{16.37}$$

It is important to stress that this property holds without use of the fact $\lambda^2 = 0$. In fact, from here on, we shall omit parameter λ in the definition of δ_{BRST}. This means that δ_{BRST} so defined is now an anticommuting object that takes a commuting field and transforms it into some anticommuting object.

To prove (16.37) one has to see that it is satisfied for each one of the fields. For example, one can easily see that acting on $\bar{c}$ it is nilpotent,

$$\delta^2_{BRST} c = \delta_{BRST} b = 0 \tag{16.38}$$

It is a little more involved to consider the case in which δ_{BRST} acts on A_μ,

$$\begin{aligned} \delta^2_{BRST} A_\mu &= \delta_{BRST}\left(\partial_\mu c + g[A_\mu, c]\right) \qquad (16.39) \\ &= -\frac{1}{2}\partial_\mu[c,c] + [\partial_\mu c, c] + [[A_\mu, c], c] - \frac{1}{2}[A_\mu, [c,c]] \end{aligned}$$

Now, the two first terms in the second line of (16.39) cancel each other. Concerning the remaining two, they are zero due to a Jacobi identity for anticommuting objects. We shall prove this just for the $SU(2)$ case but the proof can be extended to a general gauge group. One has, in components

$$\begin{aligned}\varepsilon^{abc}A^b_\mu c^c\varepsilon^{apq}c^p - \frac{1}{2}\varepsilon^{abc}A^a_\mu\varepsilon^{bpq}c^pc^q \;\; &= \;\; A^b_\mu c^bc^c - A^c_\mu c^bc^b - \frac{1}{2}A^b_\mu c^bc^c + \\ &\quad\; \frac{1}{2}A^b_\mu c^cc^b \end{aligned} \tag{16.40}$$

Because c^a is anticommuting, the second term in (16.40) is zero while the third and forth terms add to cancel the first one. One then has proved that $\delta^2_{BRST}A_\mu = 0$. We leave as a problem to show that $\delta^2_{BRST}\bar{c} = \delta^2_{BRST}c = 0$.

Once nilpotency is proved, it is easy to see that L_{eff} is BRST invariant. Of course the Yang-Mills term in L_{eff} remains invariant under BRST transformation, which is a particular case of a gauge transformation,

$$\delta_{BRST}\mathrm{tr}F^2_{\mu\nu} = 0 \tag{16.41}$$

Concerning the remaining two terms in (16.32), let us note that they can be written as the BRST transformed of a given functional. Indeed, consider

$$\delta_{BRST}(\bar{c}F[A]) = (\delta_{BRST}c)F[A] - \bar{c}\frac{\delta F[A]}{\delta A_\mu}\delta_{BRST}A_\mu \tag{16.42}$$

(Here we have taken into account that $\bar{c}$ and δ_{BRST} anticommute). Now, using (16.35) we have,

$$-\delta_{BRST}(\bar{c}F[A]) = bF[A] + \bar{c}\frac{\delta F[A]}{\delta A}D_\mu c \tag{16.43}$$

We can then write

$$L_{eff} = \frac{1}{4}\mathrm{tr}F^2_{\mu\nu} - \mathrm{tr}\delta_{BRST}(\bar{c}F[A]) \tag{16.44}$$

and then

$$\delta_{BRST}L_{eff} = \frac{1}{4}\mathrm{tr}\delta_{BRST}F^2_{\mu\nu} - \mathrm{tr}\delta^2_{BRST}(\bar{c}F[A]) = 0 \tag{16.45}$$

Generalization of the BRST method

Having learnt that the gauge fixed effective Lagrangian results from the addition to the original Lagrangian of term which can be written as the BRST transformed of some functional, we can think in the extension of the BRST method in the following sense:

Given a Lagrangian with a certain symmetry (not necessarily gauge symmetry), we will need to fix this symmetry in order not to overcount in the path-integral equivalent configurations. Otherwise, one should get a divergent non physical answer. To do this one proceeds as follows

- Given the (infinitesimal) symmetry transformation law for the original field A, with parameter ϵ, one defines a new transformation law δ_{BRST} coppied from it but replacing ϵ by an anticommuting field called a ghost, having the same tensorial character as the parameter of the original symmetry.

- Having introduced the ghost field (which we call $c(x)$), one introduces an "anti-ghost" field $\bar{c}(x)$ and an auxiliary field $b(x)$ and completes the transformation law so as to ensure that $\delta^2_{BRST} = 0$.

- The effective (gauge fixed) Lagrangian L_{eff} is constructed from the original symmetric one L by adding the BRST transformed of an <u>an arbitrary functional</u> depending in principle on all fields,

$$L_{eff}[A, b, \bar{c}, c] = L[A] + \delta_{BRST}\mathcal{F}[A, b, \bar{c}, c] \qquad (16.46)$$

- The properly gauge fixed partition function is the defined as

$$Z = \int \mathcal{D}A\mathcal{D}b\mathcal{D}\bar{c}\mathcal{D}c \exp(-\int d^4x L_{eff}[A, b, \bar{c}, c] \qquad (16.47)$$

Note that this prescription leads to results quite more general than those obtained in the original Faddeev-Popov scheme. In particular, if one choses appropriately $\mathcal{F}$, one can obtain effective Lagrangians that depend on ghost through polynomials of degree higher than 2. For example, if one takes

$$\mathcal{F} = b\bar{c}c\bar{c}F[A] \qquad (16.48)$$

one gets terms quartic in the ghosts that will never produce, by integration, a Faddeev-Popov determinant but that, nevertheless, fix the gauge correctly. This is exploited for example, in quantization of theories containing gravity.

How are we shure that this general prescription is as correct as that we have proved, based in using $\mathcal{F} = \bar{c}F[A]$? To see this we will prove (under certain assumptions) that the v.e.v. of the BRST transformed of an arbitrary functional (a BRST exact term) satisfies the important property

$$\langle \delta_{BRST} G \rangle = 0 \tag{16.49}$$

From this, one can show that the v.e.v.'s of a "physical operator" calculated with Lagrangians differing in a BRST exact term coincide.

To prove (16.49) , consider the quantity

$$\langle G \rangle = \int \mathcal{D}\Phi \exp(-S_{eff}[\Phi])G[\Phi] \tag{16.50}$$

where Φ represents all fields: original "phsical" fields, ghost fields, auxiliary fields.

Consider the change of variables

$$\Phi \rightarrow \Phi' = \Phi + \lambda \delta_{BRST} \Phi \tag{16.51}$$

By assumption, the action is invariant under BRST transformations and then remains unchanged.

If we also assume that the integration measure is BRST invariant, then the only change in the integrand comes from the change in G,

$$\langle G \rangle = \int \mathcal{D}\Phi' \exp(-S_{eff}[\Phi'])\,(G[\Phi'] - \lambda \delta_{BRST} G[\Phi']) \tag{16.52}$$

or

$$\langle G \rangle = \langle G \rangle - \lambda \langle \delta_{BRST} G \rangle \tag{16.53}$$

from which (16.49) follows.

Now, it is easy to prove that

$$\int \mathcal{D}\Phi \exp(-S_{eff}[\Phi]) = \int \mathcal{D}\Phi \exp(-S_{eff}[\Phi] + \delta_{BRST} H) \tag{16.54}$$

One just expands in powers of $\delta_{BRST}H$ the r.h.s. and uses (16.49) to prove that all terms except the first one vanish. This completes the proof of our assertion concerning the fact one can use an arbitrary BRST exact form to fix the symmetry.

Problem 1: Prove eq.(16.10),

$$\{D_k E_k^a, \partial_i A_i^b\} = \partial_i D_i^{ab} \delta^{(3)}(\vec{x} - \vec{y}) \tag{16.55}$$

Problem 2: Prove that

$$\delta^2_{BRST}\bar{c} = \delta^2_{BRST}c = 0$$

Problem 3: Prove the identity (16.86)

$$\int \mathcal{D}\bar{c}\mathcal{D}c \exp\Big(-\int d^4x(\bar{c}\mathcal{M}c + \bar{c}\theta + \bar{\theta}c)\Big) = \det\mathcal{M}\exp\Big(-\int d^4x d^4y \bar{\theta}\mathcal{M}^{-1}\theta\Big)$$

BRST symmetry and Noether charge

Up to here we have seen that the the gauge symmetry can be fixed by adding to the original gauge-invariant Lagrangian an arbitrary BRST exact term $\delta_{BRST}H[\Phi]$

$$Z = \int \mathcal{D}\Phi \exp(-S[\Phi] + \delta_{BRST}H[\Phi]) \tag{16.56}$$

where we indicate the set of all fields (the original A_μ gauge field, the auxiliary field implementing the gauge condition and the ghost and anti-ghost fields) as Φ. Physical quantities do not depend on the particular choice of H since we have also seen that

$$\langle \delta_{BRST}G[\Phi]\rangle = 0 \tag{16.57}$$

for arbitrary $G[\Phi]$.

Since the (nilpotent) BRST transformation δ_{BRST} is a symmetery of the action and also of the integration measure, there should be a charge

Q_{BRST} conserved both at the classical and at the quantum level. This charge is Grassman odd and should generate BRST transformations,

$$\begin{aligned}\lambda\delta_{BRST}\Phi &= \phi^{\lambda} - \Phi = \exp\left(-i\lambda Q_{BRST}\right)\Phi\exp\left(i\lambda Q_{BRST}\right) - \Phi \\ &= -i\lambda[Q_{BRST}, \Phi]\end{aligned} \tag{16.58}$$

The nilpotency condition $\delta^2_{BRST} = 0$ reads in terms of Q_{BRST}

$$\lambda'\lambda\delta^2_{BRST} = [\lambda' Q_{BRST}, [\lambda Q_{BRST}, \Phi]] = 0 \tag{16.59}$$

or

$$\lambda'\lambda\delta^2_{BRST} = \lambda'\lambda Q^2_{BRST}\Phi - (\lambda'\lambda + \lambda\lambda')Q_{BRST}\Phi Q_{BRST} + \Phi Q^2_{BRST}\lambda\lambda' = 0 \tag{16.60}$$

Now, since $\lambda\lambda' + \lambda'\lambda = 0$ we see that the condition

$$Q^2_{BRST} = 0 \tag{16.61}$$

ensures nilpotency at the BRST charge level.

In terms of the BRST charge, eq.(16.57) reads

$$\langle\alpha|[Q_{BRST}, G]|\alpha\rangle = 0 \tag{16.62}$$

In order for this to hold for all BRST changes it is necessary that

$$Q_{BRST}|\alpha\rangle = 0 \tag{16.63}$$

for any physical state $|\alpha\rangle = |\text{phys}\rangle$. That is, *the physical states are the kernel of the nilpotent operator Q_{BRST}*. Or, in other words,

$$\exp\left(i\lambda Q_{BRST}\right)|\text{phys}\rangle = |\text{phys}\rangle \tag{16.64}$$

That is *Physical states are BRST invariant.*

The Noether charge for Yang-Mills theory

Given the effective gauge fixed Yang-Mills Lagrangian

$$\begin{aligned}L_{eff} &= L_{YM} + \delta_{BRST}\mathcal{F} = L_{YM} + \delta_{BRST}(\text{tr}\bar{c}F[A]) \\ &= L_{YM} + \text{tr}(bF) + \text{tr}(\bar{c}\mathcal{M}c)\end{aligned} \tag{16.65}$$

which is invariant under the BRST transformation

$$\Phi \to \Phi + \lambda \delta_{BRST} \Phi \tag{16.66}$$

with global parameter λ, we can construct the associated Noether charge by promoting (16.66) to a local transformation,

$$\Phi \to \Phi' = \Phi + \lambda(x) \delta_{BRST} \Phi \tag{16.67}$$

Although the Yang-Mills Lagrangian L_{YM} is still invariant under (16.67) since the transformation corresponds to a particular gauge transformation, the complete effective Lagrangian is not. Consider for simplicity the Lorentz gauge condition,

$$F[A] = \partial_\mu A_\mu \tag{16.68}$$

giving

$$\mathcal{M} = \partial_\mu D_\mu \tag{16.69}$$

Then, the fact that $\lambda = \lambda(x)$ implies that a term linear in $\partial_\mu \lambda$ is generated when the Faddeev-Popov term is variated,

$$\delta L_{FP} = L'_{FP} - L_{FP} = \mathrm{tr}(\bar{c} \partial_\mu D_\mu c)' - \mathrm{tr}(\bar{c} \partial_\mu D_\mu c) \tag{16.70}$$

or, integrating by parts at the action level,

$$\delta L_{FP} = -\mathrm{tr}(\partial_\mu \bar{c} D_\mu c)' + \mathrm{tr}(\partial_\mu \bar{c} D_\mu c) \tag{16.71}$$

Explicitly, the change of variables we are performing is given by

$$\begin{aligned} A'_\mu &= A_\mu + D_\mu(\lambda(x) c) \\ \bar{c}' &= \bar{c} + \lambda(x) b \\ c' &= c + \frac{1}{2}\lambda(x)[c, c] \\ b' &= b \end{aligned} \tag{16.72}$$

Then, we have for δL_{FP}

$$\delta L_{FP} = -b(\partial_\mu \lambda) D_\mu c - \frac{1}{2} \partial_\mu \bar{c} (\partial_\mu \lambda)[c, c] \tag{16.73}$$

Concerning the term implementing the gauge condition,

$$\delta L_{GF} = (b\partial_\mu A_\mu)' - (b\partial_\mu A_\mu) = b\partial_\mu\partial_\mu(\lambda c) - b\partial_\mu(\lambda\partial_\mu c) = -\partial_\mu b(\partial_\mu\lambda)c \tag{16.74}$$

Putting this together, we have

$$\delta_{\lambda(x)}\mathcal{F} = -(\partial_\mu\lambda)\left(bD_\mu c + \frac{1}{2}(\partial_\mu\bar{c})[c,c] + (\partial_\mu b)c\right) \tag{16.75}$$

Performing this change at the path-integral level and assuming that the integration measure, which was invariant under global BRST changes is also invariant for the local case, we have

$$Z = \int \mathcal{D}\phi \exp\left(-S_{eff} + \int d^4x(\partial_\mu\lambda)\left(bD_\mu c + \frac{1}{2}(\partial_\mu\bar{c})[c,c] + (\partial_\mu b)c\right)\right) \tag{16.76}$$

or, integrating by parts in the term containing λ,

$$Z = \int \mathcal{D}\phi \exp\left(-S_{eff} - \int d^4x\lambda\partial_\mu\left(bD_\mu c + \frac{1}{2}(\partial_\mu\bar{c})[c,c] + (\partial_\mu b)c\right)\right) \tag{16.77}$$

Sin Z cannot depend on λ, we obtain after differentiation,

$$\frac{\delta\log Z}{\delta\lambda} = 0 = \langle\partial_\mu\left(bD_\mu c + \frac{1}{2}(\partial_\mu\bar{c})[c,c] + (\partial_\mu b)c\right)\rangle \tag{16.78}$$

so that we can identify the BRST Noether current as

$$J_\mu = bD_\mu c + \frac{1}{2}(\partial_\mu\bar{c})[c,c] + (\partial_\mu b)c \tag{16.79}$$

so that the BRST charge is given by

$$Q_{BRST} = \int d^3x\left(bD_0 c + \frac{1}{2}(\partial_0\bar{c})[c,c] + (\partial_0 b)c\right) \tag{16.80}$$

BRST symmetry and Ward identities

How does one uses this BRST invariance in order to prove renormalizability of gauge theories? We shall only sketch the procedure here. The interester reader should look, for example, the Itzykson-Zuber book for details [39]. The strategy is to obtain Ward identities exploting precisely the properties of BRST transformations. One then uses these Ward identities to establish relations between Green functions and this in turn ensures cancellations of infinities order by order in perturbation theory.

As a simple example, consider the following trivial identity,

$$0 = \int \mathcal{D}b\mathcal{D}\bar{c}\mathcal{D}c\mathcal{D}A_\mu \bar{c} \exp\left(-S_{eff} + \int d^4x j_\mu A^\mu\right) \tag{16.81}$$

where

$$S_{eff} = \int d^4x L_{eff} \tag{16.82}$$

and j^μ is an external source. Identity (16.81) is valid because

$$\int \mathcal{D}\bar{c}\mathcal{D}c\bar{c} \exp(-\mathrm{tr} \int d^4x \bar{c}\mathcal{M}c) = 0 \tag{16.83}$$

Now, we perform an infinitesimal BRST transformations as a change of variables, assuming that the measure is BRST invariant[6]. After the transformation we have

$$\begin{aligned} 0 &= \int \mathcal{D}b\mathcal{D}\bar{c}\mathcal{D}c\mathcal{D}A_\mu\, \bar{c} \exp\left(-S_{eff} + \int d^4x j_\mu \left(A^\mu + D_\mu c\lambda\right)\right) + \\ & \int \mathcal{D}b\mathcal{D}\bar{c}\mathcal{D}c\mathcal{D}A_\mu\, b\lambda \exp\left(-S_{eff} + \int d^4x j_\mu A^\mu\right) \end{aligned} \tag{16.84}$$

so that, to order λ one has

$$\begin{aligned} 0 &= \int \mathcal{D}b\mathcal{D}\bar{c}\mathcal{D}c\mathcal{D}A_\mu \left(b(x) + \int d^4y \bar{c}(x) j_\mu(y) D_\mu c(y)\right) \\ & \exp\left(-S_{eff} + \int d^4x j_\mu A^\mu\right) \end{aligned} \tag{16.85}$$

Now, the second term in (16.85) can be easily integrated over the ghost variables. Indeed, from the identity

$$I = \int \mathcal{D}\bar{c}\mathcal{D}c \exp\left(-\int d^4x(\bar{c}\mathcal{M}c + \bar{c}\theta + \bar{\theta}c)\right) = \det\mathcal{M} \exp\left(-\int d^4x d^4y \bar{\theta}\mathcal{M}^{-1}\theta\right) \tag{16.86}$$

[6]In fact, once BRST invariance is discovered at the level of the classical action, it is consistent to define a quantum measure respecting this invariance, as it should respect gauge invariance, Lorentz invariance, etc).

one gets, by differentiation with respect to the anticommuting sources θ and $\bar{\theta}$

$$- \left.\frac{\delta^2 I}{\delta\theta\delta\bar{\theta}}\right|_{\theta=\bar{\theta}=0} = \det \mathcal{M}\mathcal{M}^{-1} \tag{16.87}$$

Using this result in (16.85) one gets

$$\begin{aligned} 0 &= \int \mathcal{D}b\mathcal{D}A_\mu \left(\det \mathcal{M}b(x) + \int d^4y j_\mu(y) D^y_\mu \left(det\mathcal{M}\mathcal{M}^{-1}(x,y)\right)\right) \\ &\quad \exp\left(-S_{eff} + \int d^4x j_\mu A^\mu\right) \end{aligned} \tag{16.88}$$

We shall now integrate out the auxiliary field b. for the integrand which is linear in b we use

$$\begin{aligned} \int \mathcal{D}bb \exp(-\int d^4x bF[A]) &= \frac{\delta}{\delta s} \int \mathcal{D}b \exp(-\int d^4x b(F[A]+s)\Big|_{s=0} \\ &= \frac{\delta}{\delta s} \int \delta(F[A]+s)\Big|_{s=0} \end{aligned} \tag{16.89}$$

Now, using the alternative representation for the delta function

$$\delta(F[A]+s) = \lim_{\alpha\to 0} \exp\left(-\frac{1}{2\alpha}\int d^4x (F[A]+s)^2\right) \tag{16.90}$$

we get (we omit the explict indication that the $\alpha \to 0$ limit should be taken at the end of calculations)

$$\int \mathcal{D}bb \exp(-\int d^4x bF[A]) = \frac{F[A]}{2\alpha} \exp\left(-\frac{1}{2\alpha}\int d^4x F[A]^2\right) \tag{16.91}$$

Putting all this together we end with

$$\begin{aligned} &\int \mathcal{D}A_\mu \left(\det \mathcal{M}[\mathcal{A}]\frac{1}{2\alpha}F[A] + \int d^4y j_\mu(y) D^y_\mu[A] \left(det\mathcal{M}[\mathcal{A}]\mathcal{M}^{-1}[A;x,y]\right)\right) \\ &\qquad \exp\left(-S_{eff}[j]\right) = 0 \end{aligned} \tag{16.92}$$

or

$$\begin{aligned} &\left(\det \mathcal{M}[\frac{\delta}{\delta j}]\frac{1}{2\alpha}F[\frac{\delta}{\delta j}] + \int d^4y j_\mu(y) D^y_\mu[\frac{\delta}{\delta j}] \left(det\mathcal{M}[\frac{\delta}{\delta j}]\mathcal{M}^{-1}[\frac{\delta}{\delta j};x,y]\right)\right) \\ &\qquad Z[j] = 0 \end{aligned} \tag{16.93}$$

Here we called

$$S_{eff}[j] = S_{eff} + \int d^4 x j_\mu A_\mu$$

and

$$Z[j] = \int \mathcal{D} fields \exp(-S[j])$$

This identity, obtained for the first time by Lee and Zinn-Justin [40] allows to obtain relations between n point functions. Indeed, differentiation of $Z[j]$ with respect to j n times gives the n-point function Γ^n. Now, if one differentiatiates identity (16.93) one gets relations among certain functionals containing Γ^n. This in turn allows to obtain relations between the renormalization constants which show that infinities cancel out, a result that can be then regarded as a consequence of gauge invariance (which, in this approach là ancienne, is behind BRST invariance)

Clase 17

Computing fermion determinants

In 1958 Scwhinger [41] proposed a toy model to study a possible mechanism for giving a mass to a gauge field without breaking the gauge symmetry. The phenomenon was called *dynamical mass generation* and the model, called now the Schwinger model, has been since then used as a laboratory for studying a plethora of phenomena. In 1974, 't Hooft [29] considerered the non-Abelian extension of the model inventing the $1/N$ expansion to solve it.

The Schwinger model is Quantum Electrodynamics of massless fermions in two dimensional space time. Dynamics is defined by the Lagrangian

$$L = \frac{1}{4} F^{\mu\nu}_{\mu\nu} + \bar{\psi}(i\partial\!\!\!/ + eA\!\!\!/)\psi \tag{17.1}$$

Peculiarities of Dirac matrices in two-dimensional space-time make this model exactly soluble. However, if a mass term for fermions is included, exact solubility is lost and the only possibility to study the model is by making an expansion in powers of the fermion mass. The same happens for the non-Abelian version of (17.1). The fermionic Lagrangian can be also solved in the massless case or, what is the same, the fermion determinant can be computed exaclty. But this was only realized in the eighties [44]-[45].

Before going on into the path-integral solution of the Schwinger model, it should be stressed that calling the gauge field a photon and saying taht *due to the dynamical mass generation phenomenon* the photon acquires a mass is somehow an abuse of language: in 2 dimensional space-time there ar no transverse directions and then the gauge field cannot be considered as transverse, as it is the case in real world. Moreover, being the concept of rotations absent in 1 space dimensions, the notion of spin is artificial and then considering ψ a fermion field with spin 1/2 can be seen as a caprice. In fact, in two space-time dimensions there exist a prescription (called bosonization) which allows to find a bosonic equivalent for any fermion model.

Let us now enumerate the 2 dimensional space-time peculiarities that will be exploited for computing exactly fermion determinants. First, in $d = 2$ dimensions a general $U(1)$ gauge field can be always written in the form

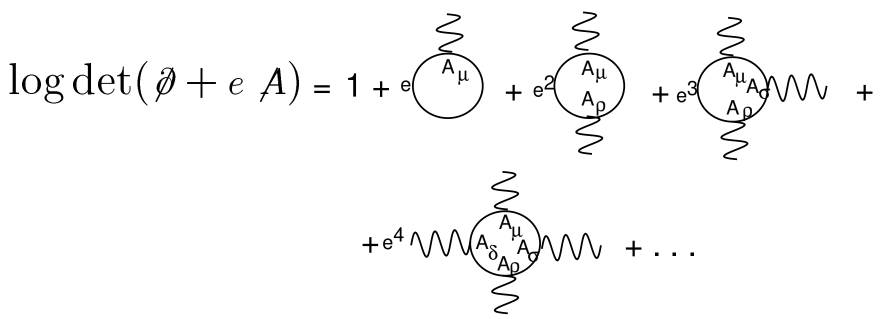
$$eA_\mu = \varepsilon_{\mu\nu}\partial_\mu\phi + \partial_\mu\eta \tag{17.2}$$

The e in the l.h.s. has been included for convenience. Note that in $d = 2$, $[e] = m$. From here on we shall work in Euclidean space (although all calculations can be performed in Minkowski space). Of course η represents a gauge degree of freedom. Indeed, if one computes the field strength one gets

$$F_{\mu\nu} = \varepsilon_{\mu\nu}\Box\phi \tag{17.3}$$

This shows that the "photon" has just one degree of freedom associated to ϕ.

A second peculiarity that makes understandable why the model can be exactly solved is the following. We have seen that a determinant has in QFT an expansion which can be represented as follows



The wriggled external lines represent A_μ insertions on the fermion loop. All diagrams with an odd number of insertions vanish (Furry's theorem). Concerning loops with an even number of insertions, what Schwinger has shown is that because of Dirac matrices properties in

2 dimensional space time, all contributions except that with 2 insertions vanish. One then has just to compute a Feynman diagram with two fermion propagators in order to get the exact fermion determinant. Instead of doing this, we shall use what we have lernt for Fujikawa's Jacobians in order to compute this determinant.

We shall work with the following Euclidean representation for the 2×2 Dirac matrices

$$\gamma_0 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \gamma_1 = \begin{pmatrix} 0 & i \\ -i & 0 \end{pmatrix}, \qquad \gamma_5 = i\gamma_0\gamma_1 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \tag{17.4}$$

A very important property relating Dirac matrices (which in fact coincide with $SU(2)$ Pauli matrices), at the origin of solubility of two dimensional fermion models is

$$\gamma_\mu \gamma_5 = i\varepsilon_{\mu\nu}\gamma_\nu \tag{17.5}$$

This property is crucial both at the classical and at the quantum level. Indeed, consider the change of fermion variables $\psi \to \chi$ and $\bar\psi \to \bar\chi$

$$\begin{aligned} \psi(x) &= \exp\left(t\left(\gamma_5\phi + i\eta\right)\right)\chi(x) \\ \bar\psi(x) &= \bar\chi \exp\left(t\left(\gamma_5\phi - i\eta\right)\right) \end{aligned} \tag{17.6}$$

where t is a parameter ($0 \le t \le 1$) while ϕ and η are related to A_μ according to (17.2). In terms of the new variables the fermion Lagrangian takes the form

$$\begin{aligned} L_F &= \bar\psi(i\partial\!\!\!/ + eA\!\!\!/)\psi = \bar\chi \exp\left(\gamma_5\phi - i\eta\right)(i\partial\!\!\!/ + eA\!\!\!/)\exp\left(\gamma_5\phi + i\eta\right)\psi \\ &= \bar\chi(i\partial\!\!\!/ + eA\!\!\!/ + ti\partial\!\!\!/\gamma_5\phi - t\ \partial\!\!\!/\eta)\chi \end{aligned} \tag{17.7}$$

Now, using (17.5), one has

$$i\partial\!\!\!/\gamma_5\phi = -\ \tilde{\partial\!\!\!/}\phi \tag{17.8}$$

so that, using (17.2), L_F can be written as

$$L_F = \bar\chi(i\partial\!\!\!/ + (1-t)eA\!\!\!/)\chi \tag{17.9}$$

We see that the choice $t = 1$ reduces the Dirac operator to the free one,

$$L_F = \bar\chi i\partial\!\!\!/\chi \tag{17.10}$$

In terms of the new fermion variables the Maxwell term is completely decoupled from the matter term!

Now, this cannot be the complete story at the quantum level. The decoupling change of variables is a combination of a gauge and a chiral rotation. We know that, at the path-integral level, one should have to take into account the Jacobian associated with the fermion rotation and that concerning the chiral part, this Jacobian will be not trivial. Let us then study this aspect by defining

$$Z = \int \mathcal{D}\bar{\psi}\mathcal{D}\psi \exp\left(-\int d^2x L_f\right) = \det(i\partial\!\!\!/ + eA\!\!\!/) \tag{17.11}$$

or, after the change of variables (17.6) with arbitrary parameter t,

$$Z = J_5[t, \phi] \det(i\partial\!\!\!/ + (1-t)eA\!\!\!/) \tag{17.12}$$

Since Z cannot depend on t, we get, after differentiating

$$0 = \frac{d \log Z}{dt} = \frac{d \log J_5}{dt} + \frac{d}{dt} \log\det(i\partial\!\!\!/ + (1-t)eA\!\!\!/) \tag{17.13}$$

or

$$\frac{d}{dt} \log\det(i\partial\!\!\!/ + (1-t)eA\!\!\!/) = -\frac{d \log J}{dt} \tag{17.14}$$

Integrating this equation over t in the interval $[0, 1]$, we get

$$\frac{\det(i\partial\!\!\!/ + eA\!\!\!/)}{\det i\partial\!\!\!/} = \exp \int_0^1 dt \frac{d \log J_5[t, \phi]}{dt} \tag{17.15}$$

We have seen that a gauge rotation gives a trivial Jacobian so that we have to concentrate in the chiral part. For an infinitesimal chiral rotation we have obtained (see problems) the following identity for an infinitesimal chiral transformation $\delta\alpha$

$$\log J_5[\delta\alpha] = \mathrm{Tr}\,(\gamma_5 \delta\alpha) \tag{17.16}$$

If we chose $\delta\alpha = \phi dt$ we can write

$$d \log J_5[\delta\alpha] = \mathrm{Tr}\,(\gamma_5 \phi dt) \tag{17.17}$$

(Being the chiral phase ϕdt we have written the corresponding Jacobian as dJ_5). We then have

$$\frac{d\log J_5[\delta\alpha]}{dt} = \mathrm{Tr}\left(\gamma_5\phi\right) \tag{17.18}$$

We know that the trace in the r.h.s. of eq.(17.18) is ill-defined and that we have to regularize it conveniently. Using the heat-kernel approach this reduces to include the exponential of the appropriate gauge covariant Dirac operator squared. In this case, the Dirac operator has a t-dependent coupling $e(1-t)$,

$$\not{D}_t[A] = i\not{\partial} + e(1-t)\not{A} \tag{17.19}$$

We then have

$$\left.\frac{d\log J_5[\delta\alpha]}{dt}\right|_{reg} = \lim_{M\to\infty} \mathrm{Tr}\left(\gamma_5\phi\exp\left(-\frac{\not{D}_t[A]^2}{M^2}\right)\right) \tag{17.20}$$

We have already computed the r.h.s. as a problem. The answer (which now depends on t) is

$$\left.\frac{d\log J_5[\delta\alpha]}{dt}\right|_{reg} = \frac{(1-t)e}{2\pi}\int d^2x\epsilon_{\mu\nu}F_{\mu\nu}\phi \tag{17.21}$$

This can be rewritten as

$$\left.\frac{d\log J_5[\delta\alpha]}{dt}\right|_{reg} = -\frac{(1-t)e}{\pi}\int d^2x\epsilon_{\mu\nu}A_\nu\partial_\mu\phi \tag{17.22}$$

or, using (17.2),

$$\left.\frac{d\log J_5[\delta\alpha]}{dt}\right|_{reg} = -\frac{(1-t)e^2}{\pi}\int d^2x\left(A_\nu A_\nu + \frac{1}{e}A_\nu\partial_\nu\eta\right) \tag{17.23}$$

Now, one easily sees from (17.2) that

$$\Box\eta = e\partial_\mu A_\mu \tag{17.24}$$

so that one has

$$\left.\frac{d\log J_5[\delta\alpha]}{dt}\right|_{reg} = -\frac{(1-t)e^2}{\pi}\int d^2x\left(A_\nu A_\nu + A_\nu\partial_\nu\Box^{-1}\partial_\mu A_\mu\right) \tag{17.25}$$

We can then write, instead of (17.15)

$$\frac{\det(i\partial\!\!\!/ + eA\!\!\!/)}{\det i\partial\!\!\!/} = \exp\!\left(-\frac{e^2}{\pi}\int_0^1 dt(1-t)\int d^2x\left(A_\nu A_\nu + A_\nu\partial_\nu\Box^{-1}\partial_\mu A_\mu\right)\right) \tag{17.26}$$

or

$$\frac{\det(i\partial\!\!\!/ + eA\!\!\!/)}{\det i\partial\!\!\!/} = \exp\left(-\frac{e^2}{2\pi}\int d^2x A_\nu\left(\delta_{\nu\mu} + \partial_\nu\Box^{-1}\partial_\mu\right)A_\mu\right) \tag{17.27}$$

We then see that integration over fermions gives an A_μ dependent contribution that adds to the Maxwell Lagrangian so that we can write for the complete partition function of the Schwinger model

$$\begin{aligned} Z &= \int \mathcal{D}A\exp\left(-\int d^2x\frac{1}{4}F_{\mu\nu}F^{\mu\nu}\right)\int \mathcal{D}\bar\psi\mathcal{D}\psi\exp\left(-\int d^2x L_F\right) \\ &= \int \mathcal{D}A\exp\left(-\frac{1}{4}\int d^2x F_{\mu\nu}F^{\mu\nu}\right)\det(i\partial\!\!\!/ + eA\!\!\!/) \\ &= \int \mathcal{D}A\exp\left(-\frac{1}{4}\int d^2x F_{\mu\nu}F^{\mu\nu} + \frac{e^2}{2\pi}\int d^2x A_\nu\left(\delta_{\nu\mu} + \partial_\nu\Box^{-1}\partial_\mu\right)A_\mu\right) \end{aligned} \tag{17.28}$$

It is precisely the Jacobian contribution that provides here a mass to the gauge field. In fact, in terms of the "physical" field ϕ the partition function reads

$$Z = \int \mathcal{D}\phi\exp\left(-\int d^2x\phi\left(\frac{1}{2}\Box\Box + \frac{e^2}{2\pi}\Box\right)\phi\right) \tag{17.29}$$

which corresponds to a theory with propagator

$$\Delta = -\frac{1}{k^2(k^2 + (e^2/\pi)^2)} = \frac{\pi}{e^2}\frac{1}{k^2 + (e/\pi)^2} - \frac{\pi}{e^2}\frac{1}{k^2} \tag{17.30}$$

This propagator corresponds to a physical massive particle with mass $e/\sqrt{\pi}$ and a massless non-physical excitation (its propagator as an "incorrect" sign).

Clase 18

The non-Abelian fermion determinant

It is in the non-Abelian case where the path-integral approach to solving fermion determinants and the consequent resolution of the model shows its power. Of course, 2-dimensional peculiarities are fundamental but, even with its help, not much was achieved in two-dimensional gauge theories coupled to fermions in respect with exact calculations till the path-integral approach was developed.

The clue in the approach is the observation that an expression for a two-dimensional $U(1)$ gauge field like (17.2), although valid, is not of much help when extended to the non-Abelian case. Indeed, the first idea one has for handling fermions coupled to a gauge field $A_\mu^a t^a$, taking values in the Lie algebra of some gauge group G with generators t^a, is to write the obvious generalization of (17.2),

$$eA_\mu^a t^a = \varepsilon_{\mu\nu}\partial_\mu \phi^a t^a + \partial_\mu \eta^a t^a \, , \quad a = 1, 2, \ldots, \mathrm{dim}\mathcal{G} \tag{18.1}$$

$$[t^a, t^b] = i f^{abc} t^c \tag{18.2}$$

That is, one introduces 2dim$\mathcal{G}$ fields ϕ^a and η^a to represent the vector field $A_\mu^a t^a$. An appropriate gauge condition should then eliminate the gauge degrees of freedom (i.e, some components of ϕ and η).

Now, the clue in the exact evaluation of the fermion determinant in the $U(1)$ case was that an appropriate chiral rotation $\psi \to \exp(\gamma_5 \phi)\chi$ of the fermion fields cancelled out the fermion-gauge-field coupling yielding the free Dirac operator. This, at the quantum level, trivialized the fermion sector but left a non-trivial Jacobian associated to the chiral rotation,

$$
\begin{aligned}
\psi \to \exp\left(\gamma_5\phi + i\eta\right)\chi\,, \qquad & \bar{\psi} \to \bar{\chi}\exp\left(\gamma_5\phi - i\eta\right) \\
L_{Dirac} = \bar{\psi}(i\partial\!\!\!/ + eA\!\!\!/)\psi \to \bar{\chi}i\partial\!\!\!/\chi\,, \qquad & \mathcal{D}\bar{\psi}\mathcal{D}\psi \to J[A]\mathcal{D}\bar{\chi}\mathcal{D}\chi
\end{aligned}
\tag{18.3}
$$

Now, if ϕ and η take values in the Lie algebra of a non-Abelian gauge group, $i\partial\!\!\!/$ does not act anymore on exponentials in a trivial way and hence the cancellation does not take place. One could think in trying a more complicated fermion change of variables in order to decouple a gauge field of the form (18.1). The opposite view is to abandon the idea of writing A_μ as a sum of a curl and a gradient but mantain the principle of trying to decouple through a chiral rotation. So, let us consider the effect of such a transformation on the Dirac operator,

$$
\begin{aligned}
\psi(x) &= \exp\left(\gamma_5\phi + i\eta\right)\chi(x) \\
\bar{\psi}(x) &= \bar{\chi}\exp\left(\gamma_5\phi - i\eta\right)
\end{aligned}
\tag{18.4}
$$

$$
\begin{aligned}
\bar{\psi}(i\partial\!\!\!/ + eA\!\!\!/)\psi \;=\; & \bar{\chi}\Big(i\partial\!\!\!/ + \exp\left(\gamma_5\phi - i\eta\right)i\partial\!\!\!/\left(\exp\left(\gamma_5\phi + i\eta\right)\right) + \\
& e\exp\left(\gamma_5\phi - i\eta\right)A\!\!\!/\exp\left(\gamma_5\phi + i\eta\right)\Big)\chi
\end{aligned}
\tag{18.5}
$$

It is then evident that if the gauge field A_μ could be written in terms of two Algebra valued fields ϕ and η so that

$$
eA\!\!\!/ = -\Big(i\partial\!\!\!/\exp\left(\gamma_5\phi + i\eta\right)\Big)\exp\left(-\gamma_5\phi - i\eta\right) \tag{18.6}
$$

then, the two last terms in the r.h.s. of (18.5) cancel and one should effectively have the decoupling,

$$
\bar{\psi}(i\partial\!\!\!/ + eA\!\!\!/)\psi = \bar{\chi}i\partial\!\!\!/\chi \tag{18.7}
$$

One can trivially see that (18.6) trivially reduces to eq.(17.2) for the $U(1)$ case but does not leads to (18.1) in the non-Abelian case.

Of course, the point here is to determine whether an arbitrary gauge field can be always written in the form (18.6). The answer, which is affirmative, has been given in [46]-[47] and it implies working with the complexified version of the gauge group. In anycase, we shall accept this and then proceed to the decoupling change of variables (18.4) so that, as it happened for the Abelian case, one ends with

$$\det(i\partial\!\!\!/ + eA\!\!\!/) = \det(i\partial\!\!\!/)J[\phi,\eta] \tag{18.8}$$

with $J[\phi,\eta]$ the Jacobian associated with (18.4).

As in the Abelian case, in order to compute Jacobia $J[\phi,\eta]$ we consider the t-dependent transformation

$$U_t = \exp(t(\gamma_5\phi + i\eta)) \tag{18.9}$$

which leads as before to the relation

$$\det(i\partial\!\!\!/ + eA\!\!\!/) = J[\phi,\eta;t]\det D_t \tag{18.10}$$

where

$$D_t = U_t^{-1}\det(i\partial\!\!\!/ + eA\!\!\!/)U_t \tag{18.11}$$

or, differentiating with respect to t,

$$0 = \frac{d}{dt}\log J[\phi,\eta;t] + \frac{d}{dt}\log\det D_t \tag{18.12}$$

which, after integration yields to

$$\det(i\partial\!\!\!/ + eA\!\!\!/) = \det i\partial\!\!\!/ \exp\left(-2\int_0^1 dt\mathbf{A}_2(t)\right) \tag{18.13}$$

with $\mathbf{A}_2(t)$ the <u>non-Abelian</u> chiral anomaly in two-dimensions, which again can be computed from the expression

$$\mathbf{A}_2(t) = \mathrm{Tr}\ (\gamma_5\phi)|_{reg} \tag{18.14}$$

As before, we introduce the gauge invariant, heat-kernel like regularization and write

$$\mathbf{A}_2(t) = \lim_{M\to\infty}\mathrm{Tr}\left(\gamma_5\exp\left(\frac{D\!\!\!\!/_t\,D\!\!\!\!/_t}{M^2}\right) * \phi\right) \tag{18.15}$$

so that finally one has

$$\mathbf{A}_2(t) = \frac{e}{2\pi}\mathrm{Tr}\int d^2x\varepsilon^{\mu\nu}\left(\partial_\mu A_\nu^t - ieA_\mu^t A_\nu^t\right)\phi = \frac{e}{4\pi}\mathrm{Tr}\int d^2x\varepsilon^{\mu\nu}F_{\mu\nu}^t\phi \tag{18.16}$$

where we have introduced

$$\gamma_\mu A^t_\mu = -\frac{1}{e}\left(i\partial\!\!\!/ U_t\right)U_t^{-1} \tag{18.17}$$

and analogously for $F^t_{\mu\nu}$. Note that

$$A^{t=1}_\mu = A_\mu \ , \quad A^{t=0}_\mu = 0 \tag{18.18}$$

so that

$$D_{t=1} = i\partial\!\!\!/ + eA\!\!\!/ \ , \quad D_{t=0} = i\partial\!\!\!/ \tag{18.19}$$

In summary, we can write for the non-Abelian fermion determinant in the form

$$\det(i\partial\!\!\!/ + eA\!\!\!/) = \exp\left(-\frac{e}{2\pi}\mathrm{Tr}\int d^2x\int_0^1 dt\ \varepsilon^{\mu\nu}F^t_{\mu\nu}\phi\right)\det i\partial\!\!\!/ \tag{18.20}$$

This result is formally the same we got in the Abelian case. Now, in that case, it was easy to express ϕ in terms of A_μ through the relation $A_\mu = \varepsilon_{\mu\nu}\partial_\mu\phi + \partial_\mu\eta$. Moreover, since $F^t_{\mu\nu} = (1-t)F_{\mu\nu}$, the integral over t could be easily done and the simple Schwinger result was attained. Life is more complicated in the non-Abelian case. First, relation (18.6) is very non-linear so that ϕ cannot be easily written in terms of A_μ. The same happens when trying to integrate out over t. Now, things become simpler if we use the relation

$$\gamma^\mu\gamma_5 = -i\varepsilon^{\mu\nu}\gamma_\nu \tag{18.21}$$

to rewrite (18.20) in the form

$$\det(i\partial\!\!\!/ + eA\!\!\!/) = \exp\left(\frac{e}{2\pi}\mathrm{Tr}\int d^2x\int_0^1 dt\gamma_5\phi\left(i\partial\!\!\!/ A\!\!\!/^t + eA\!\!\!/^tA\!\!\!/^t\right)\right)\det i\partial\!\!\!/ \tag{18.22}$$

Then, one can exploit the identity

$$\begin{aligned}\mathrm{Tr}\int d^2x\frac{1}{2}\frac{d}{dt}A\!\!\!/^tA\!\!\!/^t &= \frac{1}{e}\mathrm{tr}\int d^2x\gamma_5 i\partial\!\!\!/\phi A\!\!\!/^t + 2\mathrm{Tr}\int d^2x\gamma_5A\!\!\!/^t\phi A\!\!\!/^t \\ &+ \frac{1}{e}\mathrm{Tr}\int d^2x(D\!\!\!\!/\,\eta) * A\!\!\!/ \end{aligned} \tag{18.23}$$

and find for (18.22)

$$\log\det(i\partial\!\!\!/ + eA\!\!\!/) = -\frac{e^2}{4\pi}\mathrm{Tr}\int d^2x\, A\!\!\!/ A\!\!\!/ + \frac{e^2}{2\pi}\mathrm{tr}\int dt \int d^2x\; \gamma_5 \phi\!\!\!/ A\!\!\!/^t A\!\!\!/^t$$
$$+ \frac{e}{2\pi}\mathrm{Tr}\int dt\int d^2x\; (D\!\!\!\!/_t\eta) A\!\!\!/^t + \log\det i\partial\!\!\!/ \qquad (18.24)$$

This is the final form for the fermion determinant for the non-Abelian gauge theory. Note that only in the first term the integral over t has been solved. This term is precisely the one that remains in the Abelian case (working in the $\eta = 0$ Lorentz gauge). It is not possible to compute the remaining t-integrals to get such a simple answer in the non-Abelian case. One can however write it in a more suggestive way. Let us consider to this end the light-cone gauge $A_+ = 0$. This implies a certain relation among ϕ and eta , which finally leads to

$$A_- = (i/e)g(x)\partial_- g^{-1}(x)\ , \quad A_- = 0$$

with

$$g(x) = \exp(2\phi(x))\ , \quad g(x,t) = \exp(2\phi(x)t)$$

Then, one can see after some algebra that [47]

$$\log\left(\frac{\det(i\partial\!\!\!/ + eA\!\!\!/)}{\det i\partial\!\!\!/}\right) = -\frac{1}{8\pi}\mathrm{Tr}\int d^2x\left(\partial_\mu g(x)^{-1}\right)(\partial_\mu g(x))$$
$$+\frac{i}{12\pi}\epsilon_{ijk}\int_B d^3y g(x,t)^{-1}(\partial_i g(x,t))\, g(x,t)^{-1}(\partial_j g(x,t))\, g^{-1}(\partial_k g(x,t)) \qquad (18.25)$$

Here we have written $d^3y = d^2xdt$ so that the integral in the second line of eq.(18.25) runs over the three dimensional manifold B, which in compactified Euclidean space can be identified with a ball with boundary S^2. Index i runs from 1 to 3. Because the determinant was computed in Euclidean space, elements g should be considered as belonging to the complexified gauge group [46]. The r.h.s. in (18.25) is known as the Wess-Zumino-Witten $W[g]$ action for a group valued field g,

$$W[g] = -\frac{1}{8\pi}\mathrm{Tr}\int d^2x\left(\partial_\mu g(x)^{-1}\right)(\partial_\mu g(x))$$
$$+\frac{i}{12\pi}\epsilon_{ijk}\int_B d^3y g(x,t)^{-1}(\partial_i g(x,t))\, g(x,t)^{-1}(\partial_j g(x,t))\, g^{-1}(\partial_k g(x,t)) \qquad (18.26)$$

A very important relation satisfied by W, first derived by Polyakov and Wiegman [45] in their calculation of the fermion determinant is the so called Polyakov-Wiegman identity,

$$W[gh^{-1}] = W[g] + W[h^{-1}] + \frac{1}{4\pi}\mathrm{Tr}\int d^2x(\partial_+ gg^{-1})(\partial_h h^{-1}) \qquad (18.27)$$

So, we have found that a Wess-Zumino term arises when computing the two-dimensional non-commutative fermion determinant in a non-Abelian gauge field background. This kind of terms, related with chiral anomalies, were introduced by Wess and Zumino in studies of gauge invariance of gauge theories and anomalies [49]. It was Polyakov and Wiegman [45] and Witten [48] who first recognized its appearence in connection with the calculation of fermion determinants in 2 dimensions.

From this result, a bosonization connection between free fermions in the fundamental reprewsentation of some non-Abelian gauge group G and the bosonic Wess-Zumino-Witten model for elements $g \in G$ can be established. This last model has many important features which make it of great interest both in String Theory and in Statistical Mechanics. In particular, it is conformal invariant and, associated to this, one can construct both a Kac-Moody and a Virasoro algebra which allow to study many properties of the model in a closed form.

Two-dimensional bosonization: the Abelian case

Bosonization is a mapping of a quantum field theory of interacting fermions onto an equivalent theory of interacting bosons. This mapping is well established in the context of $1 + 1$ dimensional theories. Less is known in higher dimensions.

Since the 70's, bosonization in two dimensional space-time constitutes one of the main tools available for the study of the non-perturbative behavior of both condensed matter systems [50] and of quantum field theories [51]. The bosonization identities, which relate the fermionic current with the topological current of a bosonic theory, can be viewed as a consequence of a non-trivial current algebra.

The extension of the bosonization technique to the case in which the fermions belong to some representation of a non-Abelian group is called v*non-Abelian bosonization*. In two dimensional space time it was first formulated by Witten [48] by comparing the current algebra for free fermions and for a bosonic sigma model with a Wess-Zumino term. Afterwards, different approaches rederived and discussed the bosonization recipe [52]-[57] but in general they were not constructive in the sense that the bosonic theory was not obtained from the fermionic one by following a series of steps that could be generalized to other cases, in particular to a possible higher-dimensional bosonization.

In contrast, more recent developments of bosonization using the path-integral approach have shown to give s systematic procedure to establish the mapping in a rather general framework where one can extend the 2 dimensional results to higher dimensions. We shall describe in what follows how one reproduces, in this approach, the non-Abelian bosonization recipe. A more thoroughful discussion can be found in [58].

We start from the (Euclidean) Lagrangian for free massless Dirac fermions in 2 dimensions

$$L = \bar{\psi}(i\partial\!\!\!/)\psi \tag{18.28}$$

Here fermions are in the fundamental representation of some group G. The corresponding generating functional reads

$$Z_{fer}[s] = \int \mathcal{D}\bar{\psi}\mathcal{D}\psi \exp\left(-\int d^2x\bar{\psi}(i\partial\!\!\!/ + s\!\!\!/)\psi\right) \tag{18.29}$$

with $s_\mu = s_\mu^a t^a$ an external source taking values in the Lie algebra of G. Differentiation with respect to s_μ yields to v.e.v.'s of products of fermion currents. For example,

$$\frac{1}{Z}\frac{\delta Z}{\delta s_\mu^a}\bigg|_{s=0} = \langle\bar{\psi}^i\gamma_\mu t_{ij}^a\psi^j\rangle = \langle j_\mu^a\rangle \tag{18.30}$$

The path-integral derivation of bosonization rules both in 2 and higher dimensions heavily relies on the invariance of the measure under local transformations of the fermion variables, $\psi \to h(x)\psi$, $\bar{\psi} \to$

$\bar{\psi}h(x)^{-1}$ with $h \in G$. As a consequence the generating functional (**??**) is automatically invariant under local transformations of the source

$$s_\mu \to s_\mu^h = h^{-1} s_\mu h + i h^{-1} \partial_\mu h \tag{18.31}$$

$$Z[s^h] = Z[s] \tag{18.32}$$

In view of this, if we perform the change of variables

$$\begin{aligned} \psi &= g(x)\psi' \\ \bar{\psi} &= \bar{\psi}' g^{-1}(x). \end{aligned} \tag{18.33}$$

$Z_{fer}[s]$ becomes

$$Z_{fer}[s] = \int \mathcal{D}\bar{\psi}\mathcal{D}\psi\mathcal{D}g \exp[-\int d^2x \bar{\psi}(i\partial\!\!\!/ + s\!\!\!/^g)\psi] \tag{18.34}$$

where an integration over g has been included since it just amounts to a change of normalization. Integrating out fermions we have

$$Z_{fer}[s] = \int \mathcal{D}g \;\det(i\partial\!\!\!/ + s\!\!\!/^g) \tag{18.35}$$

Before discussing the non-Abelian case, let us understand how the procedure works in the simpler Abelian case. We can pass from the integration over g to the integration over a vector field b_μ defined as

$$b_\mu = s_\mu^g = s_\mu + \partial_\mu \eta \;, \quad g = \exp(i\eta) \tag{18.36}$$

if we ensure that

$$\varepsilon_{\mu\nu} f_{\mu\nu}[b] = \varepsilon_{\mu\nu} f_{\mu\nu}[s] \tag{18.37}$$

Then, we write

$$Z_{fer}[s] = \int \mathcal{D}b_\mu \;\det(i\partial\!\!\!/ + b\!\!\!/)\delta[\varepsilon_{\mu\nu} f_{\mu\nu}[b] - \varepsilon_{\mu\nu} f_{\mu\nu}[s]] \tag{18.38}$$

Now, we use expression (17.27) for the fermion determinant and introduce a Lagrange multiplier a for implementing the delta-function condition,

$$\begin{aligned} Z_{fer}[s] \;\; &= \;\; \int \mathcal{D}b_\mu \mathcal{D}a \exp\left(-\frac{1}{2\pi}\int d^2x b_\nu \left(\delta_{\nu\mu} + \partial_\nu \Box^{-1}\partial_\mu\right) b_\mu\right) \\ & \exp\left(C\int d^2x a \left(\varepsilon_{\mu\nu} f_{\mu\nu}[b] - \varepsilon_{\mu\nu} f_{\mu\nu}[s]\right)\right) \end{aligned} \tag{18.39}$$

Since the integrand is gauge-invariant (under gauge transformations of the auxiliary field b), we can work in a fixed gauge. It will be convenient to use the Landau gauge in which $b_\mu = \varepsilon_{\mu\nu}\partial_\nu\phi$. One has

$$\begin{aligned} Z_{fer}[s] &= \mathcal{N}\int \mathcal{D}\phi\mathcal{D}a \exp\left(-\frac{1}{2\pi}\int d^2x\phi\Box\phi\right) \\ &\quad \exp\left(C\int d^2xa\left(\Box\phi - \varepsilon_{\mu\nu}f_{\mu\nu}[s]\right)\right) \end{aligned} \tag{18.40}$$

The integral over ϕ can be evaluated in closed form. The answer is

$$Z_{fer}[s] = \int \mathcal{D}a \exp\left(-2C\int d^2xa\varepsilon_{\mu\nu}\partial_\mu s_\nu\right)\exp\left(\frac{C^2\pi}{2}\int a\Box\Box^{-1}\Box a\right) \tag{18.41}$$

This formula can be rewritten as

$$Z_{fer}[s] = \int \mathcal{D}a \exp\left((-\frac{1}{2}\int d^2x\partial_\mu a\partial_\mu a\right)\exp\left(\frac{2}{\sqrt{\pi}}\int s_\mu\varepsilon_{\mu\nu}\partial_\nu a\right) \tag{18.42}$$

Differentiating with respect to s_μ we get

$$\langle\bar{\psi}\gamma_\mu\psi\rangle\Big|_F = \frac{2}{\sqrt{\pi}}\langle\varepsilon_{\mu\nu}\partial_\nu a\rangle\Big|_B \tag{18.43}$$

where $|_F$ ($|_B$) means that the v.e.v. is to be computed with a free fermion Lagrangian (free boson Lagrangian). This is usually written as the bosonization rule

$$\bar{\psi}\gamma_\mu\psi \to \frac{2}{\sqrt{\pi}}\varepsilon_{\mu\nu}\partial_\nu a \tag{18.44}$$

Clase 19

Non-Abelian Bosonization

Bosonization for fermions in some representation of a non-Abelia group G follows the same steps as those described in the $U(1)$ case. There are simply some complications due to the non-Abelian character of the group. In particular, it should be necessary to compute a Jacobian related to the change of variables one performs when passing from the A_μ integration to the integration over the basic variables in terms of which we write the gauge field.

Indeed, remember that in the Abelian case, we wrote a general gauge field in the form

$$eA_\mu = \varepsilon_{\mu\nu}\partial_\mu\phi + \partial_\mu\eta \tag{19.1}$$

and we ended with an integral over ϕ and η. Now, one can easily convince oneself that in the Abelian case the Jacobian $J[\phi,\eta]$,

$$\mathcal{D}A_\mu = J[\phi,\eta]\mathcal{D}\phi\mathcal{D}\eta \tag{19.2}$$

is trivial. One just has

$$\begin{aligned} eA_0 &= \partial_1\phi + \partial_0\eta \\ eA_1 &= -\partial_0\phi + \partial_1\eta \end{aligned} \tag{19.3}$$

so that $J[\phi,\eta]$ is simply

$$J[\phi,\eta] = \left|\det\begin{pmatrix} \partial_1 & \partial_0 \\ -\partial_0 & \partial_1 \end{pmatrix}\right| = \left|\det\nabla^2\right| \tag{19.4}$$

How does this change in the non-Abelian case? It is convenient to work in light-cone components and write

$$\begin{aligned} A_+ &= -ig^{-1}\partial_+ g \\ A_- &= -ih\partial_- h^{-1} \end{aligned} \tag{19.5}$$

Then, instead of (19.2) we shall have

$$\mathcal{D}A_+\mathcal{D}A_- = J[g,h]\mathcal{D}g\mathcal{D}h \tag{19.6}$$

Let us first consider for simplicity the case in which we fix the gauge to $A_- = 0$ which is equivalent to put $h = I$ The gauge fixing reduces the Jacobian we are looking for to

$$\mathcal{D}A_+ = J[g]\mathcal{D}g \tag{19.7}$$

In order to compute $J[g]$ let us note that, under an infinitesimal gauge transformation with parameter $\delta\alpha$, A_+ changes as

$$\delta A_+ = \partial_+\delta\alpha + i[A_+,\delta\alpha] \tag{19.8}$$

or

$$\delta A_+ = D_+^{adj}[A]\delta\alpha \tag{19.9}$$

Taking into account that the Haar measure $\mathcal{D}g$ is in fact a measure over some Lie algebra valued element $\alpha(x)$ used to write $g = \exp(i\alpha)$, we see, from (19.7) and (19.9) that

$$J[g] = \left|\det D_+^{adj}[A]\right| \tag{19.10}$$

Again, we have to compute the Jacobian of the Dirac operator, except that now we need it for the adjoint representation.

Now, we have learnt that determinants can be computing "by integration" of the anomaly. That is, in the fundamental, one has

$$\det D_+^{fund} = \mathcal{N}\exp\left(-\int_0^1 A_2^{fund}dt\right) \tag{19.11}$$

where

$$A_2^{fund} = \frac{e}{4\pi}\mathrm{Tr}\left(t^a t^b\right)\int d^2x\varepsilon^{\mu\nu}F_{\mu\nu}^{at}\alpha^b \tag{19.12}$$

is the anomaly arising for a t-dependent chiral transformation, which in this case we wrote as $\exp(-\gamma_5\phi t)$. The t^a's are the generators in the fundamental. We have obtained this result after regularization using $\exp(-D^{fund\,2}/M^2)$ as regulator. Any other gauge-invariant regularization yields the same answer.

Now, in the case of the adjoint, one will similarly have

$$\det D_+^{adj} = \mathcal{N}\exp\left(-\int_0^1 A_2^{adj}\right) \tag{19.13}$$

with

$$A_2^{adj} = \frac{e}{4\pi}\mathrm{Tr}\left(T^aT^b\right)\int d^2x\varepsilon^{\mu\nu}F_{\mu\nu}^{at}\alpha^b \tag{19.14}$$

arising from use of $\exp(-D^{adj\,2}/M^2)$ as regulator (We call T^a the generators in the adjoint). Then, one easily sees that

$$\det D_+^{adj} = \left(\det D_+^{fund}\right)^\kappa \tag{19.15}$$

with

$$\kappa = \frac{\mathrm{Tr}\left(T^aT^b\right)}{\mathrm{Tr}\left(t^at^b\right)} \tag{19.16}$$

the exponent κ is related to what is call in group theory the dual Coxeter number $\tilde{h}(G)$. In fact, κ coincides with $\tilde{h}(G)$ when the t^a's correspond to the group generators in the *defining* representation [59], which will be the case in what follows. By explicit calculation, one can show that κ takes the value

$$\kappa = 2C[G] \tag{19.17}$$

with $C[G]$ the quadratic Casimir in the adjoint,

$$f^{apq}f^{bpq} = C[G]\delta^{ab} \tag{19.18}$$

Since the determinant in the fundamental leads to the exponential of a Wess-Zumino-Witten action $W[g]$, we then see that in the adjoint one has

$$J[g] = \left|\det D_+^{adj}[A_+]\right| = \exp\left(2C[G]W[g]\right) \tag{19.19}$$

The same can be done by choosing as a gauge-fixing $A_+ = 0$. Using (19.5) with $g = 1$ one gets

$$J[h^{-1}] = \left|\det D_-^{adj}[A_-]\right| = \exp\left(2C[G]W[h^{-1}]\right) \tag{19.20}$$

One should be careful if one needs the complete Jacobian. It is not just the product of both $J[g]$ and $J[h^{-1}]$ since this would lead to a gauge non invariant answer. One should add a term mixing g and h so that the answer is gauge invariant. Polyakov-Wiegmann identity help us in determining the term to add. Indeed, if one writes

$$J[g,h] \;=\; \exp\left(2C[G]\left(W[g] + W[h^{-1}] + \frac{1}{4\pi}\mathrm{tr}\int d^2x g^{-1}\partial_+ g h\partial_- h^{-1}]\right)\right) \tag{19.21}$$

one gets, after use of Polyakov-Wiegmann identity,

$$J[g,h] = \exp\left(2C[G]W[gh^{-1}]\right) \tag{19.22}$$

which is a gauge invariant answer. Now, why were we allowed to add the third term in the l.h.s. of (19.21)? As we explained before, the only divergencies exhibited by Dirac operator determinants in two dimensions arise from diagrams with two A_μ insertions. Consequently, subtractions should be proportional to A_+A_- and different regularizations will correspond to changing the factor of such quadratic term. That employed in (19.21) is justified since it leads to a gauge-invariant answer.

We are now ready to attack non-Abelian bosonization. As we wrote in (18.29), the generating functional for fermions in the fundamental reads

$$Z_{fer}[s] = \int \mathcal{D}\bar\psi\mathcal{D}\psi \exp\left(-\int d^2x\bar\psi(i\partial\!\!\!/ + s\!\!\!/)\psi\right) \tag{19.23}$$

where $s_\mu = s^a_\mu t^a$ is an external source taking values in the Lie algebra of $G = SU(N)$. Then, we perform the change of variables

$$\begin{aligned} \psi &= g(x)\psi' \\ \bar\psi &= \bar\psi' g^{-1}(x). \end{aligned} \tag{19.24}$$

so that $Z_{fer}[s]$ becomes

$$Z_{fer}[s] = \int \mathcal{D}\bar\psi\mathcal{D}\psi\mathcal{D}g \exp[-\int d^2x\bar\psi(i\partial\!\!\!/ + s\!\!\!/^g)\psi] = \int \mathcal{D}g \det(i\partial\!\!\!/ + s\!\!\!/^g) \tag{19.25}$$

Then, posing again

$$s_\mu^g = b_\mu \tag{19.26}$$

and using

$$f_{\mu\nu}[b] = g^{-1} f_{\mu\nu}[s] g \tag{19.27}$$

we can trade the g integration for an integration over connections b satisfying condition (19.27). To this end we shall use the $d = 2$ identity (proven in an Appendix to this chapter)

$$\int \mathcal{D}b_\mu \mathcal{H}[b]\delta[\varepsilon_{\mu\nu}(f_{\mu\nu}[b] - f_{\mu\nu}[s])] = \int \mathcal{D}g\mathcal{H}[s^g] \tag{19.28}$$

Here $\mathcal{H}$ is any gauge invariant function. Identity (19.28) allows us to write eq.(19.25) in the form

$$Z_{fer} = \int \mathcal{D}b_\mu \Delta\delta(b_+ - s_+)\delta\left[\varepsilon_{\mu\nu}(f_{\mu\nu}[b] - f_{\mu\nu}[s])\right]\det(i\partial\!\!\!/ + b\!\!\!/)\,. \tag{19.29}$$

For convenience, we have chosen to fix the gauge using the condition $b_+ = s_+$ being Δ the corresponding Faddeev-Popov determinant.

We now introduce a Lagrange multiplier $\hat{a}$ (taking values in the Lie algebra of G) to enforce the delta function condition

$$\begin{aligned} Z_{fer}[s] \quad &= \quad \int \mathcal{D}\hat{a}\mathcal{D}b_\mu\, \Delta\delta(b_+ - s_+)\det[i\partial\!\!\!/ + b\!\!\!/] \times \\ &\qquad \exp\left(-\frac{C}{8\pi} tr \int d^2x \hat{a}\, \varepsilon_{\mu\nu}(f_{\mu\nu}[b] - f_{\mu\nu}[s])\right) \end{aligned} \tag{19.30}$$

with C a constant to be conveniently adjusted. We now write sources and the b_μ field in terms of group-valued variables,

$$s_+ = i\tilde{s}^{-1}\partial_+\tilde{s} \tag{19.31}$$

$$s_- = is\partial_- s^{-1} \tag{19.32}$$

$$b_+ = i(\tilde{b}\tilde{s})^{-1}\partial_+(\tilde{b}\tilde{s}) \tag{19.33}$$

$$b_- = i(sb)\partial_-(sb)^{-1} \tag{19.34}$$

so that the fermion determinant can be related to the Wess-Zumino-Witten action (18.24),

$$\det[i\partial\!\!\!/ + b\!\!\!/] = \exp(W[\tilde{b}\tilde{s}sb]) \tag{19.35}$$

In writing eq.(19.35) a gauge-invariant regularization is assumed so that the left and right-handed sectors enter in gauge invariant combinations. In this way, gauge transformations of the source s_μ, which as stated before, should leave the generating functional invariant, do not change the determinant. This will be the criterion we shall adopt each time determinants (always needing a regularization) has to be computed. Concerning the Jacobian for passing from the b_μ variable to the $b, \tilde{b}$ one can easily show that

$$\int \Delta\delta(b_+ - s_+)\mathcal{D}b_\mu = \int \exp(\kappa W[\tilde{b}\tilde{s}sb])\delta(\tilde{b} - I)\mathcal{D}\tilde{b}\mathcal{D}b \tag{19.36}$$

so that $Z_{fer}[s]$ becomes

$$\begin{aligned} Z_{fer}[s] \quad = \quad & \int \mathcal{D}\hat{a}\mathcal{D}b \exp\left(i\frac{C}{4\pi}tr\int d^2x(D_+[\tilde{s}]\hat{a})\, sb(\partial_- b^{-1})s^{-1}\right) \times \\ & \exp\left((1+\kappa)W[\tilde{s}sb]\right) \ . \end{aligned} \tag{19.37}$$

A convenient change of variables to pass from integration over the algebra valued Lagrange multiplier $\hat{a}$ to a group valued variable a is the one defined through

$$D_+[\tilde{s}]\hat{a} = \tilde{s}^{-1}(a^{-1}\partial_+ a)\tilde{s} \tag{19.38}$$

Calculation of the corresponding jacobian J_L

$$\mathcal{D}\hat{a} = J_L\mathcal{D}a \tag{19.39}$$

should be carefully done. Indeed, in the present approach to bosonization, it is the group valued variable a who plays the role of the boson field equivalent to the original fermion field. Since for the latter (a free fermi field) there was no local symmetry, the former should not be endowed with this symmetry. With this in mind, we shall maintain a unchanged under local transformations g while transforming $\hat{a}$, $\hat{a} \to g^{-1}\hat{a}g$, so that eq.(19.38) changes covariantly when one simultaneously changes $\tilde{s} \to \tilde{s}g$. One can easily prove that

$$J_L = \exp(\kappa W[a\tilde{s}s] - \kappa W[\tilde{s}s]) \tag{19.40}$$

so that the generating functional reads

$$Z_{fer}[s] \quad = \int \mathcal{D}a\mathcal{D}b \exp((1+\kappa)W[\tilde{s}sb]) \times \exp\left(\kappa(W[a\tilde{s}s] - W[\tilde{s}s])\right) \times \exp(-\tfrac{C}{4\pi} tr \int d^2x \tilde{s}^{-1}(a^{-1}\partial_+ a)\tilde{s}s(b\partial_- b^{-1})s^{-1}) \,. \quad (19.41)$$

If one repeatedly uses Polyakov-Wiegmann identity and chooses the up to now arbitrary constant C as

$$-C = 1 + \kappa \quad (19.42)$$

one can write $Z_{fer}[s]$ in the form

$$Z_{fer}[s] = \int \mathcal{D}a\mathcal{D}b \, \exp(W[\tilde{s}s] - W[a\tilde{s}s]) \exp\left((1+\kappa)W[a\tilde{s}sb]\right) \quad (19.43)$$

Now, the b integration can be trivially factorized this leading to

$$Z_{fer}[s] = \int \mathcal{D}a \, \exp(-W[a\tilde{s}s] + W[\tilde{s}s]) \,. \quad (19.44)$$

or, after the shift $a\tilde{s}s \to \tilde{s}as$

$$Z_{fer}[s_+] \quad = \quad \int \mathcal{D}a \, \exp(-W[a] + \frac{i}{4\pi} tr \int d^2x (s_+ a\partial_- a^{-1} + s_- a^{-1}\partial_+ a)) \times \exp(\frac{1}{4\pi} tr \int d^2x (a^{-1}s_+ a s_- - s_+ s_-)) \quad (19.45)$$

We have then arrived to the identity

$$Z_{fer}[s] = Z_{bos}[s] \quad (19.46)$$

where $Z_{bos}[s]$ is the generating function for a Wess-Zumino-Witten model. Differentiation with respect to any one of the two sources gives correlation functions in a given chirality sector. The answer corresponds to Witten's bosonization recipe [48]

$$\bar{\psi} t^a \gamma_+ \psi \to \frac{i}{4\pi} a^{-1}\partial_+ a \quad (19.47)$$

$$\bar{\psi} t^a \gamma_- \psi \to \frac{i}{4\pi} a\partial_- a^{-1} \,, \quad (19.48)$$

the l.h.s. to be computed in a free fermionic model, the r.h.s. in a Wess-Zumino-Witten model.

Anomalous gauge theories

We have seen that gauge symmetry can be preserved by the regularization prescription that renders fermion determinants finite, at the cost of having a chiral (global) anomaly. Now, when fermions are Weyl fermions (i.e., fermions with a definite chirality) there is a problem: the gauge current and the chiral symmetry coincide so that either the theory is not anomalous or there is an anomaly that affects both the gauge and the chiral symmetry. In this last case one refers to *anomalous gauge theories*.

Consider for definiteness left-handed Weyl fermions coupled to a gauge field. The Lagrangian can be written in the form

$$L = \frac{1}{4}\mathrm{tr}\,(F_{\mu\nu}F^{\mu\nu}) + \bar{\psi}D_L[A]\psi \tag{19.49}$$

where

$$D_L[A] = (i\not{\partial} + e\not{A})\frac{1-\gamma_5}{2} \tag{19.50}$$

Here, the $(1-\gamma_5)/2$ projector ensures that fermions are indeed left-handed. Then, fermions can be taken as Dirac fermions since the projector kills the right-handed components both of ψ and $\bar{\psi}$. This Lagrangian is of course gauge invariant: a phase transformation of the fermions leaves it unchanged. However, being the fermion left-handed, this phase rotation cannot be distinguished from a chiral rotation (with the adequate phase). So, the Lagrangian is both gauge and chiral invariant (locally), provided the gauge field is adequately changed. This means that, at the quantum level, if there is a chiral anomaly, there will be also a gauge anomaly! Either one finds a regularization prescription respecting invariance under phase rotation of fermions or the theory is, necessarily, anomalous.

Now, even before finding an adequate regularization scheme, one faces the problem of how to define the fermionic partition function $Z[A]$

$$Z[A] = \int \mathcal{D}\bar{\psi}\mathcal{D}\psi \exp\left(-\int d^4x\bar{\psi}D_L[A]\psi\right) \tag{19.51}$$

Indeed, for Dirac fermions, we have seen that $Z[A]$ can be taken as the determinant of the Dirac operator, i.e., as the product of its eigenvalues.

Now, when a projector is present as in (19.50), the Dirac operator does not define an eigenvalue problem. Given a left-handed Weyl fermion

$$\psi_L = \frac{1-\gamma_5}{2}\psi = \begin{pmatrix} 0 \\ \psi_- \end{pmatrix} \tag{19.52}$$

the Dirac operator $\not{D} = i\not{\partial} + e\not{A}$ which is always of the form

$$\not{D} = \begin{pmatrix} 0 & D_+ \\ D_- & 0 \end{pmatrix} \tag{19.53}$$

maps negative chirality fermions into positive chirality ones. Then, if we consider

$$D_L[A]\begin{pmatrix} 0 \\ \psi_- \end{pmatrix} = \not{D}[A]\frac{1-\gamma_5}{2}\psi = \begin{pmatrix} 0 & D_+ \\ D_- & 0 \end{pmatrix}\frac{1-\gamma_5}{2}\psi = \begin{pmatrix} D_+\psi_- \\ 0 \end{pmatrix} \tag{19.54}$$

Then, there is no eigenvalue problem associated to the projected operator. A possible way out from this conundrum is the following. When we defined $Z[A]$, we have written the fermion measure without specifying whether it was a measure over Dirac fermions or over left-handed fermions. In fact, the distinction was irrelevant: even if right-handed fermions were included, they were completely decoupled from the gauge field and can be then factored out from $Z[A]$. That is, if we write

$$\mathcal{D}\bar{\psi}\mathcal{D}\psi = \mathcal{D}\bar{\psi}_R\mathcal{D}\psi_R\mathcal{D}\bar{\psi}_L\mathcal{D}\psi_L \tag{19.55}$$

then $Z[A]$ is defined as

$$Z[A] = \int \mathcal{D}\bar{\psi}_R\mathcal{D}\psi_R \int \mathcal{D}\bar{\psi}_L\mathcal{D}\psi_L \exp\left(-\int d^4x\bar{\psi}_L D_L[A]\psi_L\right) \tag{19.56}$$

Now, if we include in this formula a free Lagrangian for the decoupled right-handed measure, the only effect will be a factor, irrelevant for computing physical quantities,

$$\begin{aligned} Z[A] \quad = \quad & \int \mathcal{D}\bar{\psi}_R\mathcal{D}\psi_R \exp\left(-\int d^4x\bar{\psi}_R i\not{\partial}\frac{1+\gamma_5}{2}\psi_L\right) \times \\ & \int \mathcal{D}\bar{\psi}_L\mathcal{D}\psi_L \exp\left(-\int d^4x\bar{\psi}_L D_L[A]\psi_L\right) \end{aligned} \tag{19.57}$$

or

$$Z[A] = \int \mathcal{D}\bar{\psi}\mathcal{D}\psi \exp\left(-\int d^4x\bar{\psi}D[A]\psi\right) \tag{19.58}$$

where

$$D[A] = i\partial\!\!\!/ + eA\!\!\!/\frac{1-\gamma_5}{2} \tag{19.59}$$

Now, $D[A]$ does have a well-defined eigenvalue problem

$$D[A]\psi_n = \lambda_n\psi_n \tag{19.60}$$

since its eigenfunctions have no more a given chirality. Hence we can write

$$Z[A] = \det D[A]|_{reg} = \prod \lambda_n\Big|_{reg} \tag{19.61}$$

But now, as a result of the change $D\!\!\!\!/[A](1-\gamma_5)/2 \to D[A]$, designed to solve the problem related to chirality flip, we have created another problem: the modified Dirac operator $D[A]$ is not a covariant derivative.

$$D[A^g] = i\partial\!\!\!/ + eA\!\!\!/^g\frac{1-\gamma_5}{2} = g^{-1}\left(i\partial\!\!\!/ + eA\!\!\!/\right)g - g^{-1}i\partial\!\!\!/\frac{1+\gamma_5}{2}g \neq g^{-1}D[A]g \tag{19.62}$$

Then, eigenvalues and, a fortioti, the determinant will be in general gauge invariant. The theory will be in principle anomalous. Indeed, under a gauge transformation we have seen that the associated Jacobian takes the form

$$J[g,A] = \frac{\det D[A]}{\det D[A^g]} \tag{19.63}$$

Only when $D[A^g] = g^{-1}D[A]g$ the resulting Jacobian will be trivial.

The existence of this Jacobian, associated to gauge (or local chiral) transformations means that the gauge current is not (covariantly) conserved at the quantum level,

$$\begin{aligned}
\left\langle D_\mu{}^{ab} j^{b\mu}\right\rangle &= \mathcal{A}^a(x) \neq 0 \\
\mathcal{A}^a(x) &= -\left.\frac{\delta J[g,A]}{\delta\alpha(x)}\right|_{\alpha=0} \\
g(x) &= \exp\left(iT^a\alpha^a(x)\right)
\end{aligned} \tag{19.64}$$

The lose of gauge invariance at the quantum level poses a serious problem. For example, renormalizability of gauge theories depends heavily on the fact that certain relations between (infinite) renormalization constants hold. Now, these relations are guaranteed by gauge invariance and if this principle is lost, renormalizability will be also lost.

One way out is to choose the gauge group so that the Jacobian is trivial. To see when this can happen, just note that, extending what we did for $U(1)$ chiral transformations, when a non-Abelian gauge transformation with generators T^a is applied to Weyl fermions, the anomaly takes the form (see problem 1) $\mathcal{A}^a \sim \mathrm{tr}\left(T^a T^b T^c\right) F^b_{\mu\nu}\tilde{F}^c_{\mu\nu}$. There are relevant Lie groups for which the trace is zero,

$SU(2)$ in its three dimensional irreducible representation.
$SO(N)\,, \quad N \geq 5\,, \quad N \neq 6$
$G_2\,, F_4\,, E_7\,, E_8$

In the standard model, based on the group $SU(3) \times SU(2) \times U(1)$ one should rely on the cancellation of the contributions to $\mathcal{A}^a$ coming from quarks and leptons. One can see that all anomalies cancel and this can be understood from the fact that $SU(3) \times SU(2) \times U(1)$ can be embedded in $SO(10)$ which is anomaly free. This cancellation was first discussed in [63].

What happens if there is no such cancellation? Can one implement some kind of consistent quantization of gauge theories? There has been a proposal by Faddeev and Shatashvily on this direction [64]. The proposal was taken seriously in [65]-[66] and will be developped now.

Consider the *complete* partition function for a (left-handed) chiral gauge theory, where extra right-handed decoupled (free) fermions have been added so as to have a well-behaved eigenvalue problem allowing to define a fermion determinant,

$$Z[A] = \int \mathcal{D}A\mathcal{D}\bar{\psi}\mathcal{D}\psi \exp\left(-\int d^4x \left(\frac{1}{4}\mathrm{tr}\left(F_{\mu\nu}F^{\mu\nu}\right) + \bar{\psi}D[A]\psi\right)\right) \tag{19.65}$$

If the fermion theory is anomalous, the normal procedure of fixing the gauge is in some sense redundant. However, we can think in writing the integration measure over gauge fields in a way inspired in the Faddeev-Popov method,

$$\mathcal{D}A = \mathcal{D}\tilde{A}\mathcal{D}g\Delta_{FP}[A] \tag{19.66}$$

Here $\tilde{A}$ is the set of all gauge fields satisfying some gauge condition F

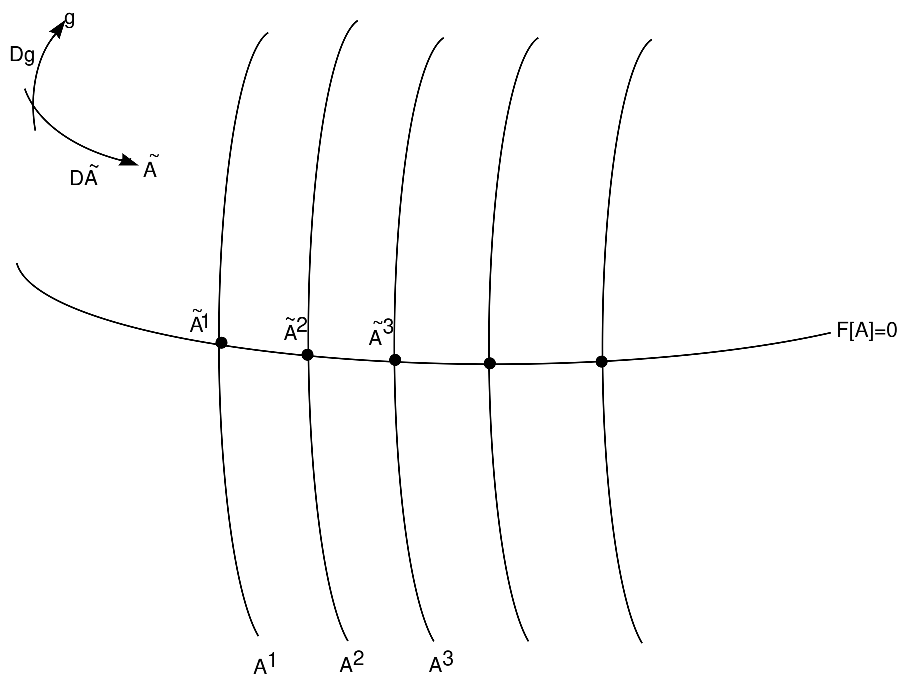
and Δ_{FP} the corresponding Faddeev-Popov Jacobian. Concerning $\mathcal{D}g$, it is the Haar measure over the gauge group elements $g(x)$. In formulae,

$$\mathcal{D}\tilde{A} = \mathcal{D}A\delta(F[A^g])Dg\Delta_{FP}[A]\,, \qquad \Delta_{FP}[A] = D_\mu \frac{\delta F[A]}{\delta A_\mu} \tag{19.67}$$

where

$$F[A] = 0 \tag{19.68}$$

is supposed to have just one solution for each orbit (no Gribov problem). Writing the measure in this form means a choice of coordinates in the gauge field space corresponding to some point in each orbit times the measure over the orbit. The Faddev-Popov determinant can then be interpreted as the natural metric in these coordinates (the equivalent to $\sqrt{g}$ (g the determinant of the space time metric) in the space-time measure.



Were the theory non-anomalous we know how to factor out the integration over the gauge group making infinites produced by gauge invariance harmless. Indeed, one changes variables in the form $A \to A' = A^{g^{-1}}$, $\psi \to \psi' = g\psi$, $\bar\psi \to \bar\psi' = \bar\psi g$ and uses that the action, the Faddeev-Popov determinant and the measure are gauge invariant and any dependence on g would disappear from (19.65). However, we know that for anomalous theories the fermion measure is not invariant so that a Fujikawa Jacobian should be included when this change is performed,

$$\begin{aligned} Z[A] =& \int \mathcal{D}A\delta(F[A^g])Dg\Delta_{FP}[A]\mathcal{D}\bar\psi\mathcal{D}\psi \exp\left(-S[A,\bar\psi,\psi]\right) \\ =& \int \mathcal{D}A'\delta(F[A'])Dg\Delta_{FP}[A']J[g^{-1},A']\mathcal{D}\bar\psi'\mathcal{D}\psi' \exp\left(-S[A',\bar\psi',\psi']\right) \end{aligned} \tag{19.69}$$

The integration over gauge degrees of freedom cannot be factored out anymore. On the contrary, a coupling between g and A_μ is provided by the Fujikawa Jacobian so that gauge degrees of freedom acquire the status of physical fields! Have we gained anything? To answer this question, let us write (19.69) in the form

$$Z = \int \mathcal{D}A\delta(F[A])\Delta_{FP}[A]\exp\left(-S_{eff}[A]\right) \tag{19.70}$$

where S_{eff} is defined as

$$\exp\left(-S_{eff}[A]\right) = \int \mathcal{D}g\mathcal{D}\bar\psi\mathcal{D}\psi J[g^{-1},A]\exp\left(-S[A,\bar\psi,\psi]\right) \tag{19.71}$$

Were the theory non-anomalous, the effective action would be just the logarithm of the fermion determinant (since the Jacobian would be 1 and the Haar measure would factor out into a trivial constant).

Remarkably, it is the pressence of the Jacobian which solves the problem of anomalies. Indeed, (19.71) can be suggestively written in the form

$$\exp\left(-S_{eff}[A]\right) = \int \mathcal{D}gJ[g,A]\det D[A] \tag{19.72}$$

or, using

$$J[g^{-1},A] = \frac{\det D[A^{g-1}]}{\det D[A]} \tag{19.73}$$

$$\exp\left(-S_{eff}[A]\right) = \int \mathcal{D}g\det D[A^{g-1}] = \int \mathcal{D}g\det D[A^g] \tag{19.74}$$

It is now evident that the effective action (remember, the fermion determinant for non-anomalous theories) is gauge invariant. Indeed, if we consider

$$\begin{aligned} \exp\left(-S_{eff}[A^h]\right) &= \int \mathcal{D}g \det D[A^{hg}] \\ &= \int \mathcal{D}(hg) \det D[A^{hg}] = \int \mathcal{D}f \det D[A^f] \\ &= \exp\left(-S_{eff}[A]\right) \end{aligned} \tag{19.75}$$

Now, if $S_{eff}[A^h] = S_{eff}[A]$, the fermion current, which is naturally defined as

$$j^\mu[A] = -\frac{\delta S_{eff}[A]}{\delta A_\mu} \tag{19.76}$$

is conserved

$$D_\mu j^\mu[A] = 0 \tag{19.77}$$

To see this, consider the quantity

$$\begin{aligned} I &= \int \mathcal{D}A \exp\left(-S_{eff} - \int d^4x j_\mu[A] A^\mu\right) \\ &= \int \mathcal{D}A \exp\left(-S_{eff} + \int d^4x \frac{\delta S_{eff}[A]}{\delta A_\mu} A^\mu\right) \end{aligned} \tag{19.78}$$

and perform a change of variables of the type one does whithin the Noether method,

$$A_\mu \to A_\mu + D_\mu \alpha \tag{19.79}$$

with α infinitesimal. Using the gauge invariance of the effective action, one gets

$$I = \int \mathcal{D}A \exp\left(-S_{eff} + \int d^4x \frac{\delta S_{eff}[A]}{\delta A_\mu}(A^\mu + D_\mu\alpha)\right) \tag{19.80}$$

Expanding in powers of α up to first order one has

$$I = I + \int \mathcal{D}A \exp\left(-S_{eff}\right) \int dx \frac{\delta S_{eff}[A]}{\delta A_\mu} D_\mu \alpha \tag{19.81}$$

or

$$0 = \int \mathcal{D}A \exp\left(-S_{eff}\right) \int dx D_\mu \frac{\delta S_{eff}[A]}{\delta A_\mu} \alpha \tag{19.82}$$

which means that

$$\langle D_\mu \frac{\delta S_{eff}[A]}{\delta A_\mu} \rangle = \langle D_\mu j^\mu \rangle = 0 \tag{19.83}$$

Appendix

We shall prove identity (19.28) used in our derivation of $d = 2$ bosonization rules,

$$\int \mathcal{D}b_\mu \mathcal{H}[b]\delta\left[\varepsilon_{\mu\nu}(f_{\mu\nu}[b] - f_{\mu\nu}[s])\right] = \int \mathcal{D}g\mathcal{H}[s^g] \tag{19.84}$$

where $\mathcal{H}$ is a gauge-invariant functional. Note that in eq.(19.84) it is implicit that b_μ should be treated as a gauge field and hence a gauge fixing is required. A convenient gauge choice is

$$b_+ = s_+ \tag{19.85}$$

so that identity (19.84) takes the form

$$\int \mathcal{D}b_+\mathcal{D}b_-\ \Delta\delta(b_+ - s_+)\mathcal{H}[b_+, b_-]\delta[\varepsilon_{\mu\nu}(f_{\mu\nu}[b] - f_{\mu\nu}[s]] = \int \mathcal{D}g\mathcal{H}[s^g] \tag{19.86}$$

with Δ the Faddeev-Popov determinant for gauge condition (19.85),

$$\Delta = detD_+^{Adj}[s_+] \tag{19.87}$$

We now prove eq.(19.86). Let us start from the l.h.s. of eq.(19.86) performing first the b_+ trivial integration and then the b_- one

$$\begin{aligned}
&\int \mathcal{D}b_+\mathcal{D}b_-\Delta\delta(b_+ - s_+)\mathcal{H}[b_+, b_-]\delta(\varepsilon_{\mu\nu}(f_{\mu\nu}[b] - f_{\mu\nu}[s]) = \\
&\Delta[s_+]\int \mathcal{D}b_-\mathcal{H}[s_+, b_-]\delta(D_+[s_+]b_- - D_+[s_+]s_-) = \\
&\frac{\Delta[s_+]}{detD_+^{Adj}[s_+]}\int \mathcal{D}b_-\mathcal{H}[s_+, b_-]\delta(b_- - s_-) = \mathcal{H}[s_+, s_-]
\end{aligned} \tag{19.88}$$

In the last line we have used the explicit form of the Faddeev-Jacobian to cancel out both determinants. Being $\mathcal{H}[s_+, s_-]$ gauge independent, we can rewrite (19.88) in the form (appart from a gauge group volume factor)

$$\int \mathcal{D}b_+\mathcal{D}b_-\Delta\delta(b_+ - s_+)\mathcal{H}[b_+, b_-]\delta\left[\varepsilon_{\mu\nu}(f_{\mu\nu}[b] - f_{\mu\nu}[s])\right] = \int \mathcal{D}g\mathcal{H}[s_+^g, s_-^g] \tag{19.89}$$

Identity (19.84) is then proven.

Problem

Consider a gauge transformation with infinitesimal parameter $\alpha^a T^a$ acting on Weyl fermions. Compute the gauge anomaly and show that

$$\mathcal{A}^a = C\mathrm{tr}\left(T^a T^b T^c\right) F^b_{\mu\nu}\tilde{F}^c_{\mu\nu}$$

Compute C.

Clase 20

The "correct" path-integration variables

Consider a scalar field $\phi(x)$. In flat space, we have associated to it a measure $\mathcal{D}\phi$ in terms of Bessel-Fourier coefficients. That is, given an expansion of the form

$$\phi(x) = \sum_n c_n \varphi_n \tag{20.1}$$

in terms of an adequate basis $\{\varphi_n\}$, we defined

$$\mathcal{D}\phi(x) = \prod_n dc_n \tag{20.2}$$

In fact there was a disregarded normalization constant relating $\mathcal{D}\phi(x)$ with $\prod_n dc_n$ but this normalization was in fact implicit when defining the gaussian integral in a univoque way. That is, given a quadratic scalar "action"

$$S = \int d^d x \phi(x) B\phi \tag{20.3}$$

with A a scalar operator, one computes the normalization path-integral

$$\mathcal{N} \equiv \int \mathcal{D}\phi \exp\left(-\int d^d x \phi(x) B\phi\right) = \det{}^{-1/2} A \tag{20.4}$$

and then, evrywhere, one takes into account appropriately $\mathcal{N}$ which, being a <u>constant</u> can be factored out trivially when computing v.e.v.'s.

Now, what happens if space-time is not flat? One should have, instead of (20.3),

$$S = \int \phi(x) B\phi\sqrt{g} d^d x \tag{20.5}$$

with

$$\det g_{\mu\nu} = g \ , \quad \det g^{\mu\nu} = g^{-1} \tag{20.6}$$

and $\mathcal{N}$ is no more a constant!

$$\mathcal{N}[g] = \int \mathcal{D}\phi \exp\left(-\int \phi(x) B\phi\sqrt{g} d^d x\right) \tag{20.7}$$

This means for example that when computing the energy-momentum tensor through derivatives with respect to $g_{\mu\nu}$, $\mathcal{N}[g]$ cannot be disregarded. Even worse, if the metric is a dynamical variable, as in gravitation or in string theory, $\mathcal{N}[g]$ cannot be factored out from the path-integral since in those cases one is integrating also over the metric.

The first to look to this problem and propose a solution was again Fujikawa in his calculation of the Weyl anomaly [60]-[61]. He proposed a simple criterium which then was improved and which can be exemplified precisely with the scalar case. Suppose that instead of taking ϕ as integration variable, one takes $\tilde{\phi}$,

$$\tilde{\phi} = g^{1/4}\phi \tag{20.8}$$

In terms of this new variable the action becomes

$$S = \int \tilde{\phi}(x) A \tilde{\phi} d^d x \tag{20.9}$$

and the defining path-integral

$$\int \mathcal{D}\tilde{\phi} \exp\left(-\int \tilde{\phi}(x) A \tilde{\phi} d^d x\right) \equiv \tilde{\mathcal{N}} \tag{20.10}$$

one has that $\tilde{\mathcal{N}}$ is truly a constant. It is important to stress that (20.8) is not a change of variables but a definition of the correct path-integral variables when a metric is present. We shall here indicate these correct variables, which are usually called Fujikawa's variables, with a tilde.

The same analysis can be performed for fermion fields. Starting from a simple quadratic action, say

$$S = \int \bar{\psi}(x) A \psi \sqrt{g} d^d x \tag{20.11}$$

one ends with Fujikawa's variables in the form

$$\begin{aligned}\tilde{\bar{\psi}} &\equiv g^{1/4}\bar{\psi} \\ \tilde{\psi} &\equiv g^{1/4}\psi \end{aligned} \tag{20.12}$$

In fact, the fermionic case is a little more subtle because of the fact that usually one defines the normalization choosing A as the free Dirac operator which already contains the metric both in the contraction of ∂_μ with the Dirac matrices and also because the Dirac matrices algebra

$$\{\gamma_\mu, \gamma_\nu\} = 2g_{\mu\nu}(x) \tag{20.13}$$

now should be carefully handled. We shall come back to this point but first analyse a simpler case, that of the path-integration of a certain vector field A_μ. For the normalization, one would naively consider

$$\begin{aligned} N[g] &= \int \mathcal{D}A_\mu \exp\left(-\int A_\mu B^{\mu\nu} A_\nu\right)\sqrt{g}d^dx \\ &= \int \mathcal{D}A_\mu \exp\left(-\int A_\mu B_{\alpha\beta} A_\nu\right) g^{\mu\alpha}g^{\nu\beta}\sqrt{g}d^dx \\ &= \det{}^{-1/2}\left(B_{\alpha\beta}g^{\mu\alpha}g^{\nu\beta}\sqrt{g}\right) \end{aligned} \tag{20.14}$$

where we have accepted that covariant components should be considered as the physical ones. Now, one has

$$\begin{aligned} \det{}^{-1/2}\left(B_{\alpha\beta}g^{\mu\alpha}g^{\nu\beta}\sqrt{g}\right) &= \det{}^{-1/2}g^{\mu\alpha} \times \det{}^{-1/2}g^{\nu\beta} \times g^{-d/4} \times \det{}^{-1/2}B_{\alpha\beta} \\ &= g^{1-d/4}\det{}^{-1/2}B_{\alpha\beta} \end{aligned} \tag{20.15}$$

We can then write, instead of (20.14),

$$g^{(-1+d/4)}\int \mathcal{D}A_\mu \exp\left(-\int A_\mu B^{\mu\nu} A_\nu\right)\sqrt{g}d^dx = \det{}^{-1/2}\left(B_{\alpha\beta}\right) \tag{20.16}$$

Then, if instead of the naive path integral variables A_μ we use

$$\tilde{A}_\mu = g^{1/4-1/d}A_\mu \tag{20.17}$$

we shall have a true metric independent integral to define the normalization

$$\int \mathcal{D}\tilde{A}_\mu \exp\left(-\int \tilde{A}_\mu B^{\mu\nu} \tilde{A}_\nu\right) g^{-1/2}d^dx = \det{}^{-1/2}\left(B_{\alpha\beta}\right) = \tilde{\mathcal{N}} \tag{20.18}$$

Analogous reasoning can be applied for tensors of higher rank. In particular, for the metric, one has that the correct integration variable is

$$\tilde{g}_{\mu\nu} = g^{1/4-1/d} g_{\mu\nu} \tag{20.19}$$

An alternative way of obtaining the correct Fujikawa variables is to consider a supersymmetric theory, which implies using a path-integral measure containing both bosons and fermions. Requiring invariance of this measure under supersymmetric transformations automatically leads to the correct variables [62].

Clase 21

The Effective Action

We shall present here an approach that allows to study properties of quantum field theories using concepts and the intuition developed in classical mechanics.

In quantum field theory the objects with the greatest physical interest are the Green functions from which the S matrix can be constructed. Perturbative calculations are usually expressed in terms of these functions, the Feynman diagram technique is specifically deviced to compute them; their analytic properties give much information about the quantum physics.

On the other hand, certain quantum phenomenon, as for example the spontaneous symmetry breaking of a classical symmetry are difficult to understand in terms of Green functions. Of course one can follow, through the analysis of Green functions, step by step, the appearence of Goldston bosons, the acquisition of mass by certain fields without lose of symmetry, etc. But the starting classical scenario in which a symmetry breaking potential provides non trivial classical minima has a difficult translation to the quantum level in terms of Green functions.

Functional methods can be used to construct objects which will be called effective quantum actions and effective potentials. As we shall see, although closely related to Green functions, they provide the necessary bridge between classical and quantum theory. It is not hard to understand why the path-integral framework, in which the classical action plays a central role, provides the privileged tool to define and compute these objects.

For simplicity, we first consider the theory of a real scalar field ϕ with mass m and a quartic self interaction. Dynamics is described, at the classical level, by the Lagrangian

$$\begin{aligned} L &= \frac{1}{2}(\partial_\mu\phi)(\partial^\mu\phi) + \frac{1}{2}m^2\phi^2 + \frac{1}{4!}\phi^4 \\ S &= \int d^4x L \end{aligned} \tag{21.1}$$

The generating functional of disconnected Green functions $Z_{disc}[j]$ ($j(x)$ is an external source) is defined in terms of path-integrals as

$$Z_{disc}[j] = \frac{1}{Z_{disc}[0]}\int \mathcal{D}\phi \exp\left(-\frac{1}{\hbar}S[j]\right) \equiv \exp\left(-\frac{1}{\hbar}W[j]\right) \tag{21.2}$$

Here we have defined the action $S[j]$ in the presence of a source j as

$$S[j] = S + \int d^4x\phi(x)j(x) \tag{21.3}$$

The expansion of $Z_{disc}[j]$ in powers of j gives disconnected Green functions G^{2n}_{disc},

$$Z_{disc}[j] = \sum_n \frac{1}{(2n)!}\int d^4x_1 \ldots d^4x_{2n} G^{2n}_{disc}(x_1, x_2, \ldots, x_{2n}) j(x_1)\ldots j(x_{2n}) \tag{21.4}$$

with

$$G^{2n}_{disc}(x_1, x_2, \ldots, x_{2n}) = \hbar^{2n} \left.\frac{\delta^{2n} Z_{disc}[j]}{\delta j(x_1)\delta j(x_2)\ldots\delta j(x_{2n})}\right|_{j=0} \tag{21.5}$$

We have introduced W in (21.2) in order to pass from disconnected to connected Green functions. Indeed, being W related to Z_{disc} through

$$W[j] = -\hbar \log Z_{disc}[j] \tag{21.6}$$

the logarith relating them ensures that all disconnected contributions are eliminated from W derivatives,

$$G^{2n}_{conn}(x_1, x_2, \ldots, x_{2n}) = -\frac{1}{\hbar} \left.\frac{\delta^{2n} W[j]}{\delta j(x_1)\delta j(x_2)\ldots\delta j(x_{2n})}\right|_{j=0} \tag{21.7}$$

We have seen that a possible approximate way to evaluate a path-integral like that defining Z_{disc} is to use the steepest descent method. To this end, one first looks for a solution $\phi_0 = \phi_0[j]$ of the classical equations of motion deriving from the action in the path-integrand,

$$\delta S[j] = 0 \longrightarrow \left(-\nabla^2 + m^2\right)\phi_0 + \frac{\lambda}{3!}\phi_0^3 + j = 0 \tag{21.8}$$

Once the classical solution is found, one proceeds to a change of variables at the quantum level and proceeds to a pertubative calculation. That is, one writes

$$\phi(x) = \phi_0[j] + \sqrt{\hbar}\varphi(x) \tag{21.9}$$

and then writes the action $S[j]$ in terms of the new variable,

$$S[j] = S[\phi_0] + \frac{\hbar}{2}\int d^4x\varphi\Delta\varphi + \frac{\hbar^{3/2}\lambda}{3!}\int d^4x\phi_0(x)\varphi^3(x) + \frac{\hbar^2\lambda}{4!}\int d^4x\varphi^4(x) \tag{21.10}$$

where

$$\Delta = -\nabla^2 + \left(m^2 + \frac{\lambda\phi_0^2[j]}{2}\right) \tag{21.11}$$

Note that Δ replaces the kinetic energy terms in the original action and contains an effective (in general x-dependent) mass since $\phi_0[j] = \phi_0[j(x)]$.

Being eq.(21.8) elliptic, a boundary condition of the form

$$\phi_0[j(\infty) = 0] = 0 \tag{21.12}$$

will lead to a unique solution. (Demanding the external source to satisfy $j(\infty) = 0$ is a natural and allowed condition).

At the path integral level the shift (21.9) has no associated Jacobian so that one has

$$\begin{aligned}\exp\left(-\frac{1}{\hbar}W[j]\right) &= \frac{1}{Z[0]}\exp\left(-\frac{1}{\hbar}S[\phi_0]\right)\int\mathcal{D}\varphi\exp\left(-\int d^4x\,(\varphi\Delta\varphi + \right.\\ &\left.\left. \frac{\hbar^{1/2}\lambda}{3!}\phi_0\varphi^3(x) + \frac{\hbar\lambda}{4!}\varphi^4(x)\right)\right)\end{aligned} \tag{21.13}$$

Up to now, no approximation has been assumed. We face however a non-quadratic path-integral over φ which cannot be computed in a closed form and hence, we have to develop some pertubative treatment in order to compute Z_{disc} . A possibility is to expand in powers of $\hbar^\alpha\lambda^\beta$ which corresponds to the factors which appear in the non-quadratic terms. This will be a good approximation both for weak coupling and in the semiclassical limit. The answer is

$$\exp\left(-\frac{1}{\hbar}W[j]\right) = \frac{\exp\left(-\frac{1}{\hbar}S[\phi_0]\right)\left(\det^{-1/2}\Delta\right)(1+O(\hbar))}{\exp\left(-\frac{1}{\hbar}S[\phi_0[j=0]]\right)\left(\det^{-1/2}\Delta_0\right)(1+O(\hbar))} \tag{21.14}$$

where

$$\Delta_0 \equiv \Delta[\phi_0[j=0]] = -\nabla^2 + m^2 \tag{21.15}$$

Since $S[\phi_0[j=0]] = 0$, we can write

$$W[j] = S[\phi_0] + \frac{\hbar}{2}\log\det\left(\frac{\Delta}{\Delta_0}\right) + O(\hbar^2) \tag{21.16}$$

We see from (21.16) that in the $\hbar \to 0$ limit, the generating functional of connected quantum Green functions coincides with the classical action evaluated at a solution of the classical equations of motion in the presence of a source.

Now, objects appearing in (21.16) are all functionals of the source, an artifact introduced to get Green functions by functional differentiation. One can rework the results so as to define objects which are not functionals of the source but of "classical" fields. This will make possible a contact with classical field dynamics. To this end, we use the so called Legendre transformation that allows to express functionals $F[f]$ depending on functions $f(x)$ in terms of functionals of $g(x) = df(x)/dx$.

Define

$$F[g] = f(x) - xg(x) \tag{21.17}$$

then

$$\frac{dF}{dg} = \frac{df}{dx}\frac{dx}{dg} - \frac{dx}{dg}g - x \tag{21.18}$$

Now, the first and second term in the l.h.s. of (21.18) cancel so that we can write

$$x = -\frac{dF}{dg} \tag{21.19}$$

and then (21.17) takes the form

$$F[g] = f(x) + g(x)\frac{dF}{dg} \tag{21.20}$$

or

$$f(x) = F[g] - g(x)\frac{dF}{dg} \tag{21.21}$$

which should be compared with (21.17)

$$F[g] = f(x) - xg(x) \rightleftharpoons f(x) = F[g] - g(x)\frac{dF}{dg} \tag{21.22}$$

We can directly use this connection identifying

$$f \mapsto W[j]\,, \qquad x \mapsto j \tag{21.23}$$

and introducing Γ and ϕ_{cl} such that

$$F \mapsto \Gamma\,, \qquad g \mapsto \phi_{cl} \tag{21.24}$$

so that

$$g = \frac{df}{dx} \mapsto \phi_{cl} = \frac{\delta W[j]}{\delta j} \tag{21.25}$$

Then we have for (21.17),

$$\Gamma[\phi_{cl}] = W[j] - \int \phi_{cl}(x)j(x)dx^4 \rightleftharpoons W[j] = \Gamma[\phi_{cl}] - \int d^4x\phi_{cl}\frac{\delta\Gamma}{\delta\phi_{cl}} \tag{21.26}$$

Now, if we solve

$$\phi_{cl} = \frac{\delta W[j]}{\delta j} \tag{21.27}$$

we have

$$j = j[\phi_{cl}] \tag{21.28}$$

and then we can write

$$\Gamma[\phi_{cl}] = W[j[\phi_{cl}]] - \int \phi_{cl}(x)j[\phi_{cl}]d^4x \tag{21.29}$$

That, is, the Legendre transfomed of W has been entirely written in terms of a field ϕ_{cl}.

What can we say about ϕ_{cl}? Already the notation cl suggests that this object is some kind of "classical" field. To investigate this, let us start from the expansion (21.16) for W,

$$W[j] = S[\phi_0[j]] + \hbar S^{(1)}[\phi_0[j]] + \hbar^2 S^{(2)}[\phi_0[j]] + \dots \tag{21.30}$$

so that

$$\frac{\delta W}{\delta j} = \frac{\delta S[\phi_0[j]]}{\delta j} + \hbar \frac{\delta S^{(1)}[\phi_0[j]]}{\delta j} + \dots \tag{21.31}$$

But this means, after (21.27) that

$$\phi_{cl} = \frac{\delta S[\phi_0[j]]}{\delta j} + \hbar \frac{\delta S^{(1)}[\phi_0[j]]}{\delta j} + \dots \tag{21.32}$$

or, using

$$\frac{\delta S}{\delta j} = \phi_0 \tag{21.33}$$

$$\phi_{cl} = \phi_0 + O(\hbar) \tag{21.34}$$

We see that ϕ_{cl} includes the classical solution to the equations of motion in the presence of a source and also quantum corrections. In order to go on in the interpretation of quantum calculations in terms of classical concepts, let us expand $\Gamma[\phi_{cl}]$, the Legendre transformed of the generating functional of connected Green functions, in powers of $\hbar$.

$$\Gamma[\phi_{cl}] = \Gamma^{(0)}[\phi_{cl}] + \hbar\Gamma^{(1)}[\phi_{cl}] + \hbar^2\Gamma^{(2)}[\phi_{cl}] + \dots \tag{21.35}$$

Using the expansion (21.30) for W and the relation (21.29) between W and Γ, one gets

$$\Gamma[\phi_{cl}] = S[\phi_0] + \hbar S^{(1)}[\phi_0] - \int d^4x\, (\phi_0(x) + O(\hbar))\, j(x) \tag{21.36}$$

To order zero in $\hbar$ we can then write,

$$\Gamma^{(0)}[\phi_{cl}] = S[\phi_0] - \int d^4x \phi_0(x) j(x) \tag{21.37}$$

Now, the second term in the right hand side of (21.37) cancels the source term in $S[\phi_0]$ so that we in fact have

$$\Gamma^{(0)}[\phi_{cl}] = \int d^4x L = \int d^4x \left(\frac{1}{2}(\partial_\mu\phi_0)(\partial^\mu\phi_0) + \frac{1}{2}m^2\phi_0^2 + \frac{1}{4!}\phi_0^4 \right) \tag{21.38}$$

We have seen (eq.(21.34)) that there is no distinction between ϕ_0 and ϕ_{cl} to order $\hbar$. Then, we can rewrite (21.38) in the form

$$\Gamma^{(0)}[\phi_{cl}] = \int d^4x \left(\frac{1}{2}(\partial_\mu \phi_{cl})(\partial^\mu \phi_{cl}) + \frac{1}{2} m^2 \phi_{cl}^2 + \frac{1}{4!}\phi_{cl}^4 \right) \tag{21.39}$$

Functional $\Gamma^{(0)}$ represents the classical action written as a functional of a classical object which we called ϕ_{cl}. But which is the precise meaning of ϕ_{cl} in terms of quantum concepts? To see this, let us rewrite formula (21.2) defining W,

$$\exp\left(-\frac{1}{\hbar} W[j]\right) = \frac{1}{Z_{disc}[0]} \int \mathcal{D}\phi \exp\left(-\frac{1}{\hbar}\int d^4x \,(L + j\phi)\right) \tag{21.40}$$

Differentiation with respect to j shows that

$$\frac{\delta W}{\delta j(x)} = \langle \phi(x) \rangle_j \tag{21.41}$$

We know from eq.(21.27) the the l.h.s. of (21.41) is nothing but ϕ_{cl} so that

$$\phi_{cl} = \phi_{cl}[j] = \langle \phi(x) \rangle_j = \phi_0[j] + \hbar \phi^{(1)} + \dots \tag{21.42}$$

or, using $\phi_0[0] = 0$,

$$\phi_{cl}[0] = \langle \phi \rangle = \hbar \phi_1 + O(\hbar^2) \tag{21.43}$$

That is, the argument ϕ_{cl} in $\Gamma[\phi_{cl}]$ can be associated, for zero source, with the vacuum expectation value of the field. But we can say more. From the definition of Γ as a Legendre transformation, eq.(21.29), we can write

$$\frac{\Gamma[\phi_{cl}]}{\delta \phi_{cl}} = \frac{\delta W[j[\phi_{cl}]]}{\delta \phi_{cl}} - j[\phi_{cl}] - \int d^4x \phi_{cl} \frac{\delta j}{\delta \phi_{cl}} \tag{21.44}$$

but since

$$\frac{\delta W[j[\phi_{cl}]]}{\delta \phi_{cl}} = \int d^4x \frac{\delta W[j[\phi_{cl}]]}{\delta j} \frac{\delta j}{\delta \phi_{cl}} \tag{21.45}$$

the first and third terms in the r.h.s. of (21.44) cancel and one is left with

$$\frac{\delta \Gamma[\phi_{cl}]}{\delta \phi_{cl}} = -j \tag{21.46}$$

or

$$\left.\frac{\delta\Gamma[\phi_{cl}]}{\delta\phi_{cl}}\right|_{j=0} = 0 \qquad (21.47)$$

So we see that the extrema of $\Gamma[\phi_{cl}]$ as a functional of ϕ_{cl} will give something that is equal to (to order $\hbar$) the v.e.v. of the quantum field ϕ. It is then clear why one calls Γ the effective action: it encodes quantum information exactly as the classical action does for classical dynamics: its minimum correspond to the v.e.v. that the field takes.

We have seen in eq.(35) that there is an expansion in Γ in powers of $\hbar$ that identifies its zeroth order with the classical action. This agrees with the interpretation above and shows that, to zeroth order, we can identify the v.e.v. of the quantum field with the extrema of the classical action. This identification legalizes, at least to the lowest order, the description of quantum spontaneous breaking of gauge symmetry as described in terms of minima of a classical potential. What about the first order, second order, etc? We have seen that, to order $\hbar$, we have for W (see eq.(21.16))

$$W^{(1)} = \hbar S^{(1)} = \hbar\frac{1}{2}\log\det\left(\frac{\Delta}{\Delta_0}\right) \qquad (21.48)$$

Then, from the relation (21.29) between W and Γ we can see that

$$\hbar\Gamma^{(1)} = \hbar S^{(1)} = \hbar\frac{1}{2}\log\det\left(\frac{\Delta}{\Delta_0}\right) \qquad (21.49)$$

So, in the computation of v.e.v.'s using the effective action method, to order zero we identify the v.e.v. with the minimum of the classical action. To order one, we have to compute the fermion determinant in the r.h.s. of (21.49) and the addition of $\Gamma^{(1)}$ to the classical axction will incorporate a contribution of order $\hbar$ to the information of the minimum of the effective action. The procedure can be systematically ameliorated. In Problem 1 we shall show that [67]

$$\Gamma^{(2)}[\phi_{cl}] = S^{(2)}[\phi_{cl}] + \frac{1}{2}\int d^4x d^4y \phi^{(1)}\Delta\phi^{(1)} - \int d^4x \phi^{(1)}\frac{\delta S}{\delta\phi_{cl}} \qquad (21.50)$$

The effective potential

Up to now we have been working with functions $\phi_{cl}(x)$ which were related to classical extrema of the action. Now, if one particularizes to constant functions, by analogy with the classical case in which the action becomes just (minus) the potential, here we shall say we are computing the effective potential. Minima of this quantum object will correspond to the quantum version of the minima of the classical potential.

So, let us put

$$\phi_{cl}(x) = \phi_{cl} = \text{constant} \tag{21.51}$$

and then define the effective potential through the identity

$$\Gamma[\phi_{cl}] = \int d^4x V_{eff}[\phi_{cl}] \tag{21.52}$$

Of course, being ϕ_{cl} constant, V_{eff} in (21.52) does not depend on x and the integration gives a divergent factor which will be identified with the volume of space-time and should be consequently extracted.

Using our previous results, we can write

$$V_{eff}[\phi_{cl}] = V_0[\phi_{cl}] + \hbar V_1[\phi_{cl}] + \hbar^2 V_2[\phi_{cl}] + \dots \tag{21.53}$$

with

$$V_0[\phi_{cl}] = \frac{m^2}{2}\phi_{cl}^2 + \frac{\lambda}{4!}\phi_{cl}^4 \tag{21.54}$$

That is, the first term in the expansion of the effective potential just gives the classical potential, the Higgs potential in the present case. Concerning V_1, from our previous calculations we have (eq.(21.49))

$$\Gamma^{(1)} = \frac{1}{2}\log\det\left(\frac{\Delta}{\Delta_0}\right) \tag{21.55}$$

with

$$\Delta = -\nabla^2 + \left(m^2 + \frac{\lambda\phi_{cl}^2}{2}\right) = \Delta_0 + \frac{\lambda\phi_{cl}^2}{2} \tag{21.56}$$

Then, we have

$$\Gamma^{(1)} = \frac{1}{2}\mathrm{tr}\log\left(\frac{\Delta}{\Delta_0}\right) = \frac{1}{2}\mathrm{tr}\log\left(1 + \frac{\lambda}{2}\Delta_0^{-1}\phi_{cl}^2\right) \tag{21.57}$$

here, we have called $\Delta_0^{-1} \equiv G_0$ the Green function for the free problem,

$$\Delta_0 G_0(x-y) = \delta^{(4)}(x-y) \tag{21.58}$$

The logarithm in (21.57) can be more precisely written as

$$\begin{aligned} \log\left(1+\frac{1}{2}\Delta_0^{-1}\phi_{cl}^2\right) &= \log\left(\delta^{(4)}(x-y)+\frac{\lambda}{2}\phi_{cl}G_0(x-y)\right) \\ &\equiv \log\left(\delta_{xy}^{(4)}+K_{xy}\right) \end{aligned} \tag{21.59}$$

We can the use the expansion

$$\log(1+y) = \sum_{n=1}(-1)^n\frac{y^n}{n} \tag{21.60}$$

and write

$$\log\left(\delta_{xy}^{(4)}+K_{xy}\right) = \sum_{n=0}\frac{1}{n+1}\left(-K_{xy}\right)^{n+1} \tag{21.61}$$

or, explicitly

$$\begin{aligned} &\mathcal{I} \equiv \log\left(\delta^{(4)}(x-y)+K(x-y)\right) \\ &\quad= \sum_{n=0}\frac{(-1)^{n+1}}{n+1}\int d^4x_1 d^4x_2 \ldots d^4x_n K(x-x_1)K(x_1-x_2)\ldots K(x_n-x) \end{aligned} \tag{21.62}$$

Then, Fourier transforming in the r.h.s.,

$$\begin{aligned} \mathcal{I} \;=\; & -\sum_{n=0}\frac{1}{n+1}\int d^4x_1 d^4x_2\ldots d^4x_n \frac{d^4k_1}{(2\pi)^4}\frac{d^4k_2}{(2\pi)^4}\cdots\frac{d^4k_n}{(2\pi)^4}\times \\ & \exp\left(ik_1(x-x_1)+ik_2(x_1-x_2)++ik_n(x_n-x)\right)\times \\ & \left(-\frac{\lambda}{2}\phi_{cl}^2\tilde{G}_0(k_1)\right)\left(-\frac{\lambda}{2}\phi_{cl}^2\tilde{G}_0(k_2)\right)\ldots\left(-\frac{\lambda}{2}\phi_{cl}^2\tilde{G}_0(k_n)\right) \end{aligned} \tag{21.63}$$

Each integral over x_i gives $(2\pi)^4\delta(k_i-k_{i+1})$ which in turn eliminates one integral over k One then ends with

$$\begin{aligned} \mathrm{tr}\mathcal{I} &= \sum_n\frac{1}{n+1}\int d^4x\frac{d^4k_1}{(2\pi)^4}\left(-\frac{\lambda}{2}\phi_{cl}^2\tilde{G}_0(k_1)\right)^{n+1} \\ &= \int d^4x\int\frac{d^4k}{(2\pi)^4}\log\left(1+\frac{\lambda}{2}\phi_{cl}^2\tilde{G}_0(k)\right) \end{aligned} \tag{21.64}$$

where "tr" has been explicitly included in the r.h.s. as an integral over x. Then, extracting from tr$\mathcal{I}$ the volume factor $\int d^4x$ we get for the first quantum correction to the classical potential,

$$V_1[\phi_{cl}] = \frac{1}{2}\int \frac{d^4k}{(2\pi)^4} \log\left(1 + \frac{\lambda}{2}\phi_{cl}^2 \tilde{G}_0(k)\right) \tag{21.65}$$

Now, since

$$\tilde{G}_0(k) = \frac{1}{k^2 + m^2} \tag{21.66}$$

we finally have

$$V_1[\phi_{cl}] = \frac{1}{2}\int \frac{d^4k}{(2\pi)^4} \log\left(1 + \frac{\lambda\phi_{cl}^2}{2(k^2 + m^2)}\right) \tag{21.67}$$

Of course, the integral over k is divergent. This is not unexpected since we are in fact computing quantum effects in a field theory and this, in general, needs renormalization. The divergence is an ultraviolet one and manifests in the first two terms in the expansion of the logarithm in powers of k^2. Since each power of k^2 is accompanied by ϕ_{cl}^2, only two counterterms proportional to ϕ_{cl}^2 and ϕ_{cl}^4 will suffice to cure infinities. We then write

$$V_1[\phi_{cl}]|_{ren} = V_1[\phi_{cl}] + A\phi_{cl}^2 + B\phi_{cl}^4 \tag{21.68}$$

We need two normalization conditions in order to fix A, B. Inspired in the form of the classical potential we chose as conditions for the complete effective potential

$$\left.\frac{d^2V_{eff}}{d\phi_{cl}^2}\right|_{\phi_{cl}=0} = m^2\,, \qquad \left.\frac{d^4V_{eff}}{d\phi_{cl}^4}\right|_{\phi_{cl}=0} = \lambda \tag{21.69}$$

These conditions are effectively defining mass and coupling constant for the renormalized theory. This imply, for V_1 the conditions

$$\left.\frac{d^2V_1}{d\phi_{cl}^2} + 2A\right|_{\phi_{cl}=0} = 0\,, \qquad \left.\frac{d^4V_1}{d\phi_{cl}^4}\right|_{\phi_{cl}=0} + 4!B = 0 \tag{21.70}$$

Using these conditions in (21.68) we get,

$$\begin{aligned} A &= -\frac{\lambda}{2}\int \frac{d^4k}{(2\pi)^4}\frac{1}{k^2+m^2} \\ B &= \frac{\lambda^2}{2}\int \frac{d^4k}{(2\pi)^4}\frac{1}{(k^2+m^2)^2} \end{aligned} \tag{21.71}$$

The final renormalized formula for V_1 is then

$$\begin{aligned} V_1[\phi_{cl}] &= \frac{1}{2}\int \frac{d^4k}{(2\pi)^4}\left(\log\left(1+\frac{\lambda\phi_{cl}^2}{2(k^2+m^2)}\right)\right. \\ &\left. -\frac{\lambda\phi_{cl}^2}{2}\frac{1}{k^2+m^2}\right)+\frac{1}{2}\left(\frac{\lambda\phi_{cl}^2}{2}\right)^2\frac{1}{(k^2+m^2)^2} \end{aligned} \tag{21.72}$$

Up to now we can then write

$$V_{eff} = \frac{m^2}{2}\phi_{cl}^2 + \frac{\lambda}{4!}\phi_{cl}^4 + \hbar V_1[\phi_{cl}] + O(\hbar^2) \tag{21.73}$$

It is clear that the effective potential incorporates quantum corrections to the classical potential so that any analysis, as for example that starting from the minima of the potential, will then take into account quantum aspects.

Problems

Problem 1: Prove that

$$\Gamma^{(2)}[\phi_{cl}] = S^{(2)}[\phi_{cl}] + \frac{1}{2}\int d^4x d^4y \phi^{(1)}\Delta\phi^{(1)} - \int d^4x \phi^{(1)}\frac{\delta S}{\delta\phi_{cl}}$$

(Hints can be found in [67])

Clase 22

More on the effective potential

We have seen that the renormalized effective potential for a real scalar field takes the form, up to $\hbar$ terms,

$$V_{eff} = \frac{m^2}{2}\phi_{cl}^2 + \frac{\lambda}{4!}\phi_{cl}^4 + \hbar V_1[\phi_{cl}] + O(\hbar^2) \tag{22.1}$$

where V_1 is defined according to

$$\Gamma_1[\phi_{cl}] = \int d^4x V_1[\phi_{cl}] \tag{22.2}$$

with

$$\Gamma_1[\phi_{cl}] = \frac{1}{2}\mathrm{Tr}\log\frac{\Delta[\phi_{cl}]}{\Delta_0} \tag{22.3}$$

this leading to

$$V_1[\phi_{cl}] = \int \frac{d^4k}{(2\pi)^4}\log\left(1 + \frac{\lambda}{2}\phi_{Cl}^2\tilde{G}_0(k)\right) \tag{22.4}$$

with $\tilde{G}_0$ the scalar propagator. The expression above has to be cured from ultraviolet divergencies. Then, after renormalization, $V_1[\phi_{cl}]$ takes the form

$$\begin{aligned} V_1[\phi_{cl}] &= \frac{1}{2}\int \frac{d^4k}{(2\pi)^4}\left(\log\left(1 + \frac{\lambda\phi_{cl}^2}{2(k^2+m^2)}\right)\right. \\ &\left. -\frac{\lambda\phi_{cl}^2}{2}\frac{1}{k^2+m^2} + \frac{1}{2}\left(\frac{\lambda\phi_{cl}^2}{2}\right)^2\frac{1}{(k^2+m^2)^2}\right) \end{aligned} \tag{22.5}$$

We see that $V_1[\phi_{cl}]$ can be written as a function of

$$x = \frac{\lambda\phi_{cl}}{2m^2} \tag{22.6}$$

in the form

$$V_1[\phi_{cl}] = m^4 H(x) \tag{22.7}$$

with

$$H(x) = \frac{1}{2}\int \frac{d^4k}{(2\pi)^4}\left(\log\left(1+\frac{x}{k^2+1}\right) - \frac{x}{k^2+1} + \frac{1}{2}\frac{x^2}{(k^2+1)^2}\right) \tag{22.8}$$

Integrals in (22.8) can be easily performed leading to

$$H(x) = \frac{1}{64\pi^2}\left((x+1)^2\log(1+x) - \frac{3}{2}x - x\right) \tag{22.9}$$

Let us now discuss whether quantum corrections to the potential can change dramatically its behavior. To start with, we shall be interested in the massless case, $m = 0$. The classical potential is given by just the $\lambda\phi^4$ term and then there is not be symmetry breaking at the classical level since the minimum of the potential is located at $\phi = 0$. Can this behavior change because of quantum corrections?

To begin with, let us observe that QFT of massless particles is affected by infrared divergencies. These divergencies have to manifest in some way in the effective potential approach and indeed they affect the renormalization conditions we have used. Indeed, in the massive case we fixed the two required counterterms imposing the conditions

$$\left.\frac{d^2V_{eff}}{d\phi_{cl}^2}\right|_{\phi_{cl}=0} = m^2 \tag{22.10}$$

$$\left.\frac{d^4V_{eff}}{d\phi_{cl}^4}\right|_{\phi_{cl}=0} = \lambda \tag{22.11}$$

leading to the following formulae for the counterterms,

$$A = -\frac{1}{2}\left.\frac{d^2V_1}{d\phi_{cl}^2}\right|_{\phi_{cl}=0}, \qquad B = -\frac{1}{4!}\left.\frac{d^4V_1}{d\phi_{cl}^4}\right|_{\phi_{cl}=0} \tag{22.12}$$

If we try to repeat this procedure for the massless case, conditions (22.10)- (22.11) change to

$$\left.\frac{d^2V_{eff}}{d\phi_{cl}^2}\right|_{\phi_{cl}=0} = 0 \tag{22.13}$$

$$\left.\frac{d^4V_{eff}}{d\phi_{cl}^4}\right|_{\phi_{cl}=0} = \lambda \tag{22.14}$$

but this produces in B an infrared divergence leading to an incorrect result. Indeed, from (22.5) and (22.14) one has

$$B = \frac{3}{2}\lambda^2 \int \frac{d^4k}{(2\pi)^4}\frac{1}{k^4} \tag{22.15}$$

Although this result does its work in the ultraviolet, it introduces an infrared divergence which cannot be cured within this choice of normalization conditions. Now, instead of (22.14), consider the condition

$$\left.\frac{d^4V_{eff}}{d\phi_{cl}^4}\right|_{\phi_{cl}=\mu\neq 0} = \lambda \tag{22.16}$$

with μ an arbitrary mass scale. Moreover, we shall maintain $m \neq 0$ till the end of calculations and then take the $m \to 0$ limit, then no infrared divergences appear and one ends with the result

$$V_1[\phi_{cl}] = \frac{1}{64\pi^2}\left(\frac{\lambda\phi_{cl}^2}{2}\right)^2\left(\log\left(\frac{\phi_{cl}^2}{\mu^2}\right) - \frac{25}{6}\right) \tag{22.17}$$

The following points should be remarked concerning this result

- Infrared divergencies appear here as singularities of the effective potential at $\phi_{cl} = 0$. They are cured introducing a mass scale.
- The first correction to the classical potential is order λ^2. Then, if one expected to pass from a trivial minimum classical potential to an effective potential with non-trivial minima, this does not happen for the pure scalar case. There is no choice of scale and coupling constant changing the minimum.

An interesting point to discuss in the present massless case (but which can also be discussed in the massive one) concerns the dependence on the mass scale μ introduced through the renormalization conditions. These mass scales acts here as the mass of the platinum-iridium cylinder kept at the International Bureau of Weights and Measure in Sèvres, near Paris. Instead of μ one can use, when writing condition (22.16), another scale μ' (say the pound of the British measure system),

$$\left.\frac{d^4V_{eff}}{d\phi_{cl}^4}\right|_{\phi_{cl}=\mu'\neq 0} = \lambda_{\mu'} \tag{22.18}$$

With the subscript μ' in $\lambda_{\mu'}$ we indicate that the coupling constant is now defined in relation to a mass scale μ' The effective potential, with this new scale is such that

$$\begin{aligned} V_1[\phi_{cl}] &= \frac{1}{64\pi^2}\left(\frac{\lambda_{\mu'}\phi_{cl}^2}{2}\right)^2\left(\log\left(\frac{\phi_{cl}^2}{\mu'^2}\right)-\frac{25}{6}\right) \\ &= \frac{1}{64\pi^2}\left(\frac{\lambda_{\mu'}\phi_{cl}^2}{2}\right)^2\left(\log\left(\frac{\phi_{cl}^2}{\mu^2}\right)+\log\left(\frac{\mu^2}{\mu'^2}\right)-\frac{25}{6}\right) \end{aligned} \tag{22.19}$$

Adding the original, classical potential, we have

$$\begin{aligned} V_{eff} &= \frac{m^2}{2}\phi_{cl}^2+\frac{\lambda_{\mu'}}{4!}\phi_{cl}^4+ \\ & \frac{1}{64\pi^2}\left(\frac{\lambda_{\mu'}\phi_{cl}^2}{2}\right)^2\left(\log\left(\frac{\phi_{cl}^2}{\mu^2}\right)+\log\left(\frac{\mu^2}{\mu'^2}\right)-\frac{25}{6}\right) \end{aligned} \tag{22.20}$$

to be compared with

$$\begin{aligned} V_{eff} &= \frac{m^2}{2}\phi_{cl}^2+\frac{\lambda}{4!}\phi_{cl}^4+ \\ & \frac{1}{64\pi^2}\left(\frac{\lambda_{\mu}\phi_{cl}^2}{2}\right)^2\left(\log\left(\frac{\phi_{cl}^2}{\mu^2}\right)-\frac{25}{6}\right) \end{aligned} \tag{22.21}$$

Let us now try to find a connection between $\lambda_{\mu'}$ and λ_μ so that (22.20) becomes (22.21). We propose

$$\lambda_{\mu'} = \lambda_\mu + C\lambda_\mu^2 \tag{22.22}$$

Plugging (22.22) into (22.20) and adjusting C so as to obtain (22.21) one gets

$$C = \frac{3}{32\pi^2} \log\left(\frac{\mu'^2}{\mu^2}\right) \tag{22.23}$$

so that the relation between the coupling constants associated to the two scales is

$$\lambda_{\mu'} = \lambda_\mu + \frac{3}{32\pi^2}\lambda_\mu^2 \log\left(\frac{\mu'^2}{\mu^2}\right) \tag{22.24}$$

We now introduce a dimensionless parameter t such that

$$t = \frac{1}{2}\log\left(\frac{\mu'^2}{\mu^2}\right) \tag{22.25}$$

With this, we can write instead of (22.24),

$$\lambda(t) = \lambda(0) + \frac{6}{32\pi^2}\lambda(0)^2 t \tag{22.26}$$

Observing that

$$\mu' = \exp(t)\mu \tag{22.27}$$

we see that the parameter t effectively changes the scale and equation (22.26) tells us how the coupling constant should be changed when the scale changes. That is why $\lambda(t)$ is called a *running coupling constant*: it runs when t runs. An important quantity that can now be introduced is the so called β-function, defined as

$$\beta \equiv \frac{d\lambda(t)}{dt} \tag{22.28}$$

To the order we are working, β is positive definite for the present scalar model,

$$\beta = \frac{3}{16\pi^2}\lambda(0)^2 \tag{22.29}$$

That is, in the weak coupling regime, the running coupling constant grows when one uses a larger scale. Now, using a larger scale implies that one uses smaller values to represent masses in such a scale. Smaller masses is equivalent to larger momenta. In fact, a scaling of the form (22.27) should be accompanied by an analogous scaling of the momenta,

$$p \to \exp(t)p \tag{22.30}$$

Then, using a larger scale implies to test larger energies. We then conclude that the running coupling constant, which can be seen as the effective coupling constant at a given energy, grows with the energy for the case of a real scalar theory. A negative β-function gives an opposite behavior: the running coupling constant becomes smaller with growing energy, precisely what is observed in the case of strong interactions: quarks are nearly free at large energies (asymptotic freedom). We then see that a field theory describing strong interactions should incorporate this property. Many theories can be discarded just because they have a positive β-function. It was a great triumph when it was discovered that Yang-Mills theory can leads to $\beta < 0$. Such an analysis is known as a Renormalization Group analysis. We shall come back to it in more detail below.

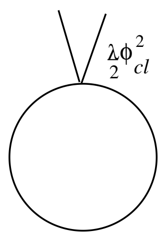
A loop expansion

Till now we have not clearly analised what kind of approximation were we performing. To tackle this problem, let us remember that the calculation of the contribution V_1 to the effective potential reduces to the evaluation of a functional determinant,

$$V_1 = \frac{1}{2}\mathrm{tr}\log\left(1 + \frac{\lambda}{2}\Delta_0^{-1}\phi_{cl}^2\right) \tag{22.31}$$

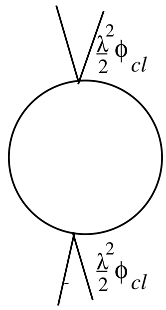
After expansion of the logarithm we ended with a series for which the first non trivial term was a loop consisting of a closed line representing the propagator Δ_0^{-1} with a ϕ_{cl}^2 insertion,

$$\frac{\lambda\phi_{cl}^2}{2}\int\frac{d^4k}{(2\pi)^4}\Delta_0^{-1}(k) \quad =$$



It is not difficult to guess the form of the following term

$$\left(\frac{\lambda\phi^2_{cl}}{2}\right)^2 \int \frac{d^4k}{(2\pi)^4}\Delta_0^{-1}(k)\Delta_0^{-1}(k) =$$



The same happens with the terms that follow: they correspond to one-loop Feynman diagrams with 3 insertions, 4 insertions, etc. Each diagram has a power of λ which corresponds to the number of insertions. We can then represent what we are doing in this computation of the effective potential by just writing the following self-explanatory formula

$$V_{eff} = \frac{\lambda}{4!}\phi^4_{cl} + \hbar\underbrace{(a[\phi_{cl}]\lambda + b[\phi_{cl}]\lambda^2 + \ldots)}_{V_1} + \hbar^2\underbrace{(\quad ? \quad)}_{V_2} + \ldots \qquad (22.32)$$

This formula shows the effective potential expressed as an expansion in powers of $\hbar$. Each term in the expansion contains all powers of λ so that the contribution say of first order in $\hbar$ is not just the result one should obtain using the habitual perturbative in λ approach but contains all orders in λ.

In order to better understand the statement above, let us observe that, given a connected diagram with L loops, I internal lines, E external lines and V vertices, one necessarily has, on the one hand

$$2I + E = 4V \qquad (22.33)$$

since to each vertex of a λ^4 theory arrive 4 lines which should be counted as a 1 if they are external or as a 1/2 if they are internal (an internal line is shared by 2 vertices).

On the other hand, the number of loops obeys

$$L = I - (V-1) \Longleftrightarrow L - 1 = I - V \qquad (22.34)$$

Indeed, by definition, the number of loops is the number of integrations over independent 4-momenta in a Feynman diagram. In principle, there are as many integrations as internal lines a diagram has. However, at each vertex one has a conservation delta function. This reduces the number of integrations in $V-1$, the -1 arising because the "last" delta function concerns external momenta and then does not reduce the number of integrations. Combining (22.33) and (22.34) one can write,

$$L = V - \frac{E}{2} + 1 \tag{22.35}$$

According to Feynman diagrams recipe one should include a $\hbar$ factor for each propagator (there is a $1/\hbar$ in the kinetic energy term of the action and the propagator is obtained from the inverse of this term). The vertices arise from the expansion of the interaction Lagrangian so that each vertex contributes with a $1/\hbar$ power to the diagram. We are considering diagrams without external legs so that we can just write

$$\text{powers of } \hbar = I - V \tag{22.36}$$

or, using (22.34),

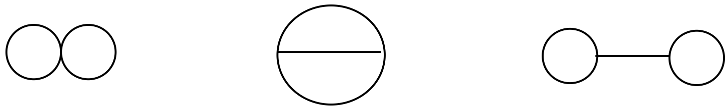
$$\text{powers of } \hbar = L - 1 \tag{22.37}$$

This counts the powers in $\hbar$ in the expansion of $\exp(-\Gamma/\hbar)$, so that the -1 appearing in (22.37) corresponds to the $1/\hbar$ appearing in the argument of the exponential. We then have for diagrams in Γ or in V_{eff},

$$\text{powers of } \hbar \text{ in a diagram} = \text{number } L \text{ of loops in the diagram}$$

Then, the approximation systematically performed within the effective potential (or effective action) approach corresponds to a loopwise expansion which gives corrections to the classical potential in powers of $\hbar$. For $\hbar = 0$ one just has the tree diagrams which represent the interaction terms in the classical potential.

Without making any new calculation, we can now infer the form of the contributions to V_2 in expression (22.32). Being an $\hbar^2$ term, it will correspond to two-loop diagrams. We can easily draw the independent two-loop diagrams arising in a $\lambda\phi^4$ theory. They are



(a) (b) (c)

Now, diagram (c) is not one particle irreducible (1PI) since with just one cut in the horizontal line it becomes disconnected. Does this kind of diagrams contribute to the effective potential or the effective action? In the same way we have seen that $Z[j]$ generates disconnected Green functions and $W[j]$ generates the connected ones, one can see that the effective action Γ generates 1PI diagrams so that only diagrams (a) and (b) will contribute to the effective potential.

To see that the statement above is true, let us consider the result in Problem 1 of the precedent class,

$$\Gamma^{(2)}[\phi_{cl}] = W^{(2)}[\phi_{cl}] + \frac{1}{2}\int d^4x d^4y \phi^{(1)}\Delta\phi^{(1)} - \int d^4x \phi^{(1)}\frac{\delta S}{\delta\phi_{cl}} \quad (22.38)$$

We shall eliminate from this expression $\phi^{(1)}$ using that

$$\phi_{cl} = \phi_0 + \hbar\phi_1 + \dots \quad (22.39)$$

then

$$\phi^{(1)} = \frac{1}{\hbar}(\phi_{cl} - \phi_0) = \frac{1}{\hbar}\left(\frac{\delta W}{\delta j} - \phi_0\right) \quad (22.40)$$

Now, since

$$\phi_{cl} = \frac{\delta W}{\delta j} = \frac{\delta S^{(0)}}{\delta j} + \hbar\frac{\delta S^{(1)}}{\delta j} + \dots \quad (22.41)$$

one has

$$\phi^{(1)} = \frac{\delta S^{(1)}}{\delta\phi_0}\frac{\delta\phi_0}{\delta j} = \frac{\delta S^{(1)}}{\delta\phi_{cl}}\Delta^{-1}[\phi_{cl}] \quad (22.42)$$

where we have used

$$\Delta\phi_0 = j \to \phi_0 = \Delta^{-1}j \quad (22.43)$$

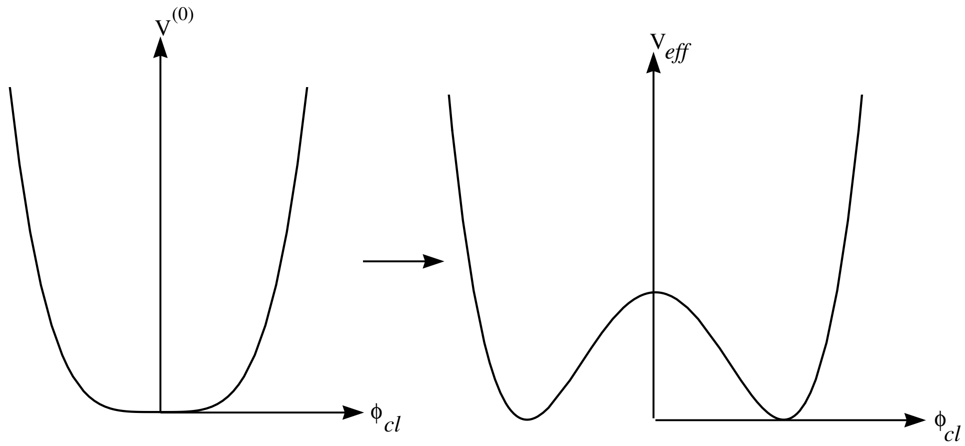
Then, from (22.42), one has for (22.38) (remember that $W^{(1)} = \hbar S^{(1)}$)

$$\Gamma^{(2)}[\phi_{cl}] = W^{(2)}[\phi_{cl}] - \frac{1}{2}\int\int d^4x d^4y \frac{\delta W^{(1)}}{\delta\phi_{cl}(x)}\Delta^{-1}\phi_{cl}(x-y)\frac{\delta W^{(1)}}{\delta\phi_{cl}(y)} \quad (22.44)$$

We see that the second term subtracts from $W^{(2)}$ precisely the non-1PI diagram (a propagator and the two bubbles, adequately normalized). The interpretation of Γ is then transparent: it is the generating functional of 1PI Green functions. This can be proved in general for each term in the expansion.

The Coleman-Weinberg phenomenon [68]

The outcome of this phenomenon can be graphically represented as follows



$V^{(0)} = \frac{\lambda}{4!}\phi_{cl}^4$
$(no\ \ \phi_{cl}^2\ \ term)$

$V_{eff} = V^0 + \hbar V^{(1)}\left[\phi_{cl}\right]$

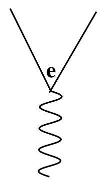
If one finds a model in which this phenomenon takes place, one can think of a process in which symmetry breaking takes place because of quantum corrections. This should be of interest, for example, when studying the physics of Grand Unified Theories for which, in different circumstances, one has different symmetries and then it should exist a dynamical mechanism producing the symmetry breaking when certain physical parameters change.

We have seen that this does not happens for the theory of a real scalar field. Coleman and Weinberg shown, in contrast, have seen that the coupling of a gauge field to a (complex) scalar does produce a dramatic effect and dynamical symmetry breaking can take place. Consider, as they did, the Abelian model

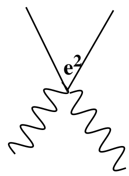
$$L = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \frac{1}{2}\left|\left(\partial\mu + ieA_\mu\right)\phi\right|^2 - \frac{\lambda}{4!}\left(\phi\phi^*\right)^2 \tag{22.45}$$

here ϕ is a charged scalar field, $\phi = \phi_1 + i\phi_2$, and A_μ and Abelian gauge field which couples to ϕ with coupling constant (charge) e. Of course one can consider *ab initio* a symmetry breaking potential but the idea is precisely to see whether quantum effects can change the classical potential with a trivial minimum to a Mexican hat potential producing symmetry breaking. Note that there are two coupling constants in the problem, e and λ. Apart from the λ vertex, one has also cubic and quartic interaction terms mixing ϕ and A_μ as represented in the Feynman rules below,

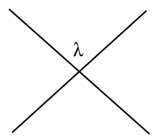
$$ieA_\mu\phi\partial_\mu\phi^* + \mathrm{cc} \longrightarrow$$



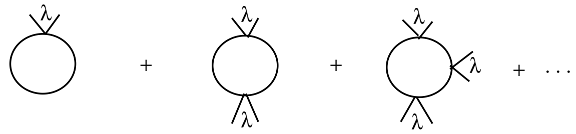
$$e^2 A_\mu{}_{\phi\phi^*} \longrightarrow$$



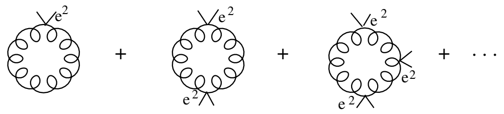
$$\frac{\lambda}{4!}\left(\phi\phi^*\right)^2 \longrightarrow$$



If we are going to compute the effective potential up to first order in $\hbar$, we have the same diagrams already studied for the pure scalar field (except that now the field is complex),



The new diagrams correspond to A_μ loops with $\phi_{cl}\phi_{cl}^*$ insertions,



It is important to note that the diagrams are identical except that, while for scalars the propagator is given by (including a $+i\epsilon$ to handle infrared divergences)

$$\Delta_\phi^{-1}(k) = \frac{1}{k^2 + i\epsilon} \tag{22.46}$$

for the gauge field one has

$$\Delta_{A_\mu}^{-1}(k) = \frac{g_{\mu\nu} + (\alpha - 1)k_\mu k_\nu}{k^2 + i\epsilon} \tag{22.47}$$

In particular, in the Feynman gauge, $\alpha = 1$ and the only difference is the $g_{\mu\nu}$ factor. One can then copy the result for one scalar field taking care of replacing the λ coupling constant by an e^2. Also, one has to take into account that one is dealing in fact with two scalars. The final answer is

$$V_{eff} = \frac{\lambda}{4!}\left(\phi_{cl}\phi_{cl}^*\right)^2 + \left(\frac{5\lambda^2}{1152\pi^2} + \frac{3e^4}{64\pi^2}\right)\left(\phi_{cl}\phi_{cl}^*\right)^2\left(\log\left(\frac{\phi_{cl}\phi_{cl}^*}{\mu^2}\right) - \frac{25}{6}\right) \tag{22.48}$$

To see how the Coleman-Weinberg phenomenon occurs, suppose that $\lambda \sim e^4$. In that case, the λ^2 term in (22.48) can be disregarded and one then has

$$V_{eff} = \frac{\lambda}{4!}\left(\phi_{cl}\phi_{cl}^*\right)^2 + \frac{3e^4}{64\pi^2}\left(\phi_{cl}\phi_{cl}^*\right)^2\left(\log\left(\frac{\phi_{cl}\phi_{cl}^*}{\mu^2}\right) - \frac{25}{6}\right) \tag{22.49}$$

Let us search for the value $\langle\phi\rangle$ that minimizes this effective potential. We have

$$\begin{aligned} \left.\frac{dV_{eff}}{d\phi_{cl}^*}\right|_{\phi_{cl}^* = \langle\phi\rangle^*} &= \frac{\lambda}{3!}\langle\phi\rangle^*\langle\phi\rangle^2 + \frac{12e^4}{64\pi^2}\langle\phi\rangle^*\langle\phi\rangle^2\left(\log\left(\frac{\phi_{cl}\phi_{cl}^*}{\mu^2}\right) - \frac{25}{6}\right) \\ &\quad + \frac{6e^4}{64\pi^2}\langle\phi\rangle^*\langle\phi\rangle^2 \end{aligned} \tag{22.50}$$

Now, since the normalization point μ^2 is arbitrary, we can chose it so as to cancel the logarithm term. This amounts to put $\mu^2 = \langle\phi\rangle^*\langle\phi\rangle$. One then has for the minima

$$\left.\frac{dV_{eff}}{d\phi^*_{cl}}\right|_{\phi^*_{cl}=\langle\phi\rangle^*} = \left(\frac{\lambda}{6} - \frac{11e^4}{16\pi^2}\right)\langle\phi\rangle^*\langle\phi\rangle^2 = 0 \tag{22.51}$$

Apart from the trivial solution $\langle\phi\rangle = 0$ there is a second possibility: a non-trivial minimum for $\langle\phi\rangle \neq 0$ whenever the following relation between coupling constants hold

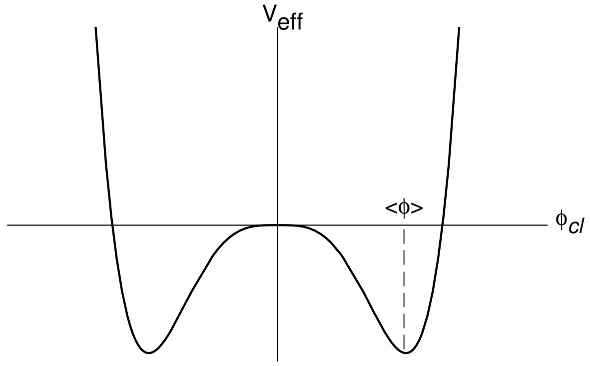
$$\lambda = \frac{33}{8\pi^2}e^4 \tag{22.52}$$

Hence, the effective potential including the first order quantum corrections supports a non-trivial minimum $\langle\phi\rangle \neq 0$ provided the coupling constant satisfy (22.52). It is important to note that we started with two dimensionless coupling constants and we ended with just one! Instead, we have gained a dimensionfull quantity, $\langle\phi\rangle$.

This phenomenon is known as *dimensional transmutation*: we traded one of the two dimensionless coupling constant (e, λ) for a dimensionfull constants, say $(e, \langle\phi\rangle)$. And we have an effective potential which has a non-trivial minimum

$$V_{eff} = \frac{3e^4}{64\pi^2}|\phi_{cl}|^2\left(\log\left(\frac{|\phi_{cl}|^2}{|\langle\phi\rangle|^2}\right) - \frac{1}{2}\right) \tag{22.53}$$

The shape of this effective potential is



Problems

Problem 1: Pass from (22.8) to (22.9).

Problem 2: Obtain (22.17).

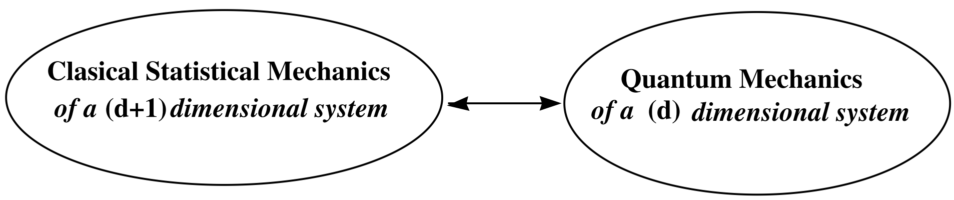
Problem 3: Obtain (22.48).

Clase 23

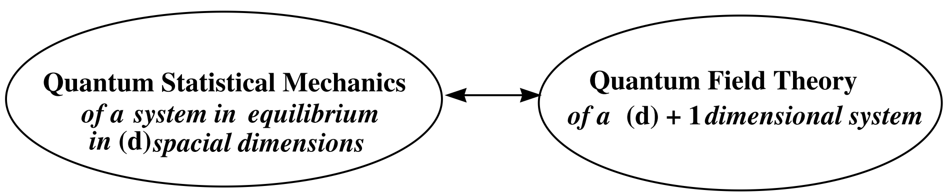
Analogies between Statistical Mechanics and Quantum Field Theory

There are two important analogies, between Statistical Mechanics and Quantum Field Theory: one for **classical** Statistical Mechanics, the second for **quantum** Statistical Mechanics. These analogies relate systems defined in different space dimensions. It is better to start with a simple drawing to clarify the connections:

FIRST ANALOGY



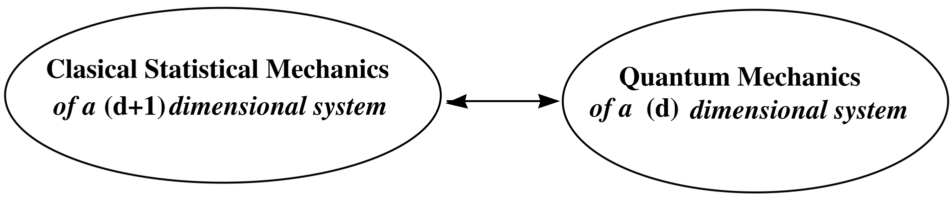
SECOND ANALOGY



When we write (d) or $(d+1)$ we indicate spacial dimensions. When we write $(d)+1$ we are indicating space-time dimensions.

Whether these analogies are just coincidences which can be exploited to attack problems in QFT using Statistical Mechanics intuition and viceversa or there are deep reasons for them, connected with properties of space-time is not clear at present. In his very nice book *Gauge Field and Strings*, Polyakov [9] advances his idea that the second option may be the correct one but he does not extends on the arguments on this.

FIRST ANALOGY



To prove this connection we can start from the right hand side, that is a certain quantum system in d dimensions and relate it with a certain statistical mechanical problem. For simplicity consider first a mechanical problem in $d=1$ space dimensions with Hamiltonian

$$H = \frac{p^2}{2m} + V(x) = p\dot{x} - L \tag{23.1}$$

where L is the corresponding Lagrangian such that

$$S = \int_0^T dtL \tag{23.2}$$

is the action of the system, defined in the interval $(0,T)$.

We have seen that the transition amplitude for the quantum mechanical system with this Hamiltonian, with initial condition $x(0)=x$

and final condition $x(T) = x'$,

$$\langle x'| \exp\left(\frac{i}{\hbar} HT\right) |x\rangle = Z^{QM}[x, x'; T, \hbar] \tag{23.3}$$

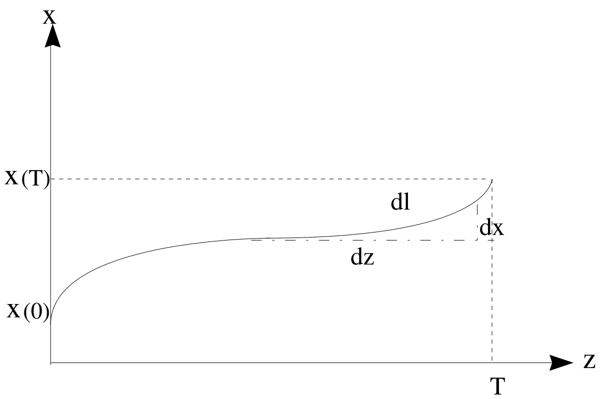
can be written as a path integral,

$$\begin{aligned} Z^{QM}[x, x'; T, \hbar] &= \int \mathcal{D}x(t) \exp\left(\frac{i}{\hbar} S[x, \dot{x}]\right) \\ &= \int \mathcal{D}x(t) \exp\left(\frac{i}{\hbar} \int_0^T dt \left(\frac{1}{2} m\dot{x}(t)^2 - V(x)\right)\right) \end{aligned} \tag{23.4}$$

Let us now pass to Euclidean space where the path-integral in (23.4) can be properly defined,

$$\begin{aligned} Z_E^{QM}[x, x'; T, \hbar] &= Z^{QM}[x, x'; iT, \hbar] = \langle x'| \exp\left(-\frac{1}{\hbar} HT\right) |x\rangle \\ &= \int \mathcal{D}x(t) \exp\left(-\frac{1}{\hbar} \int_0^T dt \left(\frac{1}{2} m\dot{x}(t)^2 + V(x)\right)\right) \end{aligned} \tag{23.5}$$

We shall now connect the treatment above, which is of quantum character, with a classical problem in 2 spatial dimensions, namely that of a string of length T and tension m which has its ends fixed at certain points x and x' and it is subject to a potential $V(x)$,



The potential energy of such a system picks two contributions, one related to the elastic potential energy which is proportional to the string elongation Δl,

$$\Delta l = \sqrt{dx^2 + dz^2} = dz\sqrt{1 + (dx/dz)^2} \approx dz + \frac{1}{2}\left(\frac{dx}{dz}\right)^2 dz \qquad (23.6)$$

The first, constant term proportional to dz can be disregarded since it just represents the arbitrariness in defining the zero of the scale. Adding to this result the potential energy associated to the potential $V(x)$, we have for the total potential energy of the string of length T

$$E_{pot} = \int_0^T dz \left(\frac{m}{2}\left(\frac{dx}{dz}\right)^2 + V(x)\right) \qquad (23.7)$$

Now, from the point of view of *classical statistical mechanics*, applying the Boltzmann principle, one has for the partition function of the system at temperature $Temp = 1/\beta$ (we are using units such that $k = 1$),

$$Z^{ClSM}[x, x'; T, \beta] = \tilde{\mathcal{N}} \sum_{\mathcal{S}} \exp\left(-\beta(E_{kin} + E_{pot})\right) \qquad (23.8)$$

The sum runs over all string configurations denoted by $\mathcal{S}$.

Since the string is fixed at the ends, the kinetic energy does not depend on the precise values of x at $z = 0$ and $z = T$. We then include its contribution in the normalization constant and write

$$Z^{ClSM}[x, x'; T, \beta] = \mathcal{N} \sum_{\mathcal{S}} \exp\left(-\beta E_{pot}\right) \qquad (23.9)$$

In fact, the sum in this expression is just the path integral over all string configurations $\mathcal{S}$ so that we can write

$$Z^{ClSM}[x, x'; T, \beta] = \int \mathcal{D}x(z) \exp\left(-\beta \int_0^T dz \left(\frac{m}{2}\left(\frac{dx}{dz}\right)^2 + V(x)\right)\right) \qquad (23.10)$$

Comparing (23.5) with (23.10), we can write

$$Z^{ClSM}[x, x'; T, \beta] = Z_E^{QM}[x, x'; T, \hbar] \qquad (23.11)$$

provided the temperature $1/\beta$ in the l.h.s. is identified with $1/\hbar$ in the r.h.s.

We have then established the identity between *the classical partition function for a string of length* T *at temperature* $1/\beta$ and *the transition amplitude for a quantum particle passing from an initial state* $|x(0)\rangle$ *to a final one* $|x(T)\rangle$ provided one identifies the temperature in the statistical mechanical system with $1/\hbar$ in the quantum mechanical one.

This was for a system in one dimension. It is trivial to pass to a field theory and establish the analogous to (23.11). As an example, one has for a scalar field in d spatial dimensions

$$Z^{ClSM} = Z_E^{QFT} \tag{23.12}$$

where

$$Z_E^{ClSM} = \int \mathcal{D}\phi \exp\left(-\beta \int d^d x \left(\frac{1}{2}(\nabla\phi)^2 + V[\phi]\right)\right) \tag{23.13}$$

and

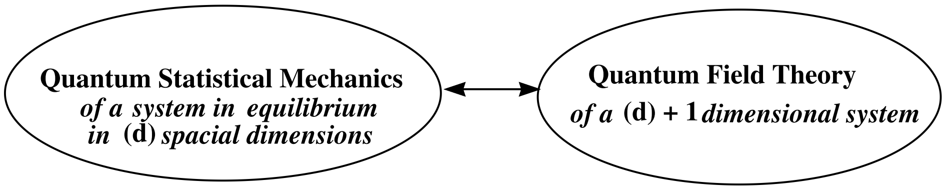
$$Z_E^{QFT} = \int \mathcal{D}\phi \exp\left(-\frac{1}{\hbar} \int d^d x \left(\frac{1}{2}(\nabla\phi)^2 + V[\phi]\right)\right) \tag{23.14}$$

We can then establish the following dictionary:

Dictionary

ClSM	*QFT*
space d	space $(d-1)$ + time 1
Temperature $1/\beta$	$1/\hbar$
Boltzmann sum over conf.	Path-integral over field conf.
Energy	Euclidean action
Transfer matrix	Hamiltonian
Deffects, dislocations	Instantons

SECOND ANALOGY



As before, we consider for simplicity a scalar theory with Hamiltonian $H = H[\phi, \pi]$. In the operator approach, $\phi = \phi(\vec{x}, t)$ is a Heisenberg operator and $\pi = \pi(\vec{x}, t)$ its conjugate momentum. We learnt how to compute the transition amplitude between a state $\phi(\vec{x}, t = 0) = |\phi_0\rangle$ at $t = 0$ and $\phi(\vec{x}, t = T) = |\phi_T\rangle$ at $t = T$ as a path-integral,

$$\langle\phi_T|\exp\left(-\frac{i}{\hbar}HT|\phi_0\rangle\right) = \mathcal{N}\int\mathcal{D}\phi\mathcal{D}\pi\exp\left(\frac{i}{\hbar}\int_0^T dt\int d^{d-1}x(\pi\dot{\phi} - H)\right) \tag{23.15}$$

The integral extends over all field configurations such that that one has at the boundaries in time that $\phi(\vec{x}, 0) = \phi_0$ and $\phi(\vec{x}, T) = \phi_T$.

Let us now consider the following renaming of parameters and variables

$$\frac{iT}{\hbar} = \beta\,, \qquad \frac{it}{\hbar} = \tau \tag{23.16}$$

Eq. (23.15) becomes

$$\langle\phi_T|\exp\left(-\beta H\right)|\phi_0\rangle = \mathcal{N}\int\mathcal{D}\phi\mathcal{D}\pi\exp\left(\int_0^\beta d\tau\int d^{d-1}x(\frac{i}{\hbar}\pi\frac{d\phi}{d\tau} - H)\right) \tag{23.17}$$

At this point we shall put $\hbar = 1$ and then recuperate $\hbar$ at the end just by dimensional analysis.

For Hamiltonians that are quadratic in the momenta we can integrate out π getting (see the end of this lecture)

$$\langle\phi_T|\exp\left(-\beta H\right)|\phi_0\rangle = \mathcal{N}(\beta)\int\mathcal{D}\phi\exp\left(-\int_0^\beta d\tau\int d^{d-1}xL_E(\phi, \partial_\mu\phi)\right) \tag{23.18}$$

with L_E the Euclidean Lagrangian for the model. The factor $\mathcal{N}(\beta)$ arises from integration over momenta and we have made explicit its dependence on β.

Now, a crucial point: let us consider the case in which initial and final states coincides, $|\phi_0\rangle = |\phi_T\rangle$ and call $\{\phi_0\}$ the set of all fields satisfying that $\phi(\vec{x}, 0) = \phi(\vec{x}, T)$. Moreover, let us sum in the l.h.s. of (23.18) over all such configurations, which means we are computing a trace,

$$\sum_{\{\phi_0\}} \langle\phi_0| \exp(-\beta H) |\phi_0\rangle = \mathcal{N}(\beta) \int_{\{\phi_0\}} \mathcal{D}\phi \exp\left(-\int_0^\beta d\tau \int d^{d-1}x L_E(\phi, \partial_\mu\phi)\right) \tag{23.19}$$

We recognize in the l.h.s. of (23.19) the Quantum Partition Function for a mechanical statistical system in $d-1$ dimensional space, at $T = 1/\beta$. We can then rewrite (23.19) in the form

$$Z = \operatorname{tr} \exp(-\beta H) = \mathcal{N}(\beta) \int_{periodic} \mathcal{D}\phi \exp\left(-\int_0^\beta d\tau \int d^{d-1}x L_E(\phi, \partial_\mu\phi)\right) \tag{23.20}$$

where the subscript *periodic* indicates that we have to integrate over all field configurations which are periodic in τ in the interval $(0, \beta)$.

The two following limits are of interest

- $T \to 0$ ($\beta \to \infty$): the r.h.s. corresponds to a quantum field theory in d dimensional Euclidean space-time.
- $T \to \infty$ ($\beta \to 0$): the r.h.s. corresponds to a quantum field theory in $d-1$ Euclidean space-time.

In particular, this means that a 4-dimensional QFT at $T = \infty$ is effectively described by a QFT in 3 dimensions.

Another point to remark concerns the fact that the in the classical limit $\hbar \to 0$ the quantum Hamiltonian becomes the classical one and then the l.h.s. in (23.20) becomes the Gibbs-Boltzmann partition function. Please note that evrywhere we have written $\beta = 1/T$ because we were taking units such that the Boltzmann constant $k = 1$. Otherwise, we have to write $\beta = 1/(kT)$.

It is interesting to observe that for charged fields, together with a conserved charge density N we can introduce the chemical potential μ in the form

$$
\begin{aligned}
Z &= \operatorname{tr}\exp\left(-\beta(H-\mu N)\right)\\
& \mathcal{N}(\beta)\int_{periodic}\mathcal{D}\phi\exp\left(-\int_0^\beta d\tau\int d^{d-1}x(L_E(\phi,\partial_\mu\phi)-\mu N)\right)
\end{aligned}
\tag{23.21}
$$

So, we can proceed to do calculations in Quantum Statistical mechanics exactly as we did for Quantum Field Theory, except that we have to consider periodic boundary conditions in τ. To see how this affects computations, let us consider the case of a $3+1$ dimensional scalar theory with Lagrangian

$$
L=\frac{1}{2}\left(\partial_\mu\phi\right)^2-\frac{m^2}{2}\phi^2-\frac{\lambda}{4!}\phi^4 \tag{23.22}
$$

What happens with Feynman propagators when computing some quantity using a perturbative expansion? It is precisely at this level that one should take care of periodic boundary conditions. The quadratic (free) action that appears in the partition functions is

$$
S_0=\frac{1}{2}\int_0^\tau\int d^3x\left((\partial_0\phi)^2+(\partial_i\phi)^2-m^2\phi^2\right) \tag{23.23}
$$

In order to derive the propagator, one has to Fourier transform the quadratic action. Because of periocid boundary conditions in the τ variable, one will have a Fourier series associated with this variable while, as usual, Fourier integrals for the spacial ones. One then writes

$$
\phi(\vec{x},\tau)=\frac{1}{\beta}\sum_{n=-\infty}^{\infty}\int\frac{d^3k}{(2\pi)^3}\exp\left(i\vec{k}\vec{x}+i\omega_n\tau\right)\phi_n(\vec{k}) \tag{23.24}
$$

with

$$
\omega_n=\frac{2\pi n}{\beta} \tag{23.25}
$$

so that the action (23.23) becomes

$$
S_0=-\frac{1}{2\beta}\sum_{n=-\infty}^{\infty}\int\frac{d^3k}{(2\pi)^3}(\omega_n^2+\vec{k}^2+m^2)\phi_n(\vec{k})\phi_{-n}(-\vec{k}) \tag{23.26}
$$

so that we can immediatly infer the propagator which now reads for a scalar field

$$\Delta(\omega_n, \vec{k}; m) = \frac{1}{\omega_n^2 + \vec{k}^2 + m^2} \tag{23.27}$$

Feynman rules have now to be modified accordingly

$$\begin{array}{rcl} \displaystyle\int \frac{d^4k}{(2\pi)^4} & \longrightarrow & \displaystyle\frac{i}{\beta}\sum_n \int \frac{d^3k}{(2\pi)^3} \\ k_0 & \longrightarrow & i\omega_n \\ (2\pi)^4\delta^{(4)}(k_1 + k_2 + \ldots) & \longrightarrow & -i(2\pi)^3\beta\delta_{\omega_1+\omega_2+\ldots,0}\delta^{(3)}(\vec{k}_1 + \vec{k}_2 + \ldots) \end{array} \tag{23.28}$$

We can start by considering the free case for wich the partition function becomes exactly integrable

$$Z_0 = \mathcal{N}(\beta)\int \mathcal{D}\phi \exp\left(-S_0\right) = \mathcal{N}(\beta)\mathrm{det}^{-1/2}D \tag{23.29}$$

with the Fourier transformed of the operator D given by

$$\tilde{D} = \omega_n^2 + \vec{k}^2 + m^2 \tag{23.30}$$

with

$$\omega_n^2 = \frac{4\pi^2 n^2}{\beta^2} \tag{23.31}$$

Then, we have for the partition function

$$\log Z_0 = \log \mathcal{N}(\beta) - \frac{1}{2}\log\det D \tag{23.32}$$

or, applying what we learnt when computing effective potentials

$$\log Z_0 = \log \mathcal{N}(\beta) - \frac{1}{2}\int\sum_n \frac{d^3k}{(2\pi)^3}\log\left(\omega_n^2 + \vec{k}^2 + m^2\right) \tag{23.33}$$

Let us compute the sum in (23.33)

$$\sum \log\left(\omega_n^2 + \omega_{\vec{k}}^2\right) = \sum_n \int_{1/\beta^2}^{\omega_{\vec{k}}^2} dx^2 \frac{1}{\omega_n^2 + x^2} + \sum_n \log(\omega_n^2 + \frac{1}{\beta^2}) \tag{23.34}$$

where we have called

$$\omega_{\vec{k}}^2 = \vec{k}^2 + m^2 \tag{23.35}$$

We shal see that the second term in (23.34) cancels $\mathcal{N}(\beta)$. Concerning the first one, we shall use the "Regge trick",

$$\sum_n \int_{1/\beta^2}^{\omega_{\vec{k}}^2} dx^2 \frac{1}{\omega_n^2 + x^2} = \frac{\beta}{2} \int d\omega \cot\left(\frac{\beta\omega}{2}\right) \frac{1}{\omega^2 + x^2} \tag{23.36}$$

where one choses the function $\cot(\beta\omega/2)$ so that its poles at $\omega = 2\pi n/\beta$ are such that they reproduce, via the Cauchy theorem, the sum in the l.h.s.. Then, one can first perform the integral over x^2, then the integral over ω finally obtaining (see Problem 1)

$$\begin{aligned} \log Z_0 &= \int \frac{d^3k}{(2\pi)^3} \log \operatorname{cosech}\left(\frac{\beta\omega_{\vec{k}}}{2}\right) + C \\ &= \int \frac{d^3k}{(2\pi)^3} \left(-\frac{\beta\omega_{\vec{k}}}{2} - \log(1 - \exp(-\beta\omega_{\vec{k}})\right) + C \end{aligned} \tag{23.37}$$

which gives the correct result for the partition function of an ideal Bose gas, included the zero point energy.

We can recapitulate some important steps in our derivation:

- There is an i in front of the momentum π in the path-integral over phase space defining the partition function . After integration over momentum, this leads to a $-$ sign in front of an Euclidean Lagrangian.

- The requierement of periodicity over the finite range of the τ integration changes integrations over $p_0 = energy$ (for example in Feynman diagrams) into energy sums. For bosons this amounts to replace p_0 by $2\pi n/\beta$ and sum over integer n instead of integrating over p_0.

- Although we have not yet discussed the case of fermions, we can advance the answer on how to treat them at finite temperature: the only change is that the correct conditions at the borders of the $(0, \beta)$ interval should be in this case anti-periodic (in contrast with the periodic boundary conditions for bosons).

- Last, but not least and paraphrasing R.Carver [69]

  What we talk about when we talk about Quantum Field Theory at finite temperature?

  Consider the field ϕ in a $\lambda\phi^4$ theory. As a quantum field theory, the scalar field ϕ describes a particles of spin zero and mass m with a quartic self-interaction of intensity λ. Now, the system of particles described by this QFT can be considered in thermal equilibrium at temperature $T = 1/\beta$. That is, the infinite system described by the $\lambda\phi^4$ model is supposed to provide a heat-bath that puts the ϕ particles at temperature T. Then, static properties of the system can be studied using the partition function (23.20)

Calculation of $\mathcal{N}(\beta)$

The temperature dependent coefficient $\mathcal{N}(\beta)$ arises from the integration over momenta in the phase space path-integral,

$$\mathrm{tr}\exp(-\beta H) = \mathcal{N} \int \mathcal{D}\phi\mathcal{D}\pi \exp\left(-\int_0^\beta d\tau(\pi\dot{\phi} - H)\right) \tag{23.38}$$

with

$$H = \frac{\pi^2}{2} + \frac{1}{2}(\vec{\nabla}\phi)^2 + m^2\phi^2 \tag{23.39}$$

As we did when introducing the path-integral in quantum mechanics, let us divide the “time” τ in intervals $\epsilon = 2\pi N/\beta$ so that, after integrating over π we end with

$$\begin{aligned}\mathrm{tr}\exp(-\beta H) = \mathcal{N} \lim_{N\to\infty}\int_{-\infty}^{\infty}\prod_{j=1}^{N}\frac{d\phi_j}{\sqrt{\epsilon}}\exp\Bigg(-\frac{1}{2}\sum_j\int d^3x\left(\frac{1}{\epsilon}(\phi_j - \phi_{j-1})^2 +\right.\\ \left.\epsilon(\vec{\nabla}\phi)^2 + \epsilon m^2\phi^2\right)\Bigg)\end{aligned} \tag{23.40}$$

or, transforming Fourier

$$\mathrm{tr}\exp(-\beta H) = \mathcal{N} \lim_{N\to\infty}\int_{-\infty}^{\infty}\prod_{j=1}^{N}\frac{d\tilde{\phi}_j}{\sqrt{\epsilon}}\exp\Bigg(-\frac{1}{2}\sum_j\int\frac{d^3k}{(2\pi)^3}\times$$

$$\left(\frac{1}{\epsilon}(\tilde{\phi}_j(\vec{k}) - \tilde{\phi}_{j-1}(\vec{k}))^2 + \epsilon\omega_{\vec{k}}^2\tilde{\phi}^2(\vec{k}) + \epsilon m^2\tilde{\phi}^2(k)\right)\Bigg) \tag{23.41}$$

At this point we recognize the path-integral for a system of oscillators with frequency ω_k^2 which we have already computed in the first part of this course. We just copy the answer

$$\begin{aligned}\operatorname{tr}\exp(-\beta H) = \lim_{N\to\infty}\Bigg(\int_{-\infty}^{\infty}\prod_{j=1}^{N} d\tilde{\phi}_j\sqrt{\frac{\omega_k}{\sinh\left(\beta\omega_k\right)}}\times \\ \exp\left(-\frac{2\omega_k\tilde{\phi}_j(\vec{k})}{\sinh\left(\beta\omega_k\right)}\right)(\cosh(\beta\omega_k)-1)\Bigg)^N\end{aligned} \tag{23.42}$$

or

$$\operatorname{tr}\exp(-\beta H) = \int\frac{d^3k}{(2\pi)^3}\log\operatorname{cosech}(\frac{\beta\omega_k}{2} + C \tag{23.43}$$

from which we confirm that $\mathcal{N}(\beta)$ cancels precisely the sum in (21.24) so that we conclude that

$$\mathcal{N}(\beta) = -\sum_n\log\left(\omega_n^2 + \frac{1}{\beta^2}\right) \tag{23.44}$$

Problems

Problem 1: Pass from (23.36) to (23.37).

Clase 24

The effective potential at finite temperature

It is not difficult to repeat the calculation that we have already done for the effective potential for a $\lambda\phi^4$ theory but now at finite temperature. As explained, one has just to take into account periodic boundary conditions in "time" in the interval $(0,\beta)$ this amounting to the change $\int d^4k \to \sum_n \int d^3k$ in Feynman diagrams.

We shall concentrate in a massless $\lambda\phi^4$ theory. For the one loop correction to the potential, we have just to copy formulæ(20.1)-(20.2), but writing the integral over energy as a sum and not including, for the moment, counterterms that will have to be adjusted appropriatedly,

$$V_{eff} = \frac{\lambda}{4!}\phi_{cl}^4 + \hbar V_1[\phi_{cl}] + O(\hbar^2) \tag{24.1}$$

where

$$V_1[\phi_{cl}] \;=\; \frac{1}{2\beta}\sum_{n=-\infty}^{\infty}\int\frac{d^3k}{(2\pi)^3}\left(\log\left(1+\frac{\lambda\phi_{cl}^2}{2(\omega_n^2+\vec{k}^2)}\right)\right) \tag{24.2}$$

or,

$$V_1[\phi_{cl}] = \frac{1}{2\pi}\int\frac{d^3k}{(2\pi)^3}\sum_{n=-\infty}^{\infty}\log\left(E_{\vec{k}}^2+\frac{4\pi^2n^2}{\beta^2}\right) + \text{constant} \tag{24.3}$$

where

$$E_{\vec{k}}^2 = \vec{k}^2 + \frac{\lambda\phi_{cl}^2}{2} \tag{24.4}$$

and we have included in the constant term a divergent (β-dependent) sum which will be determined in what follows. The sum in (24.3) is also divergent but if one differentiates with respect to $E_{\vec{k}}$ it becomes convergent,

$$\frac{1}{2}\frac{d}{dE}\sum_{n=-\infty}^{\infty}\log(n^2+E^2) = \sum_{n=-\infty}^{\infty}\frac{E}{n^2+E^2} = -\frac{1}{2E}+\frac{\pi}{2}\coth(\pi E) \quad (24.5)$$

Once the sum is performed, we can integrate over E; one can then see that the resulting divergence precisely cancels that in the constant in (24.3) so that one finally obtains

$$V_1[\phi_{cl}] = \int \frac{d^3k}{(2\pi)^3}\left(\frac{E_{\vec{k}}}{2} + \frac{1}{\beta}\log\left(1-\exp(-\beta E_{\vec{k}})\right)\right) \quad (24.6)$$

The first term in (24.6) can be written as an integral over d^4k,

$$\begin{aligned} \int \frac{d^3k}{(2\pi)^3}\frac{E_{\vec{k}}}{2} &= \frac{1}{2}\int \frac{d^3k}{(2\pi)^3}\left(\vec{k}^2 + \frac{\lambda\phi_{cl}}{2}\right) \\ &= \frac{1}{2}\int \frac{d^4k}{(2\pi)^4}\log\left(k_0^2+\vec{k}^2+\frac{\lambda\phi_{cl}^2}{2}\right) \end{aligned} \quad (24.7)$$

so that we can write (we put $\hbar = 1$)

$$\begin{aligned} V_{eff} &= \frac{\lambda}{4!}\phi_{cl}^4 + \frac{1}{2}\int\frac{d^4k}{(2\pi)^4}\log\left(k^2+\frac{\lambda\phi_{cl}^2}{2}\right) + \\ &\quad \frac{1}{\beta}\int\frac{d^3k}{(2\pi)^3}\log\left(1-\exp(-\beta E_{\vec{k}})\right) \end{aligned} \quad (24.8)$$

The first line in (24.8) is just the result we got at $T = 0$ while the second one concentrates the effect of finite temperature. We can then write, to the one loop order,

$$\begin{aligned} V_{eff} &= V_{eff}(T=0) + \frac{1}{2\pi^2\beta}\int k^2 dk \log\left(1-\exp(-\beta E_{\vec{k}}\right) \\ &= V_{eff}(T=0) + \frac{1}{2\pi^2\beta^4} I[\frac{\lambda\phi_{cl}\beta^2}{2}] \end{aligned} \quad (24.9)$$

where we have defined

$$I[y] = \int_0^\infty dx x^2 \log\left(1 - \exp\left(-\sqrt{x^2 + y^2}\right)\right) \tag{24.10}$$

It is easy to compute $I(y)$ the limit of very high temperatures in which $T \to \infty$ or $\phi_{cl}^2 \beta \to 0$. One finds

$$I[y] = -\frac{\pi^4}{45} + \frac{\pi^2}{12} y^2 + +O(y^4) \tag{24.11}$$

so that

$$V_{eff}[T \gg 1] = V_{eff}[T = 0] + \frac{\lambda}{48\beta^2}\phi_{cl}^2 - \frac{\pi^2}{90\beta^4} + O((\phi_{cl}\beta)^4) \tag{24.12}$$

The second term in the r.h.s. of (24.12) is a temperature dependent mass-term generated at the one loop level. We can then expect to have a dynamical symmetry breakdown à la Coleman-Weinberg, this time a T-dependent effect. Writing explicitly the $T = 0$ effective potential, we have

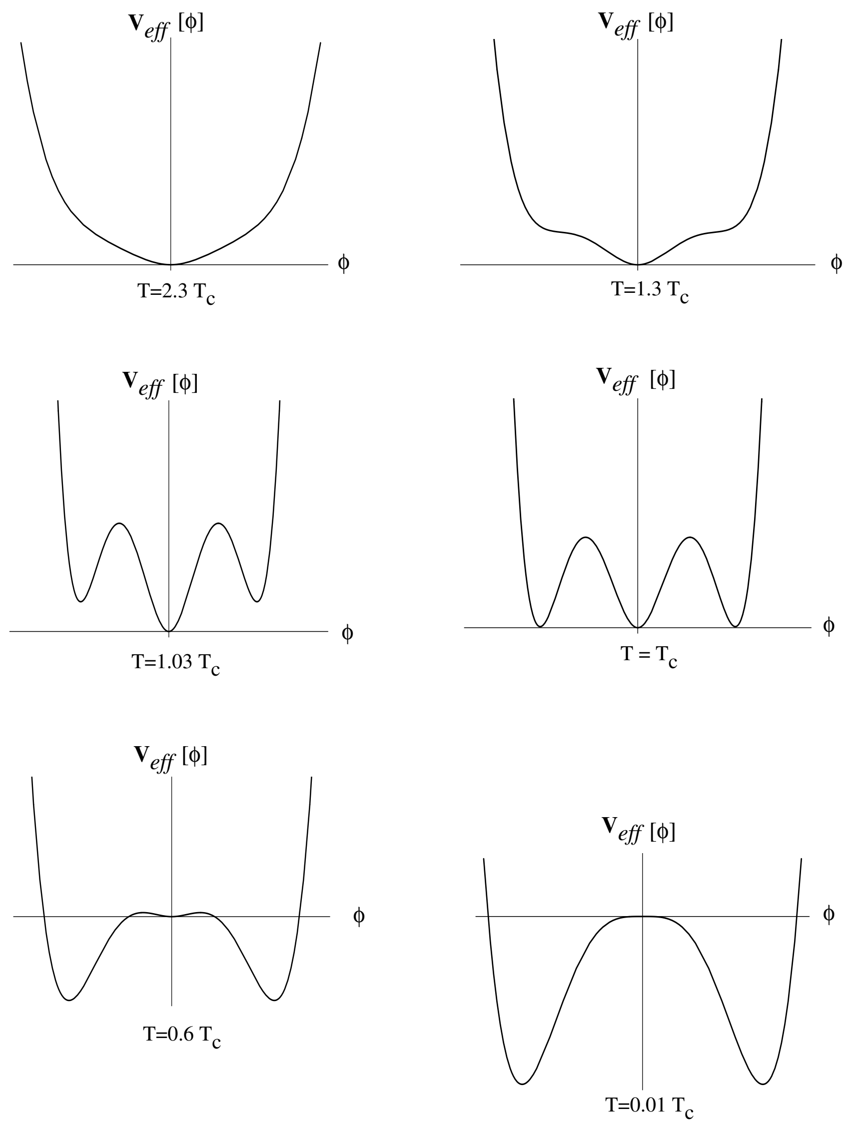
$$V_{eff} = \frac{\lambda}{4}\phi_{cl}^4 + \frac{1}{8\pi^2}\left(\frac{\lambda\phi_{cl}^2}{2}\right)^2\left(\log\left(\frac{\phi_{cl}^2}{\langle\phi\rangle^2}\right) - \frac{25}{6}\right) + \frac{\lambda}{48}T^2\phi_{cl}^2 \tag{24.13}$$

We see that, at high temperatures, the classical potential $\lambda\phi_{cl}^4$ and the $T^2\phi_{cl}^2$ correction dominate and the potential behaves as an harmonic one with a trivial minimum $\langle\phi\rangle = 0$. But as the temperature decreases, the effect of the term containing the logarithm becomes more important and finally, for sufficiently low temperature the potential develops non-trivial minima. To determine the "critical" temperature T_c such that $\langle\phi\rangle \neq 0$ we compute

$$\left.\frac{dV_{eff}}{dT}\right|_{\phi_{cl}} = 0 \Rightarrow T_c = \frac{3\lambda}{4\pi^2}\langle\phi\rangle^2 \tag{24.14}$$

Then, as can be seen in the figure in the next page, for $T > T_c$ the potential is just a symmetric one with a trivial minimum but, as the temperature decreases and reaches T_c the potential develops a double minima structure and for $T \leq T_c$ we have symmetry breaking.

Before leaving this calculation let us emphasize that no temperature-dependent infinite renormalizations were necessary here. Using the expression of the effective potential renormalized at $T = 0$ we automatically obtained a finite result. This is not particular of the present $\lambda\phi^4$ theory: in general, divergencies in field theories at finite temperature are temperature independent and then the substractions coincide with those of the $T = 0$ renoprmalized theory.



Gauge theories at finite temperature

As usual, subtleties related to gauge invariance and gauge fixing reflect in the treatment of a gauge field theory at finite temperature. We consider the Yang-Mills Lagrangian

$$L = \frac{1}{4}\mathrm{tr}F_{\mu\nu}F^{\mu\nu} \tag{24.15}$$

Since we want to present a proper derivation of the partition function of this system within the path-integral approach, we shall start from the Hamiltonian H formulation of the theory. We write H in the form H%

$$H = \int d^3x \frac{1}{2}\left(e^2(E^a)^2 + \frac{1}{e^2}(B^a)^2\right) \tag{24.16}$$

where e is the YM coupling constant,

$$E^{a\,i} = \frac{1}{e}F^{aoi} \qquad B^{a\,i} = \frac{e}{2}\varepsilon^{ijk}F^{ajk} \tag{24.17}$$

with $a = 1, 2, \ldots, \dim G$. As usual, it will be convenient to work in the physical $A_0 = 0$ gauge and take $\vec{A}^a$ and $\vec{E}^a$ as canolnically conjugate variables, with $\vec{B}^a = \vec{B}^a(\vec{A})$

Given the Hilbert space spanned by $\vec{A}^a$, we consider to states $\vec{A}_I(\vec{x}) = \vec{A}(0, \vec{x})$ and $\vec{A}_F(\vec{x}) = \vec{A}(\beta, \vec{x})$ and write the transition amplitude

$$\begin{aligned} &\langle \vec{A}_F(\vec{x})| \exp(-\beta H)|\vec{A}_I(\vec{x})\rangle \\ &\quad = \int D\vec{A} \exp\left(-\frac{1}{e^2}\int_0^\beta d\tau \int d^3x \left((\dot{\vec{A}}^a)^2 + (\vec{B}^a)^2\right)\right) \end{aligned} \tag{24.18}$$

this formula can be though as the result of naively integrating over $\vec{E}^a$ in the path-integral recipe for transition amplitudes.

In order to compute the partition function, we have to put $\vec{A}_F = \vec{A}_I = \vec{A}$ and sum over all such $\vec{A}$,

$$\begin{aligned} Z &= \sum_{A_F = A_I} \langle \vec{A}| \exp(-\beta H)|A\rangle \\ &= \sum_{A_F = A_I} \int D\vec{A} \exp\left(-\frac{1}{e^2}\int_0^\beta d\tau \int d^3x \left((\dot{\vec{A}}^a)^2 + (\vec{B}^a)^2\right)\right) \end{aligned} \tag{24.19}$$

We have learnt that (24.19) does not tell all the story: because we are dealing with a gauge theory, the Gauss law on $\vec{E}^a$ has to be imposed using an appropriate Lagrange multiplier. Now the Gauss law constraint commutes with the Hamiltonian, and this means it does not change in time. Thus, one should imposes the constraint just at "one time" (say $\tau = \beta$) and everything will work. Since the generator of time independent gauge transformations is

$$\Omega(\Lambda) = \exp\left(-i\int d^3x \Lambda(\vec{x}) D\vec{E}\right) \tag{24.20}$$

the constraint can be implemented just by including an integral over Λ, say at $\tau = 0$. One should indistinctly then write $\Lambda(\vec{x})$ or $\Lambda(0, \vec{x})$.

$$Z = \sum_{A_F = A_I} \int D\Lambda(\vec{x}) D\vec{A}(\tau, \vec{x})$$
$$\exp\left(-\frac{1}{e^2}\int_0^\beta d\tau \int d^3x \left((\dot{\vec{A}}^a)^2 + (\vec{B}^a)^2\right)\right) \exp\left(-i\int d^3x \Lambda(\vec{x}) D\vec{E}\right) \tag{24.21}$$

Now, we can define a time independent gauge group element $g(\vec{x})$ in the form

$$g(\vec{x}) = \exp(i\Lambda(\vec{x})) \tag{24.22}$$

and then, intead of including the second exponential in (24.21), we can instead write

$$Z = \sum_{A_F = A_I^g} \int D\Lambda(\vec{x}) D\vec{A}(\tau, \vec{x})$$
$$\exp\left(-\frac{1}{e^2}\int_0^\beta d\tau \int d^3x \left((\dot{\vec{A}}^a)^2 + (\vec{B}^a)^2\right)\right) \tag{24.23}$$

Here we have written

$$|A_i(\vec{x}, 0\rangle = \mathcal{G}^{-1} |A_i^g(\vec{x}, 0\rangle \tag{24.24}$$

where we used the fact that the Gauss law is the generator of gauge transformations to write

$$\mathcal{G} = \exp\left(-i\int d^3x \Lambda(\vec{x}) D\vec{E}\right) \tag{24.25}$$

Since the Lagrange multiplies depens only on $\vec{x}$ and we have written $\Lambda(\vec{x})$. Then, it cannot be naively interpreted as the zeroth component of the gauge field which should be $A_0(\tau\vec{x})$. However, we can imposed the constraint redundantly, at all times (i.e., in the whole $(0,\beta)$ interval). Then, if we perform an appropriate gauge transformation, the twist in the PBC disappears at the price of a shift $\dot{\vec{A}}^2 \to \left(\dot{\vec{A}} - D\Lambda(\tau,x)\right)^2$ generated when integrating over $\vec{E}$. Calling $\Lambda(\tau,\vec{x}) \equiv A_0(\tau,\vec{x})$, one then ends with the usual formula for the Lagrangian version of the partition function,

$$Z = \sum_{A_\mu(\beta,\vec{x})=A_\mu(0,\vec{x})} \int DA_\mu \exp\left(-\frac{1}{4}\int_0^\beta d\tau \int d^3x F_{\mu\nu}F^{\mu\nu}\right) \tag{24.26}$$

We were not very careful in handling the constraint since we didn't added corresponding variables leading to the gauge-fixed result. But this part of the treatment is exactly the same as in the $T=0$ case so we know we should end with a Faddeev-Popov fixed form of (24.26). One has just to be careful since the Faddeev-Popov determinant δ_{FP} is a product of eigenvalues of certain operator acting on a space satisfying periodic boundary conditions. Consequently, when represented through ghost fields, these fields should be anticommuting but satisfying periodic boundary conditions.

Another important point (relevant for non-Abelian gauge theories admitting instanton solutions) is that in our derivation, it was natural to impose that A_0 vanishes at spatial infinity. Now, if instead we demand that $A_0(\vec{x},\tau) \to$constant as $x \to \infty$, the constraint will change its character implying that the theory is projected onto a given charge sector. This will be important in instanton calculations at finite temperature [70].

Fermions at finite temperature

We shall now discuss how to write the partition function for a system of relativistic fermions at finite temperature using the path-integral. We shall consider free massive fermions in $d=4$ space-time dimensions but all our discussion extends trivially to $d \neq 4$.

The Lagrangian density will be, as usual, written as

$$L = \bar{\psi}(i\partial\!\!\!/ + m)\psi \qquad (24.27)$$

and the Euclidean partition function for the $T = 0$ theory takes the form

$$Z = \int \mathcal{D}\bar{\psi}\mathcal{D}\psi \exp\left(-\int d^4xL\right) \qquad (24.28)$$

Integration over fermion fields should be done à la Berezin. That is, using an appropriate basis, one expands fermion fields using anticommuting Bessel-Fourier coefficients, in such a way that one reproduces within-the path-integral framework, the results obtained using the canonical approach where fermion fields obey canonical anticommutation relations.

In order to use (24.28) to compute the partition functions for this free fermions at finite temperature, we can follow all steps already explained for bosons except by one point, related to the boundary conditions that one has to impose in order to show that Z coincides with the trace of $\exp(-\beta\hat{H})$. Let us leave for the moment this point, loosely writing

$$Z = \mathrm{tr}\exp(-\beta\hat{H}) = \int \mathcal{D}\bar{\psi}\mathcal{D}\psi \exp\left(-\int_0^\beta d\tau \int d^3xL\right) \qquad (24.29)$$

Then, once the adequate boundary conditions are specified, one can control whether the formula is correct or not just by computing termodynamical quantities using this expression and comparing the answer with that obtained by the usual procedure in quantum statistical mechanics.

Associated with the global $U(1)$ invariance of Lagrangian (24.27) there is the well known conserved Noether charge

$$Q = i\int d^3x\bar{\psi}\gamma_0\psi \qquad (24.30)$$

The factor "i" arises here because when passing from Minkowski to Euclidean space-time $\gamma_0 \to i\gamma_0$. For consistency, it is important to keep trace of this "i" in what follows.

At the statistical mechanics level, one enforces charge conservation introducing a Lagrange multiplier, known in this context as the chemical potential μ. One then writes, instead of (24.29)

$$\begin{aligned} Z &= \operatorname{tr}\exp(-\beta(\hat{H}-\mu\hat{Q})) \\ & \int \mathcal{D}\bar{\psi}\mathcal{D}\psi \exp\left(-\int_0^\beta d\tau \int d^3x \left(\bar{\psi}(i\partial\!\!\!/ + m + i\mu\gamma_0)\psi\right)\right) \end{aligned} \tag{24.31}$$

Note that the charge Q as given by (24.30) is an integral over space. In order to include a τ integration, we can write

$$Q = \frac{i}{\beta}\int_0^\beta d\tau \int d^3x\bar{\psi}\psi \tag{24.32}$$

and this justifies the way in which the Lagrange multiplier appears in the r.h.s. of (24.31). One can appreciate at this point why it is important to keep the "i" factor multiplying the chemical potential. One can in principle identify $i\mu$ with a constant A_0 component of a gauge potential. But a constant potential can always be gauge away by an appropriate gauge transformation: there could not be any observable associated with A_0 since the associated F_{0i} is identically zero. Now, because the "i" is present, the appropriate gauge rotation, which we shall call g_0, is not in fact a $U(1)$ rotation,

$$g_0 = \exp(-\tau\mu) \tag{24.33}$$

Hence, the path-integral gives a sensible formulation of quantum statistical mechanics including the case in which one introduces the chemnical potential (there was some confusion around this point, clarified in ([71]).

Let us discuss now the boundary conditions necessary to get the correct trace in (24.31). In the canonical formalism, what one should call as the <u>termal</u> Green function for a bosonic theory is defined as

$$G_{bos}(\vec{x},\vec{y};\tau,0) \equiv \frac{1}{Z}\operatorname{tr}\left(\exp(\hat{\rho})\mathcal{T}_\tau\hat{\phi}(\vec{x},\tau)\hat{\phi}(\vec{y},0)\right) \tag{24.34}$$

where the statistical density matrix is given by

$$\rho = \exp(-\beta(\hat{H} - \mu\hat{Q})) \tag{24.35}$$

and $\mathcal{T}_\tau$ is the imaginary time ordering operator which, in the path-integral approach is authomatic,

$$\mathcal{T}_\tau\left(\hat{\phi}(\tau_1)\hat{\phi}(\tau_2)\right) = \hat{\phi}(\tau_1)\hat{\phi}(\tau_2)\theta(\tau_1-\tau_2) + \hat{\phi}(\tau_2)\hat{\phi}(\tau_1)\theta(\tau_2-\tau_1) \tag{24.36}$$

while for fermions one has

$$G_{fer}(\vec{x},\vec{y};\tau,0) \equiv \frac{1}{Z}\mathrm{tr}\left(\exp(\hat{\rho})\mathcal{T}_\tau\hat{\psi}(\vec{x},\tau)\hat{\bar{\psi}}(\vec{y},0)\right) \tag{24.37}$$

and

$$\mathcal{T}_\tau\left(\hat{\psi}(\tau_1)\hat{\bar{\psi}}(\tau_2)\right) = \hat{\psi}(\tau_1)\bar{\psi}\hat{(}\tau_2)\theta(\tau_1-\tau_2) - \hat{\bar{\psi}}(\tau_2)\hat{\psi}(\tau_1)\theta(\tau_2-\tau_1) \tag{24.38}$$

We have to ensure that the answer obtained within the usual statistical calculation for Fermi-Dirac statistic is obtained by a path-integral formula of the form

$$\begin{aligned} G_{fer}(\vec{x},\vec{y};\tau,0) \quad &= \quad \frac{1}{Z}\int \mathcal{D}\bar{\psi}\mathcal{D}\psi\psi(\vec{x},\tau)\bar{\psi}(\vec{y},0)\,\times \\ & \qquad \exp\left(-\int_0^\beta d\tau\int d^3x\left(\bar{\psi}(i\partial\!\!\!/ + m + i\mu\gamma_0)\psi\right)\right) \end{aligned} \tag{24.39}$$

Analogously, one has for bosons

$$G_{bos}(\vec{x},\vec{y};\tau,0) = \frac{1}{Z}\int \mathcal{D}\phi\phi(\vec{x},\tau)\phi(\vec{y},0)\exp\left(-\int_0^\beta d\tau\int d^3xL_{bos}\right) \tag{24.40}$$

Let us start with the bosonic case. Going back to (24.34), we can easily prove the following property

$$G_{bos}(\vec{x},\vec{y};\tau,0) = G_{bos}(\vec{x},\vec{y};\tau,\beta) \tag{24.41}$$

If this is true, in order to hold also when Green functions are calculated using path-integrals according to (24.40) one needs

$$\phi(\vec{y},0) = \phi(\vec{y},\beta) \tag{24.42}$$

that is, one needs periodic boundary conditions for boson fields.

In order to prove propery (24.41) we note that, since $\tau > 0$, the chronological order can be made explicit,

$$\begin{aligned} G_{bos}(\vec{x},\vec{y};\tau,0) &= \frac{1}{Z}\mathrm{tr}\left(\exp(\hat{\rho})\hat{\phi}(\vec{x},\tau)\hat{\phi}(\vec{y},0)\right) \\ &= \frac{1}{Z}\mathrm{tr}\left(\hat{\phi}(\vec{y},0)\exp(\hat{\rho})\hat{\phi}(\vec{x},\tau)\right) \\ &= \frac{1}{Z}\mathrm{tr}\left(\exp(\hat{\rho})\exp(-\hat{\rho})\hat{\phi}(\vec{y},0)\exp(\hat{\rho})\hat{\phi}(\vec{x},\tau)\right) \end{aligned} \tag{24.43}$$

In the second line of (24.43) we have used the cyclic property of the trace. Now we can write

$$\exp(-\hat{\rho})\hat{\phi}(\vec{y},0)\exp(\hat{\rho}) = \phi(\vec{y},\beta) \tag{24.44}$$

since $\exp(\hat{\rho})$ as defined in (24.35) is just the evolution operator for "time" β (when there is a conserved charge). One then has

$$G_{bos}(\vec{x},\vec{y};\tau,0) = \frac{1}{Z}\mathrm{tr}\left(\exp(\hat{\rho})\hat{\phi}(\vec{y},\beta)\hat{\phi}(\vec{x},\tau)\right) = G_{bos}(\vec{x},\vec{y};\tau,\beta) \tag{24.45}$$

which proves (24.41).

The proof in the fermionic case is the same except that one has to take into account the relative minus sign in the chronological product (24.38) for fermions. Then, instead of identity (24.41), one can easily prove, following the same steps as in the bosonic case that

$$G_{fer}(\vec{x},\vec{y};\tau,0) = -G_{fer}(\vec{x},\vec{y};\tau,\beta) \tag{24.46}$$

this in turn implying that, at the path-integral level, one should impose

$$\psi(\vec{y},0) = -\psi(\vec{y},\beta) \tag{24.47}$$

Problem

Problem 1 : Derive eq. (24.7).

Problem 2 : Prove relation (24.46).

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